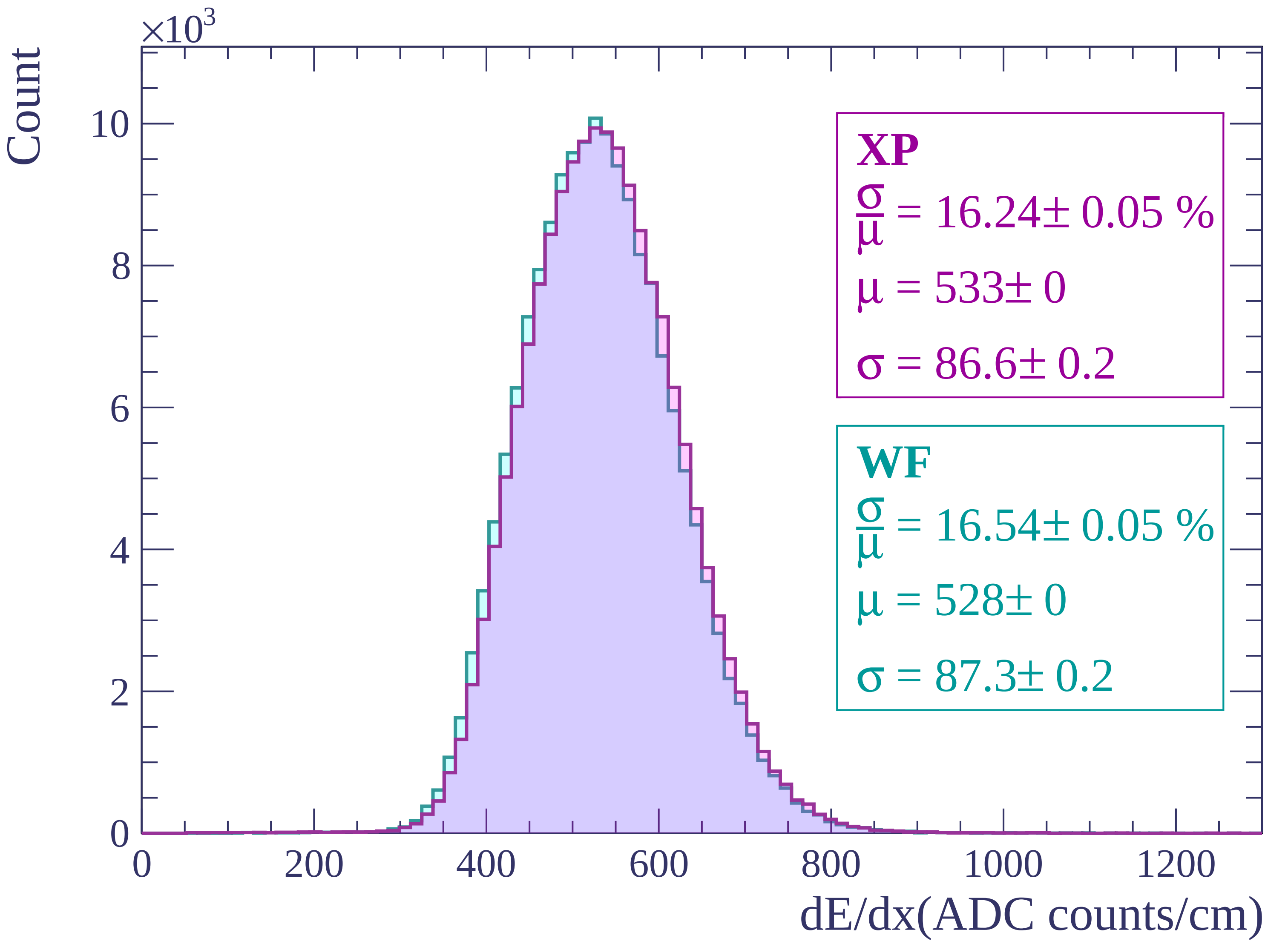
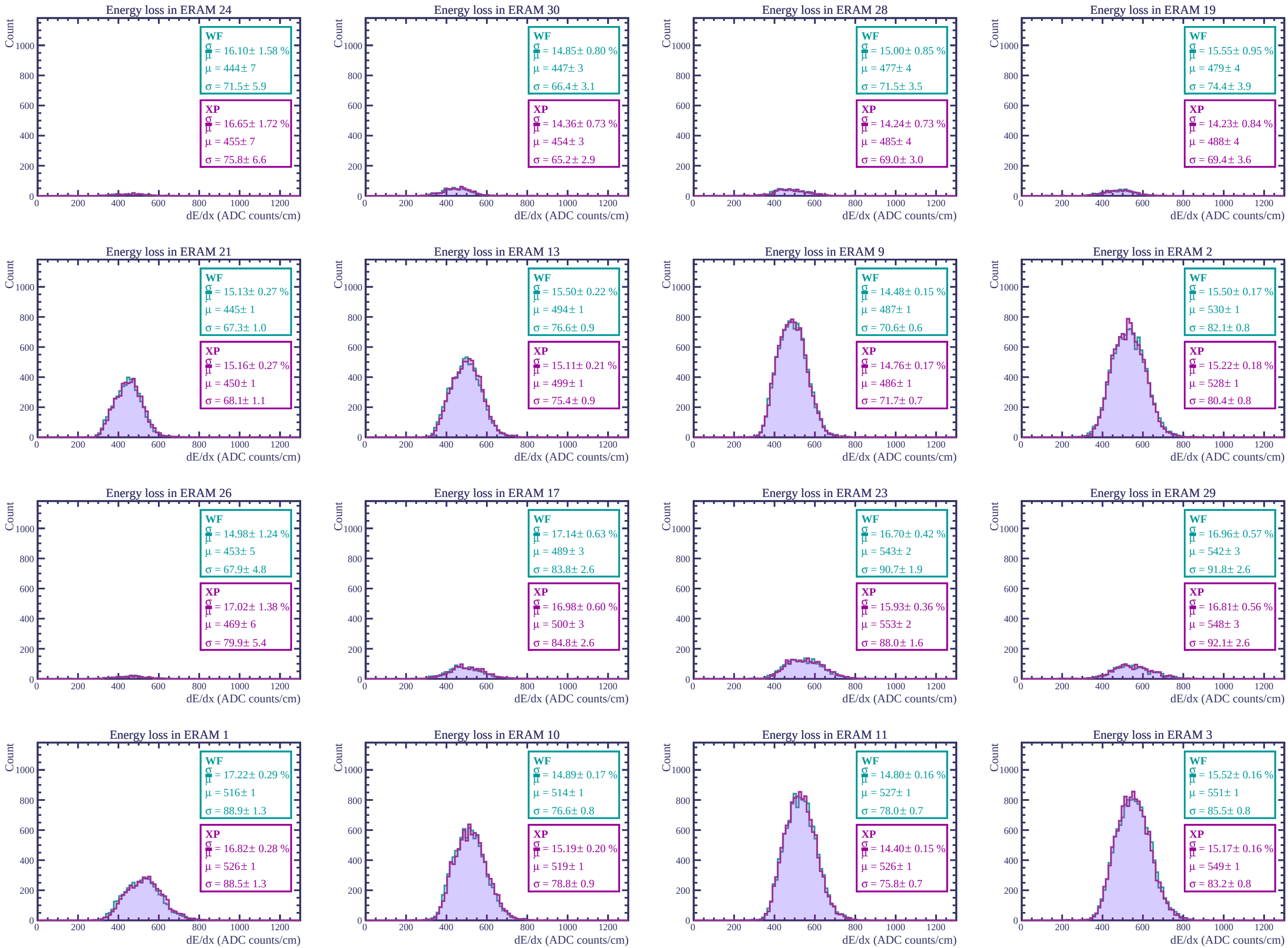
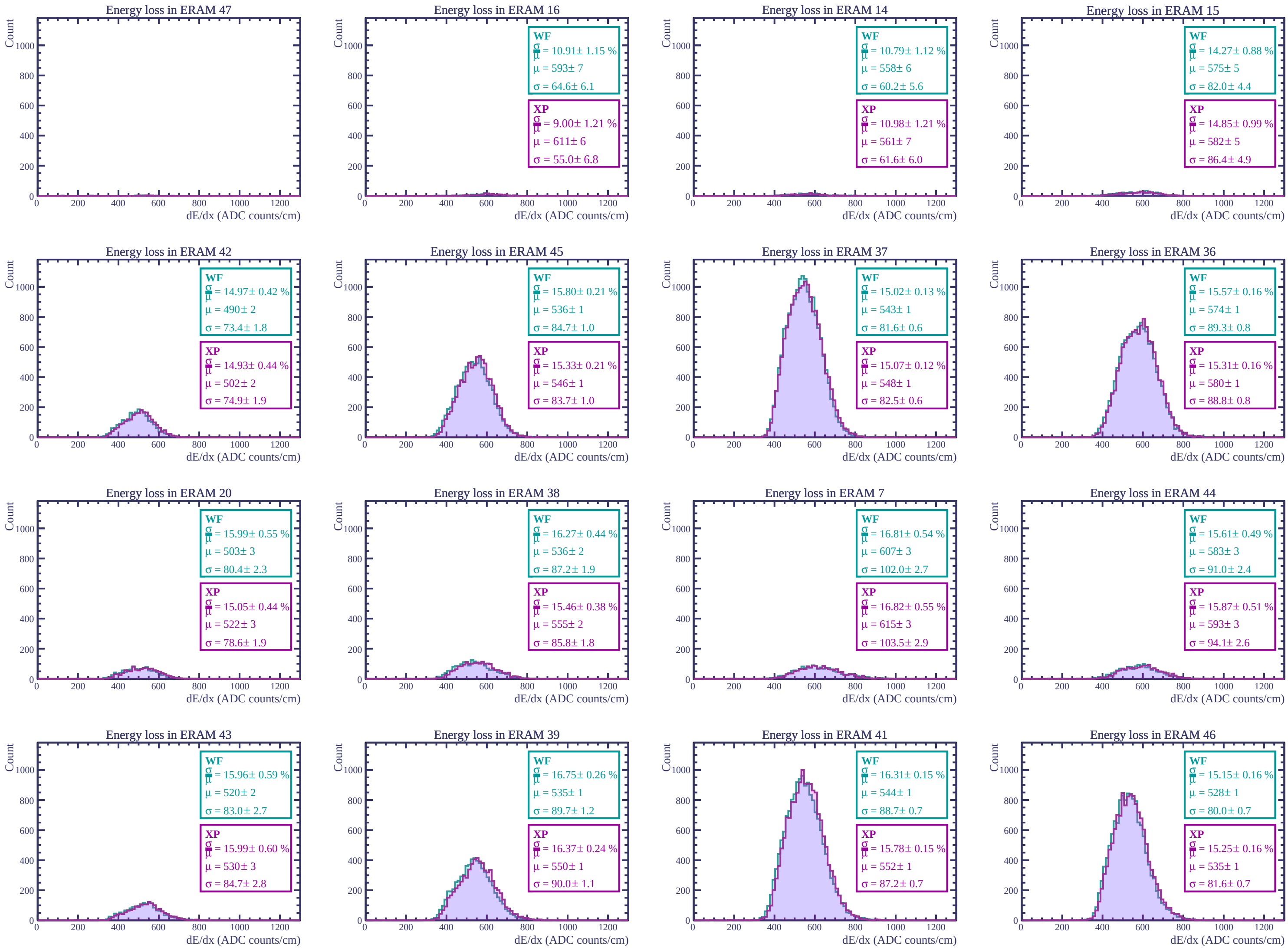
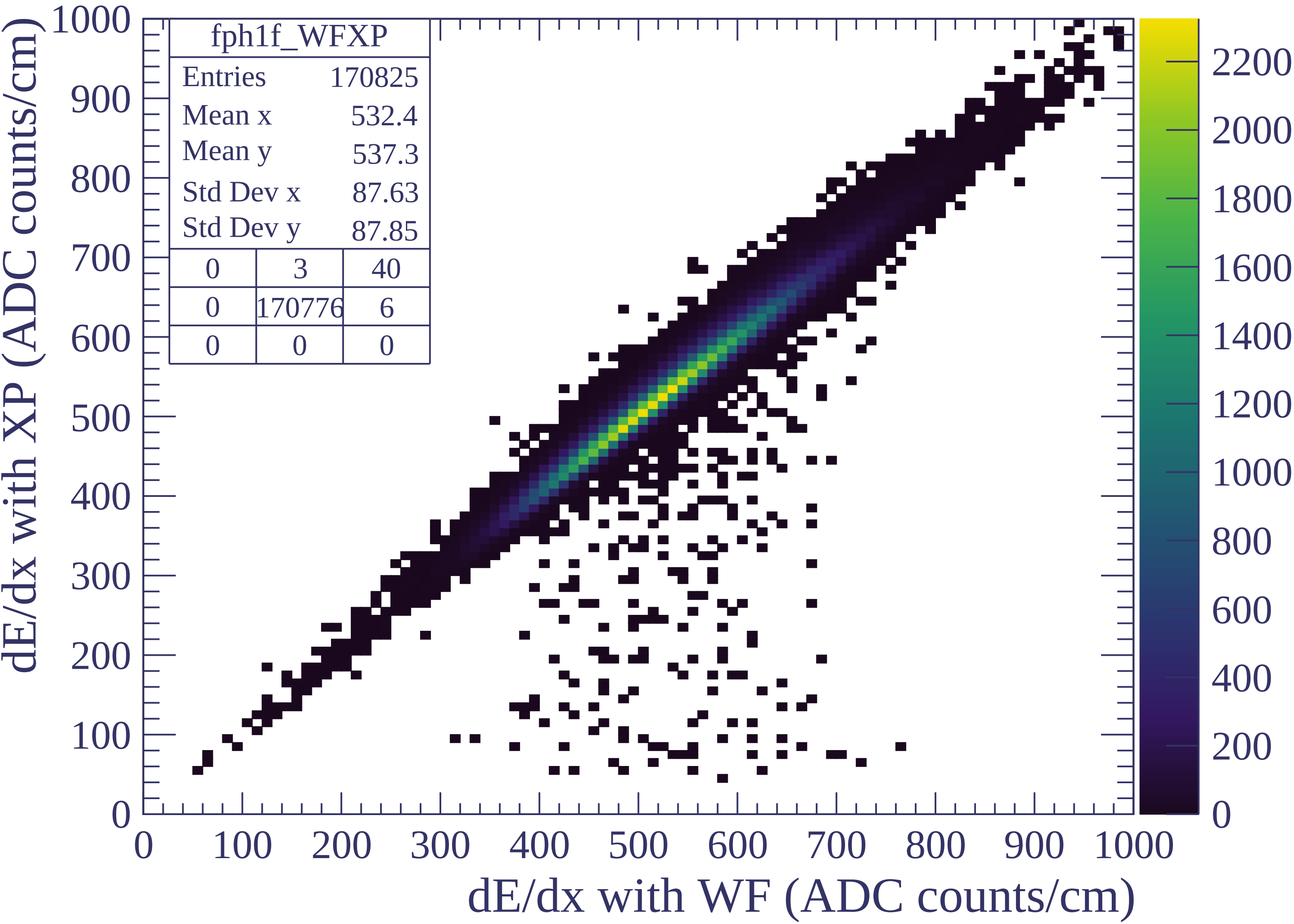


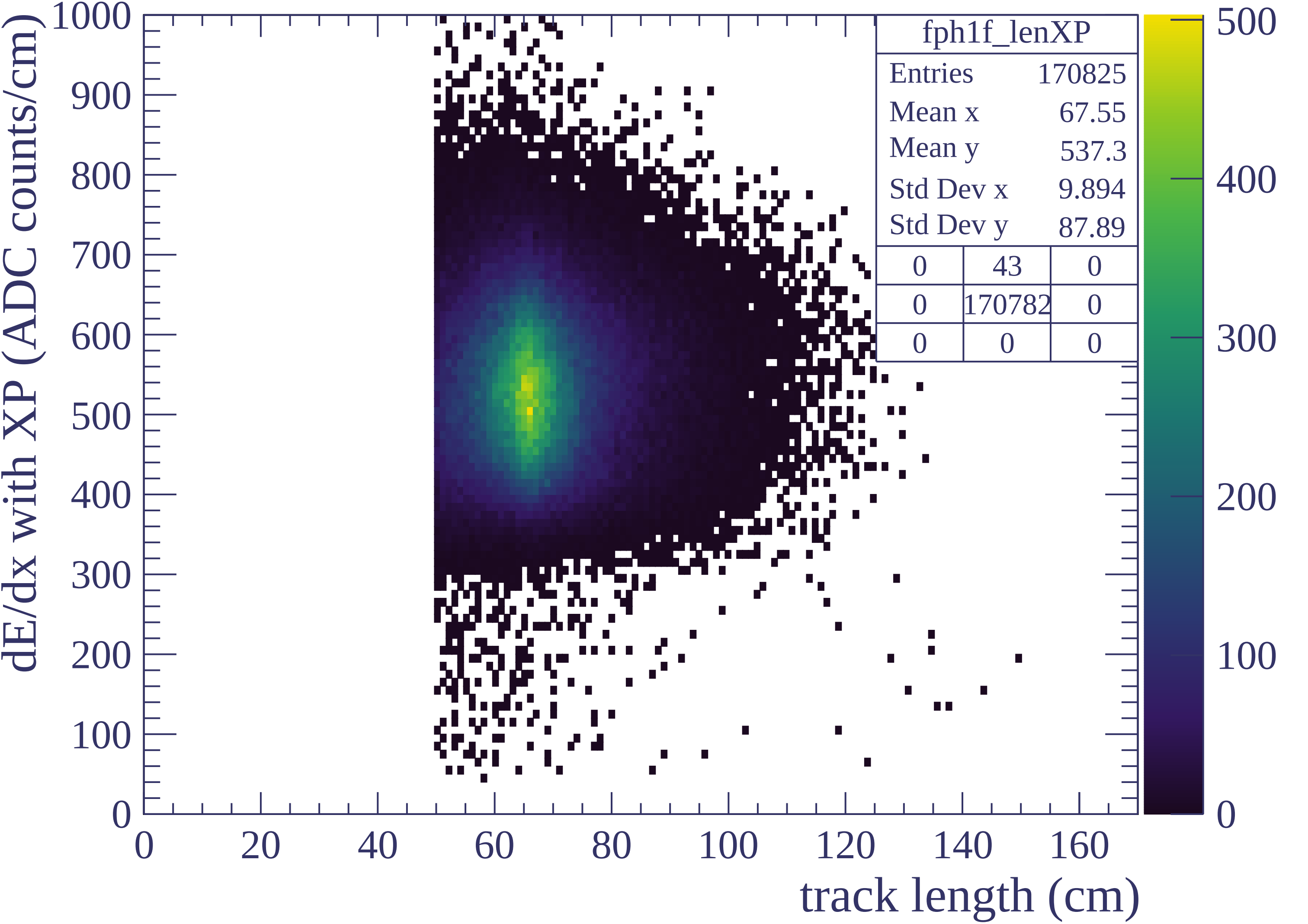
Count
$\times 10^3$
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 16.24± 0.05 %
μ = 533± 0
σ = 86.6± 0.2
WF
$\frac{\sigma}{\mu}$ = 16.54± 0.05 %
μ = 528± 0
σ = 87.3± 0.2


Energy loss in ERAM 24
WF σ/μ = 16.10± 1.58 % μ = 444± 7 σ = 71.5± 5.9
XP σ/μ = 16.65± 1.72 % μ = 455± 7 σ = 75.8± 6.6
Energy loss in ERAM 30
WF σ/μ = 14.85± 0.80 % μ = 447± 3 σ = 66.4± 3.1
XP σ/μ = 14.36± 0.73 % μ = 454± 3 σ = 65.2± 2.9
Energy loss in ERAM 28
WF σ/μ = 15.00± 0.85 % μ = 477± 4 σ = 71.5± 3.5
XP σ/μ = 14.24± 0.73 % μ = 485± 4 σ = 69.0± 3.0
Energy loss in ERAM 19
WF σ/μ = 15.55± 0.95 % μ = 479± 4 σ = 74.4± 3.9
XP σ/μ = 14.23± 0.84 % μ = 488± 4 σ = 69.4± 3.6
Energy loss in ERAM 21
WF σ/μ = 15.13± 0.27 % μ = 445± 1 σ = 67.3± 1.0
XP σ/μ = 15.16± 0.27 % μ = 450± 1 σ = 68.1± 1.1
Energy loss in ERAM 13
WF σ/μ = 15.50± 0.22 % μ = 494± 1 σ = 76.6± 0.9
XP σ/μ = 15.11± 0.21 % μ = 499± 1 σ = 75.4± 0.9
Energy loss in ERAM 9
WF σ/μ = 14.48± 0.15 % μ = 487± 1 σ = 70.6± 0.6
XP σ/μ = 14.76± 0.17 % μ = 486± 1 σ = 71.7± 0.7
Energy loss in ERAM 2
WF σ/μ = 15.50± 0.17 % μ = 530± 1 σ = 82.1± 0.8
XP σ/μ = 15.22± 0.18 % μ = 528± 1 σ = 80.4± 0.8
Energy loss in ERAM 26
WF σ/μ = 14.98± 1.24 % μ = 453± 5 σ = 67.9± 4.8
XP σ/μ = 17.02± 1.38 % μ = 469± 6 σ = 79.9± 5.4
Energy loss in ERAM 17
WF σ/μ = 17.14± 0.63 % μ = 489± 3 σ = 83.8± 2.6
XP σ/μ = 16.98± 0.60 % μ = 500± 3 σ = 84.8± 2.6
Energy loss in ERAM 23
WF σ/μ = 16.70± 0.42 % μ = 543± 2 σ = 90.7± 1.9
XP σ/μ = 15.93± 0.36 % μ = 553± 2 σ = 88.0± 1.6
Energy loss in ERAM 29
WF σ/μ = 16.96± 0.57 % μ = 542± 3 σ = 91.8± 2.6
XP σ/μ = 16.81± 0.56 % μ = 548± 3 σ = 92.1± 2.6
Energy loss in ERAM 1
WF σ/μ = 17.22± 0.29 % μ = 516± 1 σ = 88.9± 1.3
XP σ/μ = 16.82± 0.28 % μ = 526± 1 σ = 88.5± 1.3
Energy loss in ERAM 10
WF σ/μ = 14.89± 0.17 % μ = 514± 1 σ = 76.6± 0.8
XP σ/μ = 15.19± 0.20 % μ = 519± 1 σ = 78.8± 0.9
Energy loss in ERAM 11
WF σ/μ = 14.80± 0.16 % μ = 527± 1 σ = 78.0± 0.7
XP σ/μ = 14.40± 0.15 % μ = 526± 1 σ = 75.8± 0.7
Energy loss in ERAM 3
WF σ/μ = 15.52± 0.16 % μ = 551± 1 σ = 85.5± 0.8
XP σ/μ = 15.17± 0.16 % μ = 549± 1 σ = 83.2± 0.8
Count
dE/dx (ADC counts/cm)

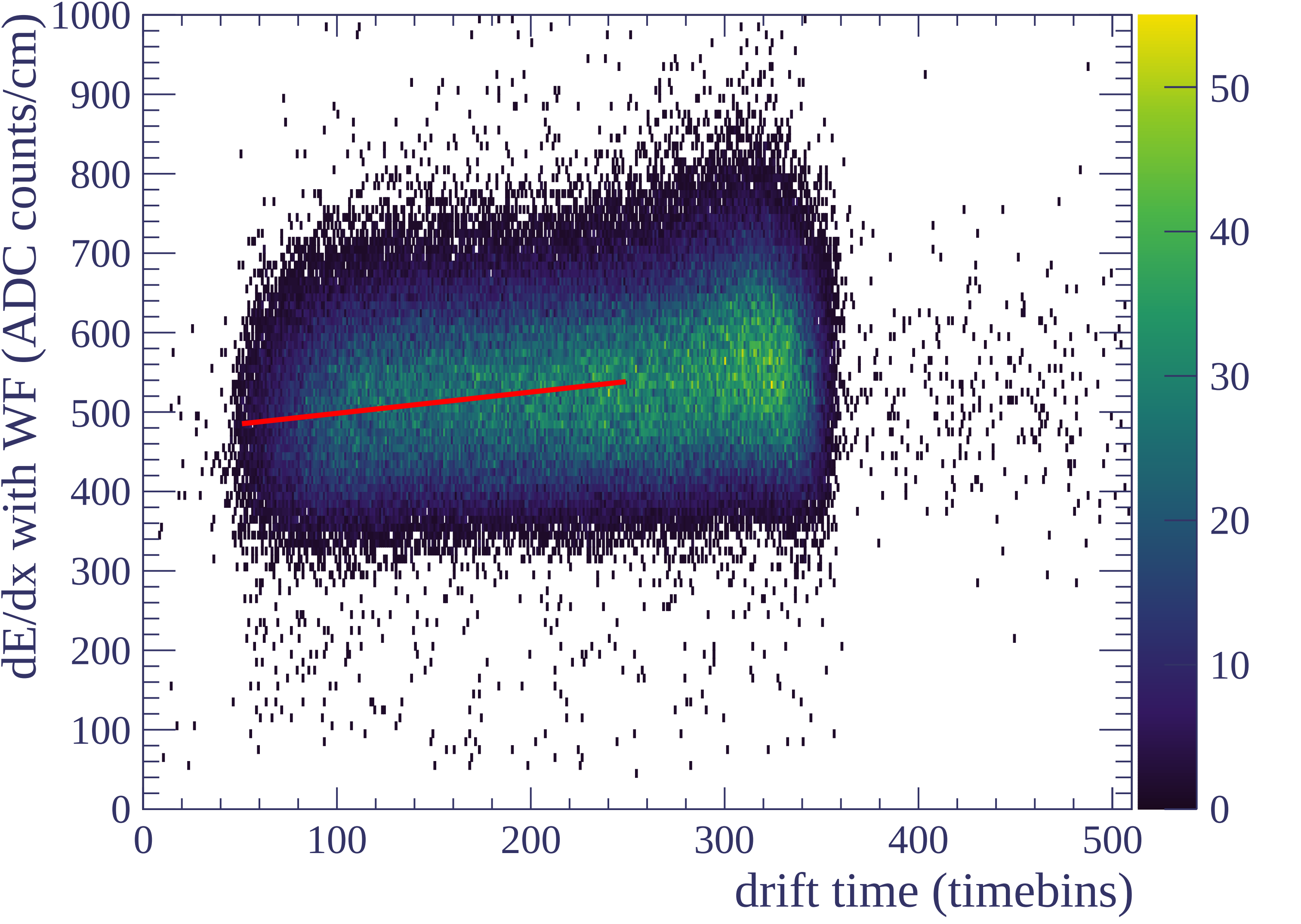


Energy loss in ERAM 47
Energy loss in ERAM 16
WF σ/μ = 10.91± 1.15 % μ = 593± 7 σ = 64.6± 6.1
XP σ/μ = 9.00± 1.21 % μ = 611± 6 σ = 55.0± 6.8
Energy loss in ERAM 14
WF σ/μ = 10.79± 1.12 % μ = 558± 6 σ = 60.2± 5.6
XP σ/μ = 10.98± 1.21 % μ = 561± 7 σ = 61.6± 6.0
Energy loss in ERAM 15
WF σ/μ = 14.27± 0.88 % μ = 575± 5 σ = 82.0± 4.4
XP σ/μ = 14.85± 0.99 % μ = 582± 5 σ = 86.4± 4.9
Energy loss in ERAM 42
WF σ/μ = 14.97± 0.42 % μ = 490± 2 σ = 73.4± 1.8
XP σ/μ = 14.93± 0.44 % μ = 502± 2 σ = 74.9± 1.9
Energy loss in ERAM 45
WF σ/μ = 15.80± 0.21 % μ = 536± 1 σ = 84.7± 1.0
XP σ/μ = 15.33± 0.21 % μ = 546± 1 σ = 83.7± 1.0
Energy loss in ERAM 37
WF σ/μ = 15.02± 0.13 % μ = 543± 1 σ = 81.6± 0.6
XP σ/μ = 15.07± 0.12 % μ = 548± 1 σ = 82.5± 0.6
Energy loss in ERAM 36
WF σ/μ = 15.57± 0.16 % μ = 574± 1 σ = 89.3± 0.8
XP σ/μ = 15.31± 0.16 % μ = 580± 1 σ = 88.8± 0.8
Energy loss in ERAM 20
WF σ/μ = 15.99± 0.55 % μ = 503± 3 σ = 80.4± 2.3
XP σ/μ = 15.05± 0.44 % μ = 522± 3 σ = 78.6± 1.9
Energy loss in ERAM 38
WF σ/μ = 16.27± 0.44 % μ = 536± 2 σ = 87.2± 1.9
XP σ/μ = 15.46± 0.38 % μ = 555± 2 σ = 85.8± 1.8
Energy loss in ERAM 7
WF σ/μ = 16.81± 0.54 % μ = 607± 3 σ = 102.0± 2.7
XP σ/μ = 16.82± 0.55 % μ = 615± 3 σ = 103.5± 2.9
Energy loss in ERAM 44
WF σ/μ = 15.61± 0.49 % μ = 583± 3 σ = 91.0± 2.4
XP σ/μ = 15.87± 0.51 % μ = 593± 3 σ = 94.1± 2.6
Energy loss in ERAM 43
WF σ/μ = 15.96± 0.59 % μ = 520± 2 σ = 83.0± 2.7
XP σ/μ = 15.99± 0.60 % μ = 530± 3 σ = 84.7± 2.8
Energy loss in ERAM 39
WF σ/μ = 16.75± 0.26 % μ = 535± 1 σ = 89.7± 1.2
XP σ/μ = 16.37± 0.24 % μ = 550± 1 σ = 90.0± 1.1
Energy loss in ERAM 41
WF σ/μ = 16.31± 0.15 % μ = 544± 1 σ = 88.7± 0.7
XP σ/μ = 15.78± 0.15 % μ = 552± 1 σ = 87.2± 0.7
Energy loss in ERAM 46
WF σ/μ = 15.15± 0.16 % μ = 528± 1 σ = 80.0± 0.7
XP σ/μ = 15.25± 0.16 % μ = 535± 1 σ = 81.6± 0.7
Count
dE/dx (ADC counts/cm)

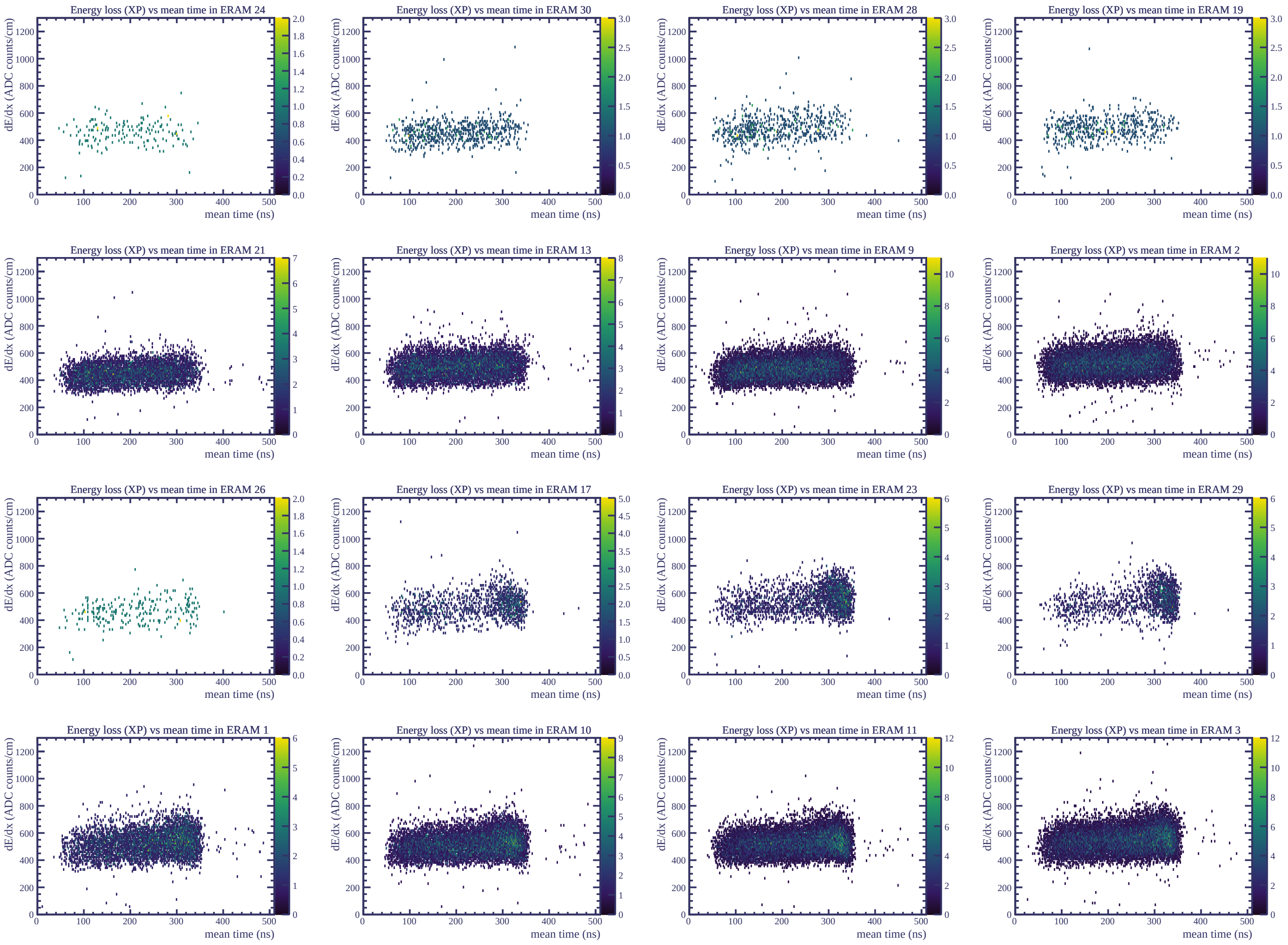

fph1f_WFXP
Entries 170825
Mean x 532.4
Mean y 537.3
Std Dev x 87.63
Std Dev y 87.85
0 3 40
0 170776 6
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
200
400
600
800
1000
1200
1400
1600
1800
2000
2200

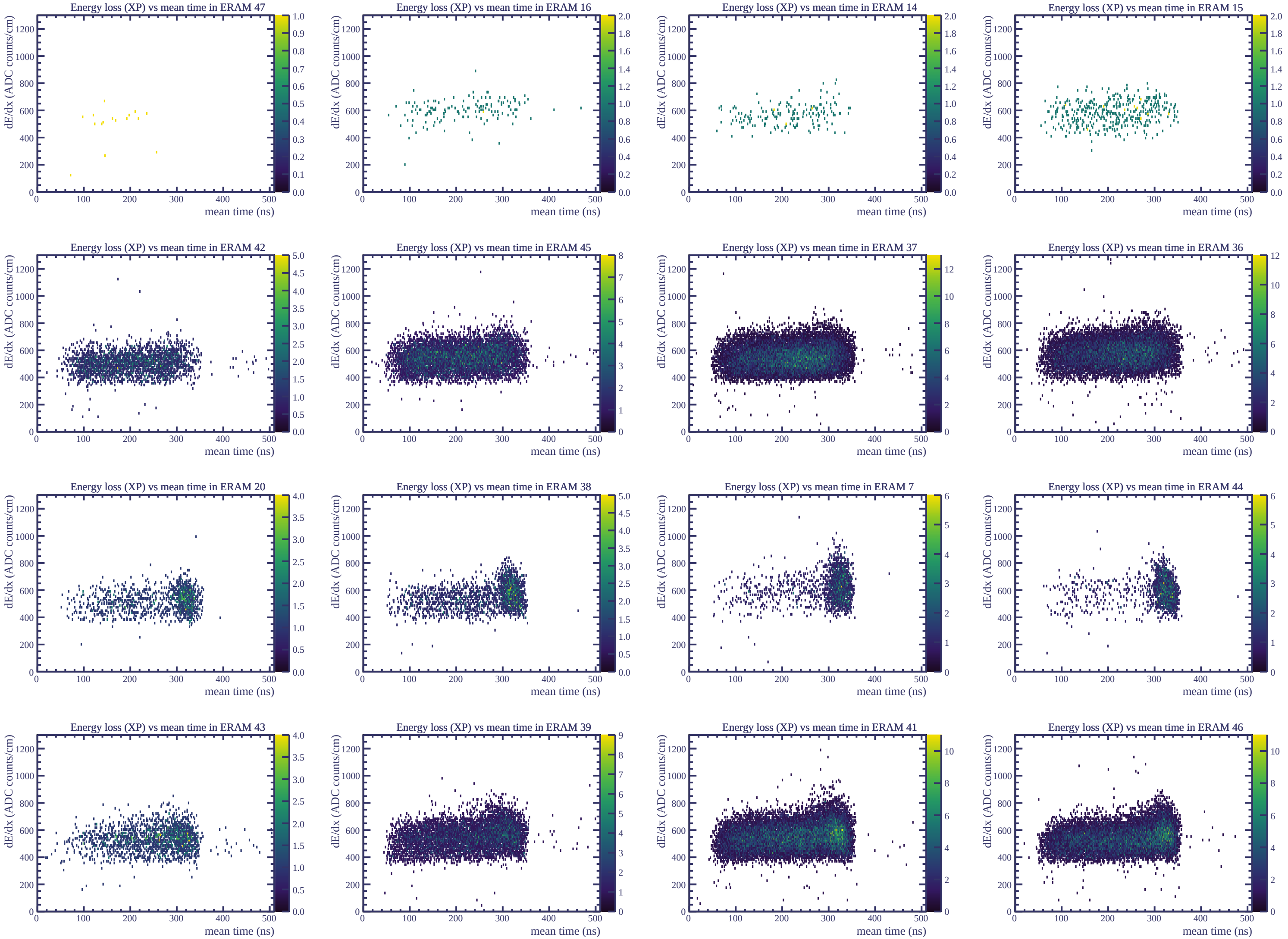


fph1f_lenXP
Entries 170825
Mean x 67.55
Mean y 537.3
Std Dev x 9.894
Std Dev y 87.89
0 43 0
0 170782 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500

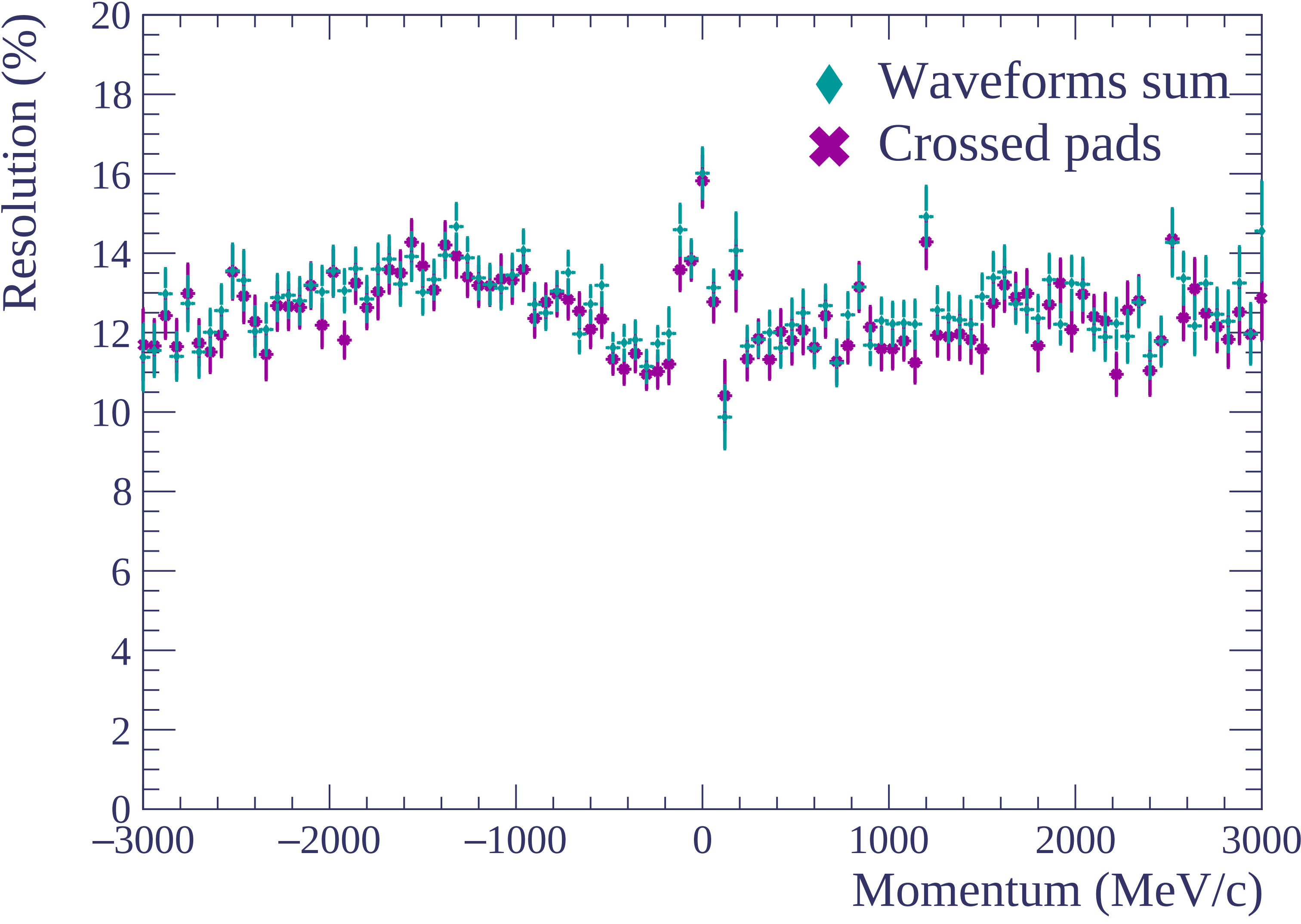


dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
10
20
30
40
50

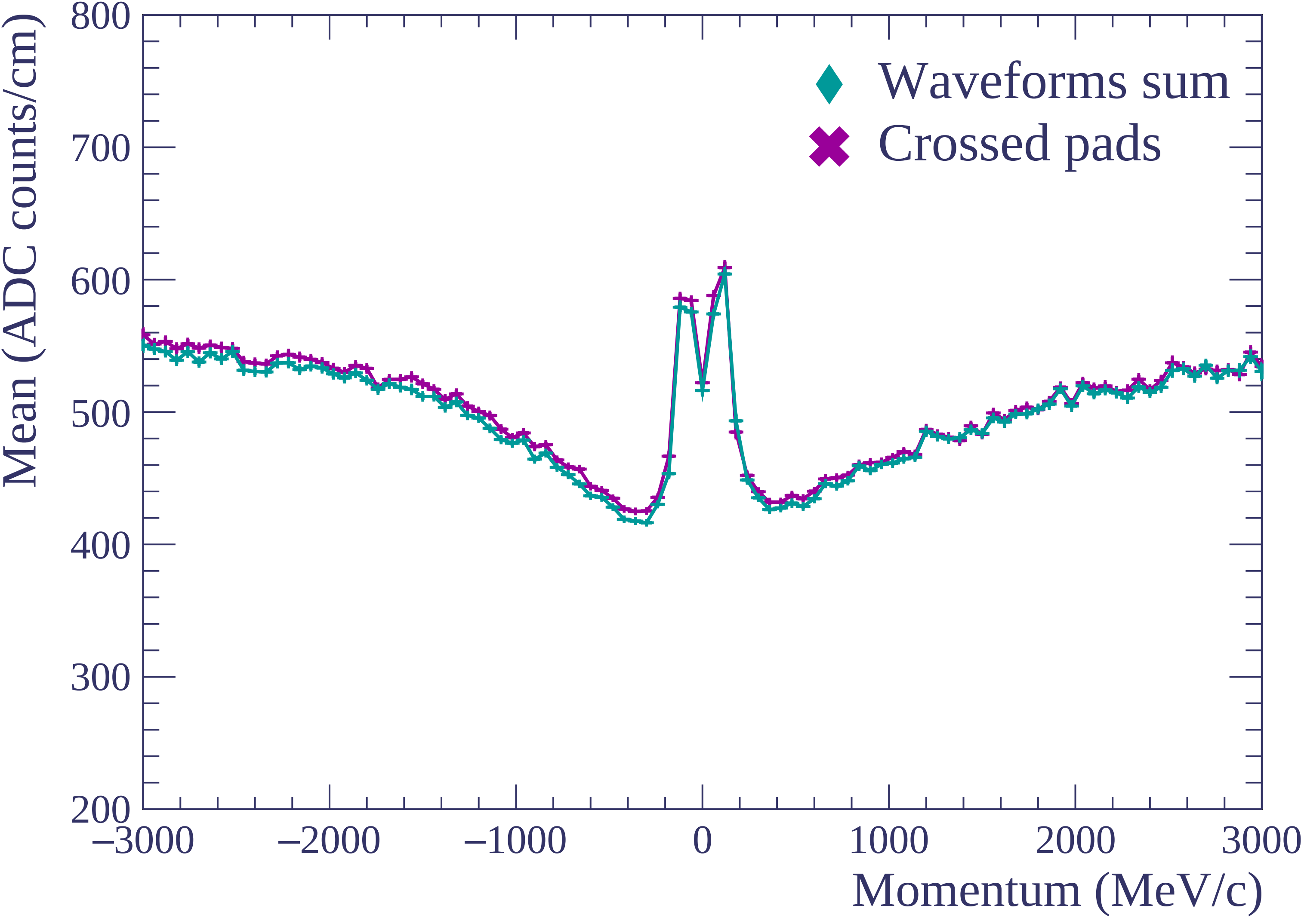

Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)



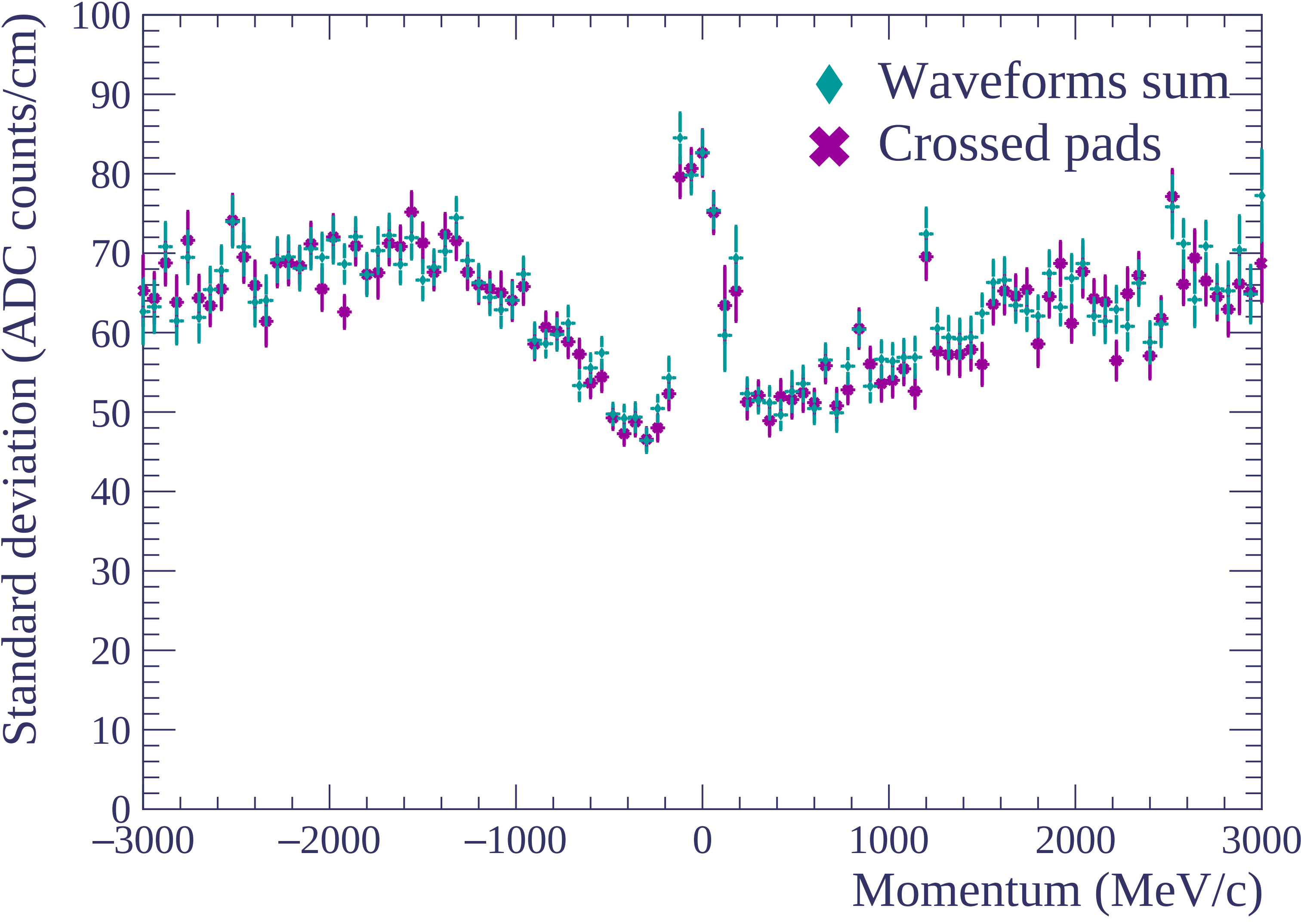
Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)



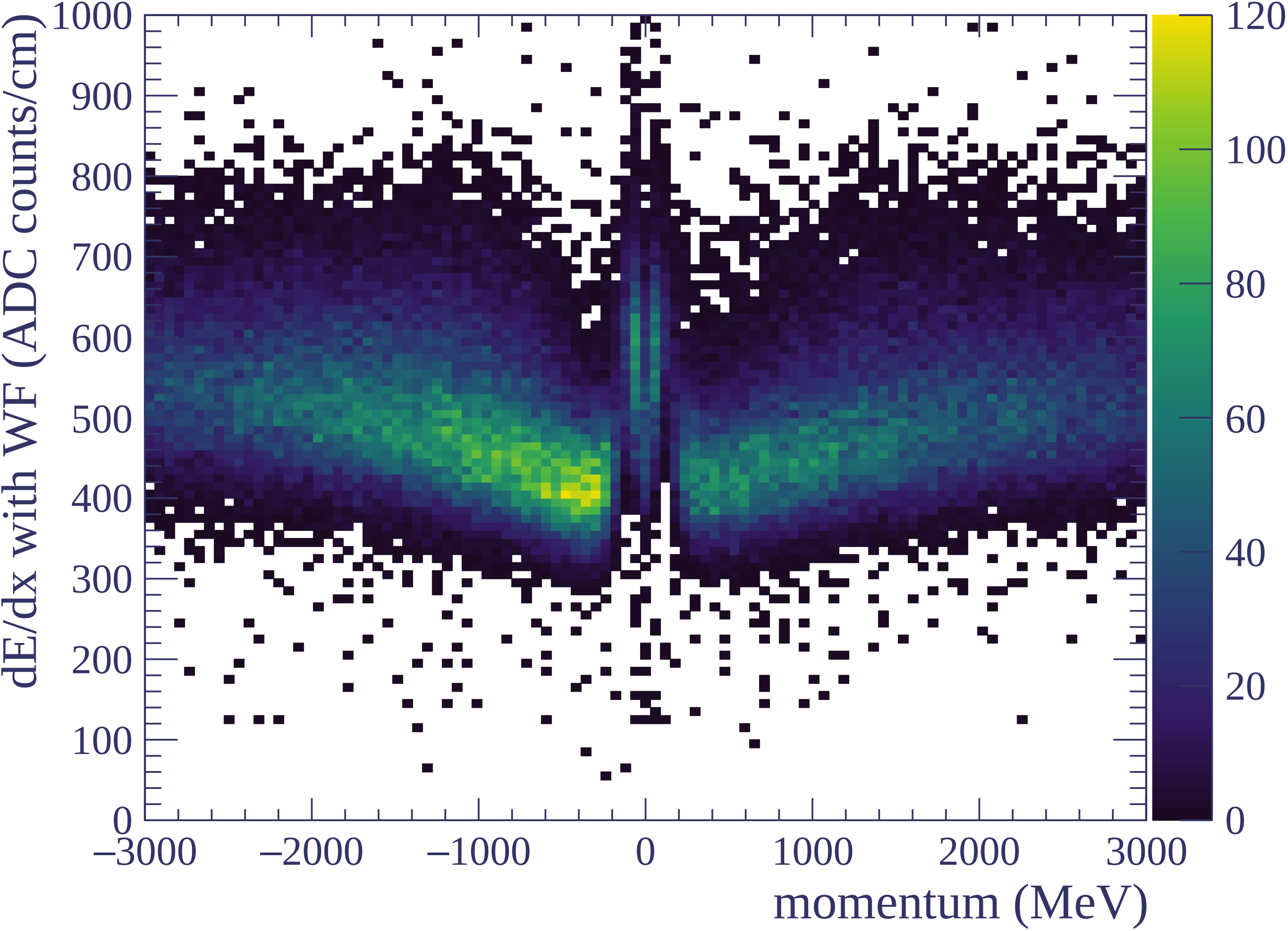
Resolution (%)
Momentum (MeV/c)
Waveforms sum
Crossed pads
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000



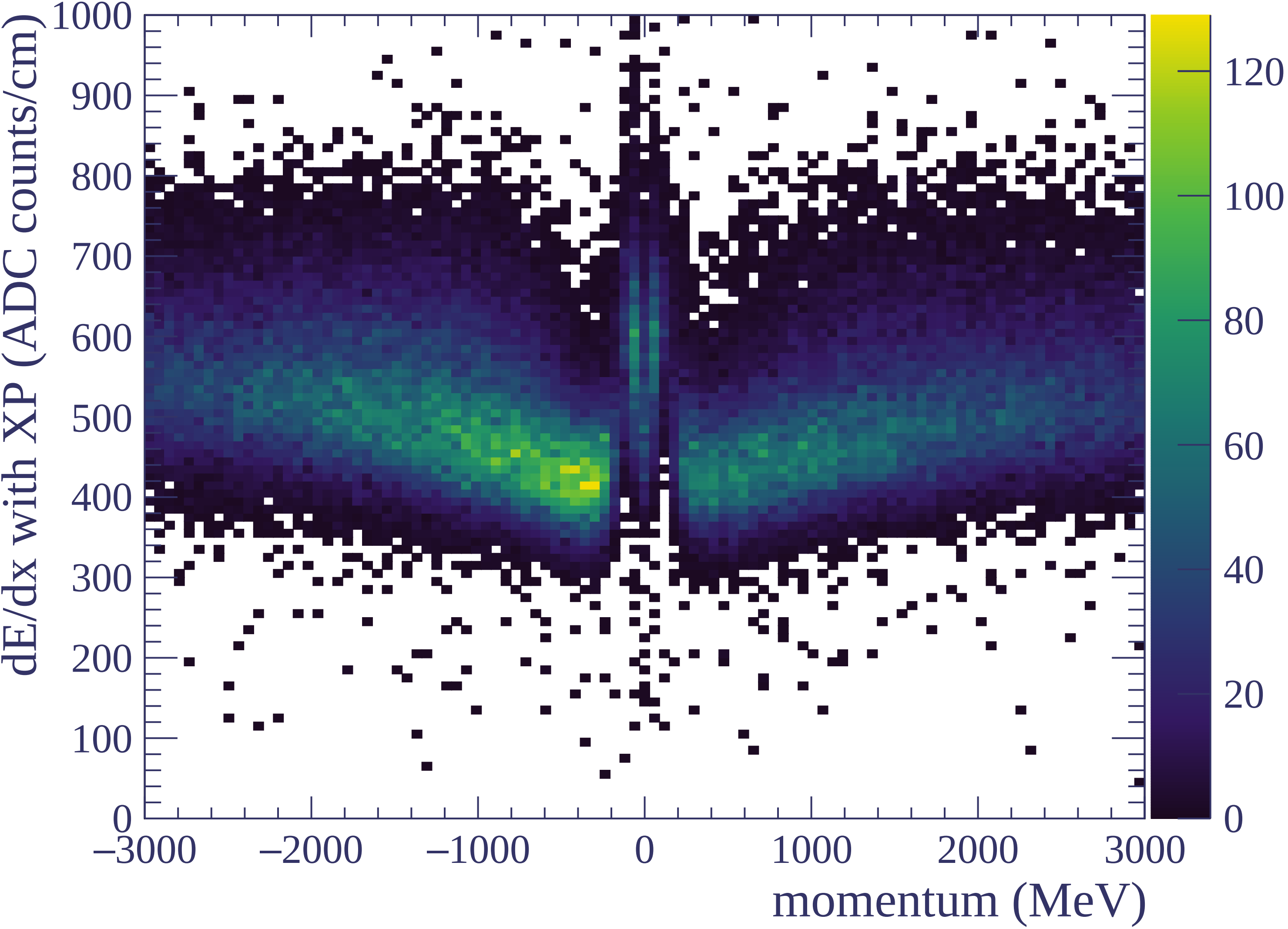
Waveforms sum
Crossed pads
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)



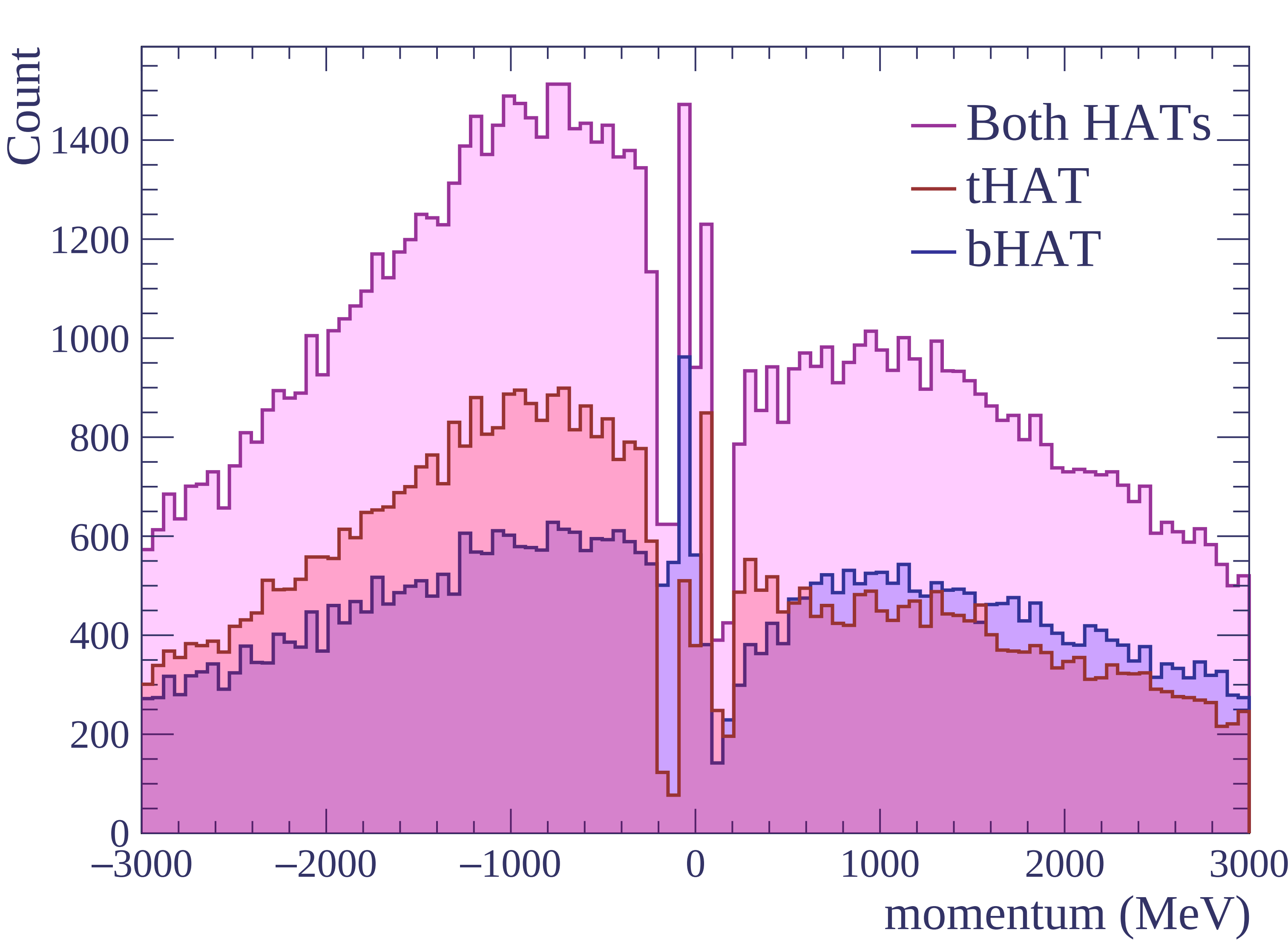
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
Waveforms sum
Crossed pads
100
90
80
70
60
50
40
30
20
10
0
−3000
−2000
−1000
0
1000
2000
3000



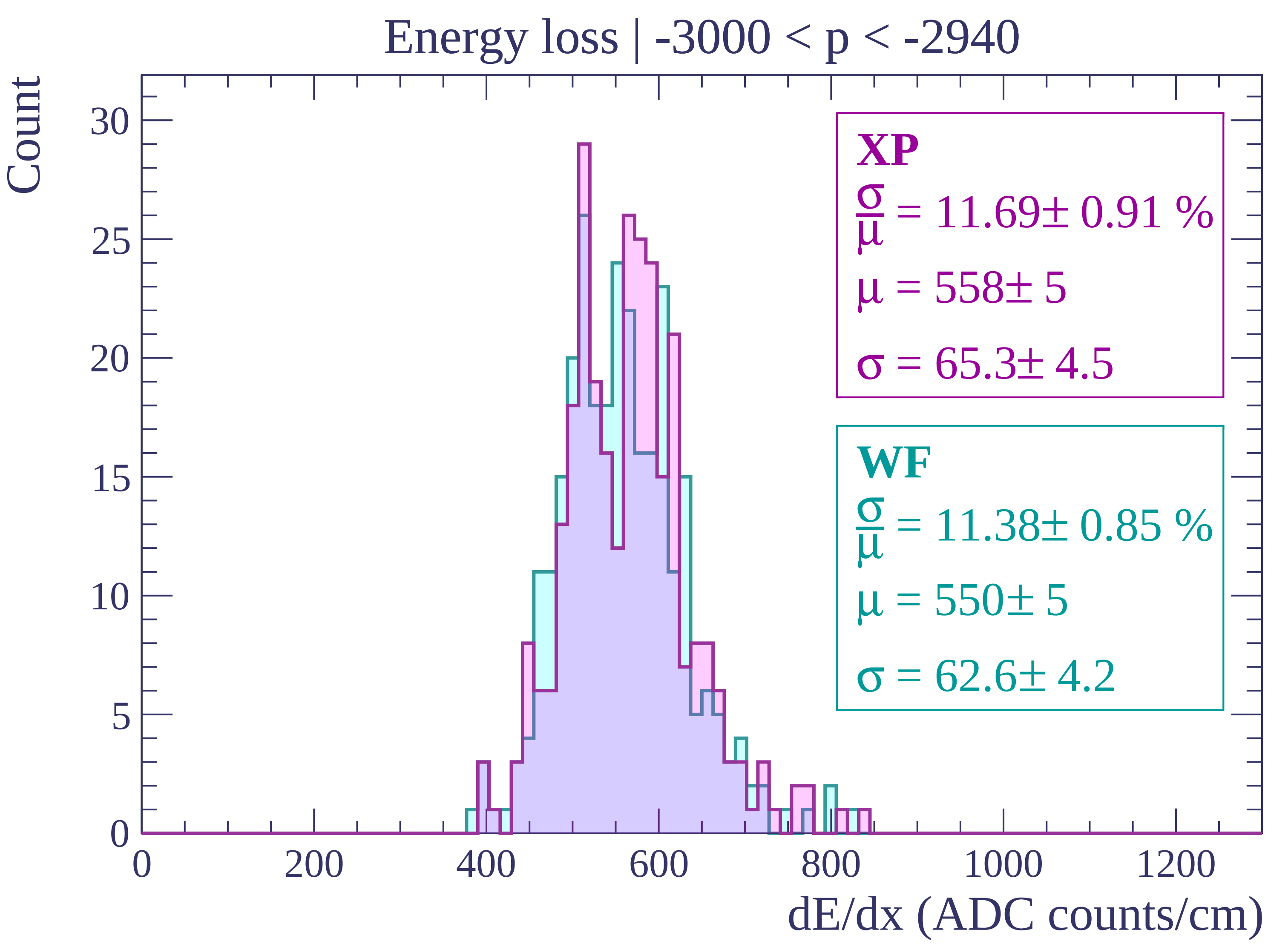
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
120
100
80
60
40
20
0



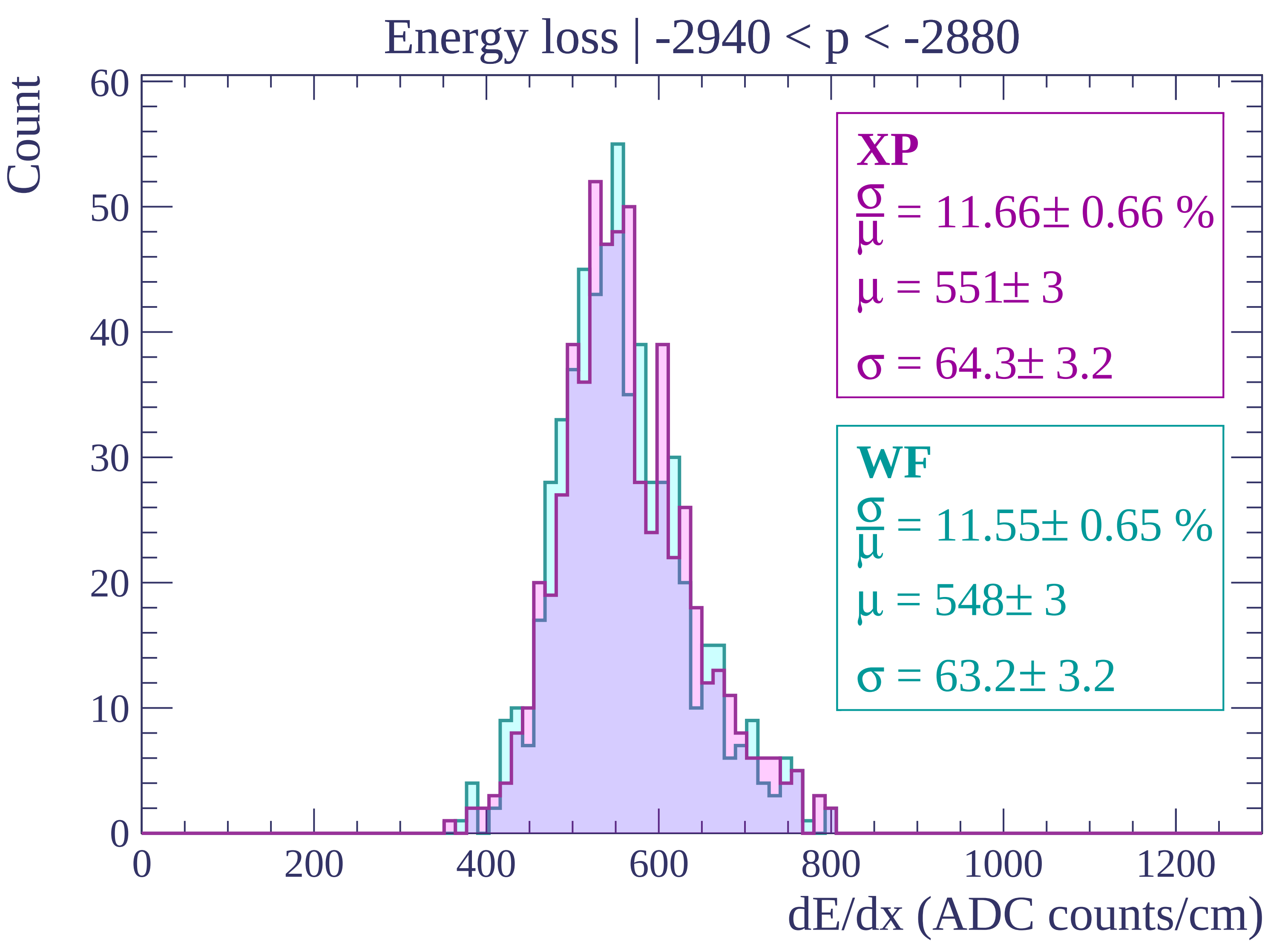
dE/dx with XP (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
0
20
40
60
80
100
120

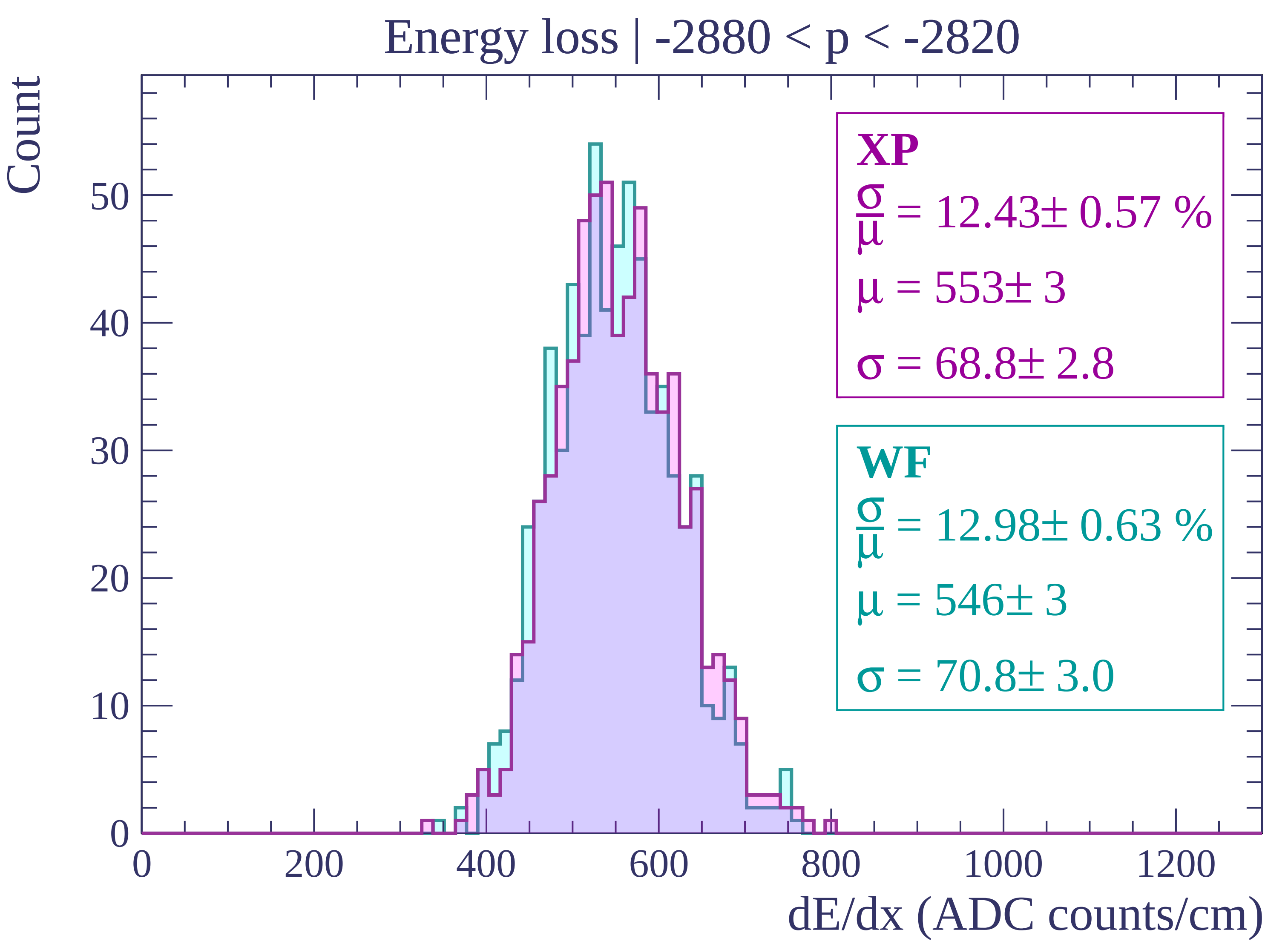


Count
0
200
400
600
800
1000
1200
1400
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

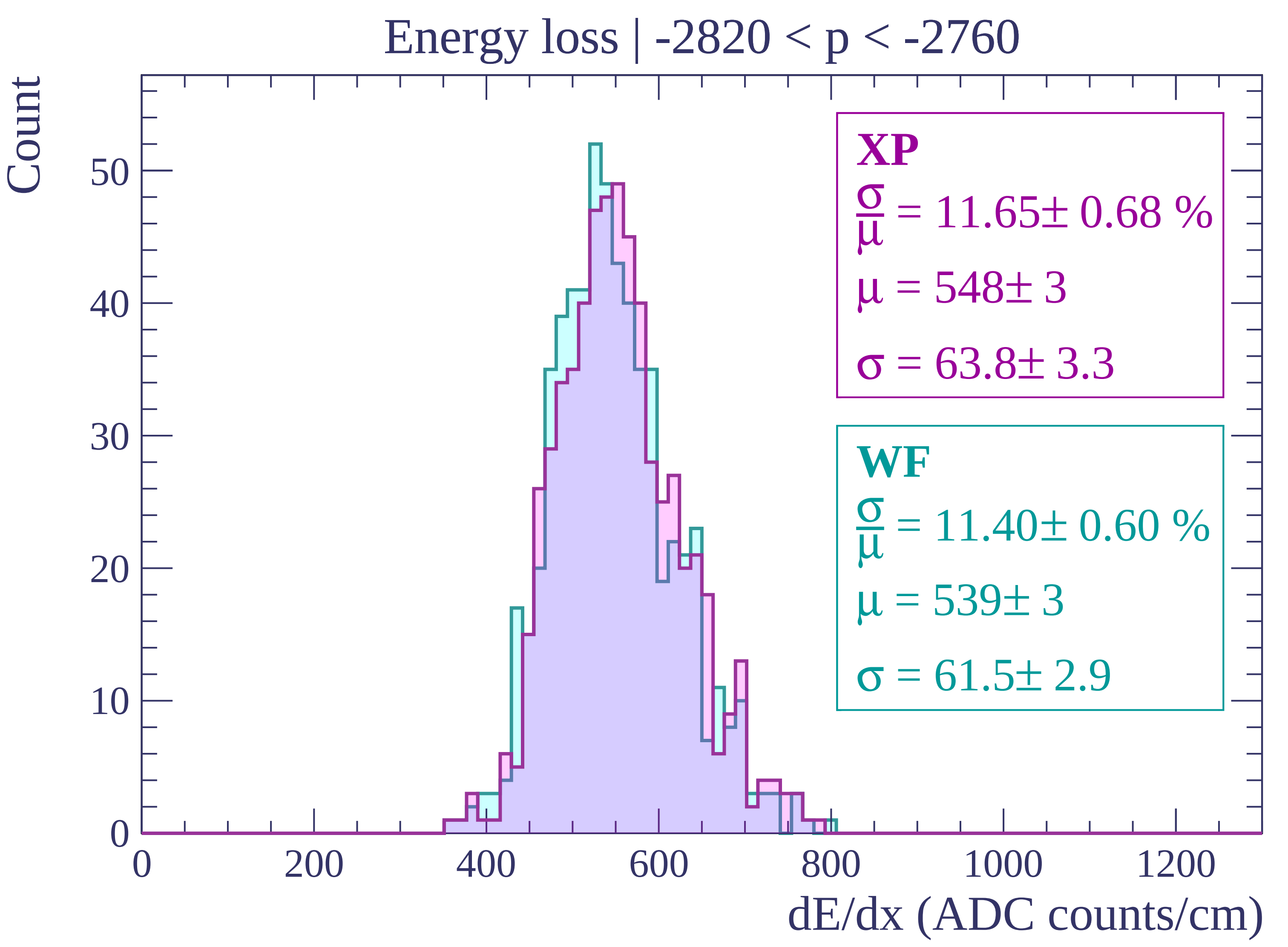


Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.69± 0.91 %
μ = 558± 5
σ = 65.3± 4.5
WF
$\frac{\sigma}{\mu}$ = 11.38± 0.85 %
μ = 550± 5
σ = 62.6± 4.2

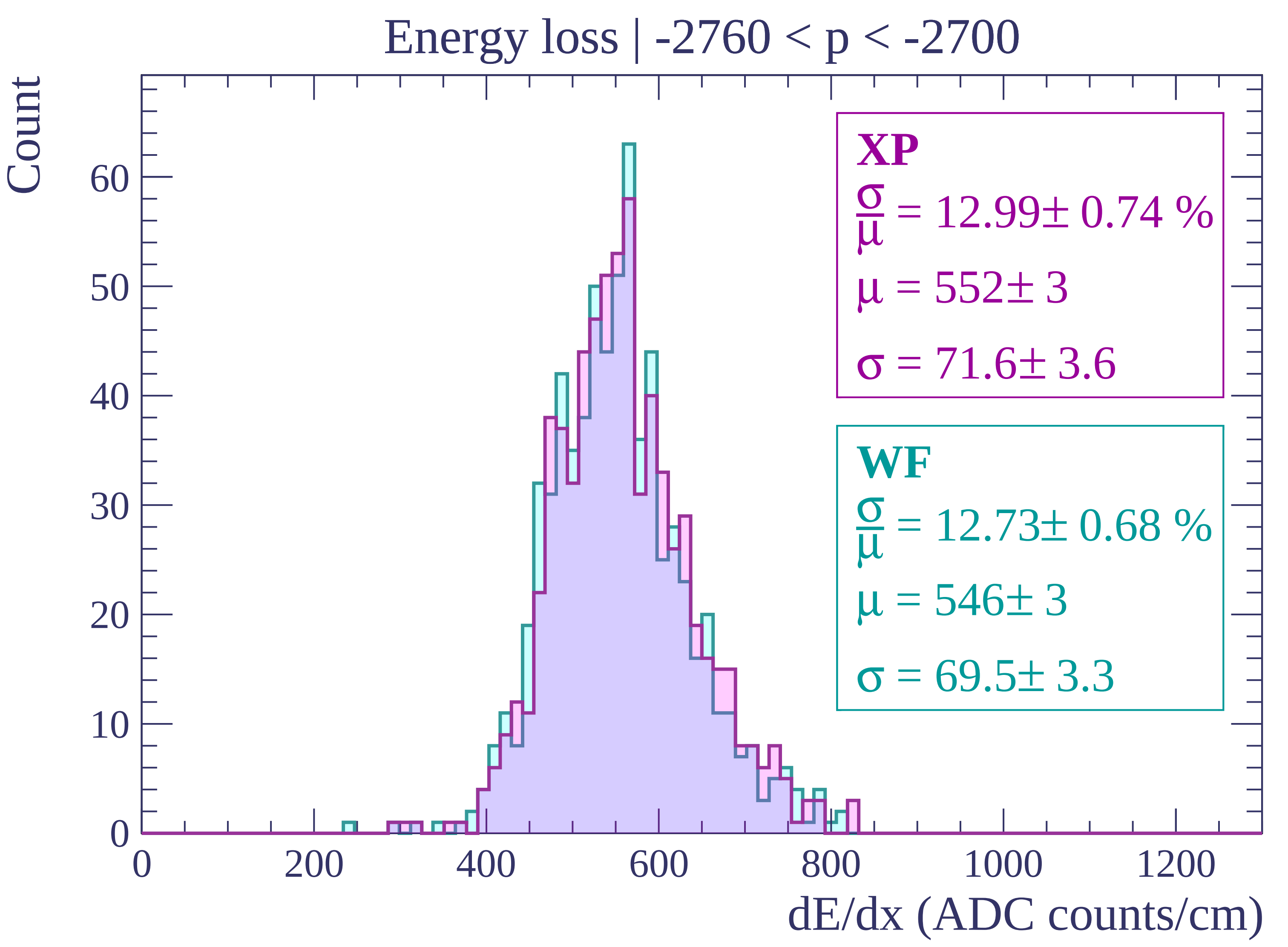

Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 11.66± 0.66 %
μ = 551± 3
σ = 64.3± 3.2
WF
σ/μ = 11.55± 0.65 %
μ = 548± 3
σ = 63.2± 3.2



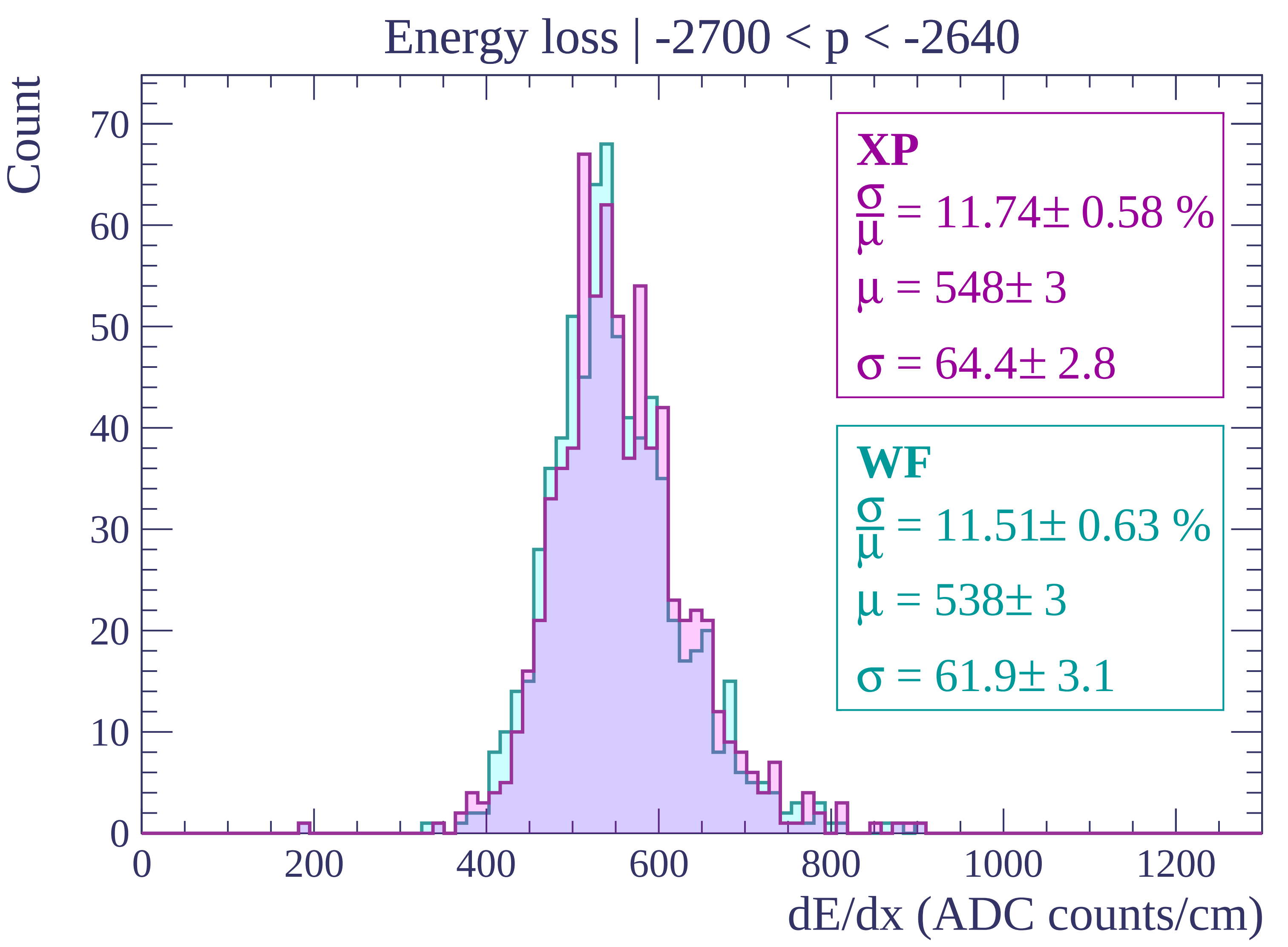
Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 12.43± 0.57 %
μ = 553± 3
σ = 68.8± 2.8
WF
σ/μ = 12.98± 0.63 %
μ = 546± 3
σ = 70.8± 3.0



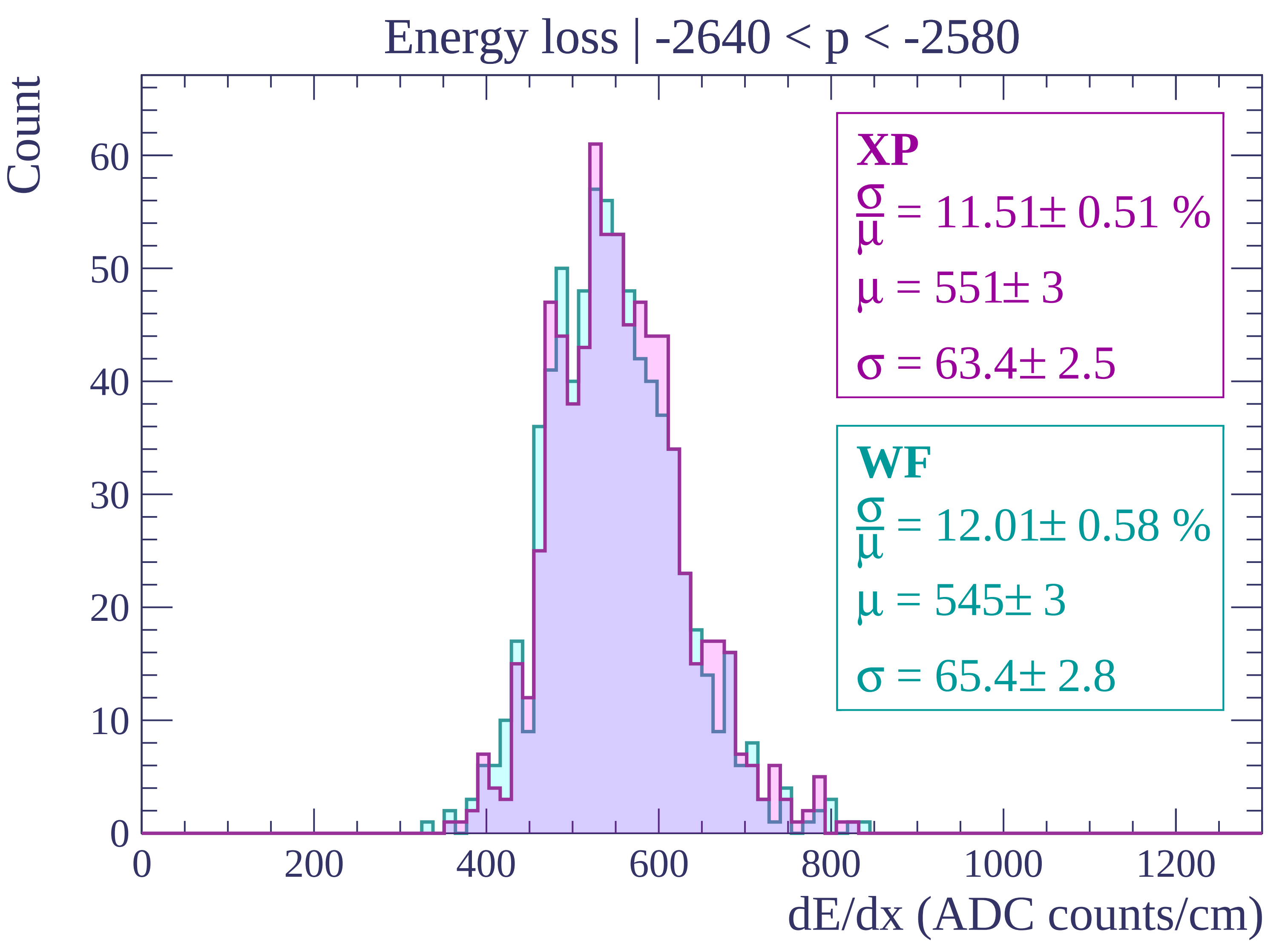
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 11.65± 0.68 %
μ = 548± 3
σ = 63.8± 3.3
WF
σ/μ = 11.40± 0.60 %
μ = 539± 3
σ = 61.5± 2.9



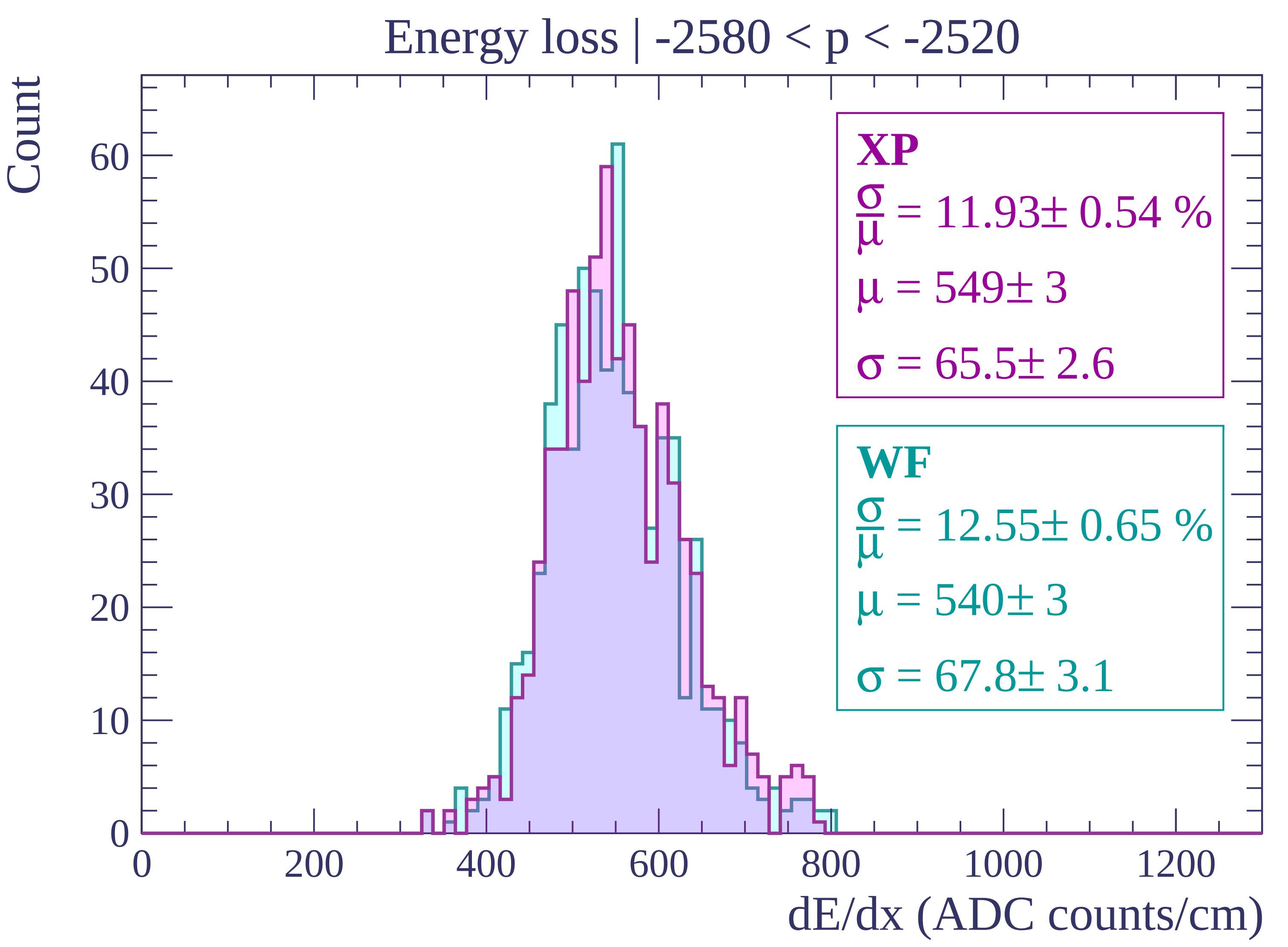
Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.99± 0.74 %
μ = 552± 3
σ = 71.6± 3.6
WF
σ/μ = 12.73± 0.68 %
μ = 546± 3
σ = 69.5± 3.3



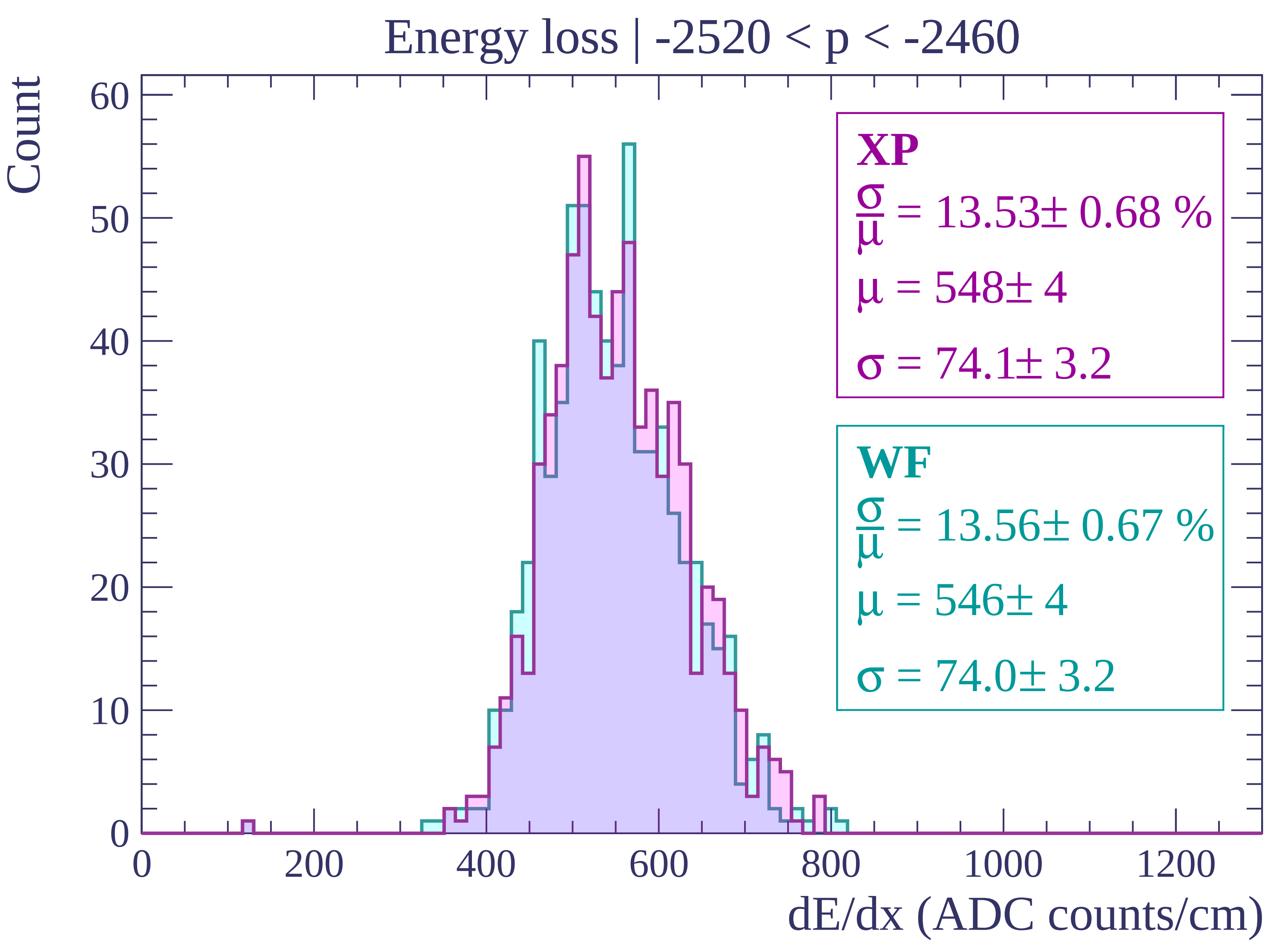
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.74± 0.58 %
μ = 548± 3
σ = 64.4± 2.8
WF
$\frac{\sigma}{\mu}$ = 11.51± 0.63 %
μ = 538± 3
σ = 61.9± 3.1



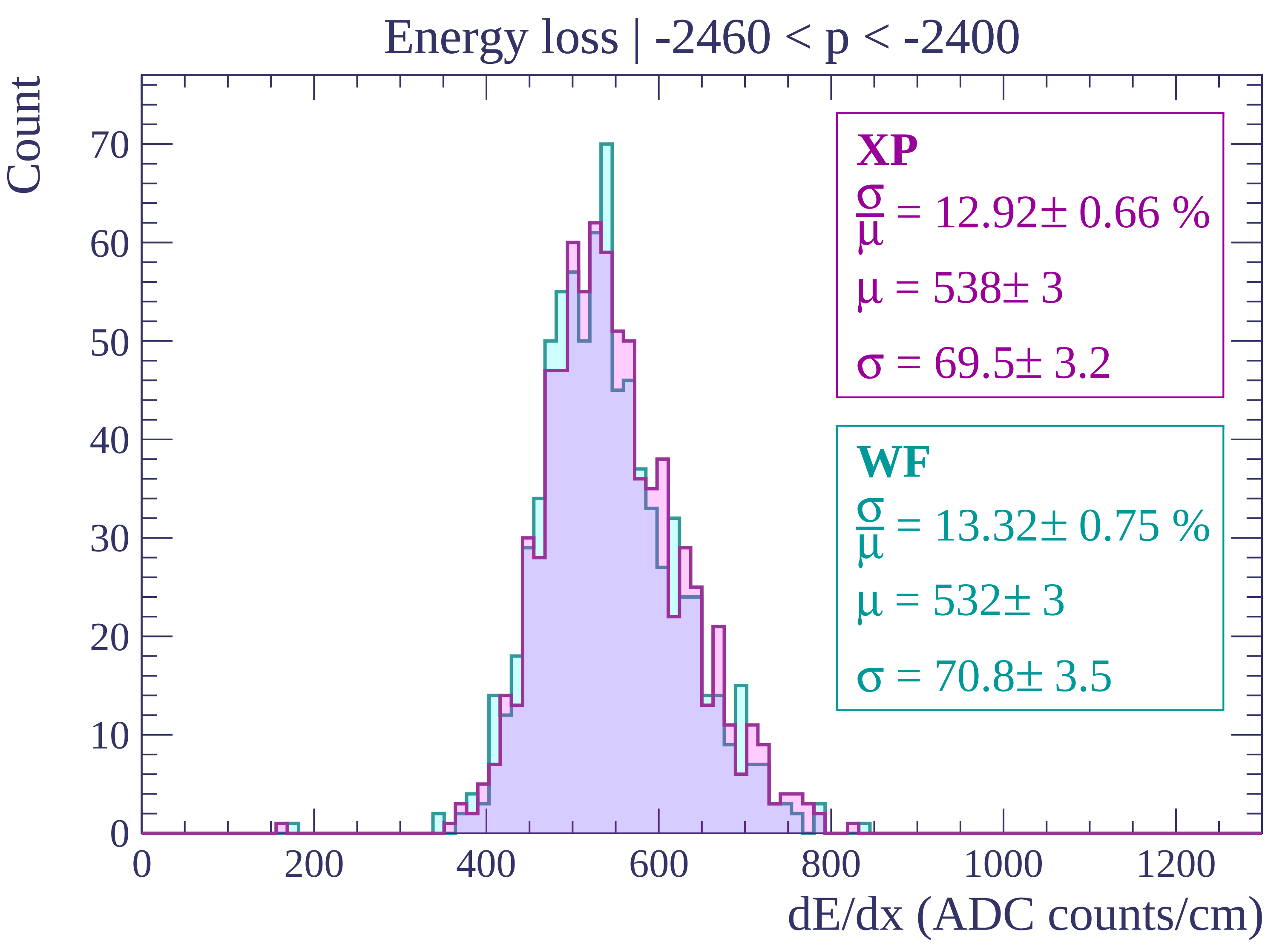
Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 11.51± 0.51 %
μ = 551± 3
σ = 63.4± 2.5
WF
σ/μ = 12.01± 0.58 %
μ = 545± 3
σ = 65.4± 2.8



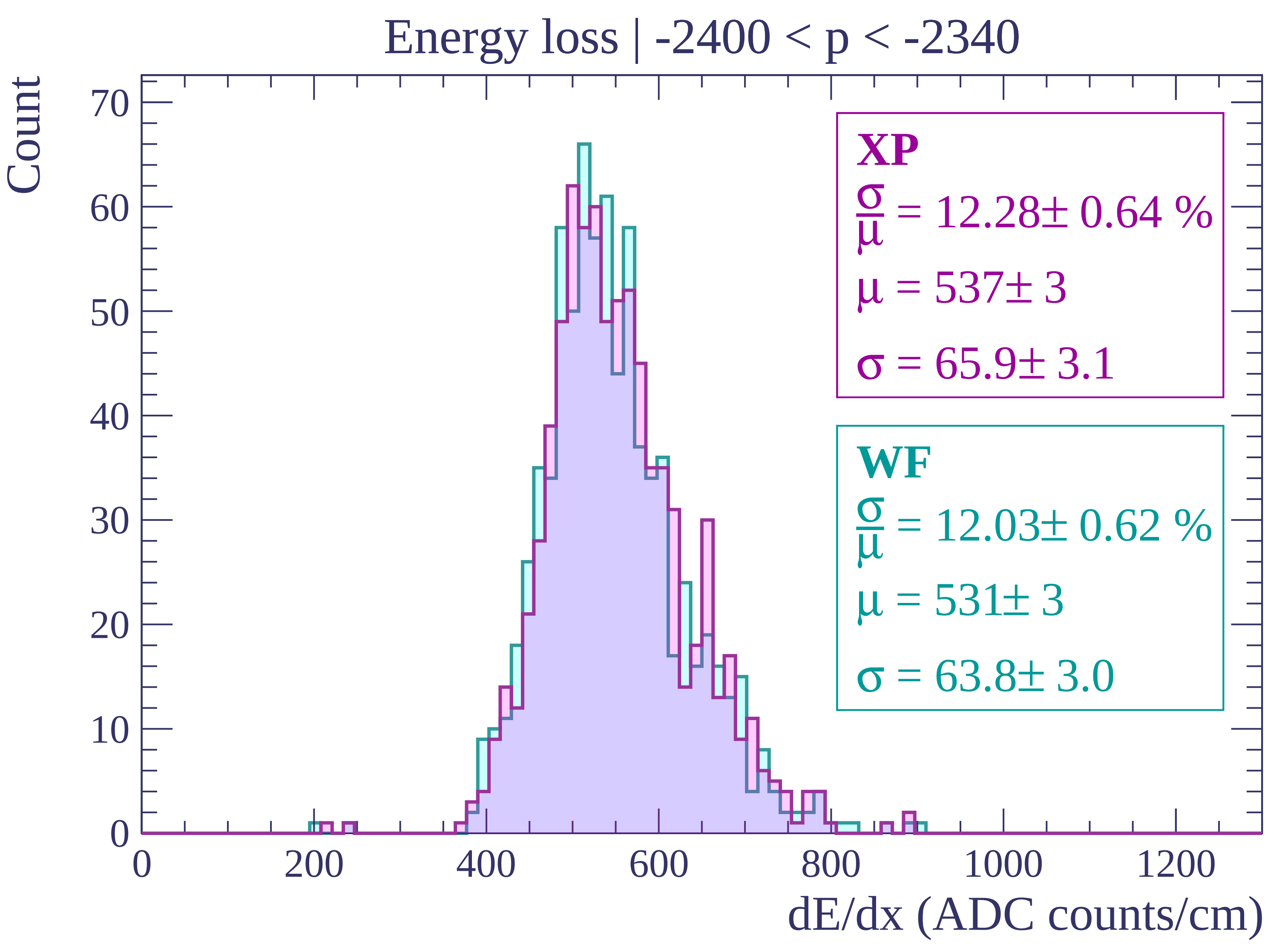
Energy loss | -2580 < p < -2520
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.93± 0.54 %
μ = 549± 3
σ = 65.5± 2.6
WF
σ/μ = 12.55± 0.65 %
μ = 540± 3
σ = 67.8± 3.1



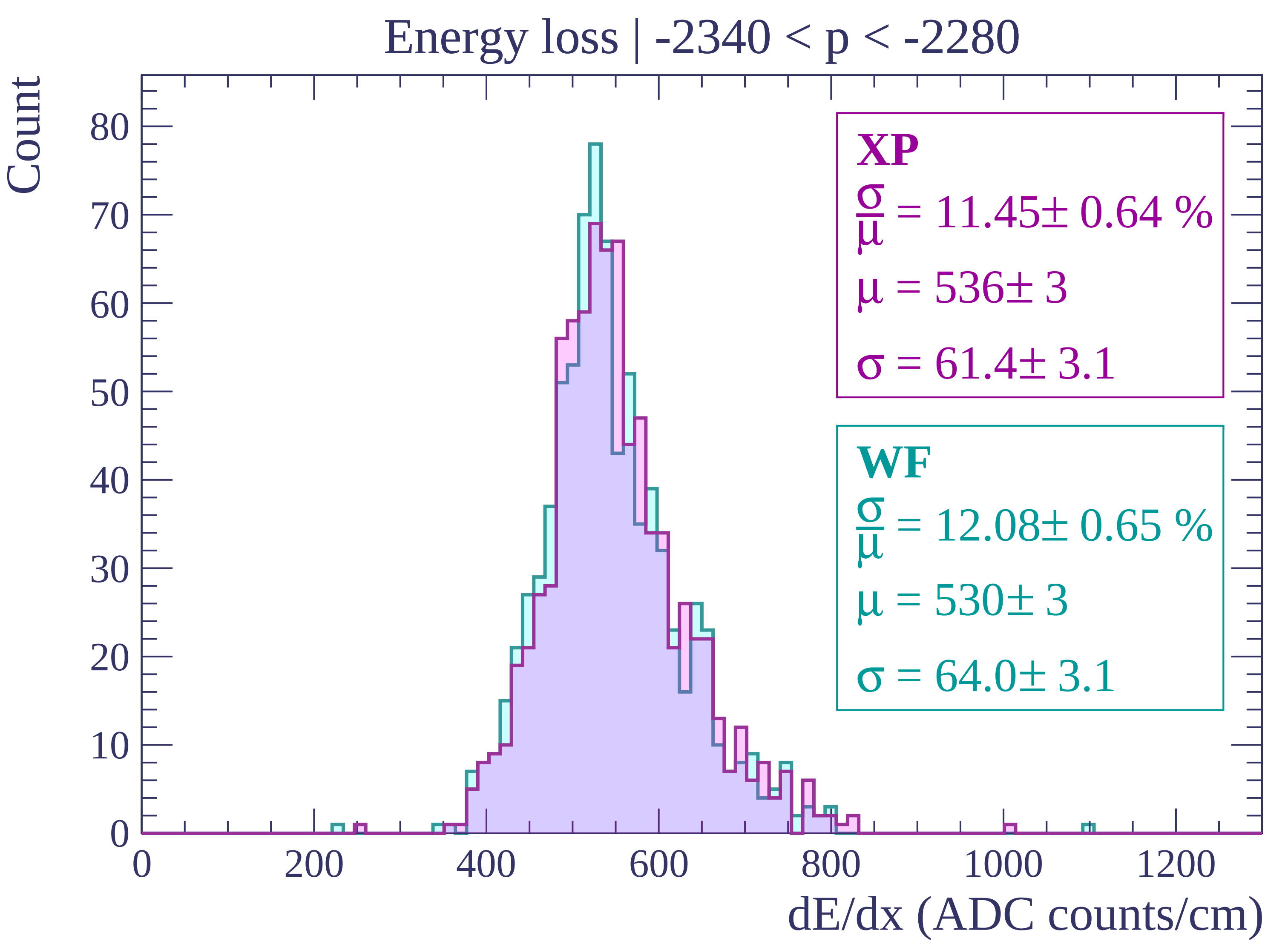
Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 13.53± 0.68 %
μ = 548± 4
σ = 74.1± 3.2
WF
σ/μ = 13.56± 0.67 %
μ = 546± 4
σ = 74.0± 3.2



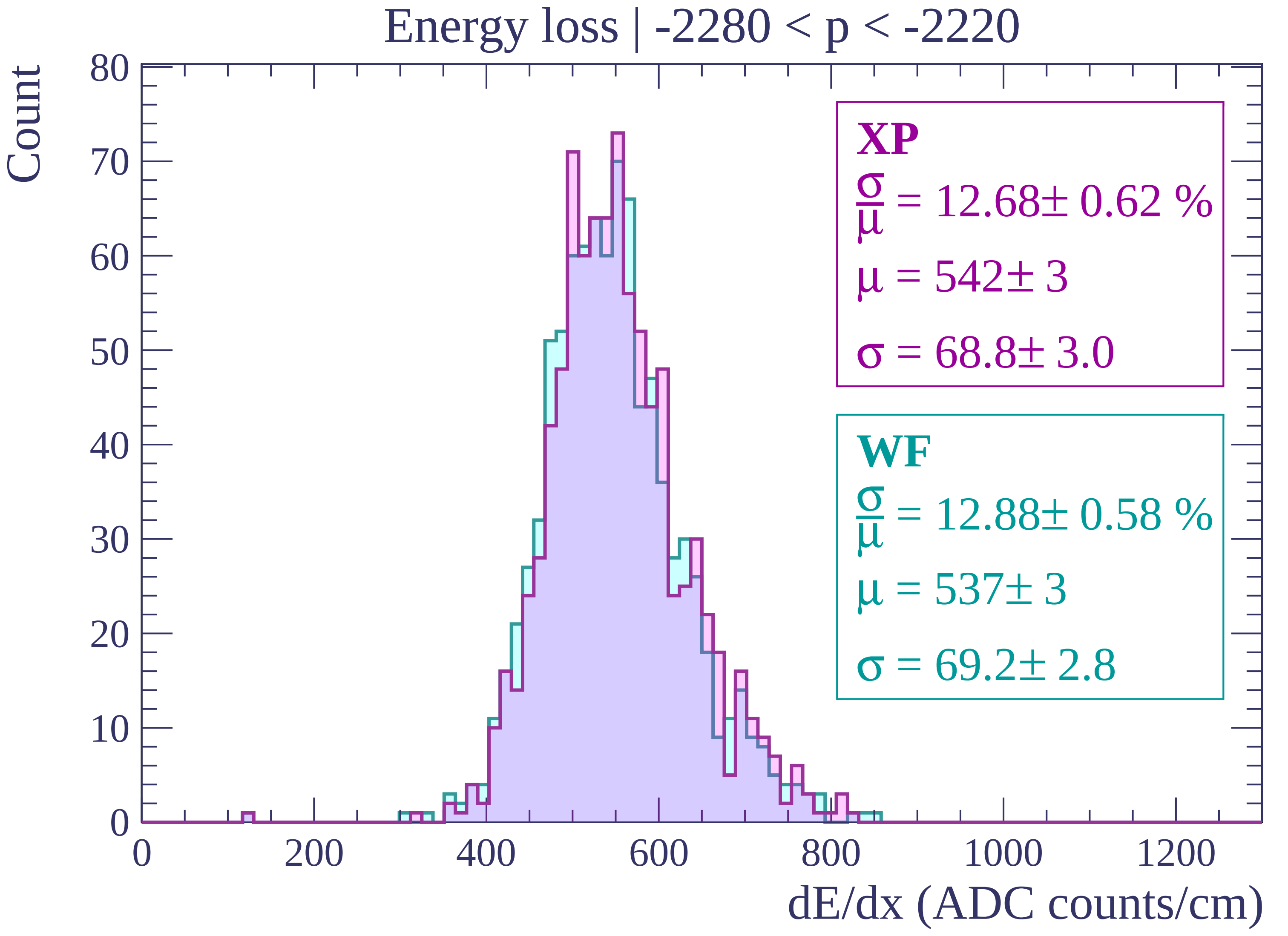
Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 12.92± 0.66 %
μ = 538± 3
σ = 69.5± 3.2
WF
σ/μ = 13.32± 0.75 %
μ = 532± 3
σ = 70.8± 3.5



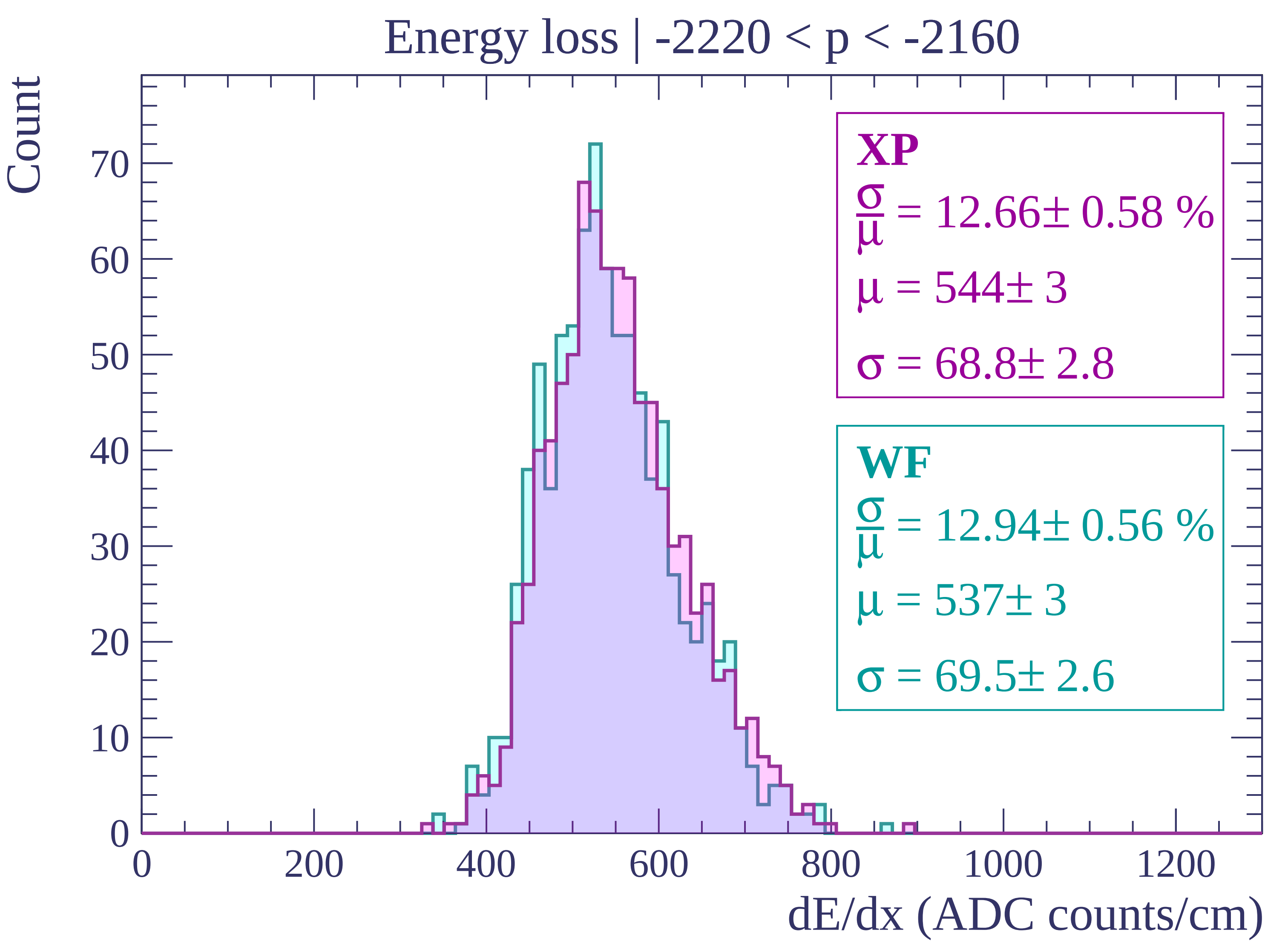
Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.28± 0.64 %
μ = 537± 3
σ = 65.9± 3.1
WF
σ/μ = 12.03± 0.62 %
μ = 531± 3
σ = 63.8± 3.0



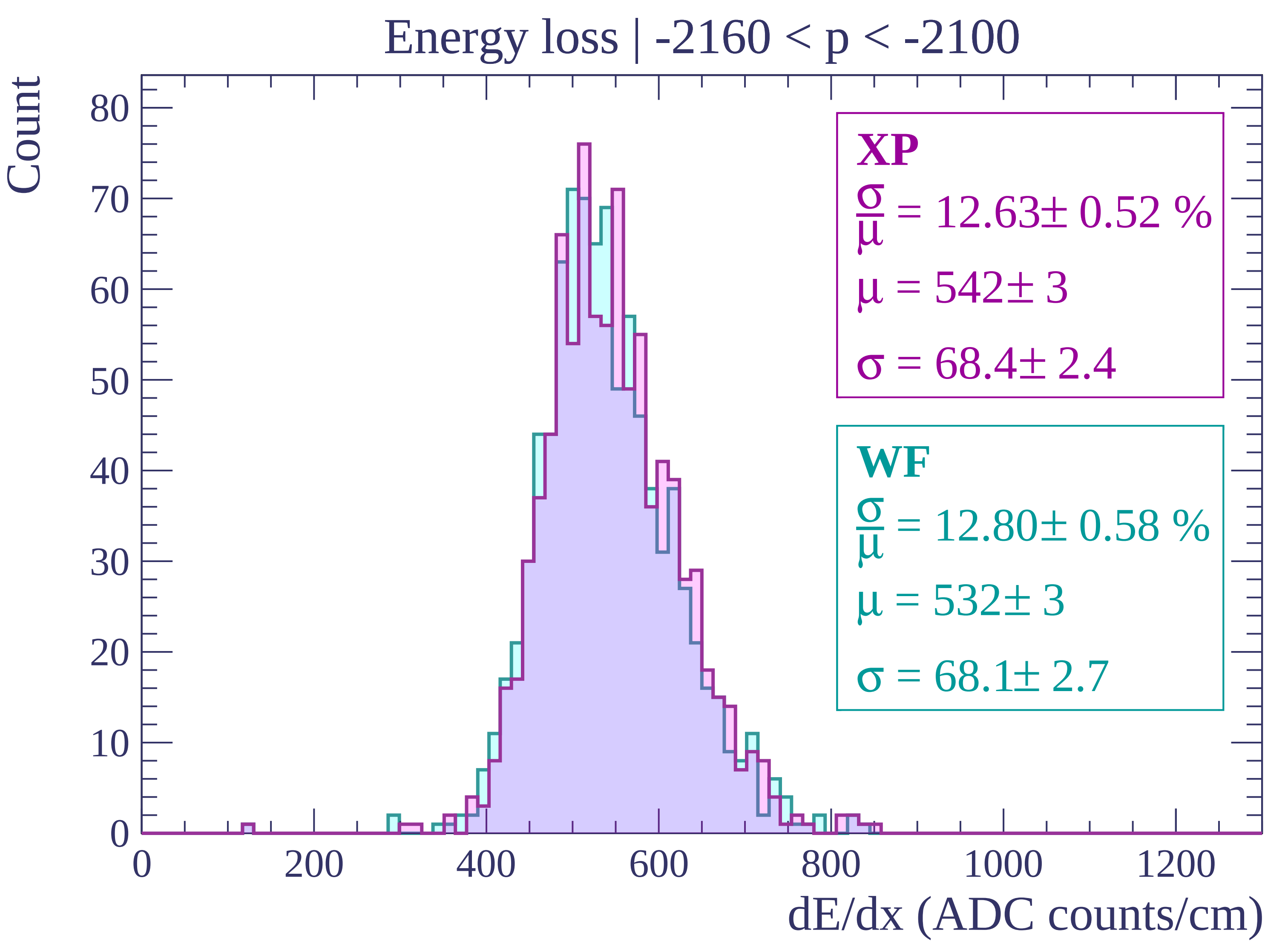
Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
σ/μ = 11.45± 0.64 %
μ = 536± 3
σ = 61.4± 3.1
WF
σ/μ = 12.08± 0.65 %
μ = 530± 3
σ = 64.0± 3.1

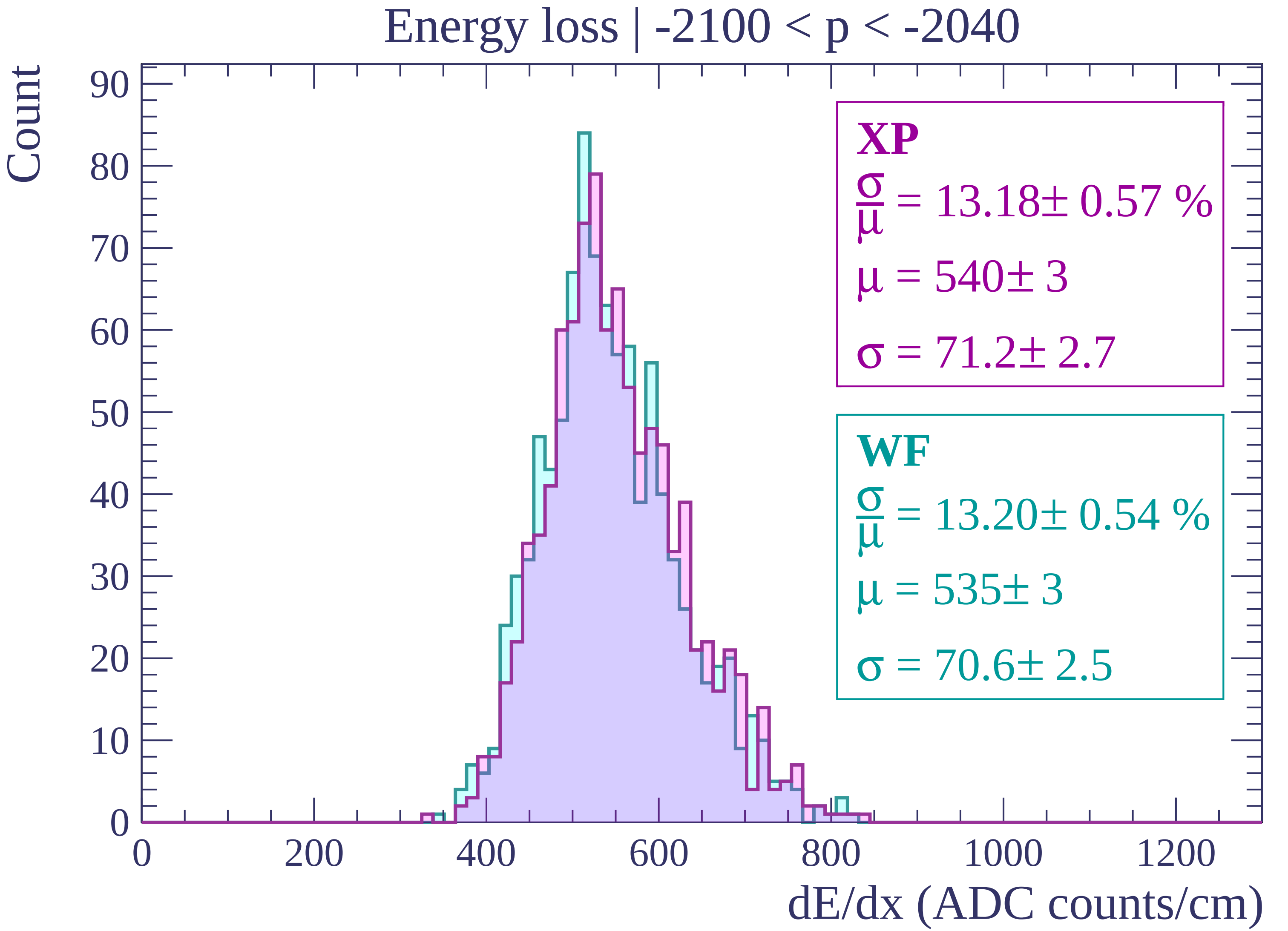


Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 12.68± 0.62 %
μ = 542± 3
σ = 68.8± 3.0
WF
σ/μ = 12.88± 0.58 %
μ = 537± 3
σ = 69.2± 2.8

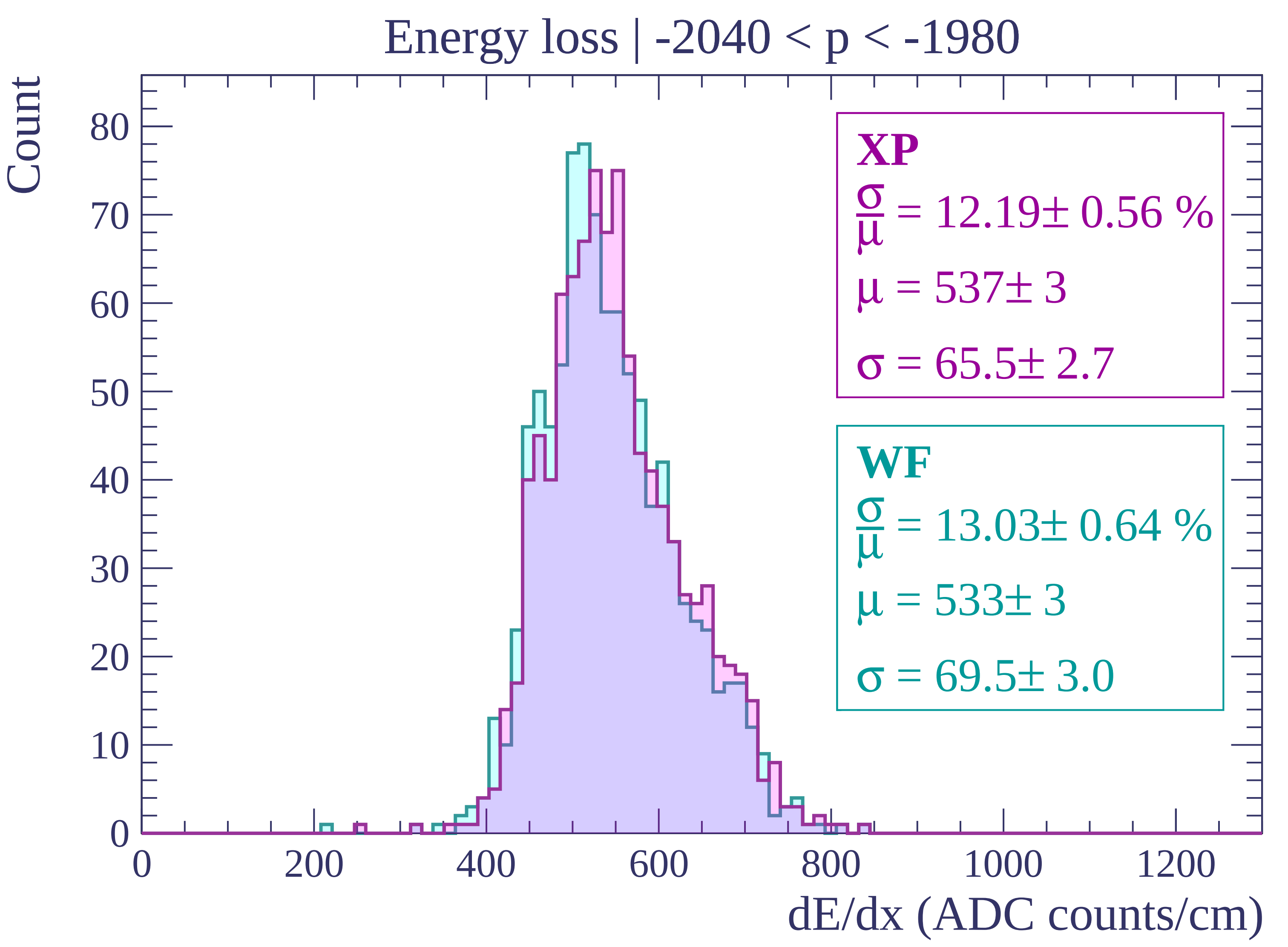


Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 12.66± 0.58 %
μ = 544± 3
σ = 68.8± 2.8
WF
σ/μ = 12.94± 0.56 %
μ = 537± 3
σ = 69.5± 2.6

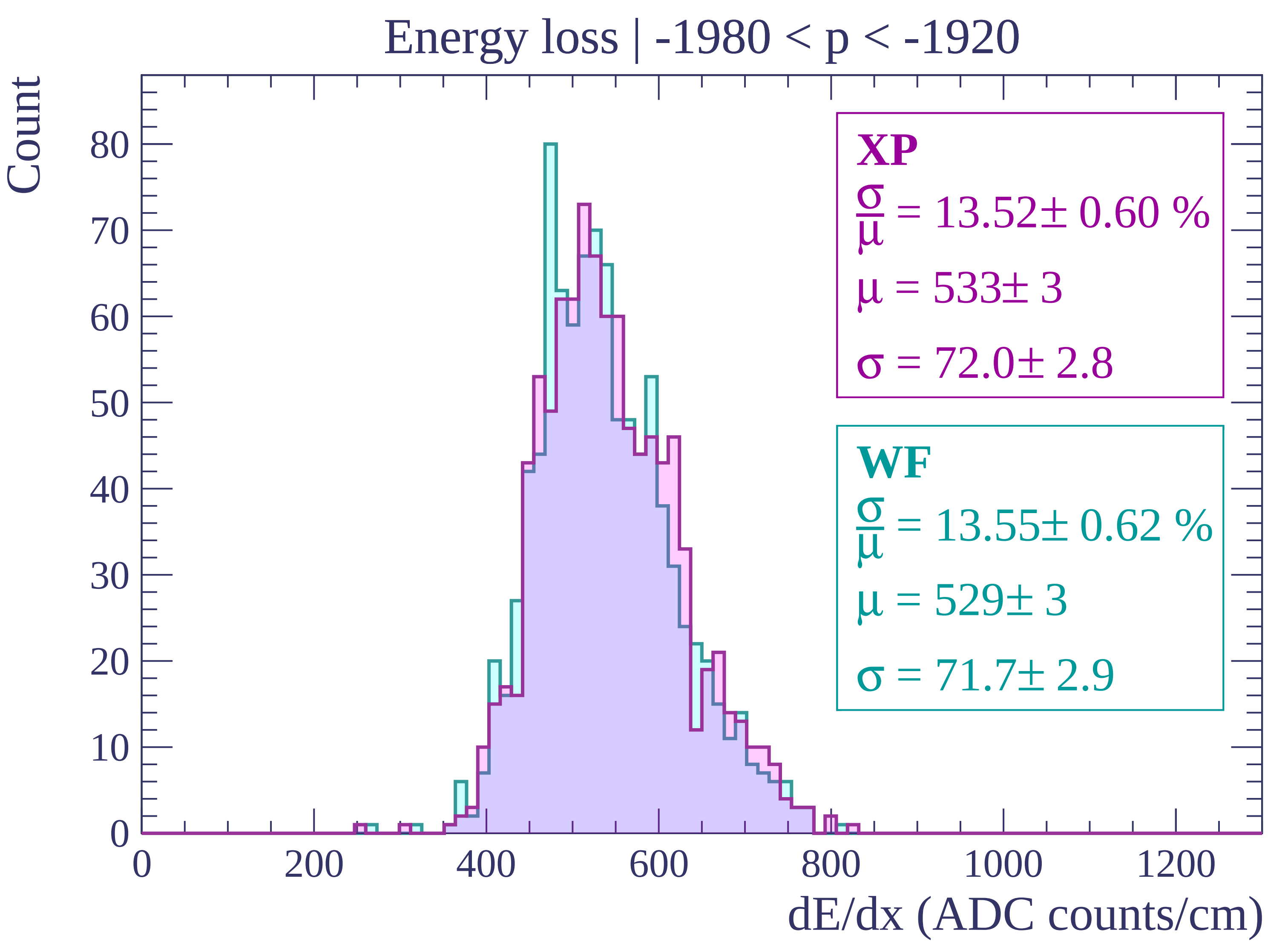

Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 12.63± 0.52 %
μ = 542± 3
σ = 68.4± 2.4
WF
σ/μ = 12.80± 0.58 %
μ = 532± 3
σ = 68.1± 2.7



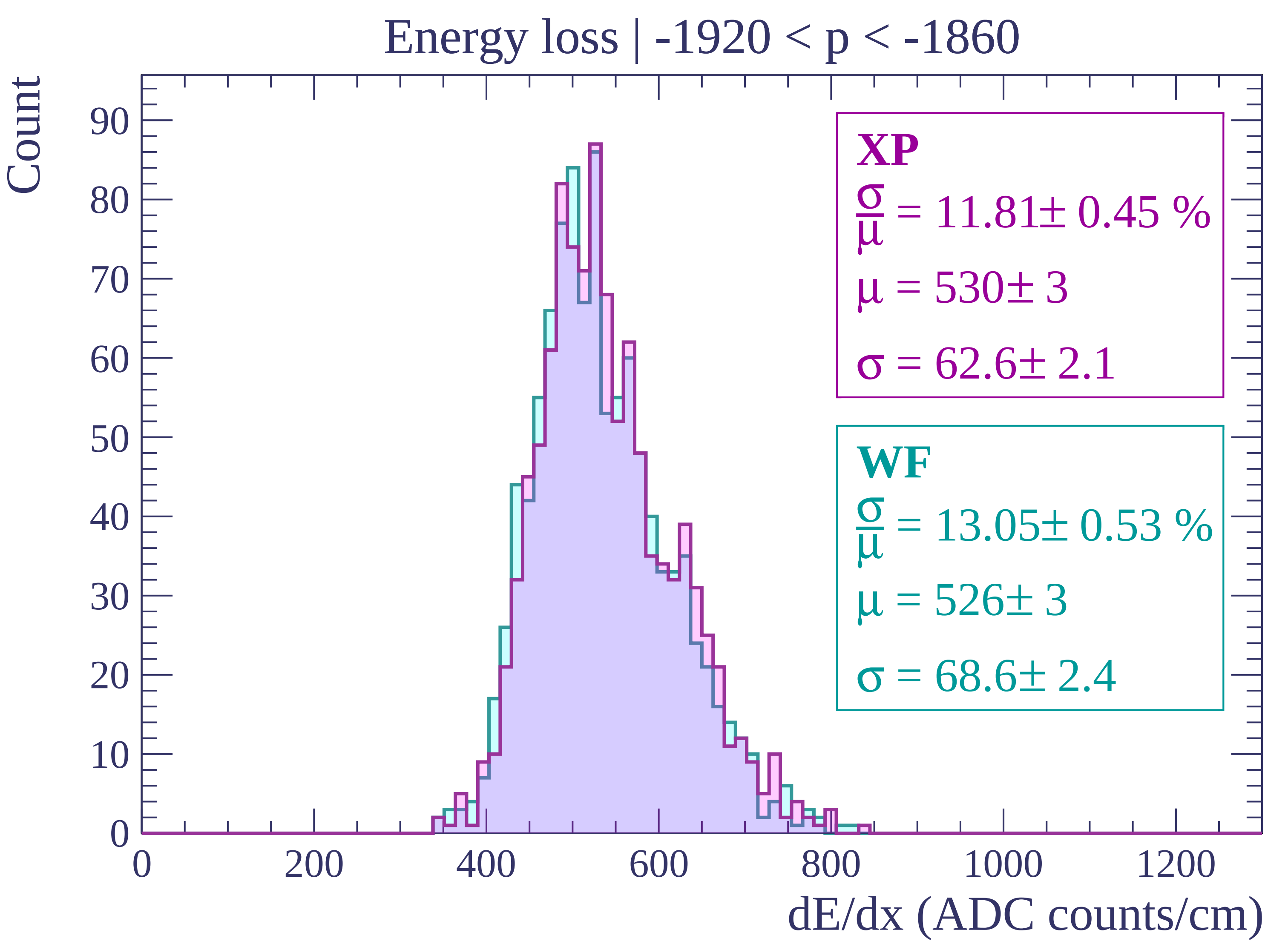
Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 13.18± 0.57 %
μ = 540± 3
σ = 71.2± 2.7
WF
σ/μ = 13.20± 0.54 %
μ = 535± 3
σ = 70.6± 2.5



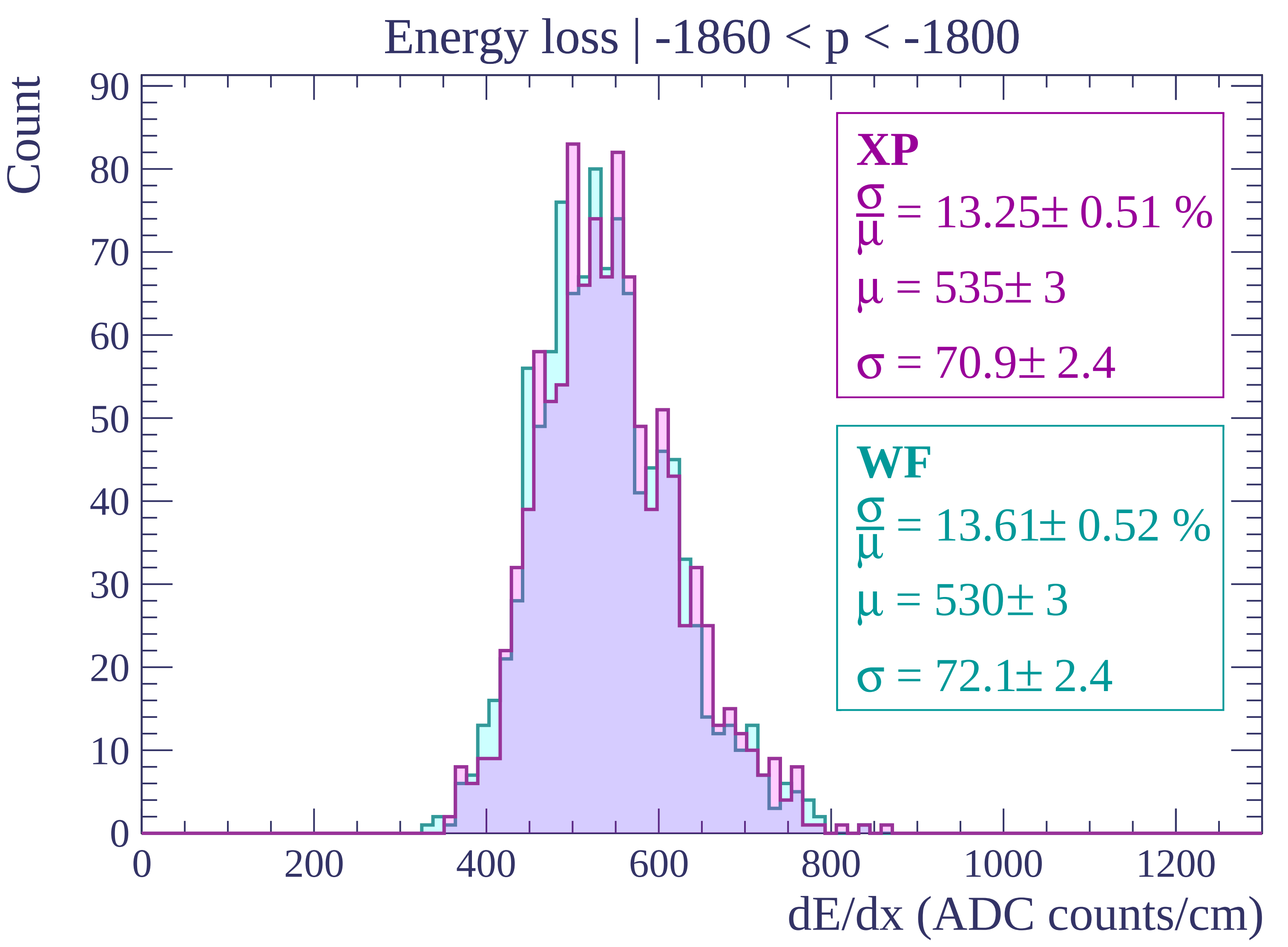
Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
$\frac{\sigma}{\mu}$ = 12.19± 0.56 %
μ = 537± 3
σ = 65.5± 2.7
WF
$\frac{\sigma}{\mu}$ = 13.03± 0.64 %
μ = 533± 3
σ = 69.5± 3.0



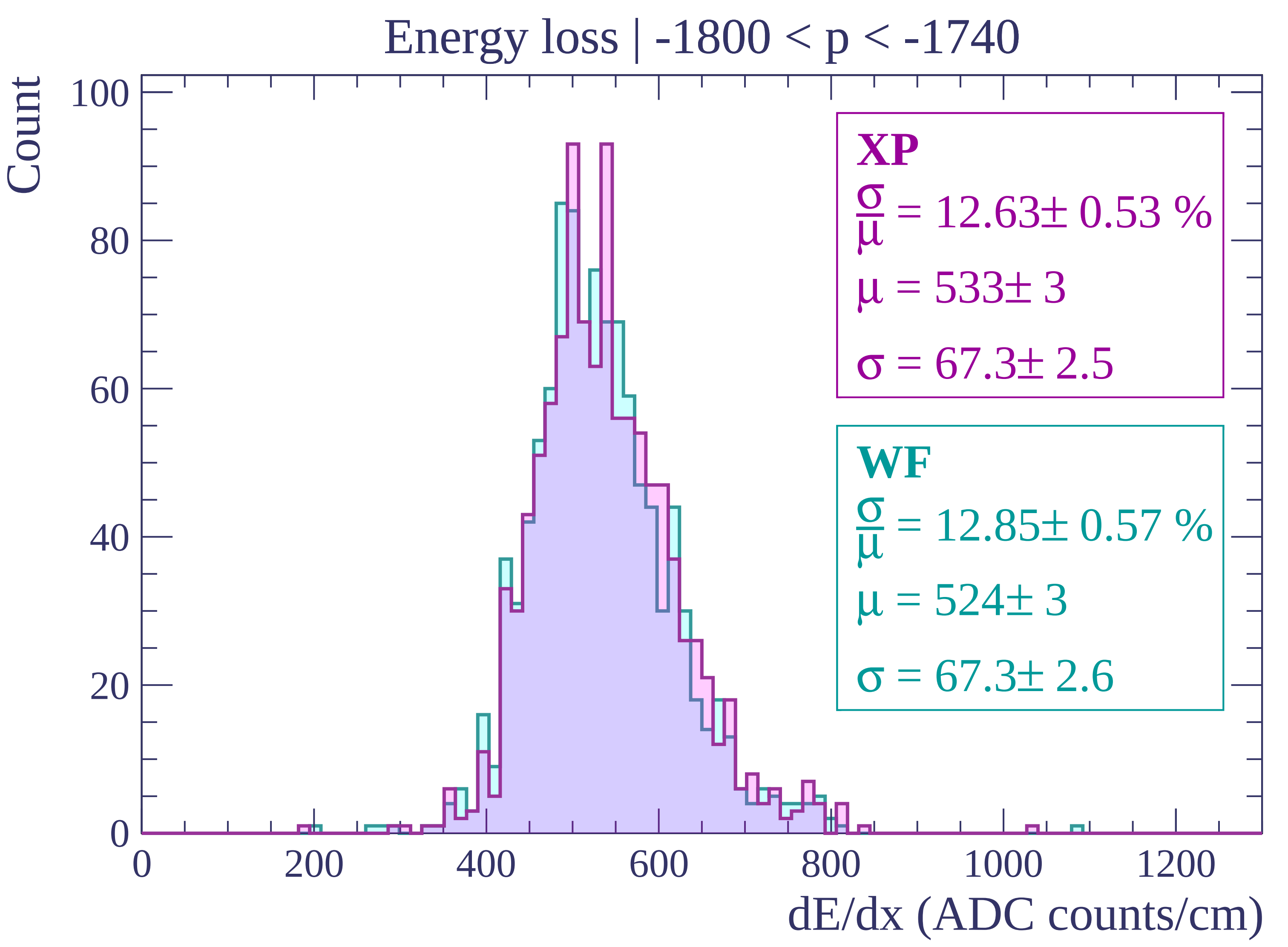
Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.52± 0.60 %
μ = 533± 3
σ = 72.0± 2.8
WF
σ/μ = 13.55± 0.62 %
μ = 529± 3
σ = 71.7± 2.9



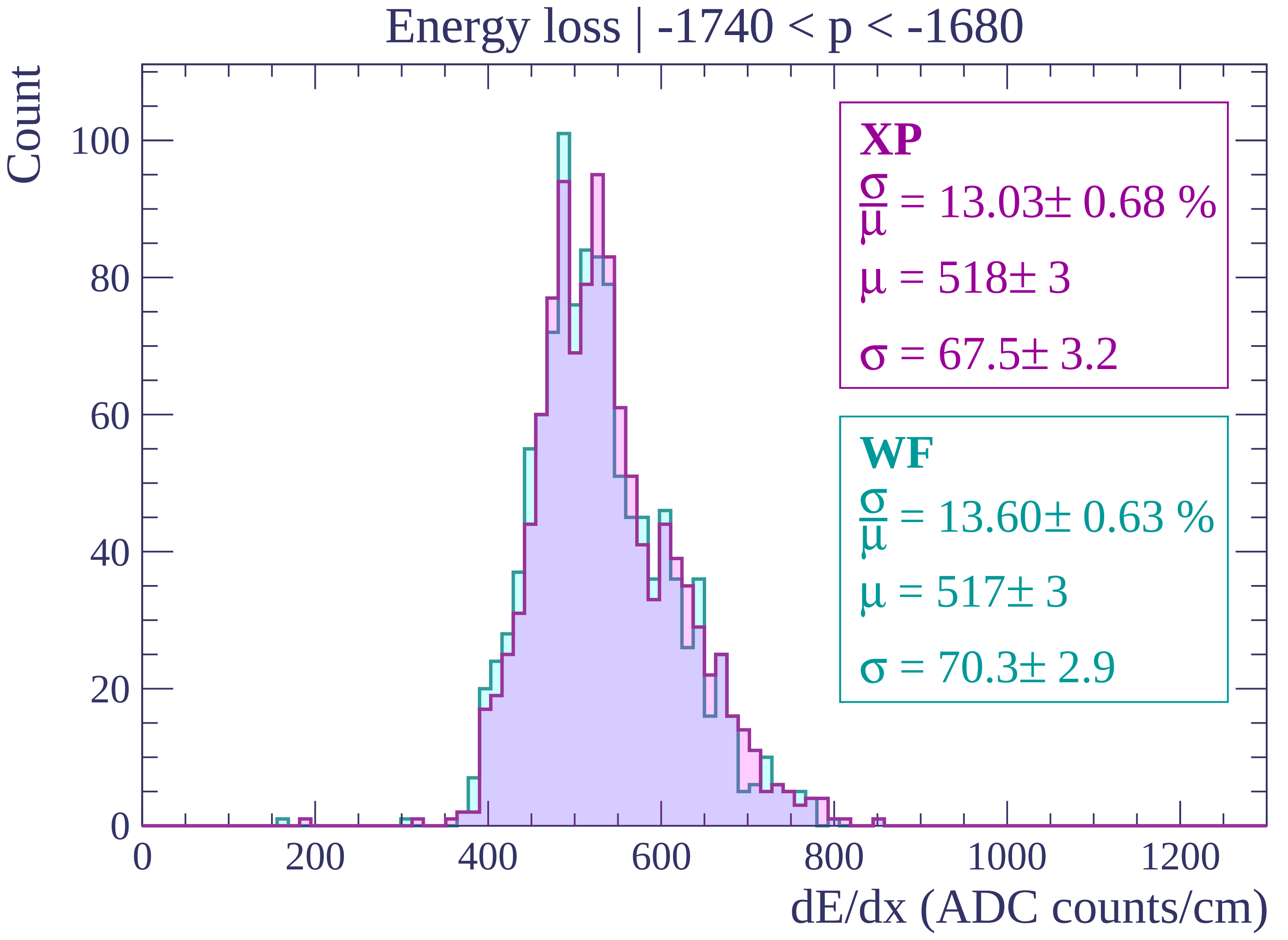
Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 11.81± 0.45 %
μ = 530± 3
σ = 62.6± 2.1
WF
σ/μ = 13.05± 0.53 %
μ = 526± 3
σ = 68.6± 2.4



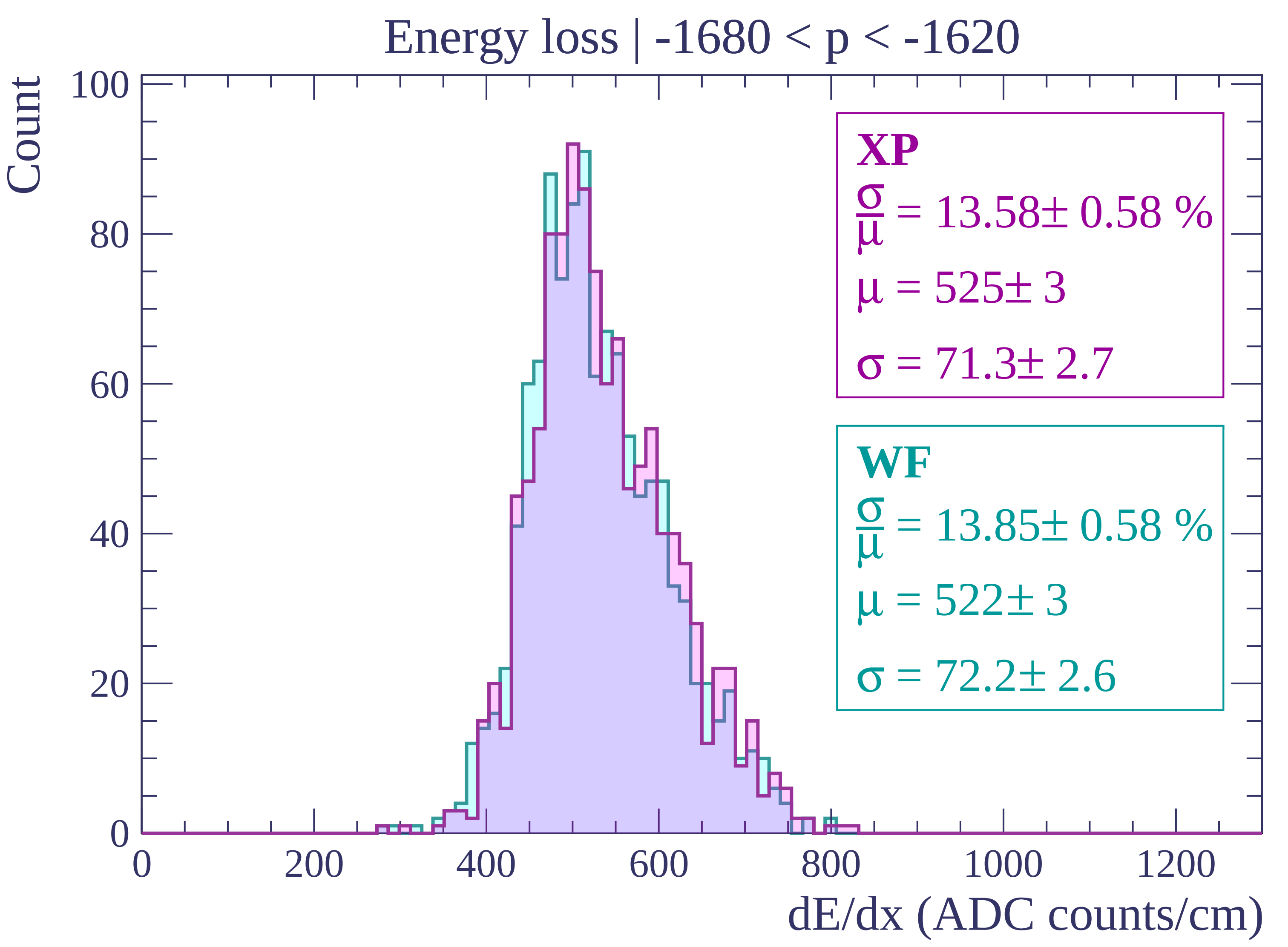
Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP
$\frac{\sigma}{\mu}$ = 13.25± 0.51 %
μ = 535± 3
σ = 70.9± 2.4
WF
$\frac{\sigma}{\mu}$ = 13.61± 0.52 %
μ = 530± 3
σ = 72.1± 2.4



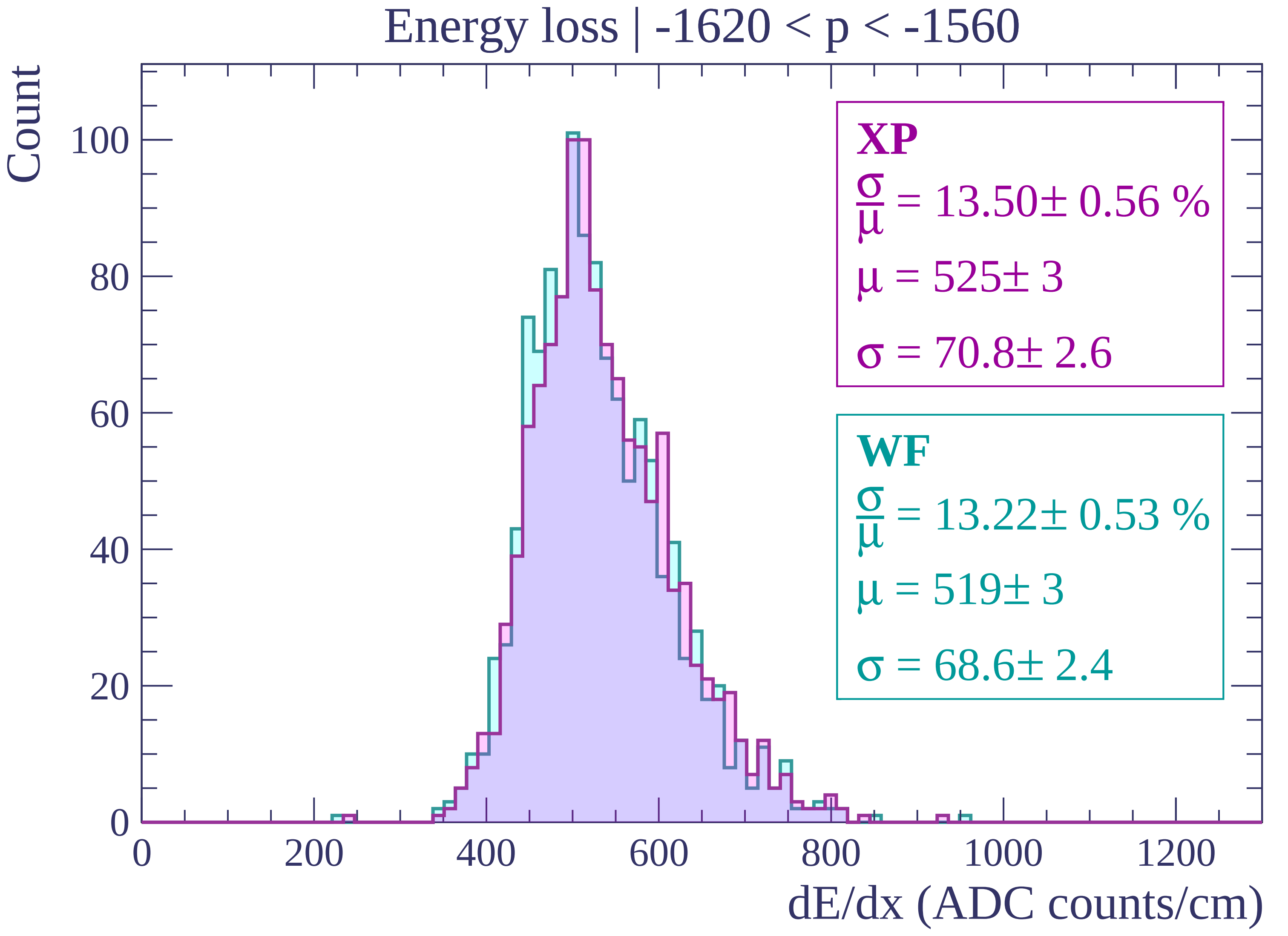
Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.63± 0.53 %
μ = 533± 3
σ = 67.3± 2.5
WF
$\frac{\sigma}{\mu}$ = 12.85± 0.57 %
μ = 524± 3
σ = 67.3± 2.6

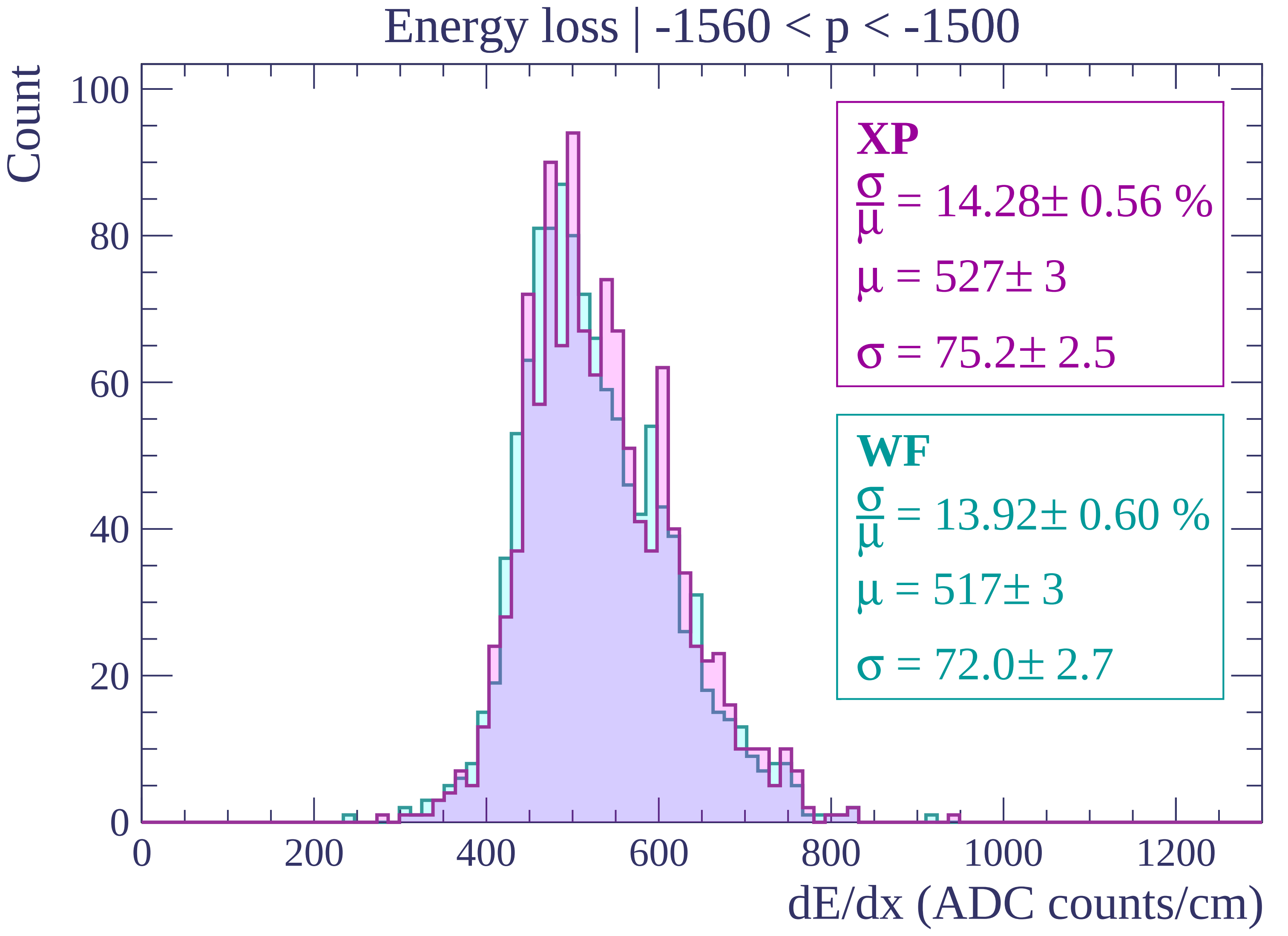


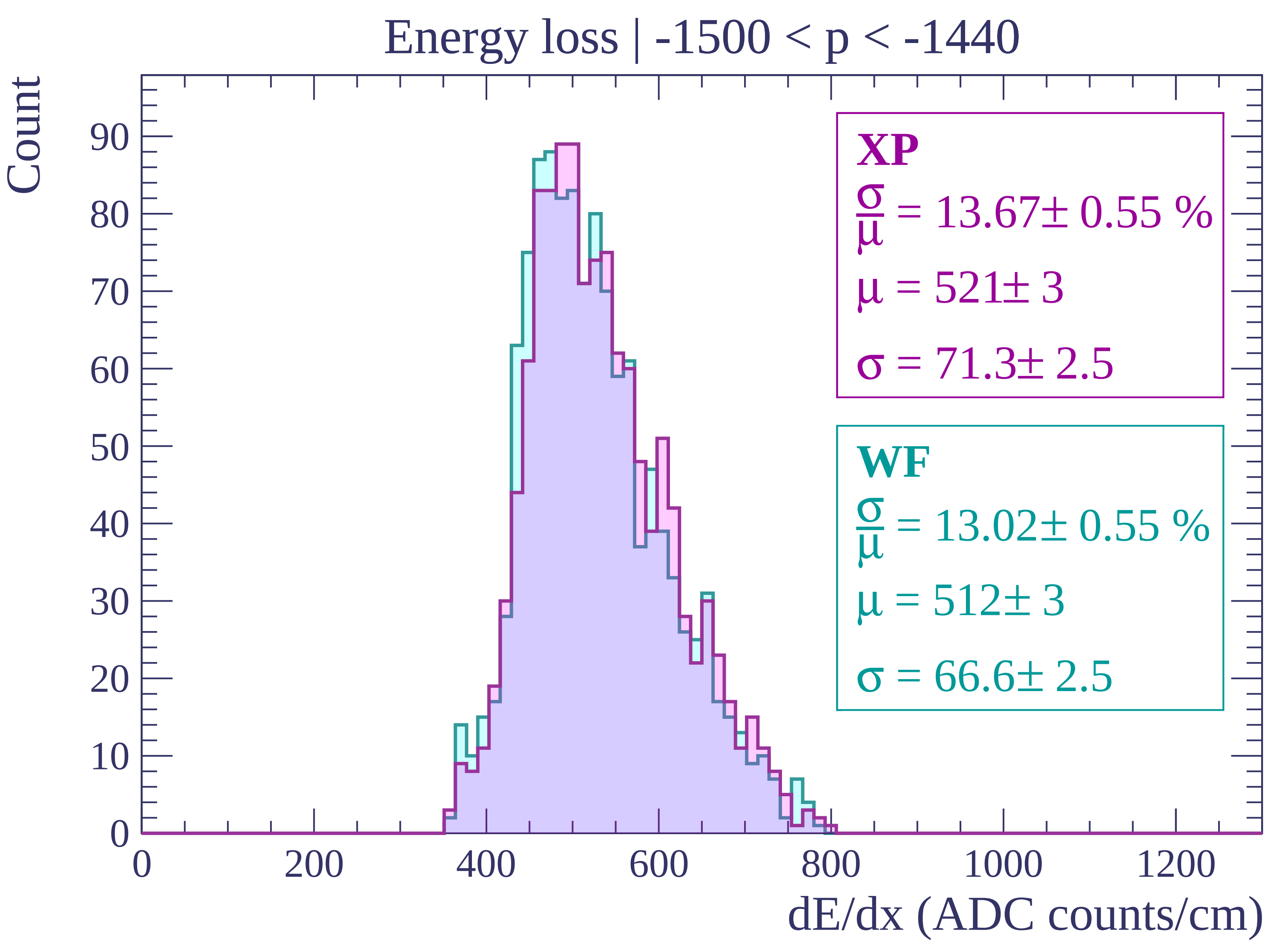
Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP
σ/μ = 13.03± 0.68 %
μ = 518± 3
σ = 67.5± 3.2
WF
σ/μ = 13.60± 0.63 %
μ = 517± 3
σ = 70.3± 2.9



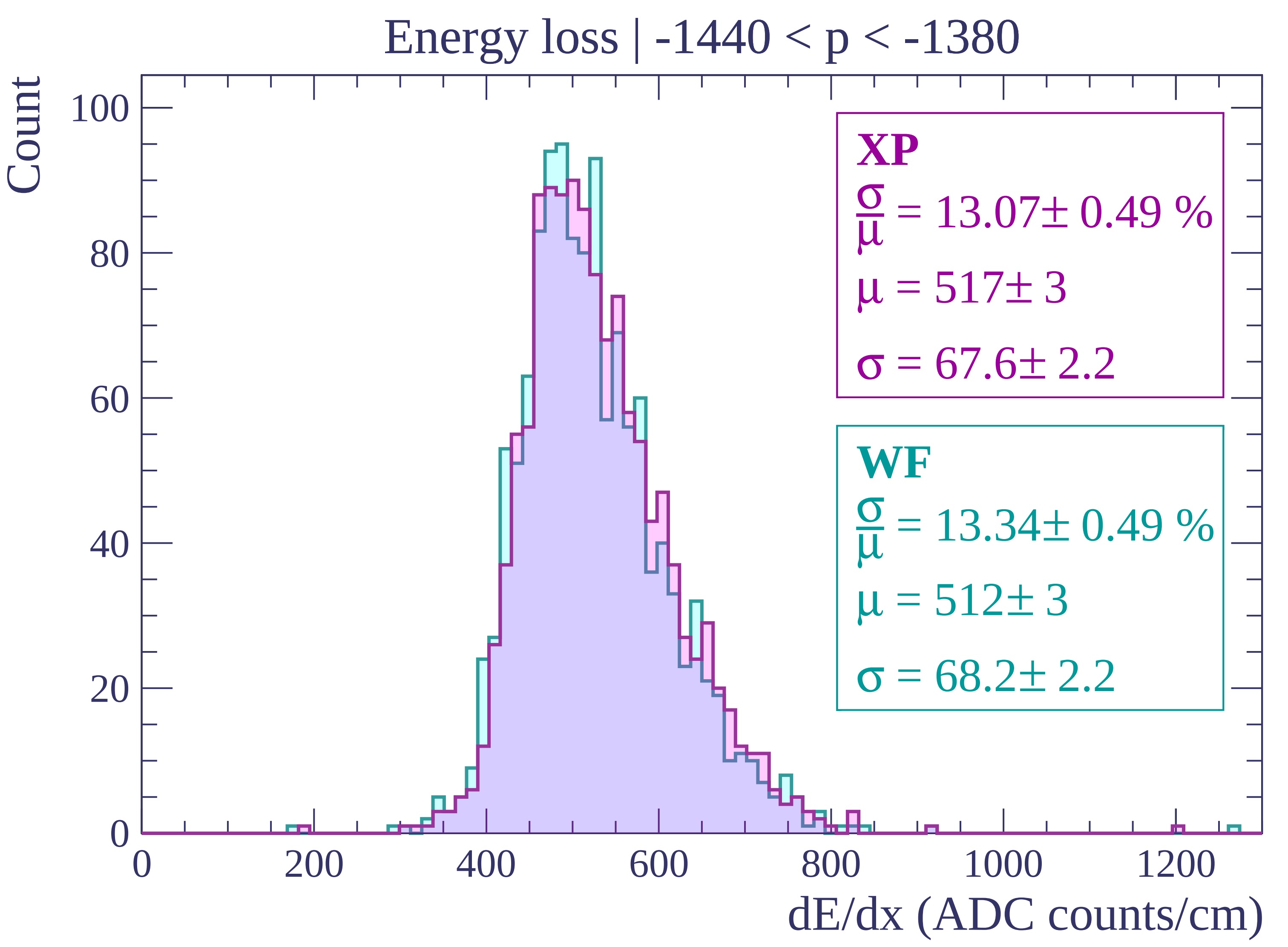
Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
$\frac{\sigma}{\mu}$ = 13.58± 0.58 %
μ = 525± 3
σ = 71.3± 2.7
WF
$\frac{\sigma}{\mu}$ = 13.85± 0.58 %
μ = 522± 3
σ = 72.2± 2.6


Energy loss | -1620 < p < -1560
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.50± 0.56 %
μ = 525± 3
σ = 70.8± 2.6
WF
σ/μ = 13.22± 0.53 %
μ = 519± 3
σ = 68.6± 2.4

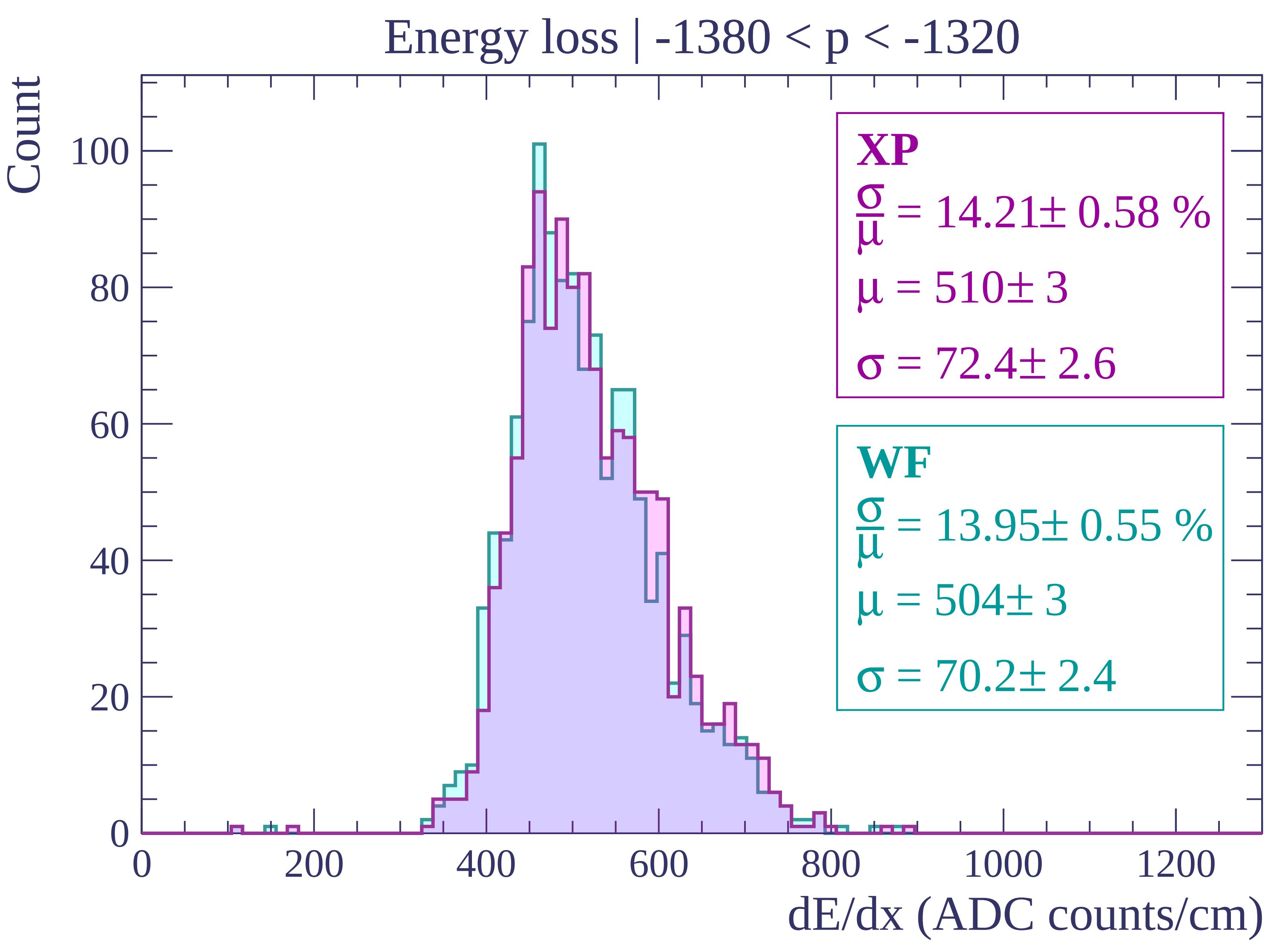

Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
$\frac{\sigma}{\mu}$ = 14.28± 0.56 %
μ = 527± 3
σ = 75.2± 2.5
WF
$\frac{\sigma}{\mu}$ = 13.92± 0.60 %
μ = 517± 3
σ = 72.0± 2.7



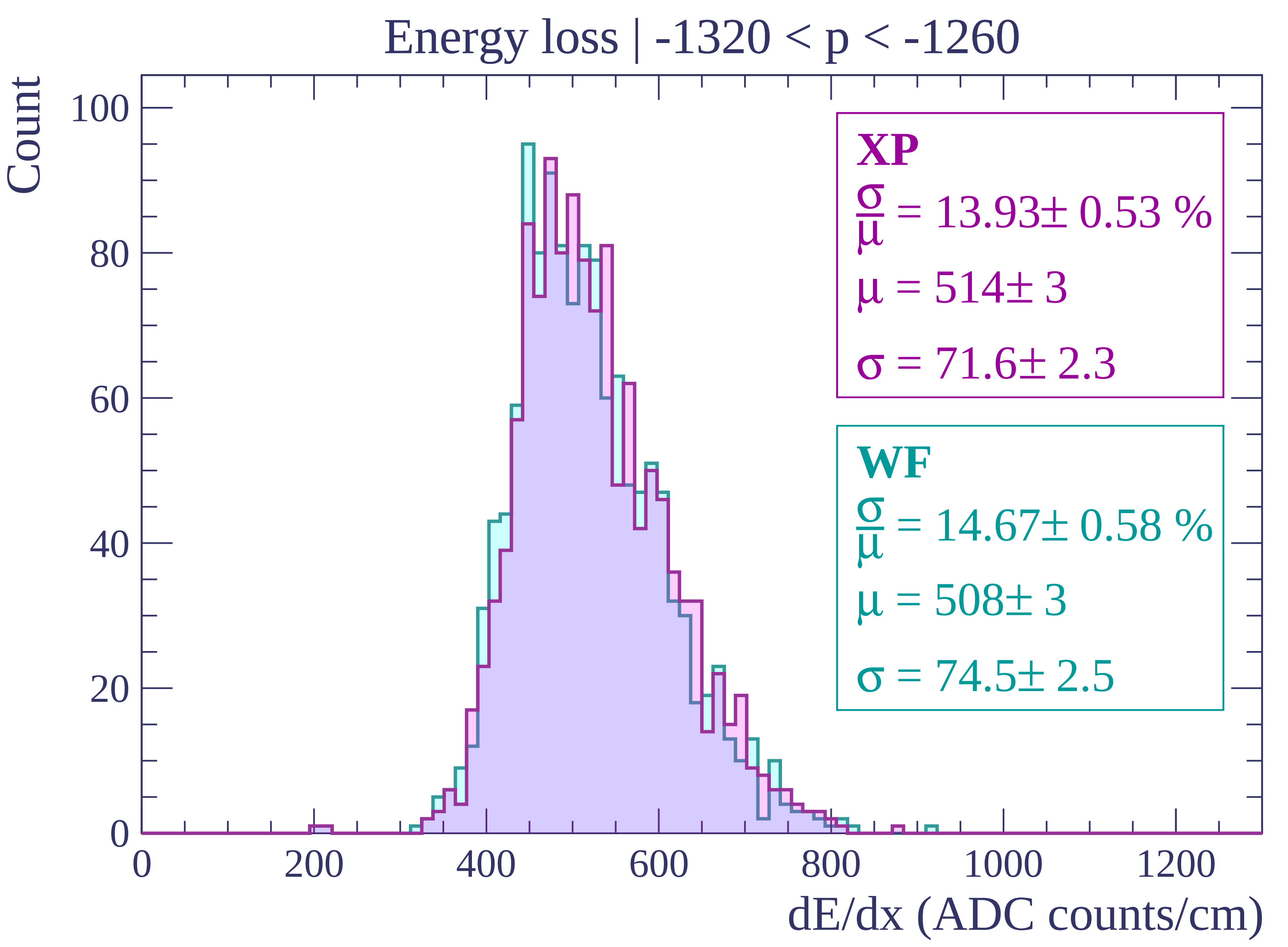
Energy loss | -1500 < p < -1440
Count
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 13.67± 0.55 %
μ = 521± 3
σ = 71.3± 2.5
WF
$\frac{\sigma}{\mu}$ = 13.02± 0.55 %
μ = 512± 3
σ = 66.6± 2.5



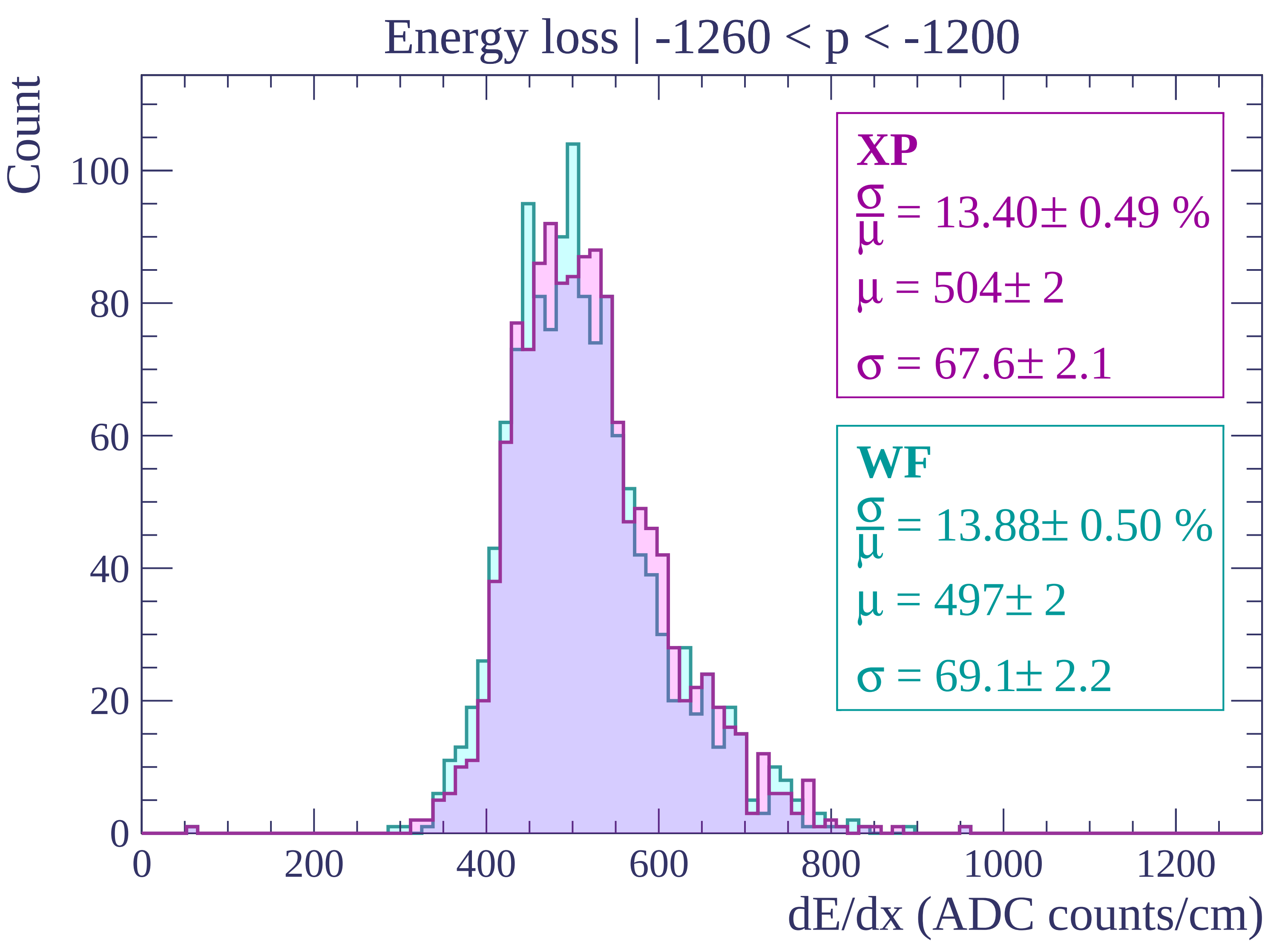
Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 13.07± 0.49 %
μ = 517± 3
σ = 67.6± 2.2
WF
$\frac{\sigma}{\mu}$ = 13.34± 0.49 %
μ = 512± 3
σ = 68.2± 2.2



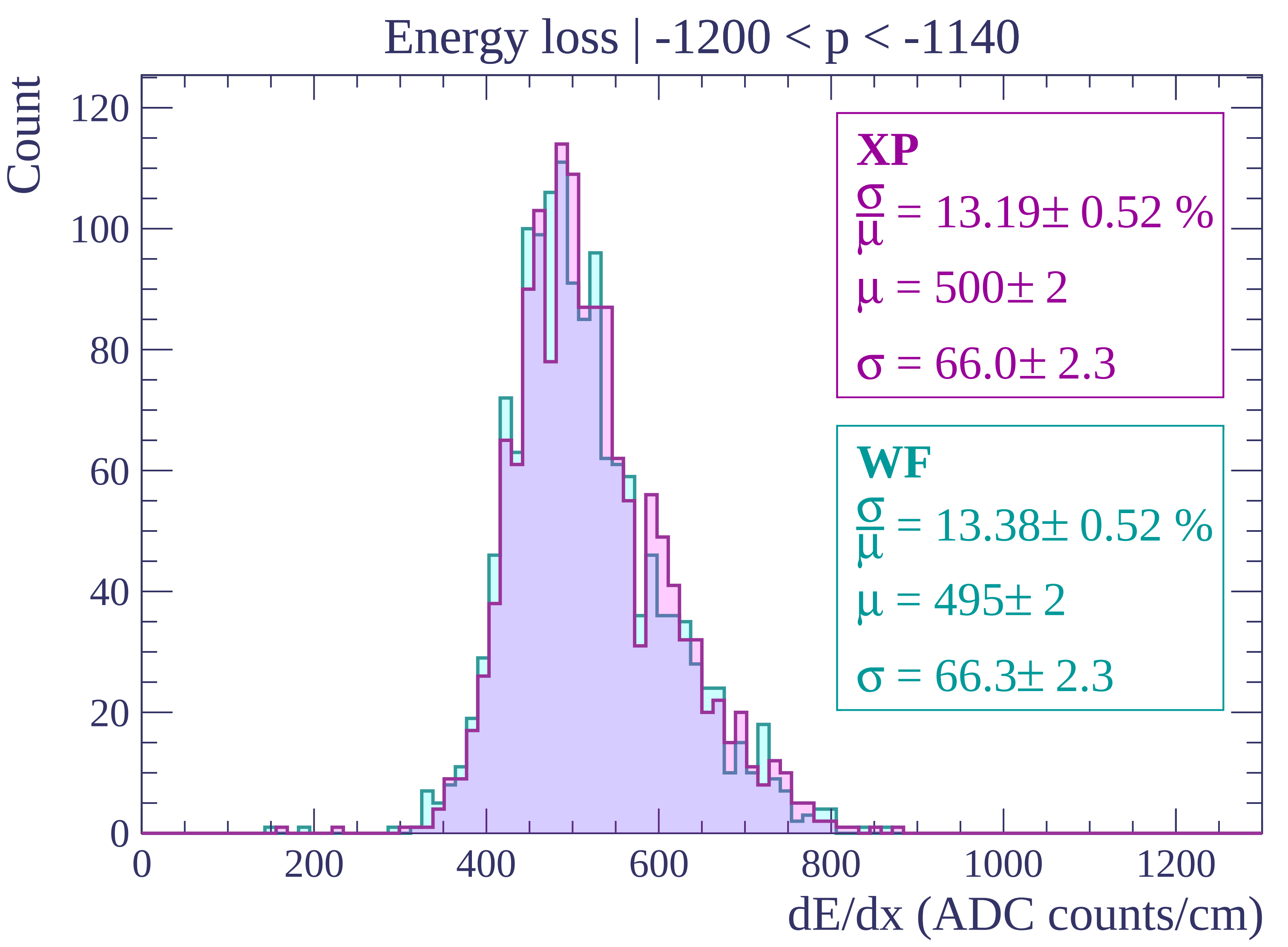
Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 14.21± 0.58 %
μ = 510± 3
σ = 72.4± 2.6
WF
σ/μ = 13.95± 0.55 %
μ = 504± 3
σ = 70.2± 2.4



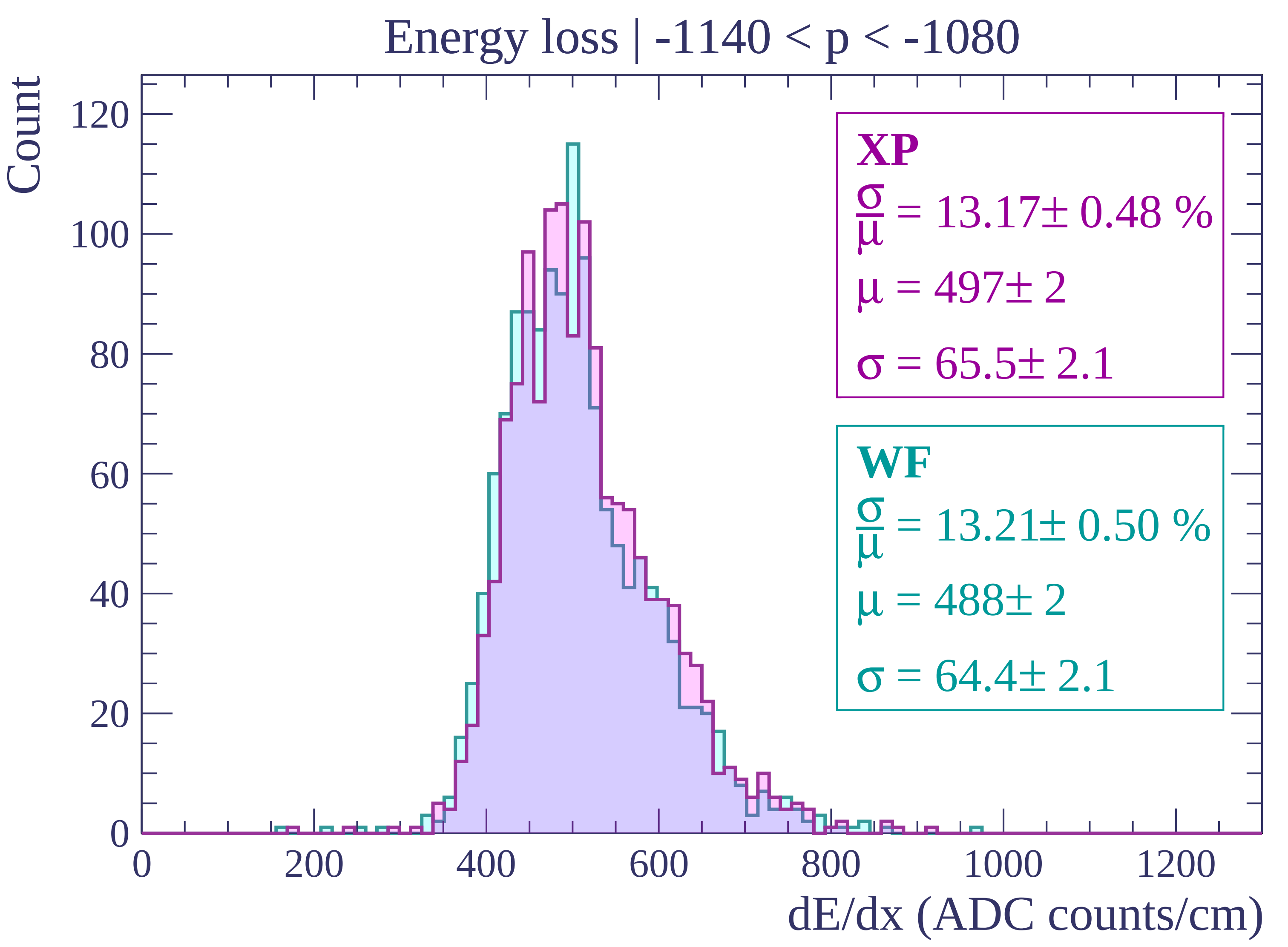
Energy loss | -1320 < p < -1260
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.93± 0.53 %
μ = 514± 3
σ = 71.6± 2.3
WF
σ/μ = 14.67± 0.58 %
μ = 508± 3
σ = 74.5± 2.5



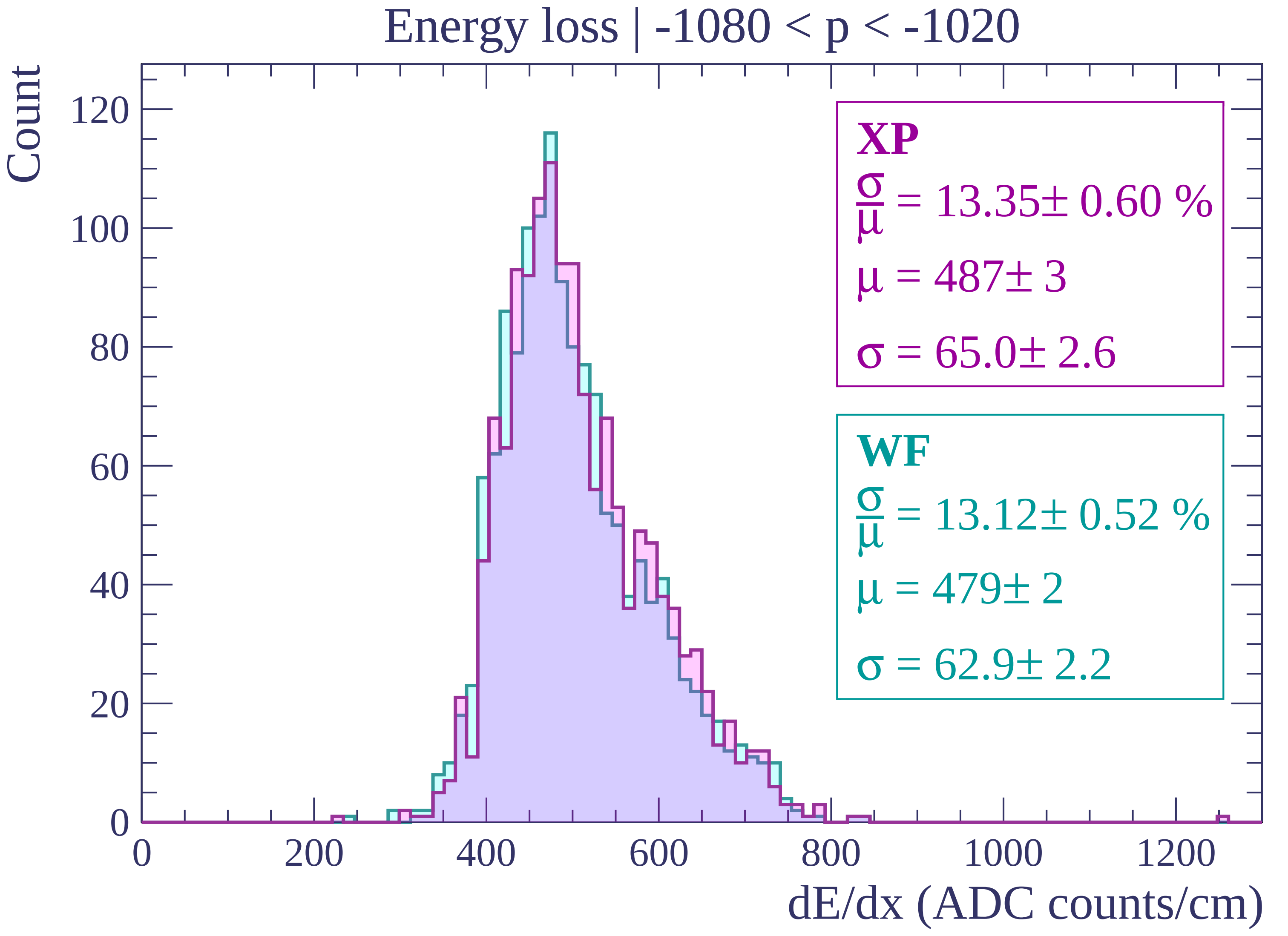
Energy loss | -1260 < p < -1200
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.40± 0.49 %
μ = 504± 2
σ = 67.6± 2.1
WF
σ/μ = 13.88± 0.50 %
μ = 497± 2
σ = 69.1± 2.2



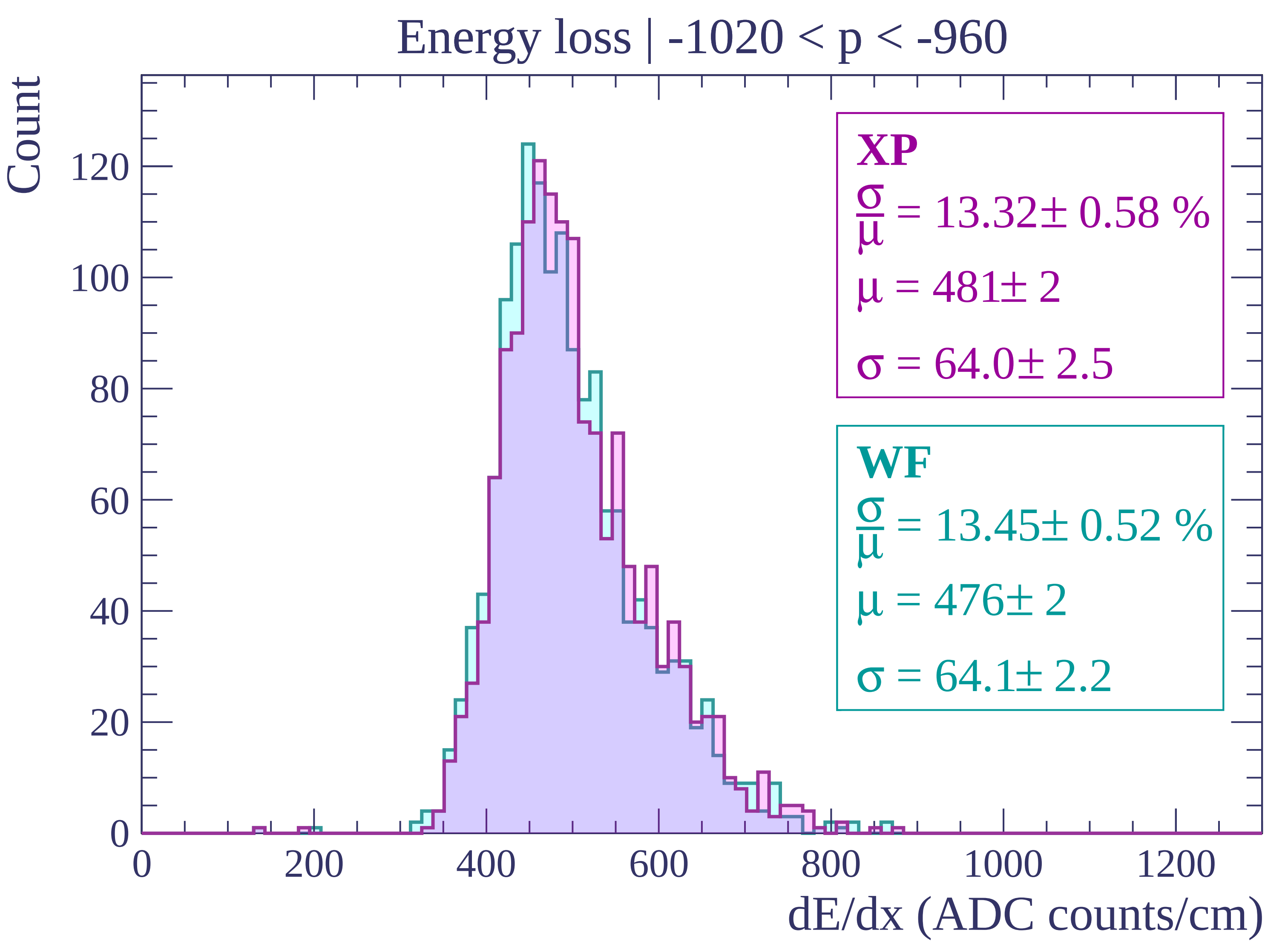
Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.19± 0.52 %
μ = 500± 2
σ = 66.0± 2.3
WF
$\frac{\sigma}{\mu}$ = 13.38± 0.52 %
μ = 495± 2
σ = 66.3± 2.3

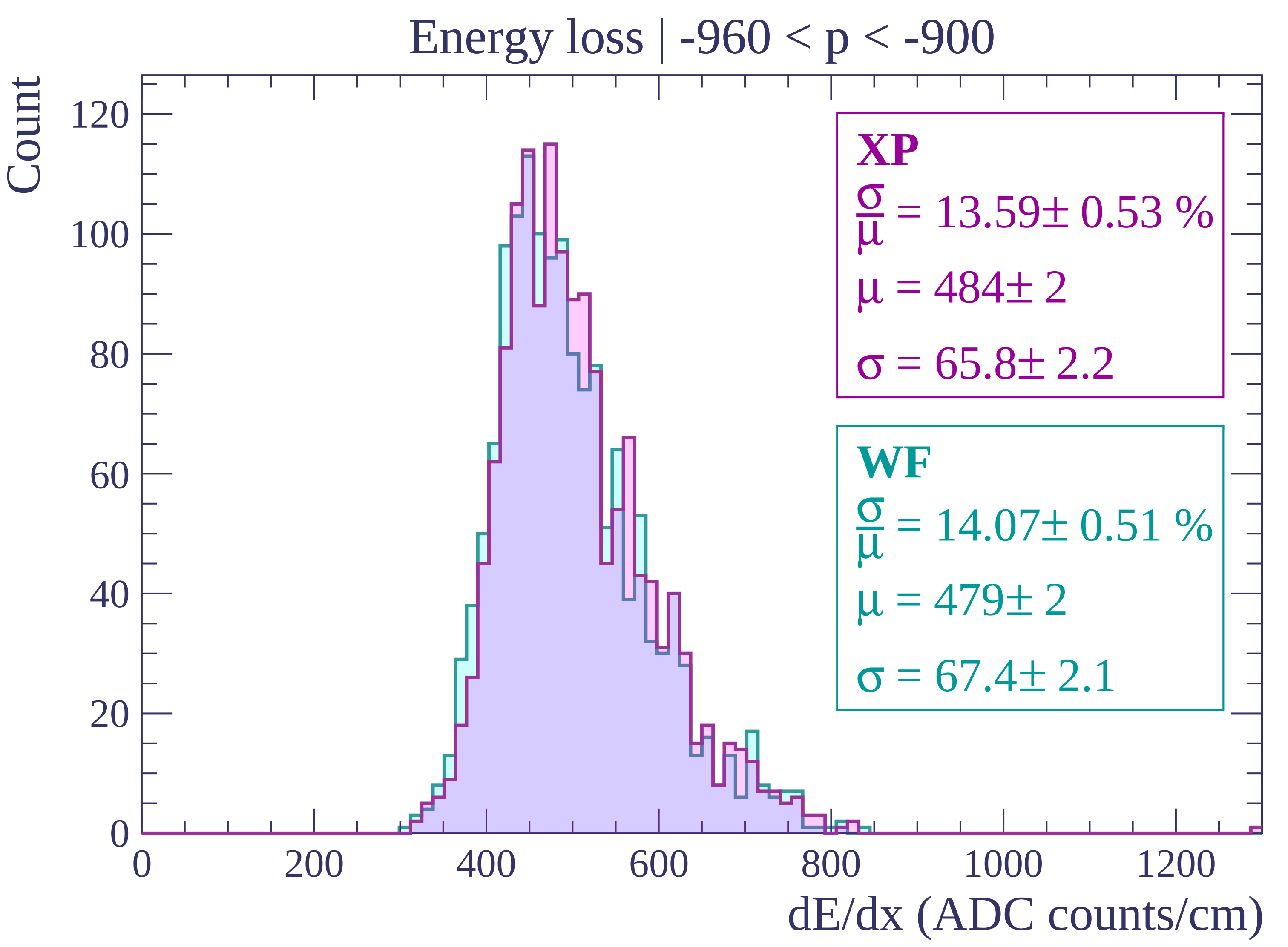


Energy loss | -1140 < p < -1080
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.17± 0.48 %
μ = 497± 2
σ = 65.5± 2.1
WF
σ/μ = 13.21± 0.50 %
μ = 488± 2
σ = 64.4± 2.1

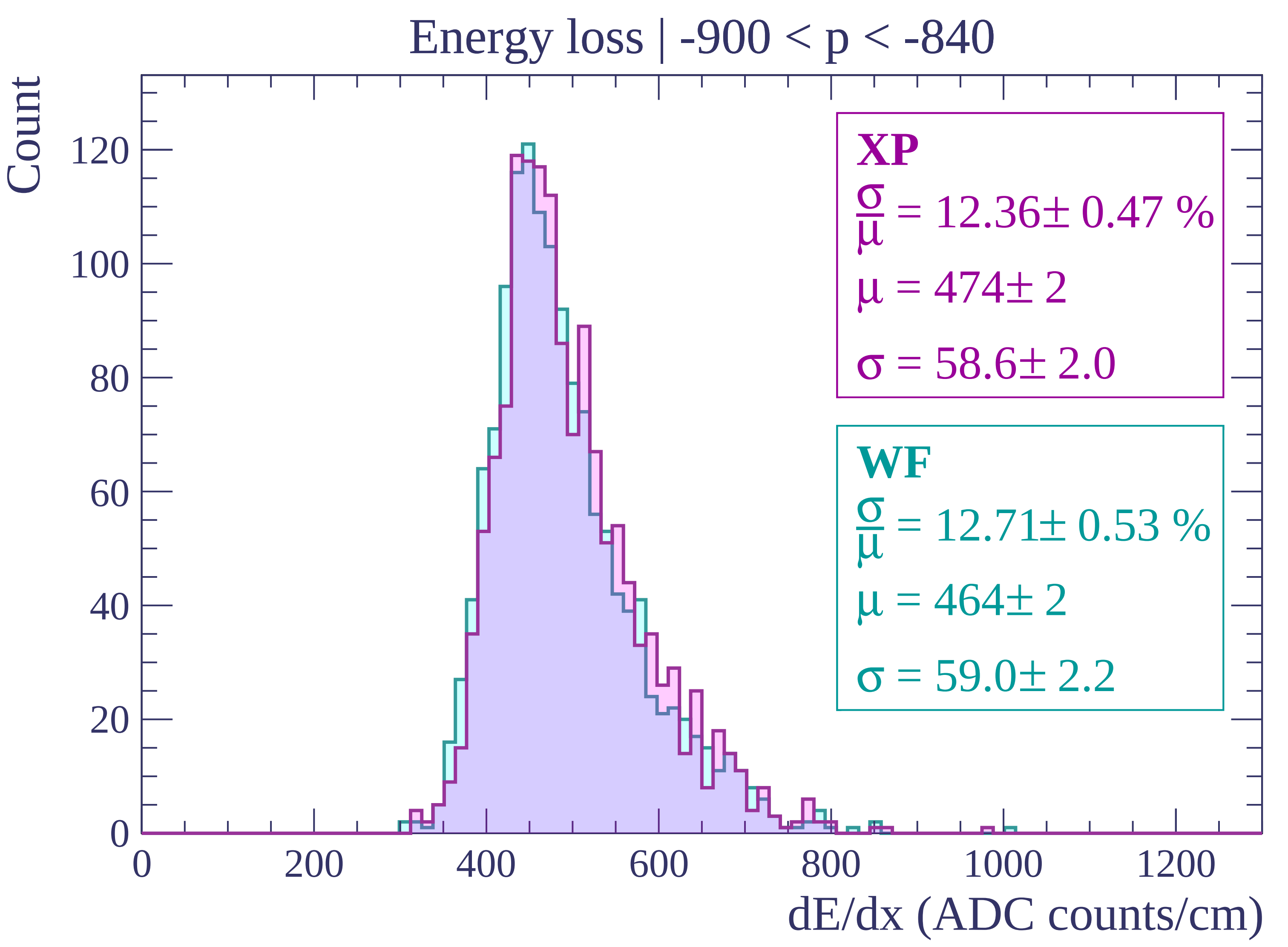


Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.35± 0.60 %
μ = 487± 3
σ = 65.0± 2.6
WF
$\frac{\sigma}{\mu}$ = 13.12± 0.52 %
μ = 479± 2
σ = 62.9± 2.2

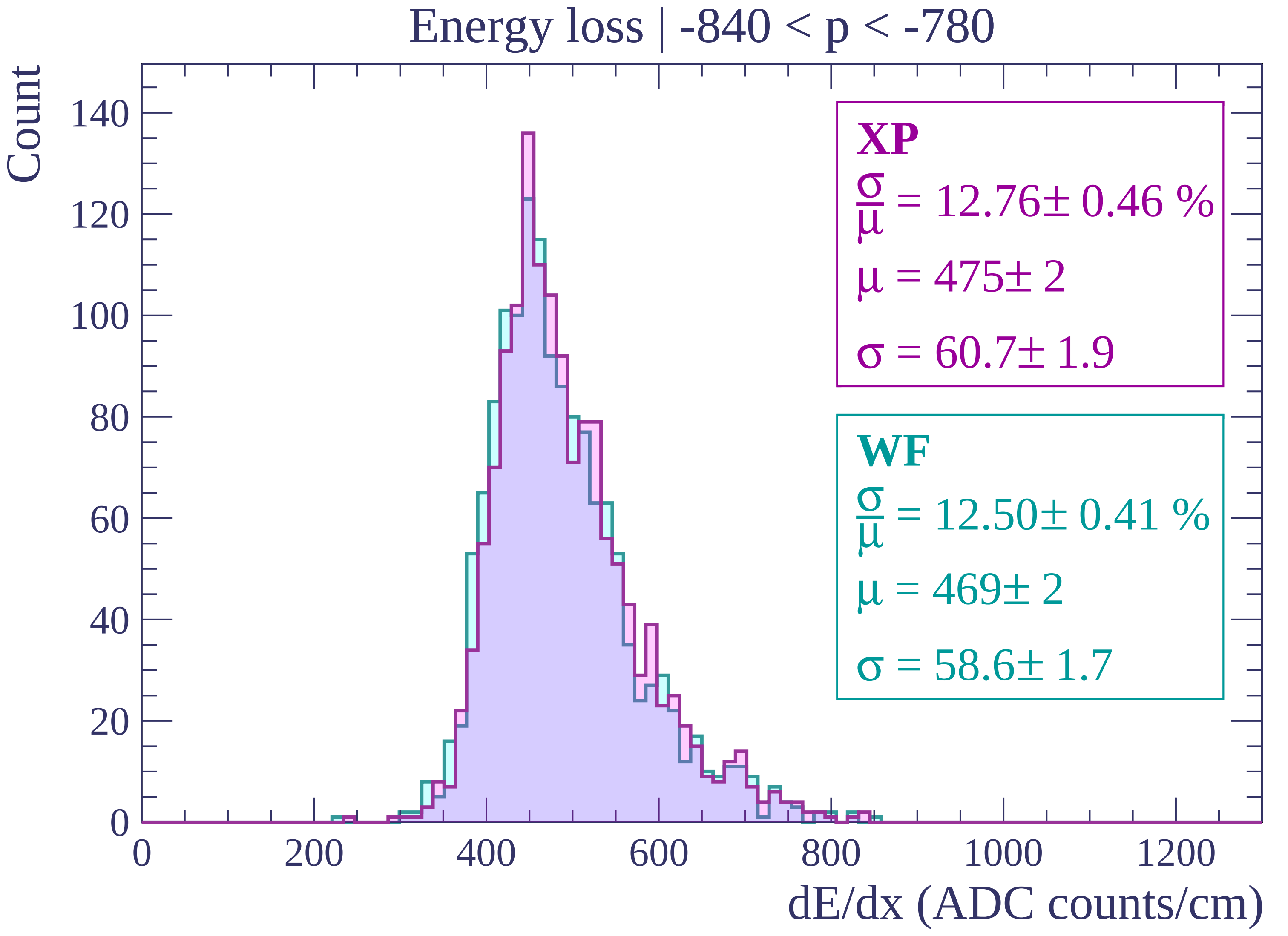

Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.32± 0.58 %
μ = 481± 2
σ = 64.0± 2.5
WF
$\frac{\sigma}{\mu}$ = 13.45± 0.52 %
μ = 476± 2
σ = 64.1± 2.2



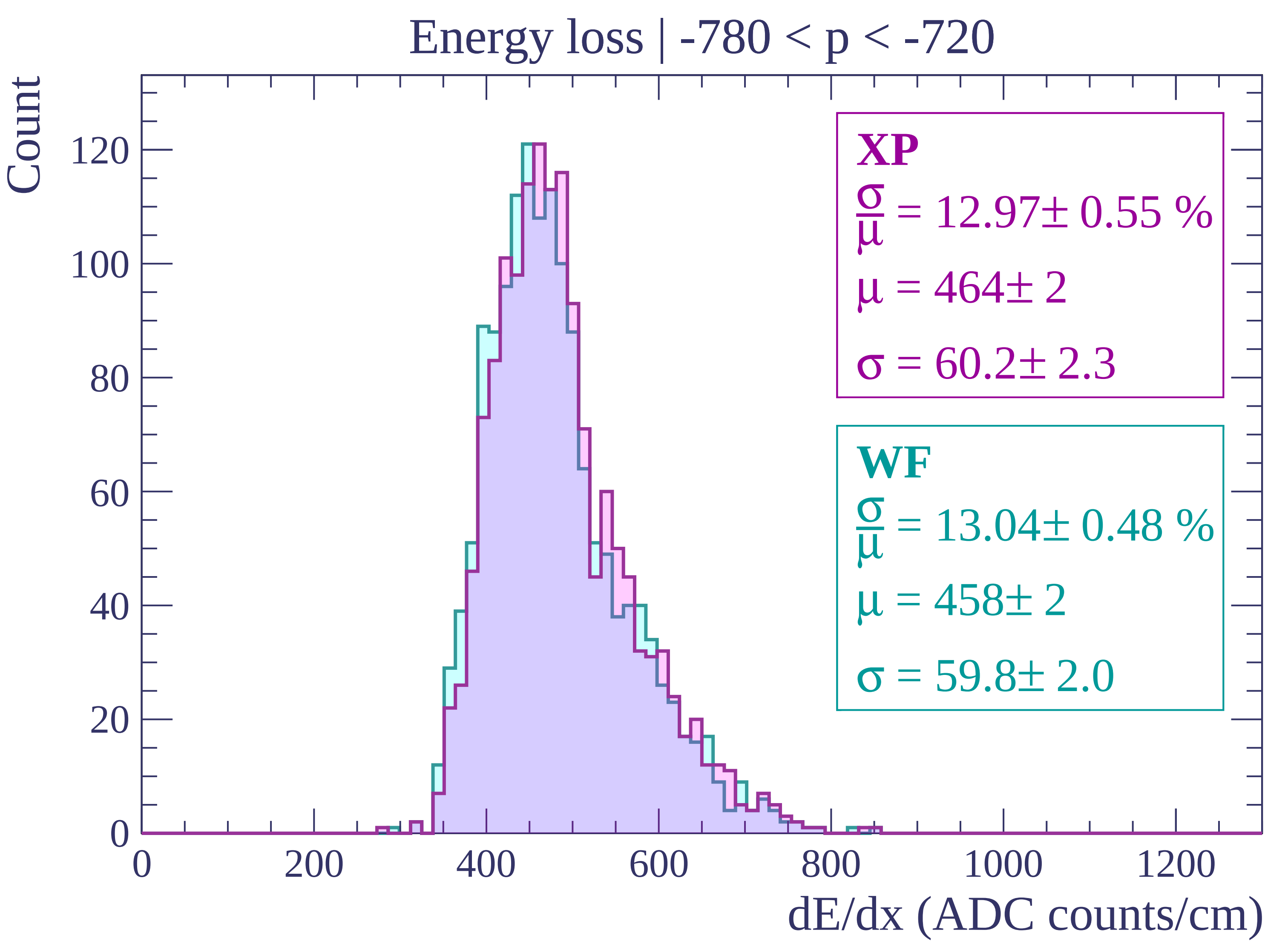
Energy loss | -960 < p < -900
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.59± 0.53 %
μ = 484± 2
σ = 65.8± 2.2
WF
σ/μ = 14.07± 0.51 %
μ = 479± 2
σ = 67.4± 2.1

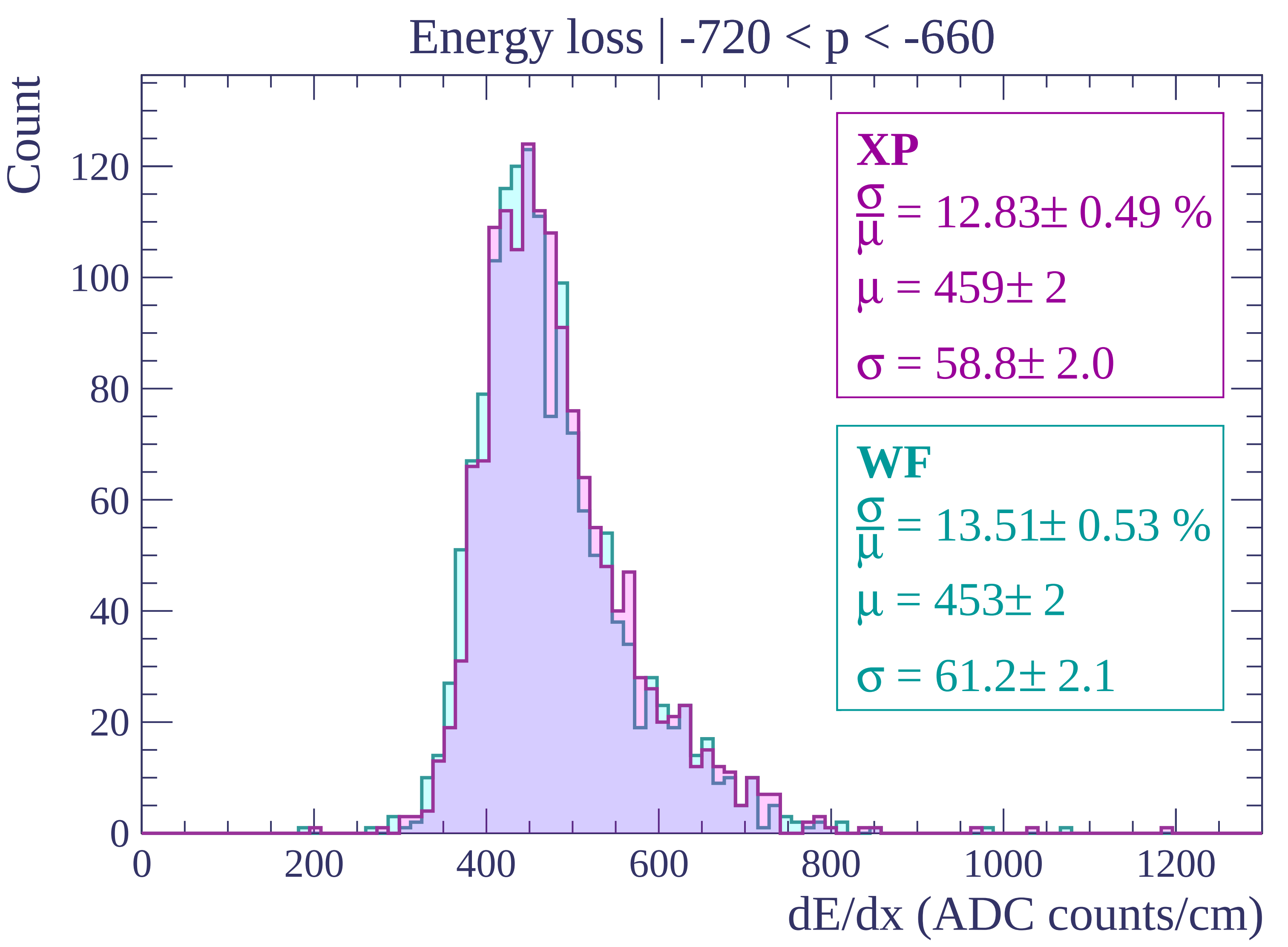


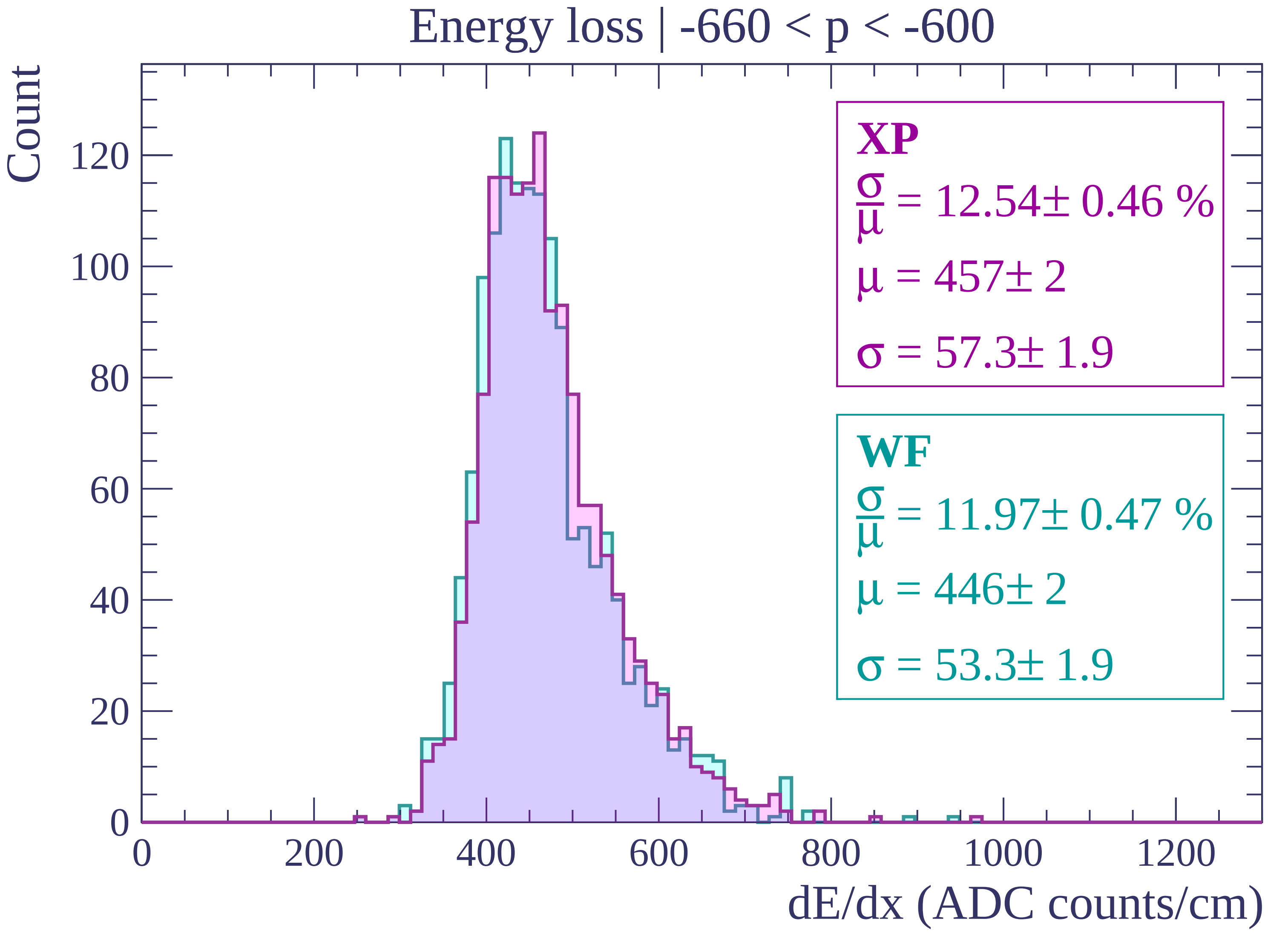
Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 12.36± 0.47 %
μ = 474± 2
σ = 58.6± 2.0
WF
σ/μ = 12.71± 0.53 %
μ = 464± 2
σ = 59.0± 2.2



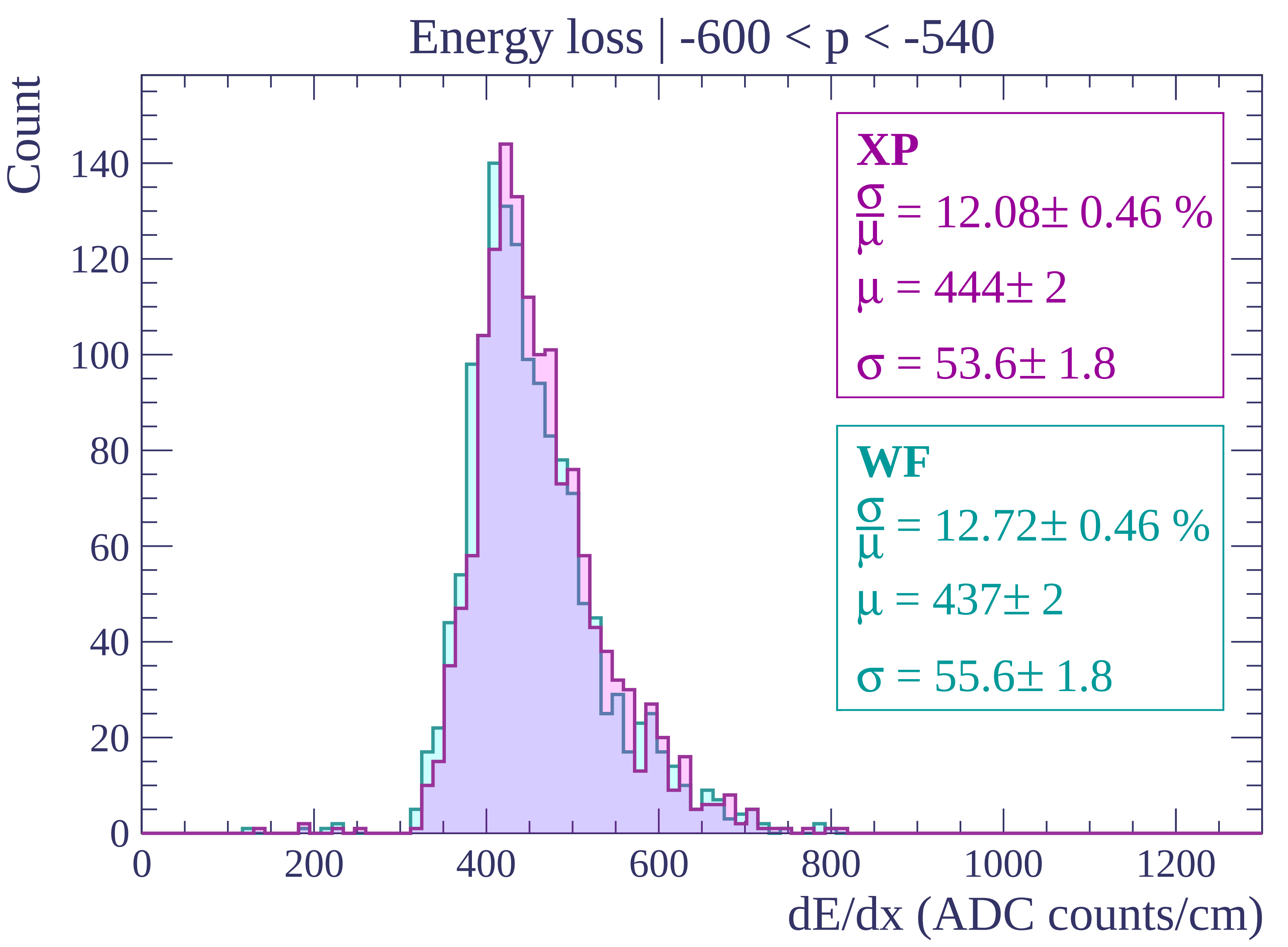
Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.76± 0.46 %
μ = 475± 2
σ = 60.7± 1.9
WF
$\frac{\sigma}{\mu}$ = 12.50± 0.41 %
μ = 469± 2
σ = 58.6± 1.7


Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 12.97± 0.55 %
μ = 464± 2
σ = 60.2± 2.3
WF
σ/μ = 13.04± 0.48 %
μ = 458± 2
σ = 59.8± 2.0

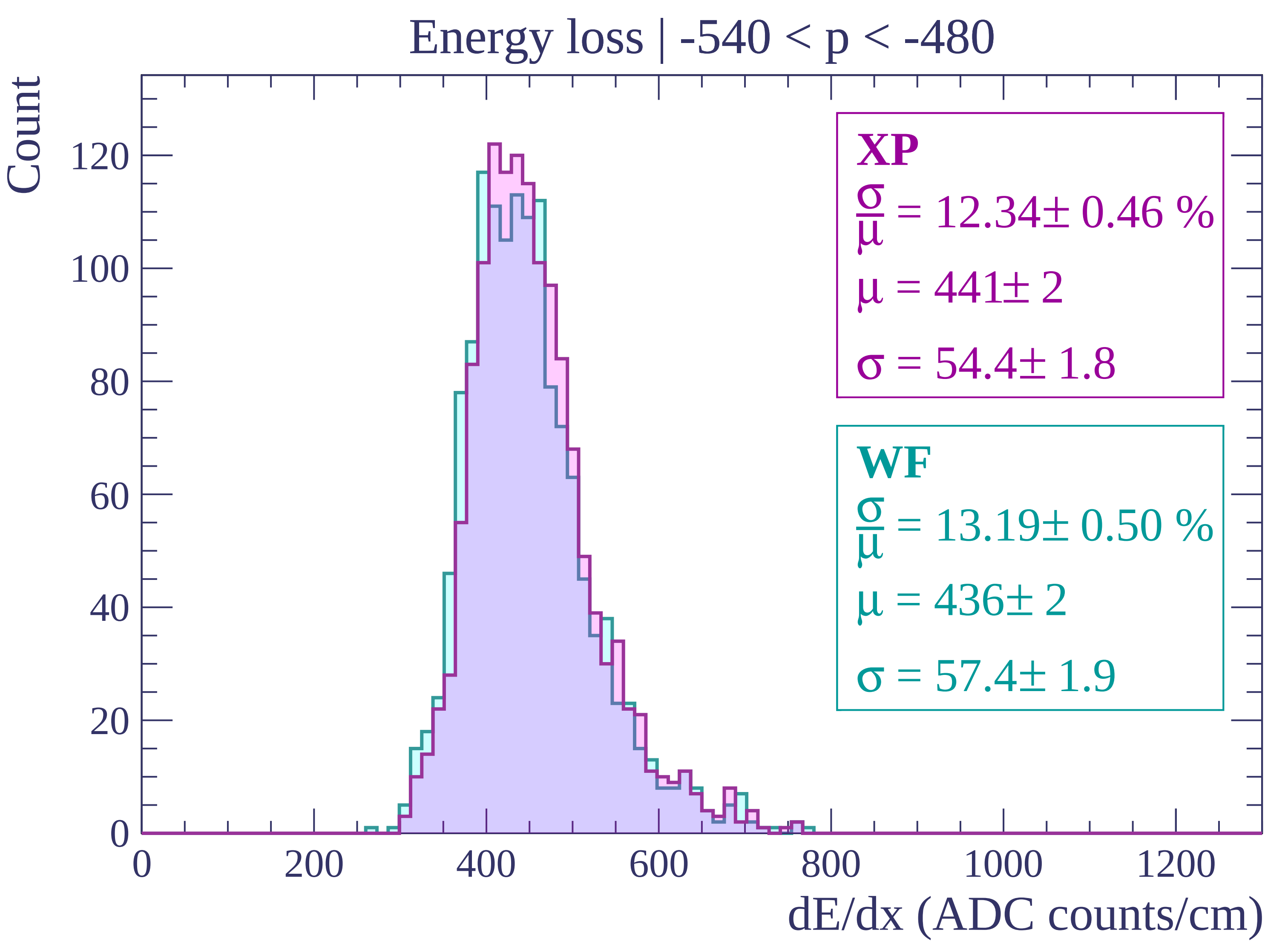

Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.83± 0.49 %
μ = 459± 2
σ = 58.8± 2.0
WF
$\frac{\sigma}{\mu}$ = 13.51± 0.53 %
μ = 453± 2
σ = 61.2± 2.1
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200



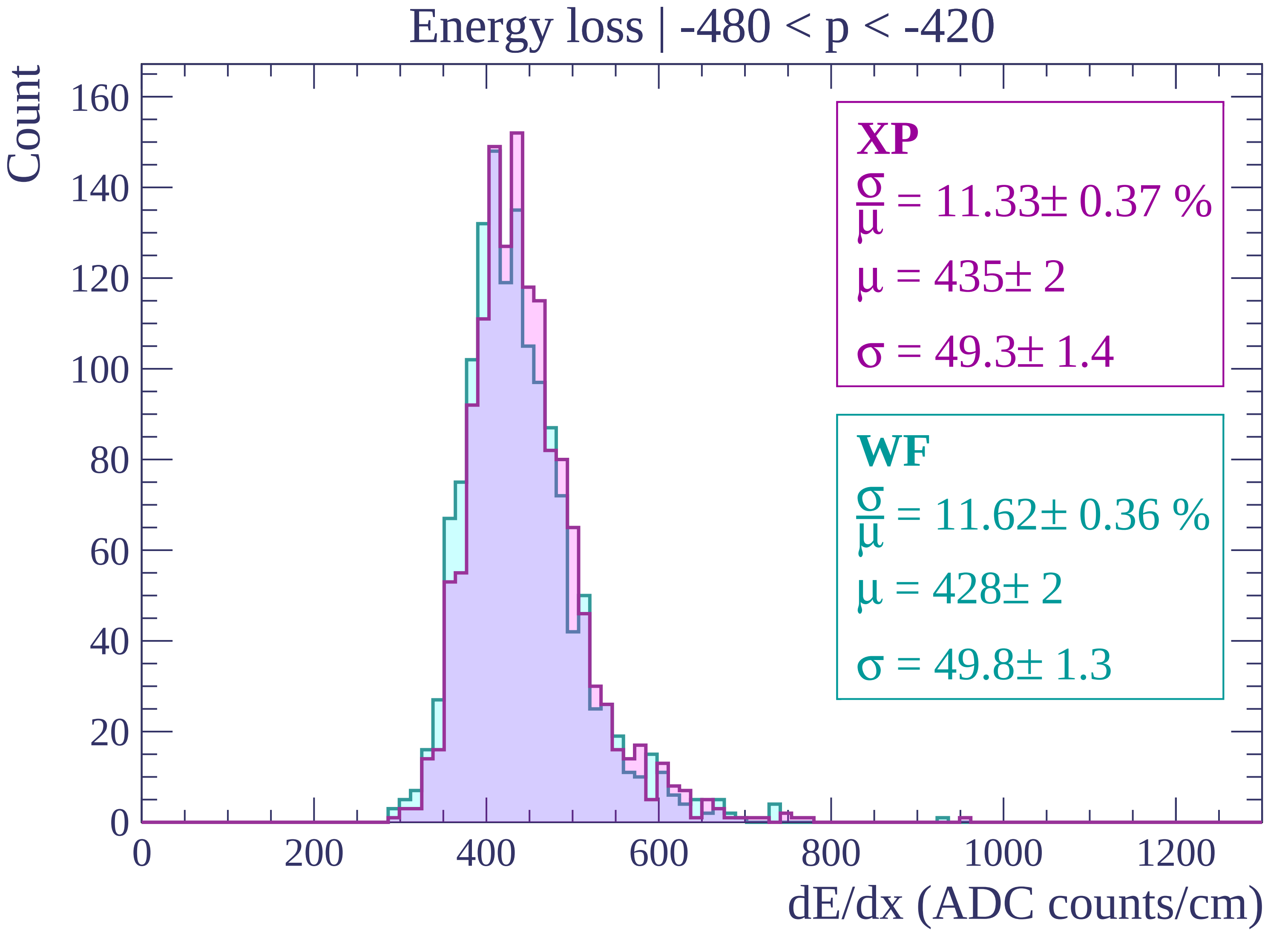
Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.54± 0.46 %
μ = 457± 2
σ = 57.3± 1.9
WF
$\frac{\sigma}{\mu}$ = 11.97± 0.47 %
μ = 446± 2
σ = 53.3± 1.9

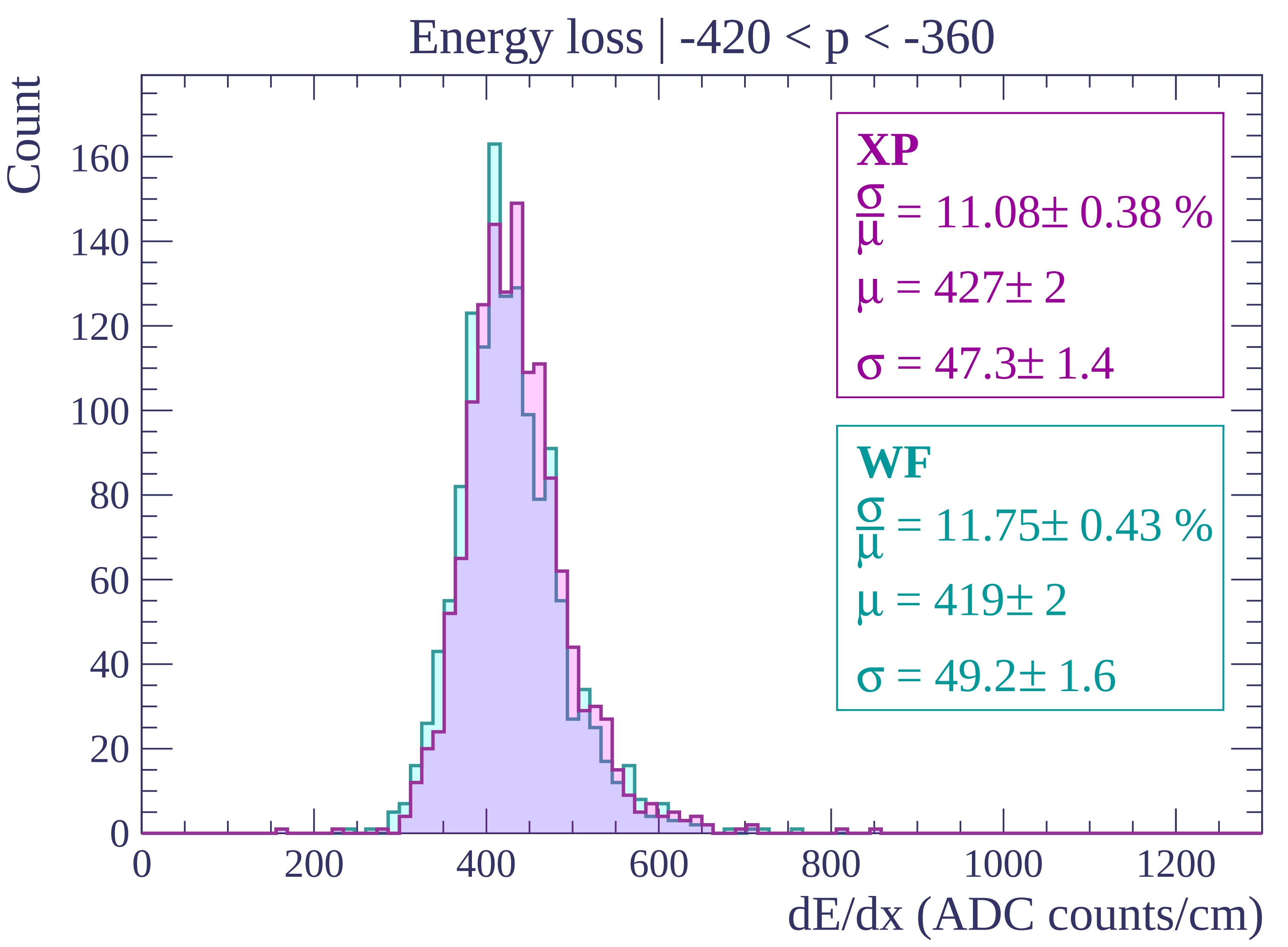


Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 12.08± 0.46 %
μ = 444± 2
σ = 53.6± 1.8
WF
σ/μ = 12.72± 0.46 %
μ = 437± 2
σ = 55.6± 1.8

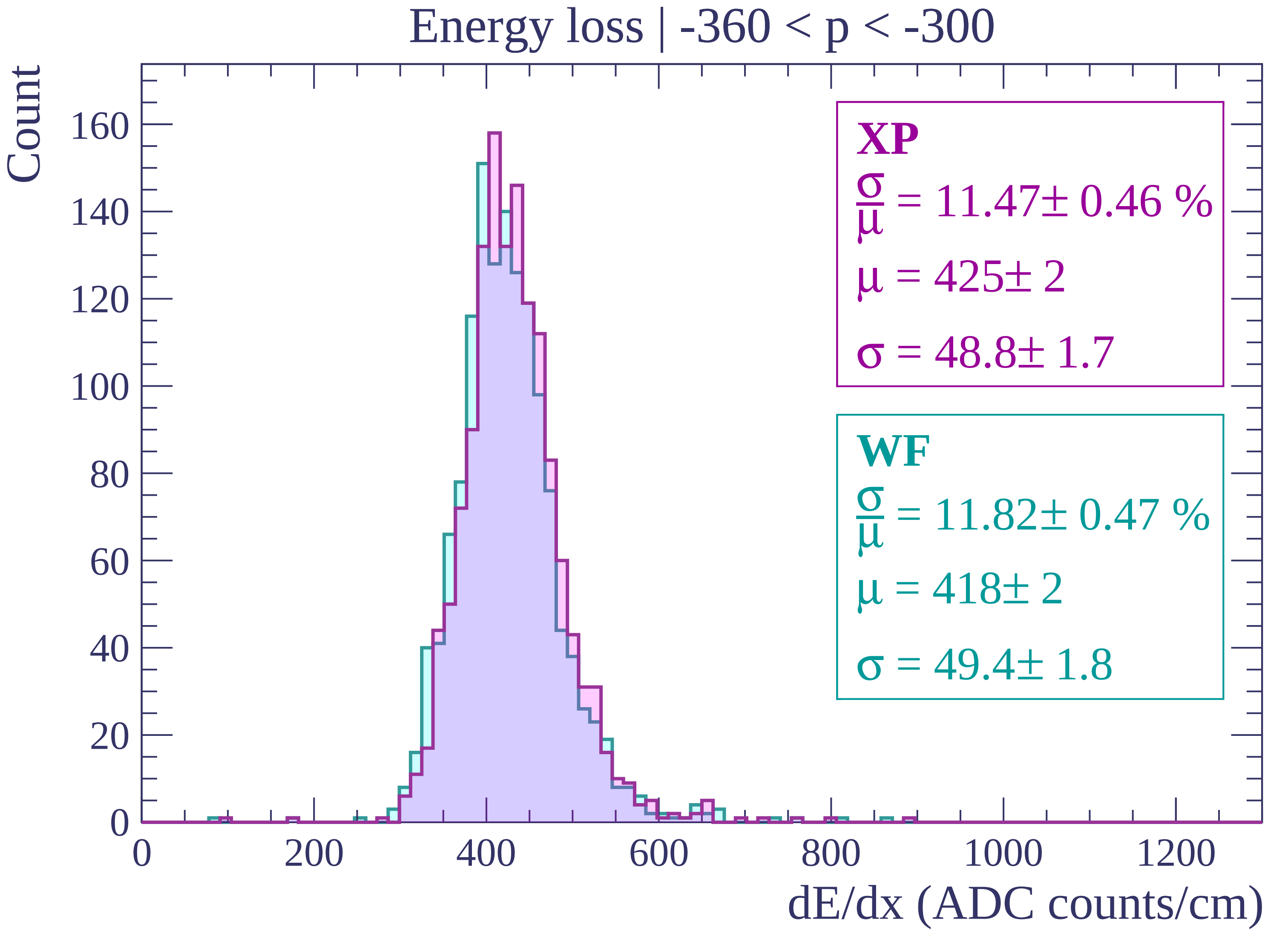


Energy loss | -540 < p < -480
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.34± 0.46 %
μ = 441± 2
σ = 54.4± 1.8
WF
σ/μ = 13.19± 0.50 %
μ = 436± 2
σ = 57.4± 1.9

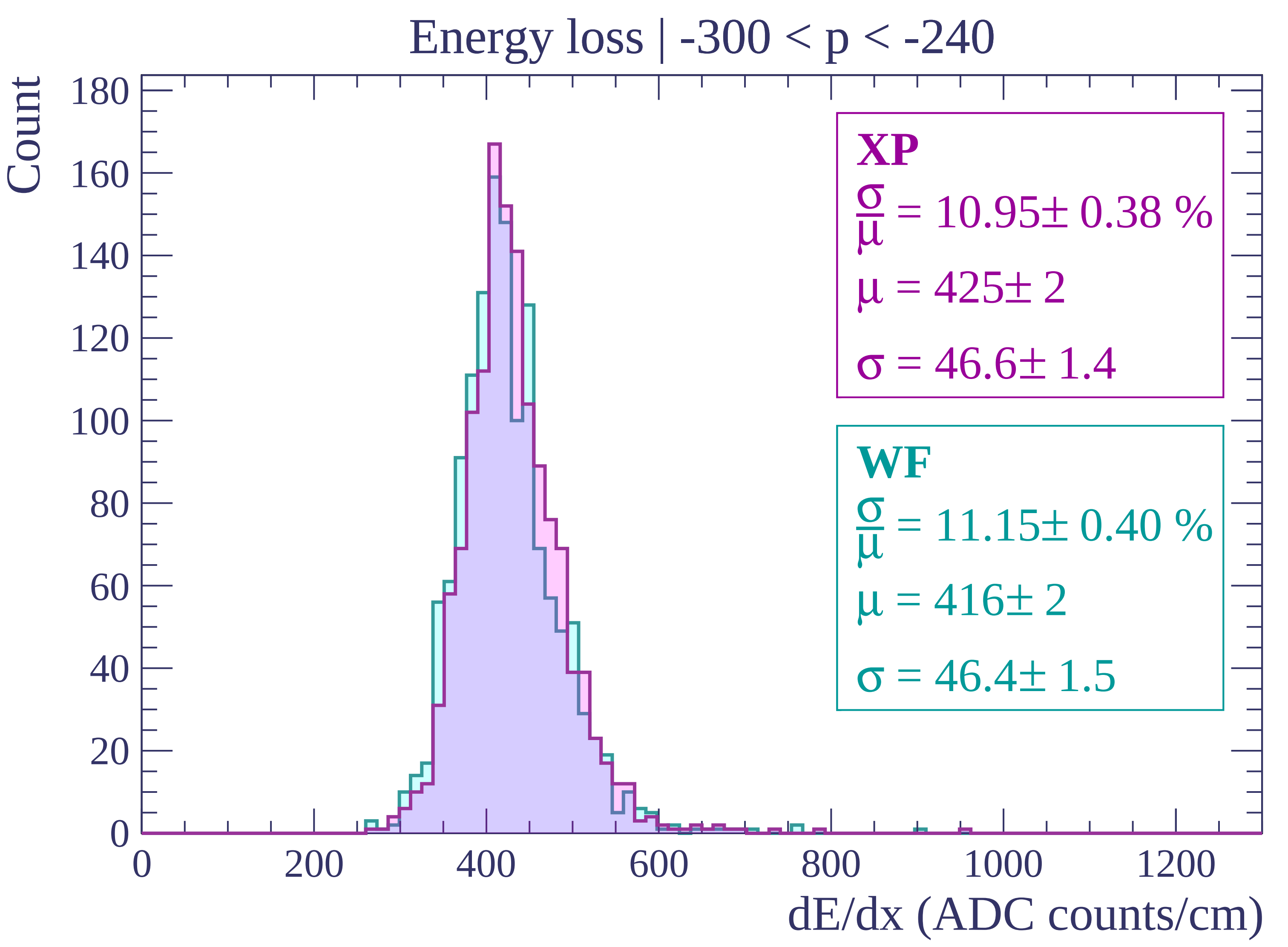

Energy loss | -480 < p < -420
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.33± 0.37 %
μ = 435± 2
σ = 49.3± 1.4
WF
σ/μ = 11.62± 0.36 %
μ = 428± 2
σ = 49.8± 1.3



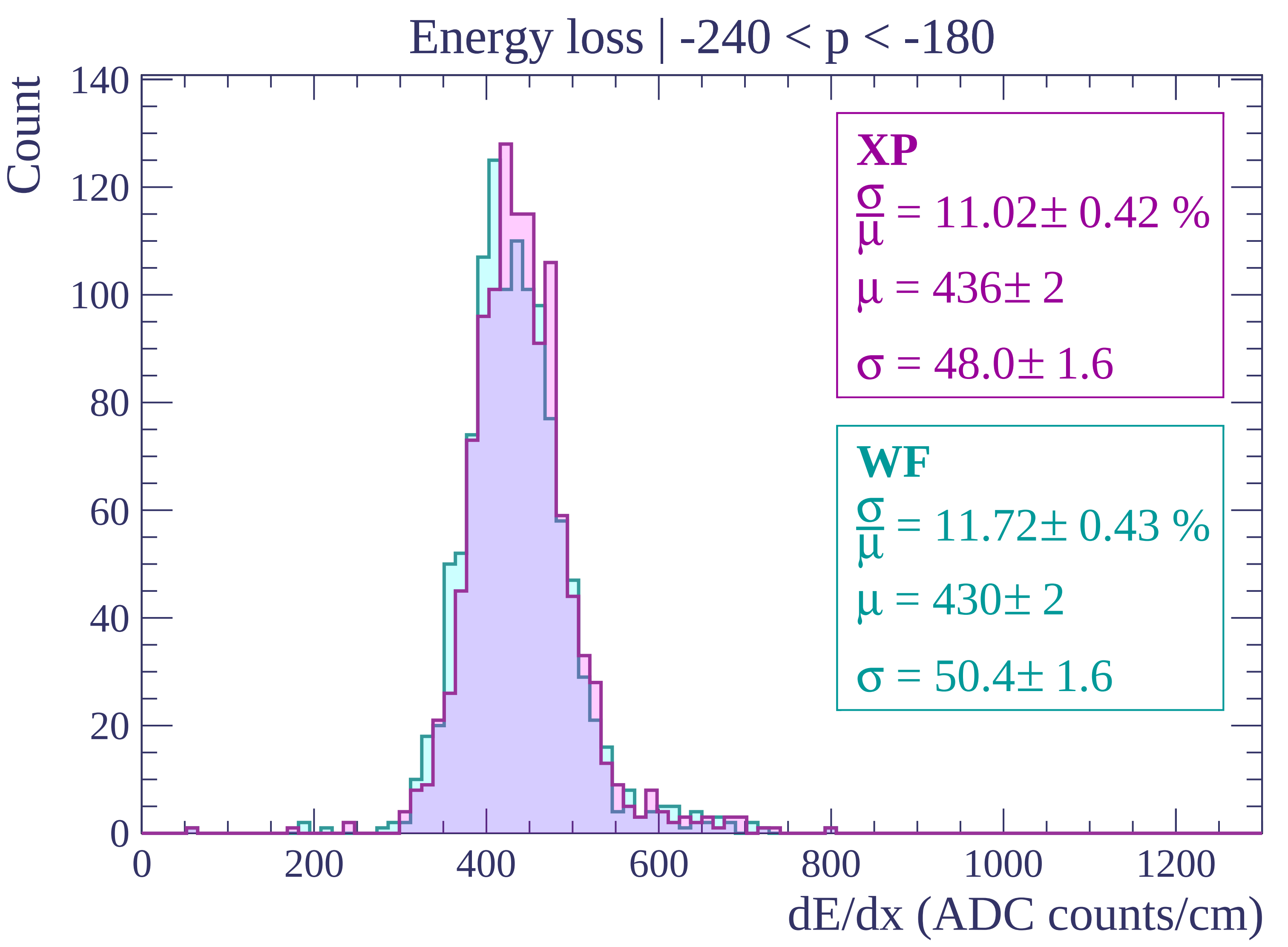
Energy loss | -420 < p < -360
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.08± 0.38 %
μ = 427± 2
σ = 47.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 11.75± 0.43 %
μ = 419± 2
σ = 49.2± 1.6



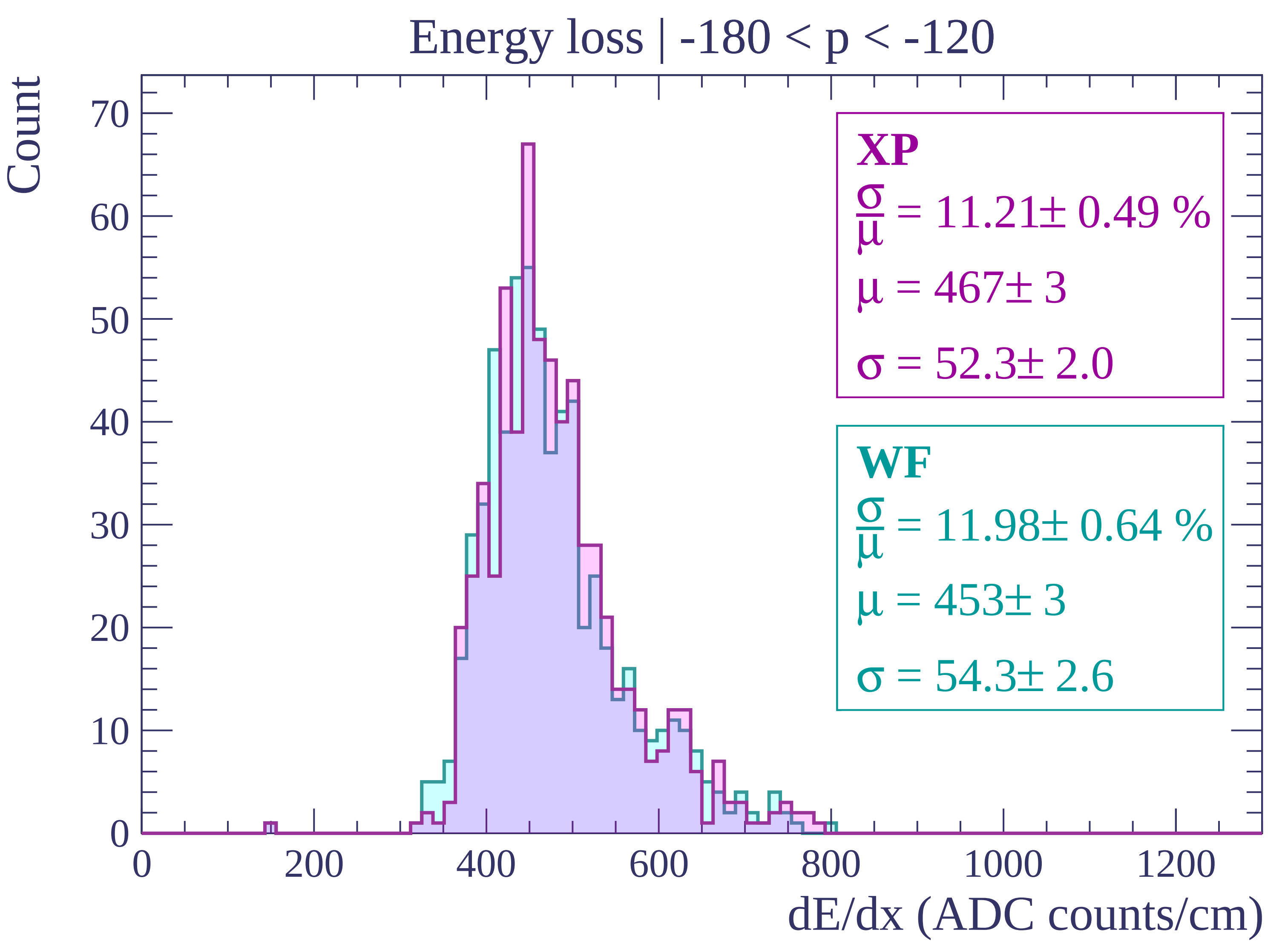
Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 11.47± 0.46 %
μ = 425± 2
σ = 48.8± 1.7
WF
σ/μ = 11.82± 0.47 %
μ = 418± 2
σ = 49.4± 1.8



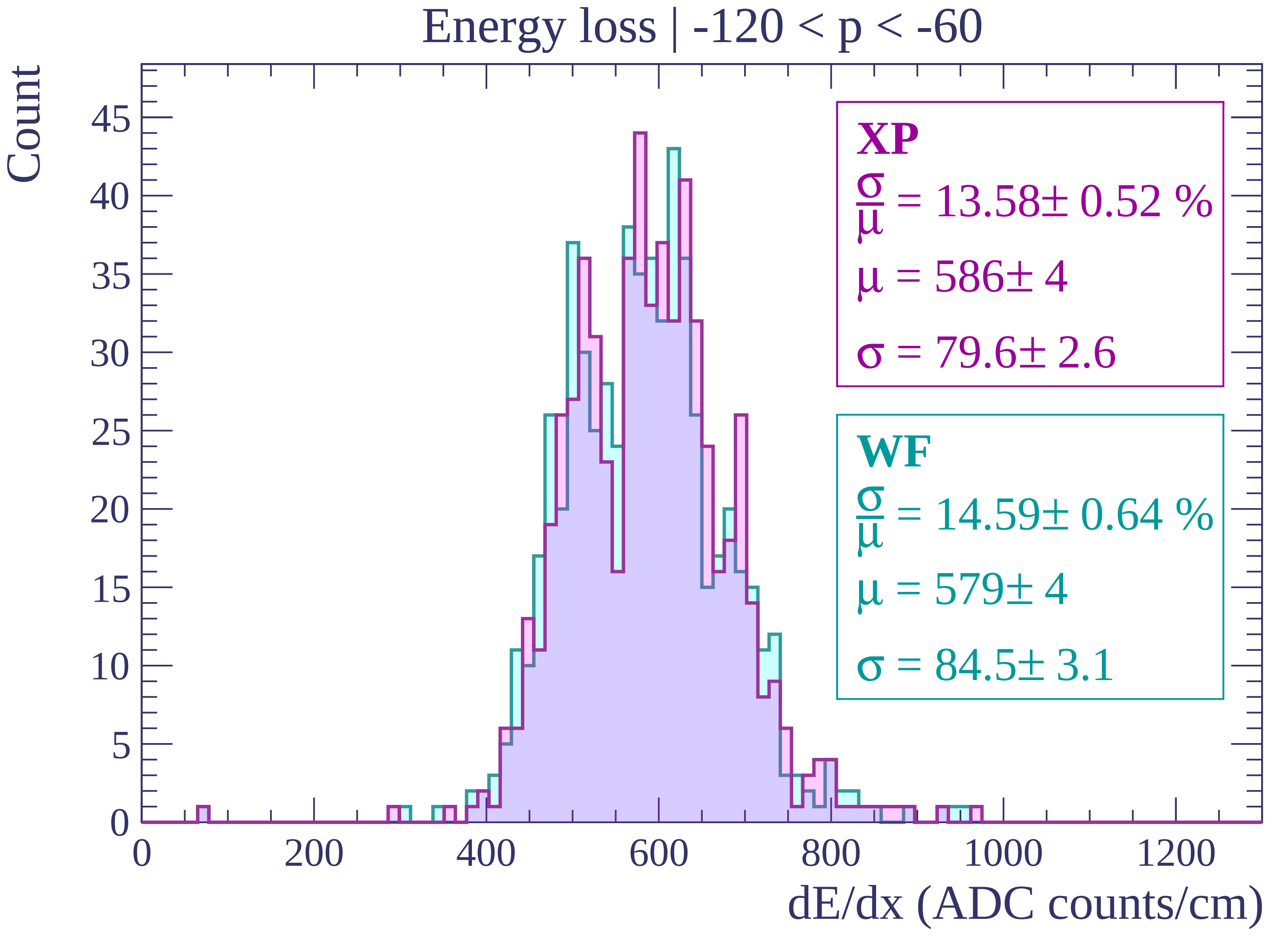
Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.95± 0.38 %
μ = 425± 2
σ = 46.6± 1.4
WF
σ/μ = 11.15± 0.40 %
μ = 416± 2
σ = 46.4± 1.5



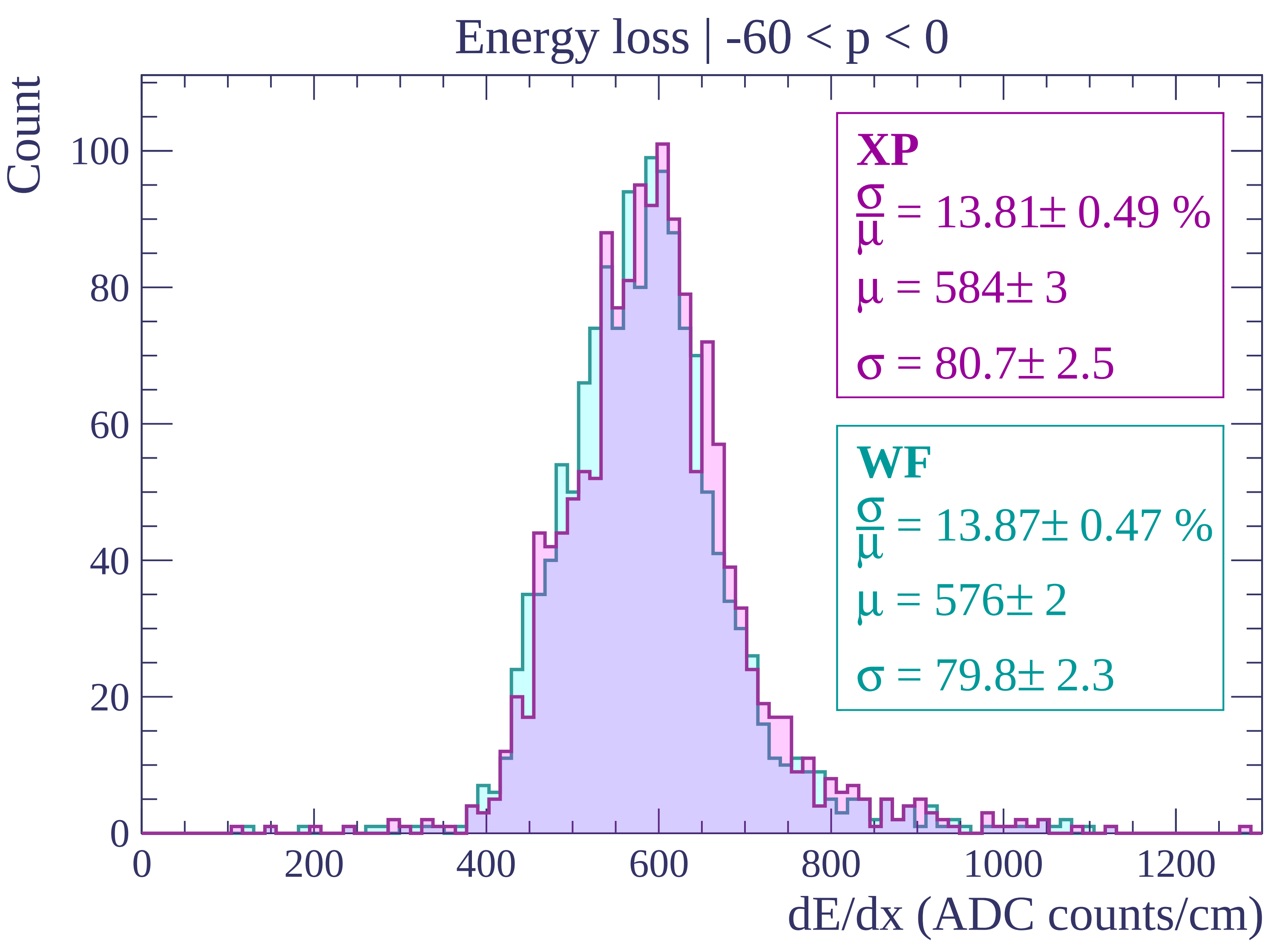
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 11.02± 0.42 %
μ = 436± 2
σ = 48.0± 1.6
WF
σ/μ = 11.72± 0.43 %
μ = 430± 2
σ = 50.4± 1.6



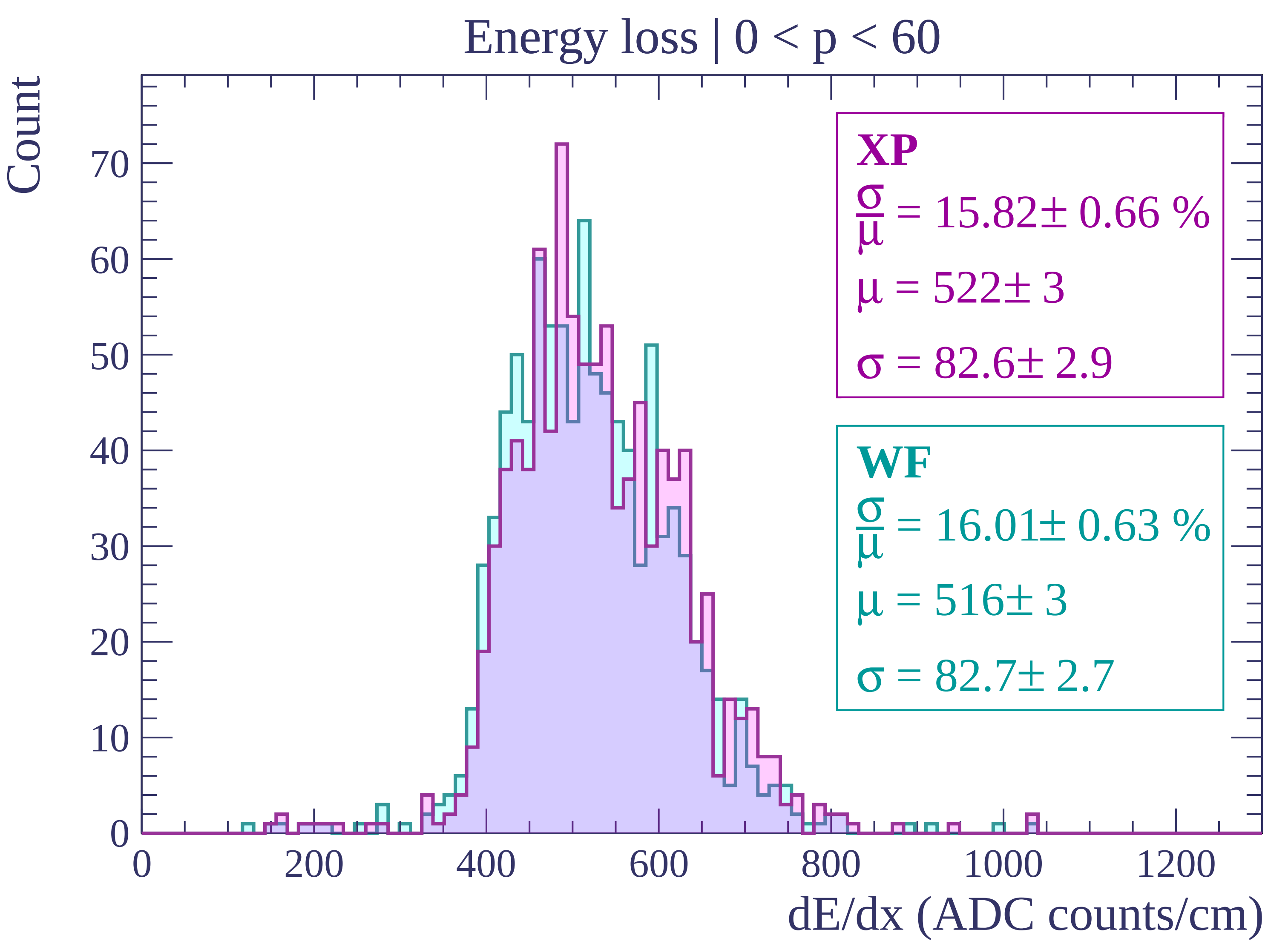
Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP
$\frac{\sigma}{\mu}$ = 11.21± 0.49 %
μ = 467± 3
σ = 52.3± 2.0
WF
$\frac{\sigma}{\mu}$ = 11.98± 0.64 %
μ = 453± 3
σ = 54.3± 2.6

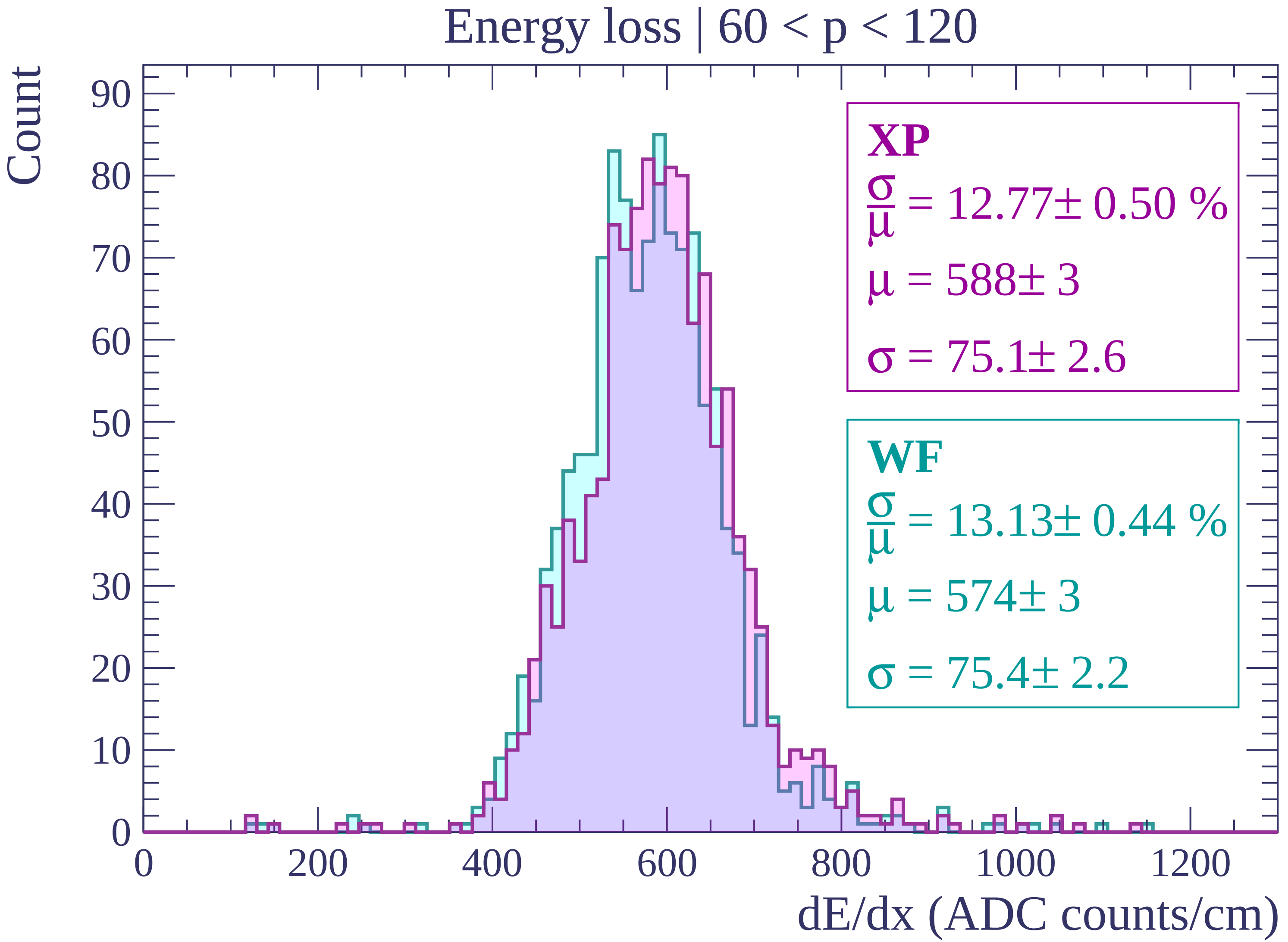


Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 13.58± 0.52 %
μ = 586± 4
σ = 79.6± 2.6
WF
σ/μ = 14.59± 0.64 %
μ = 579± 4
σ = 84.5± 3.1

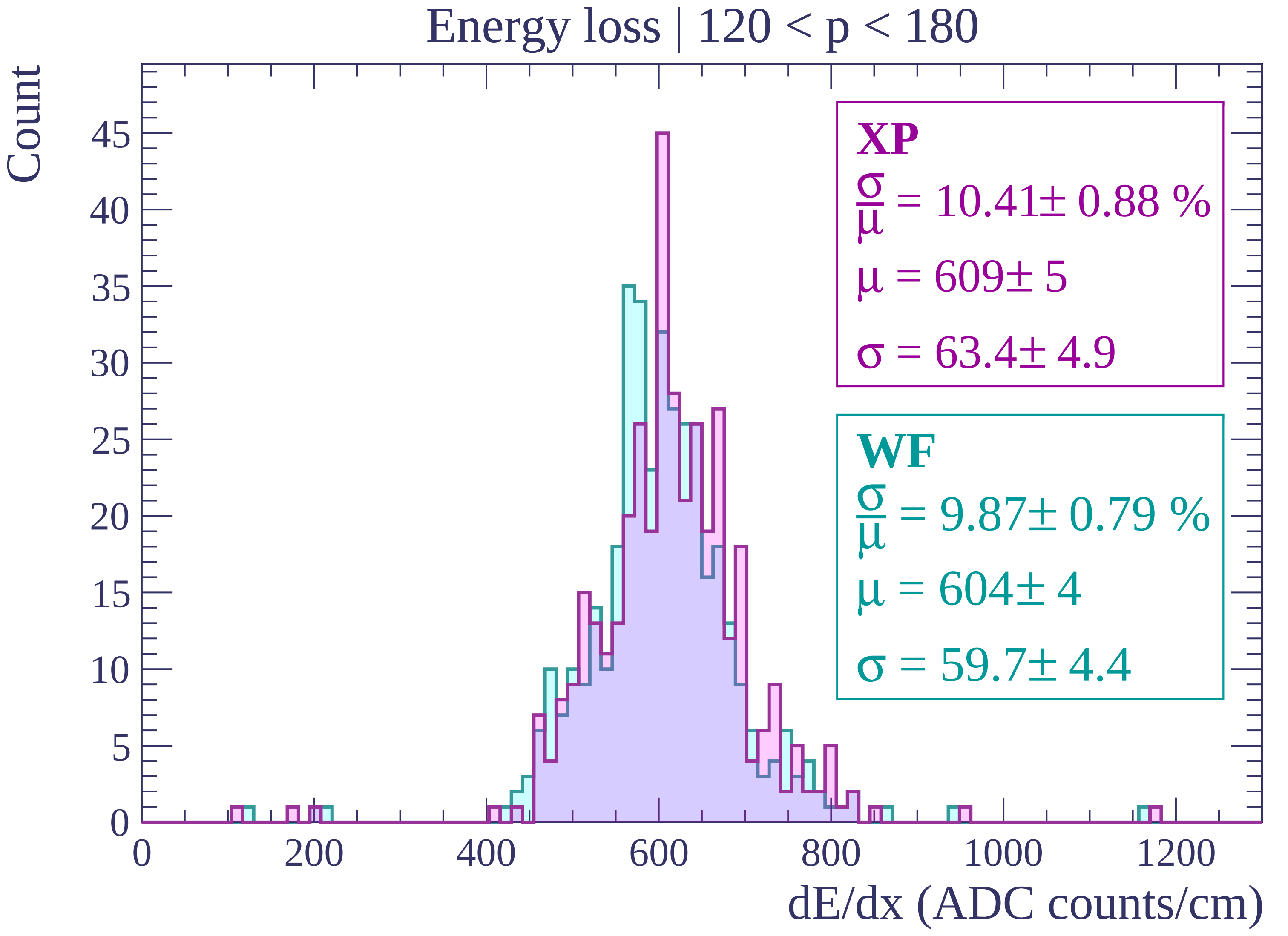


Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.81± 0.49 %
μ = 584± 3
σ = 80.7± 2.5
WF
σ/μ = 13.87± 0.47 %
μ = 576± 2
σ = 79.8± 2.3

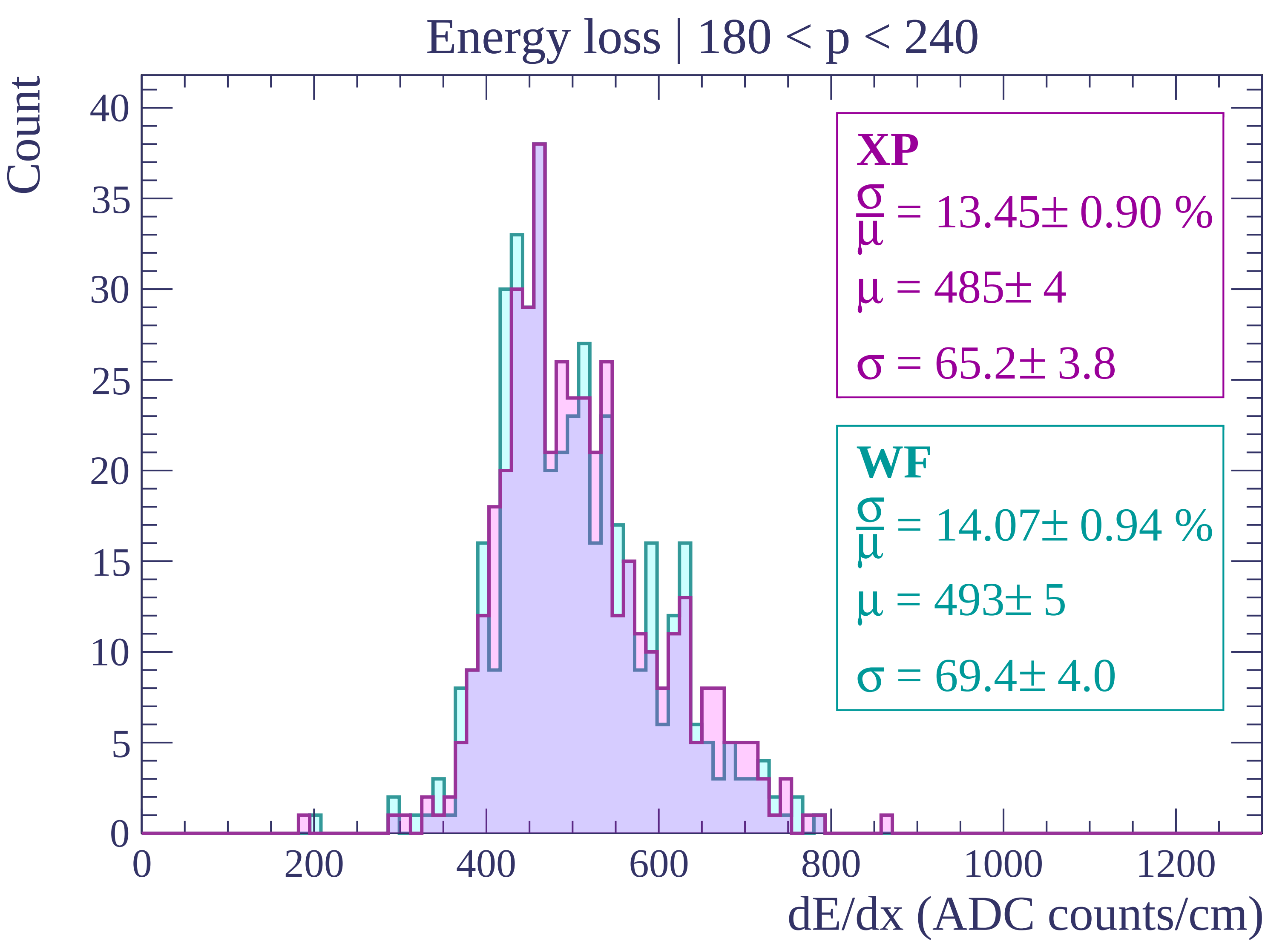

Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.82± 0.66 %
μ = 522± 3
σ = 82.6± 2.9
WF
σ/μ = 16.01± 0.63 %
μ = 516± 3
σ = 82.7± 2.7
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200



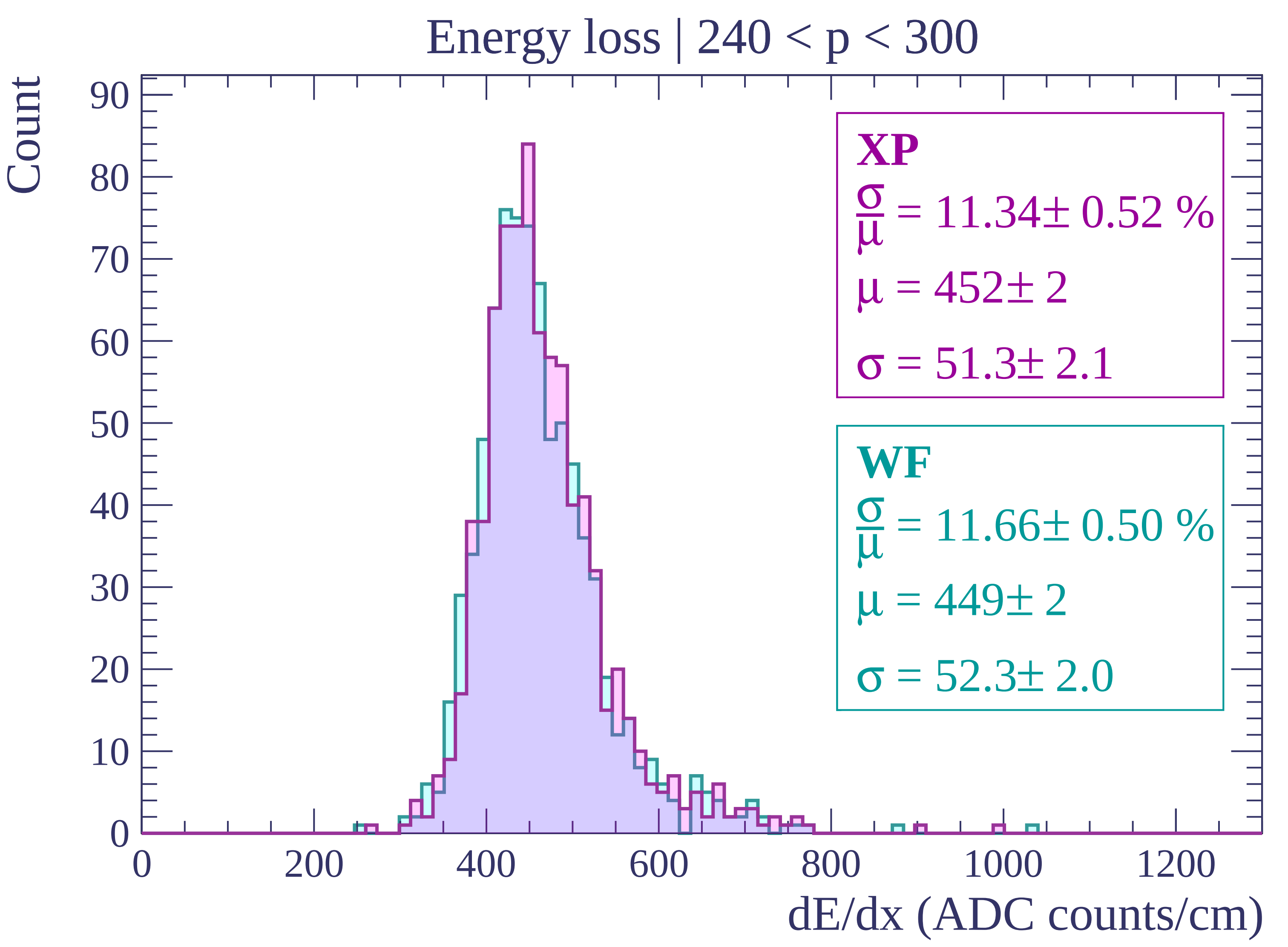
Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP
σ/μ = 12.77± 0.50 %
μ = 588± 3
σ = 75.1± 2.6
WF
σ/μ = 13.13± 0.44 %
μ = 574± 3
σ = 75.4± 2.2



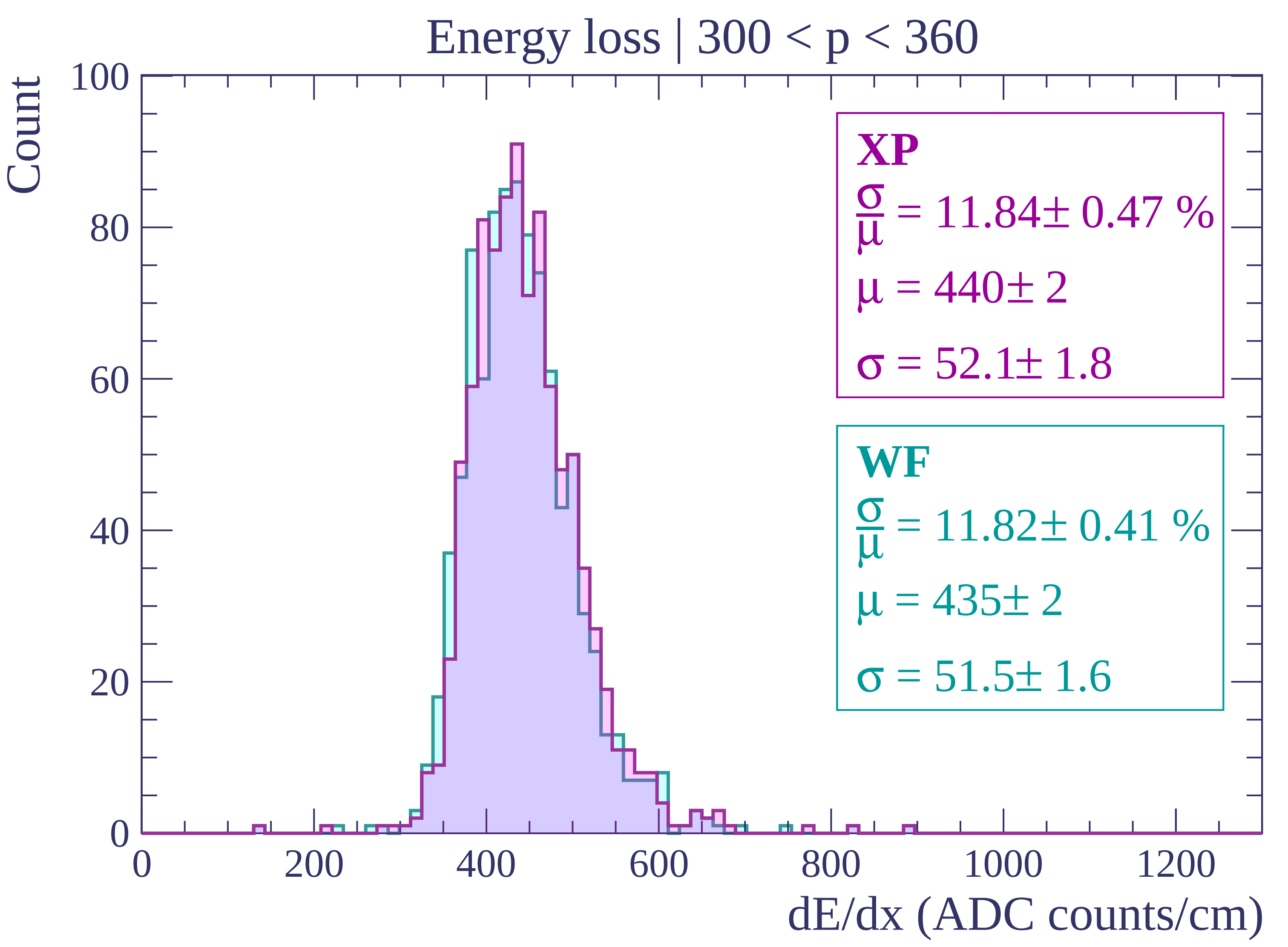
Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.41± 0.88 %
μ = 609± 5
σ = 63.4± 4.9
WF
$\frac{\sigma}{\mu}$ = 9.87± 0.79 %
μ = 604± 4
σ = 59.7± 4.4



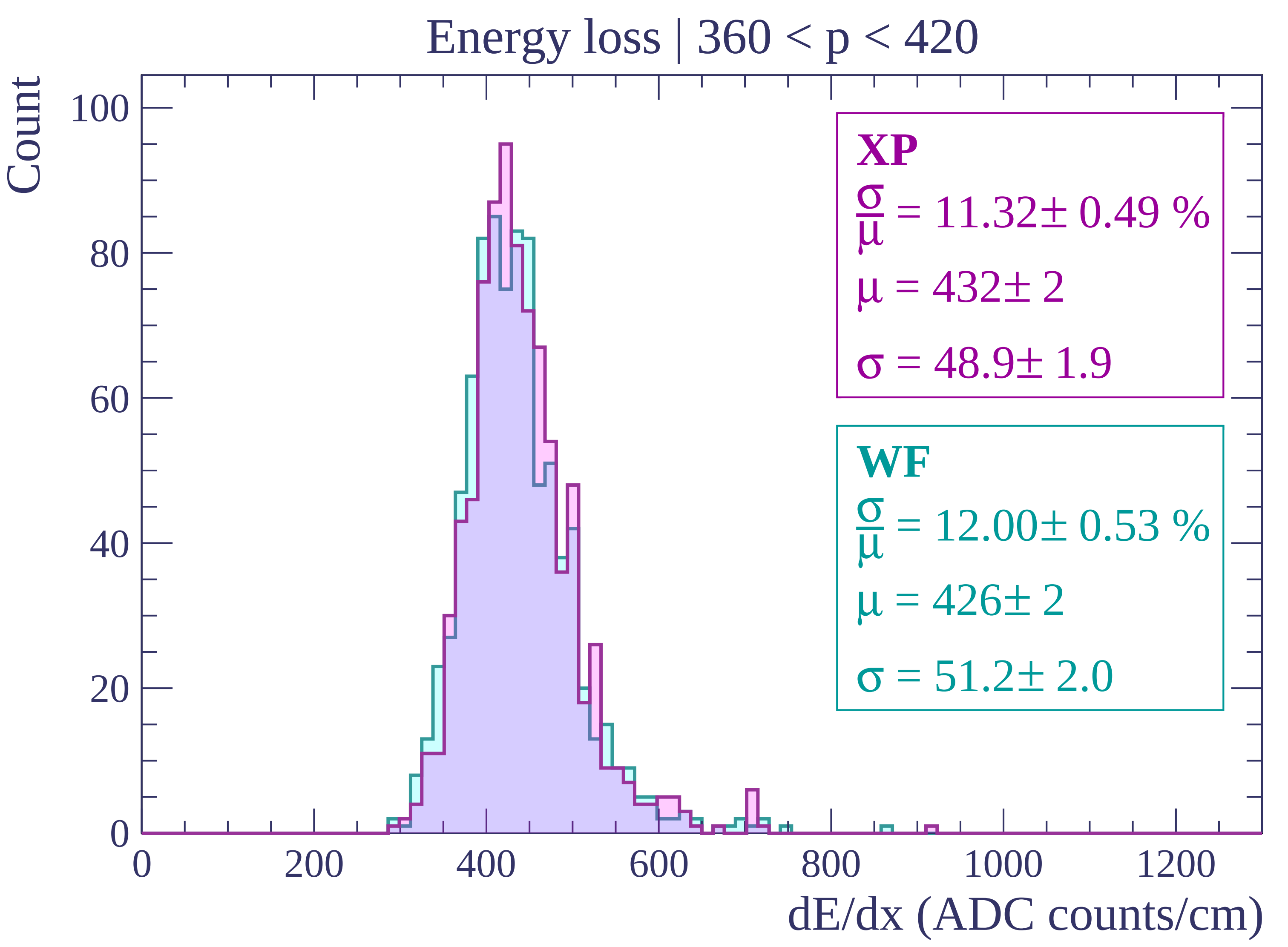
Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
σ/μ = 13.45± 0.90 %
μ = 485± 4
σ = 65.2± 3.8
WF
σ/μ = 14.07± 0.94 %
μ = 493± 5
σ = 69.4± 4.0

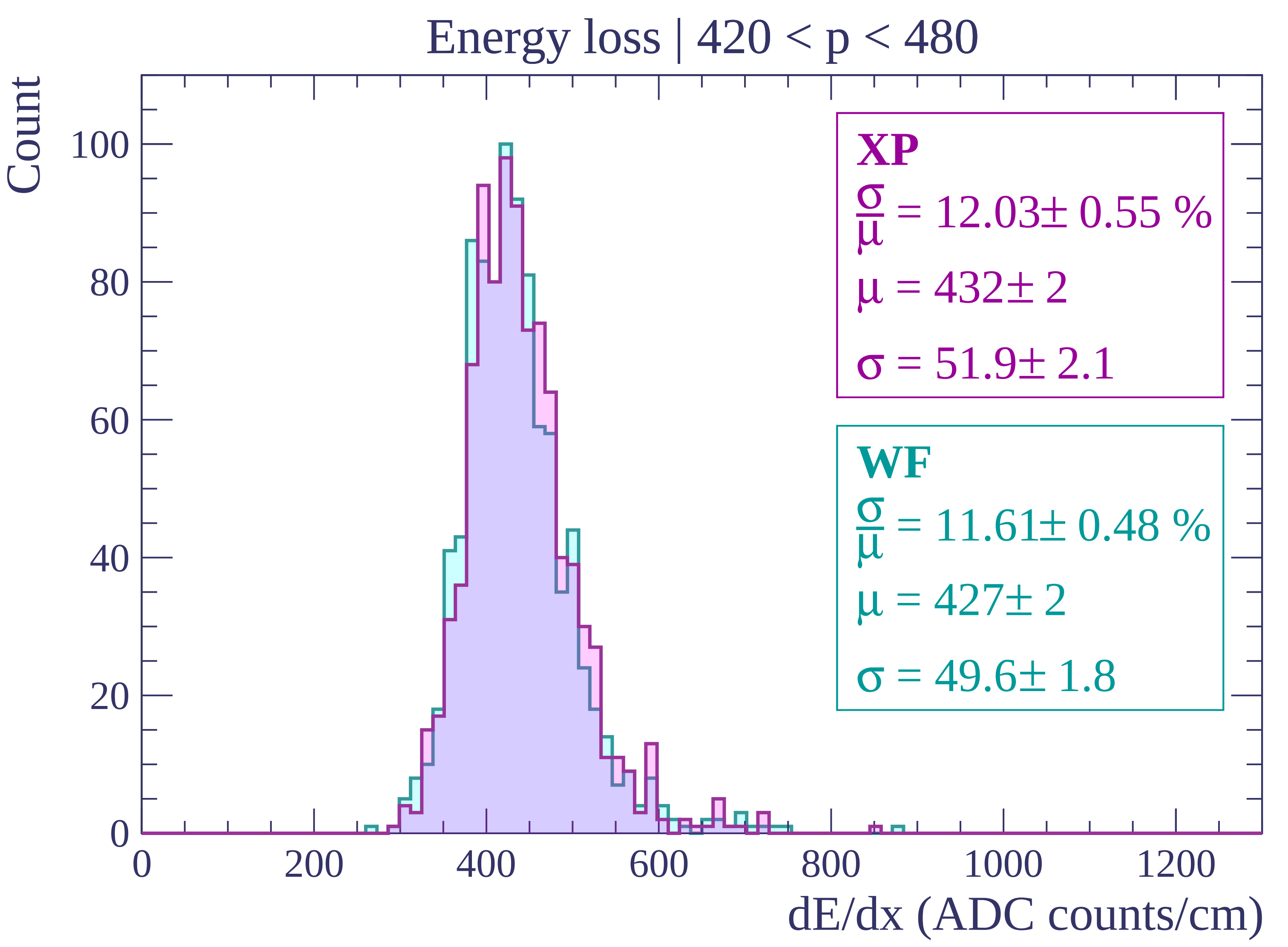


Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 11.34± 0.52 %
μ = 452± 2
σ = 51.3± 2.1
WF
σ/μ = 11.66± 0.50 %
μ = 449± 2
σ = 52.3± 2.0

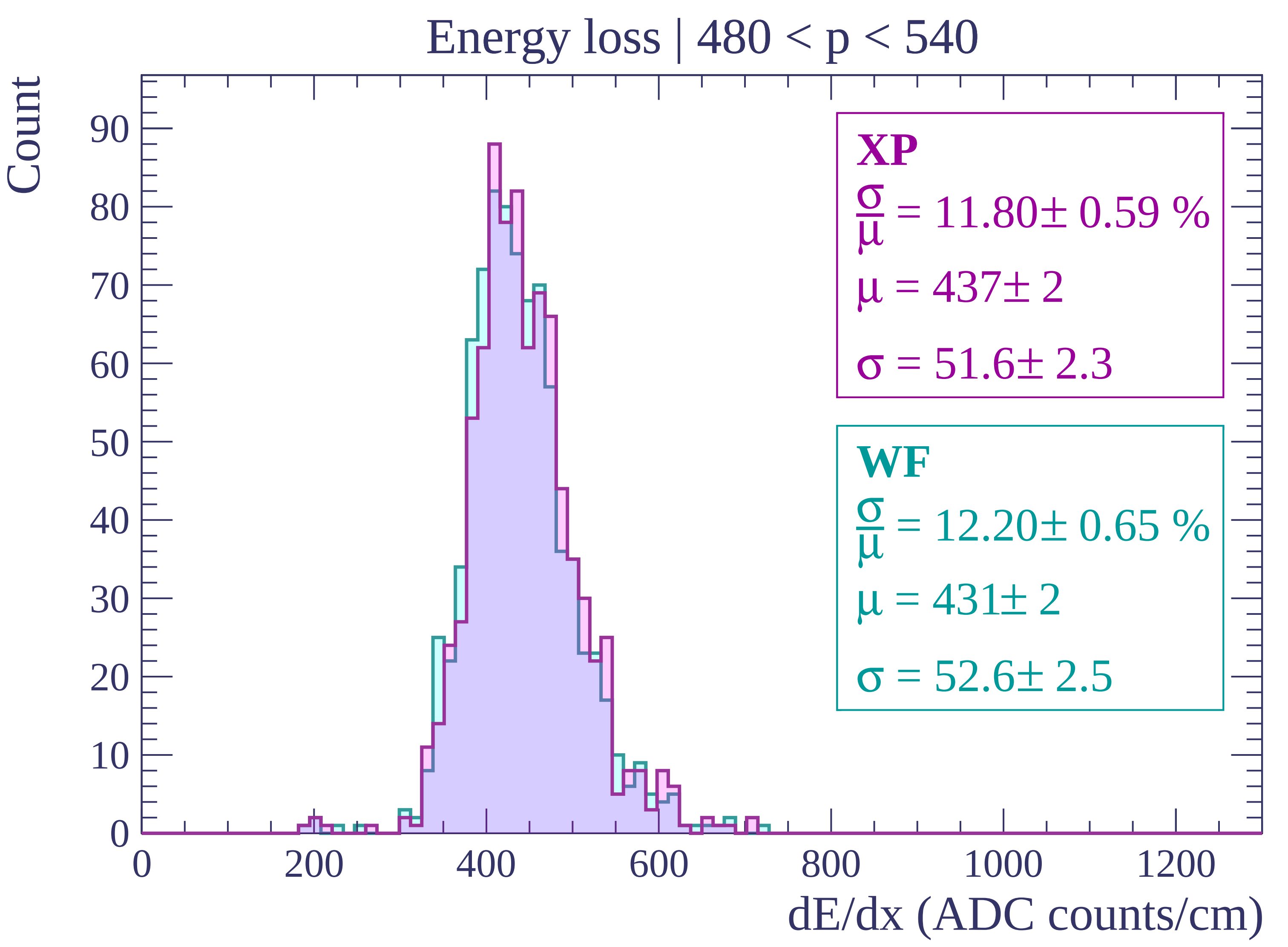


Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 11.84± 0.47 %
μ = 440± 2
σ = 52.1± 1.8
WF
σ/μ = 11.82± 0.41 %
μ = 435± 2
σ = 51.5± 1.6

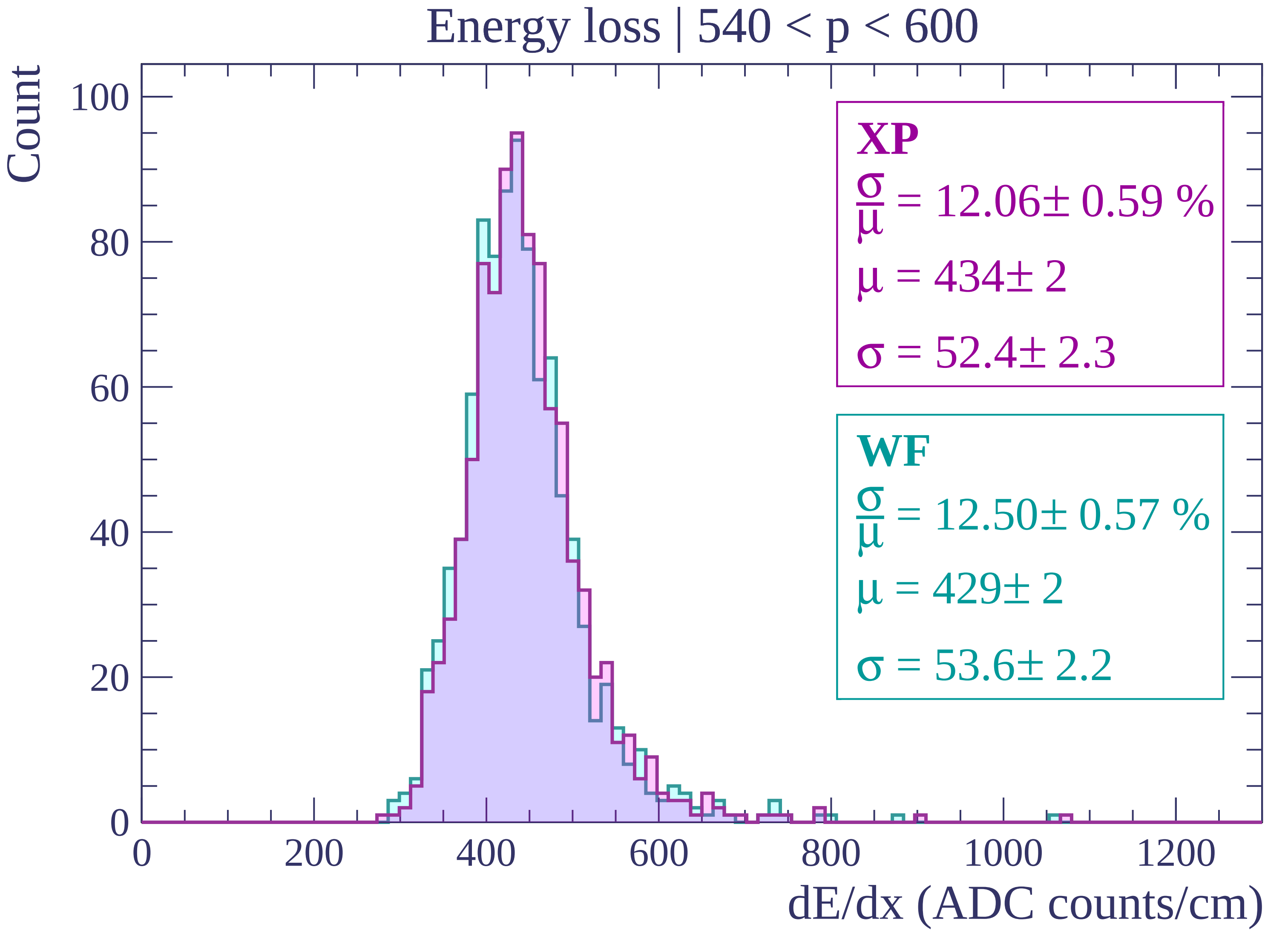

Energy loss | 360 < p < 420
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.32± 0.49 %
μ = 432± 2
σ = 48.9± 1.9
WF
$\frac{\sigma}{\mu}$ = 12.00± 0.53 %
μ = 426± 2
σ = 51.2± 2.0



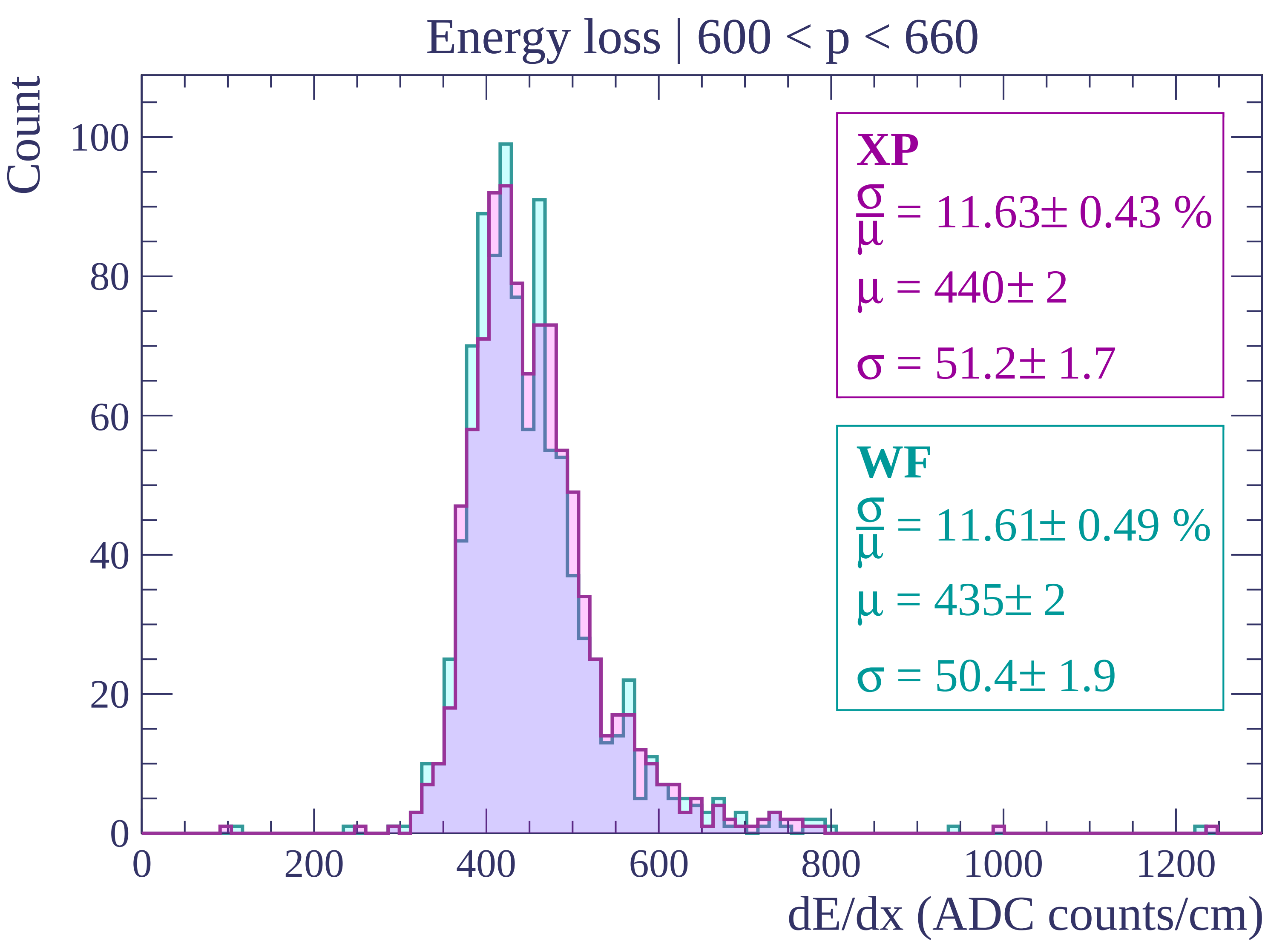
Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 12.03± 0.55 %
μ = 432± 2
σ = 51.9± 2.1
WF
σ/μ = 11.61± 0.48 %
μ = 427± 2
σ = 49.6± 1.8



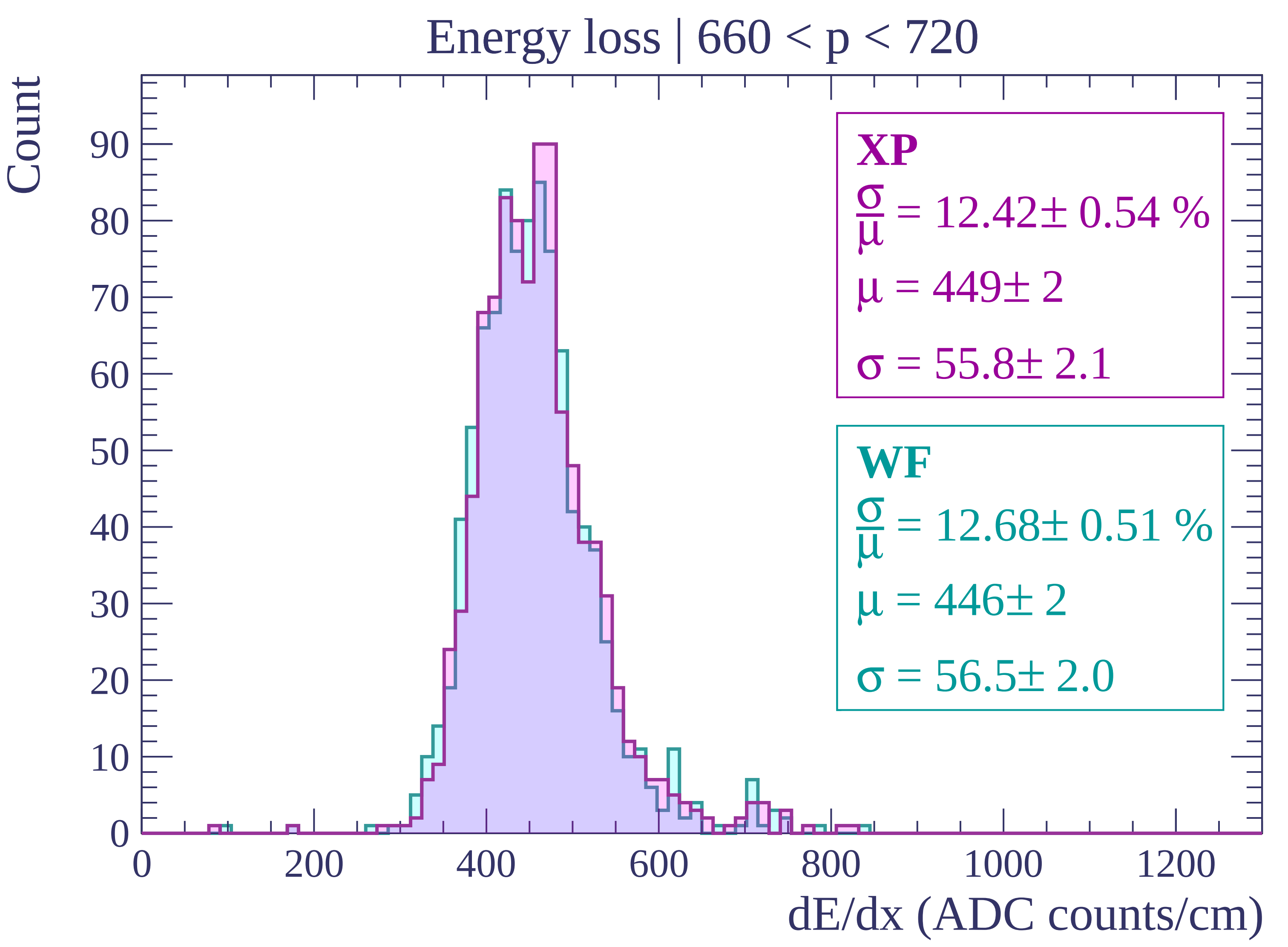
Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP
σ/μ = 11.80± 0.59 %
μ = 437± 2
σ = 51.6± 2.3
WF
σ/μ = 12.20± 0.65 %
μ = 431± 2
σ = 52.6± 2.5

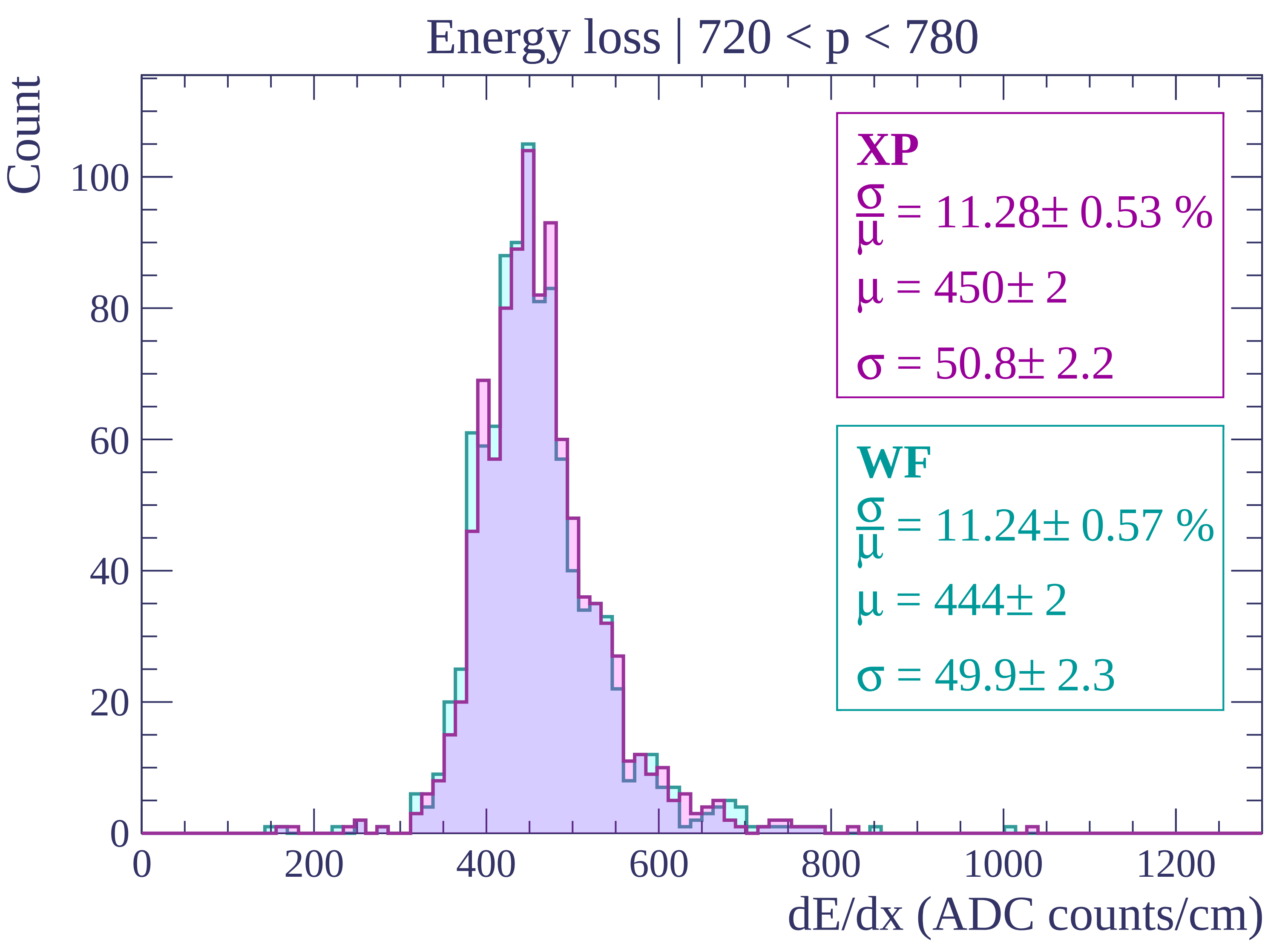


Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.06± 0.59 %
μ = 434± 2
σ = 52.4± 2.3
WF
$\frac{\sigma}{\mu}$ = 12.50± 0.57 %
μ = 429± 2
σ = 53.6± 2.2

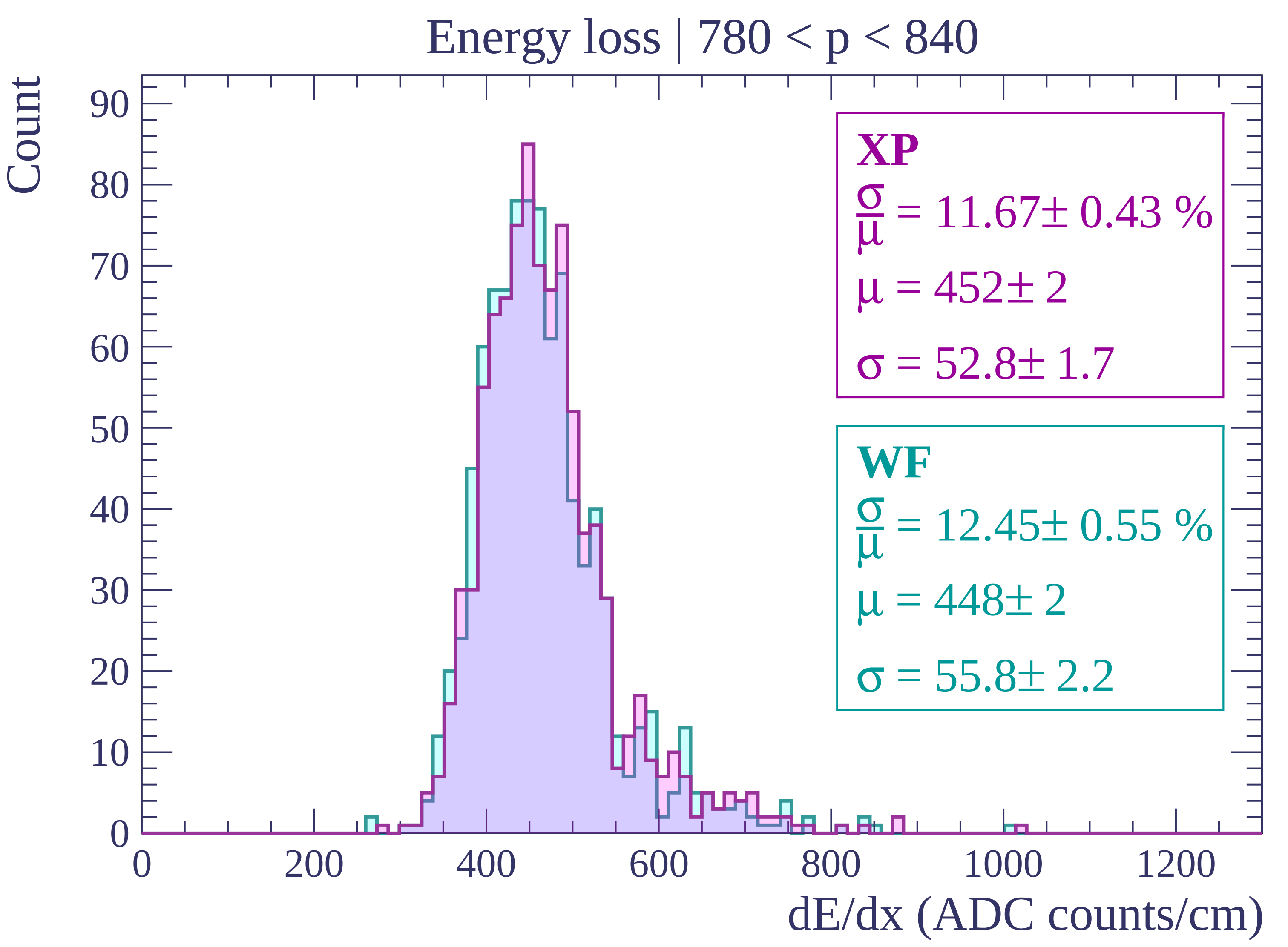


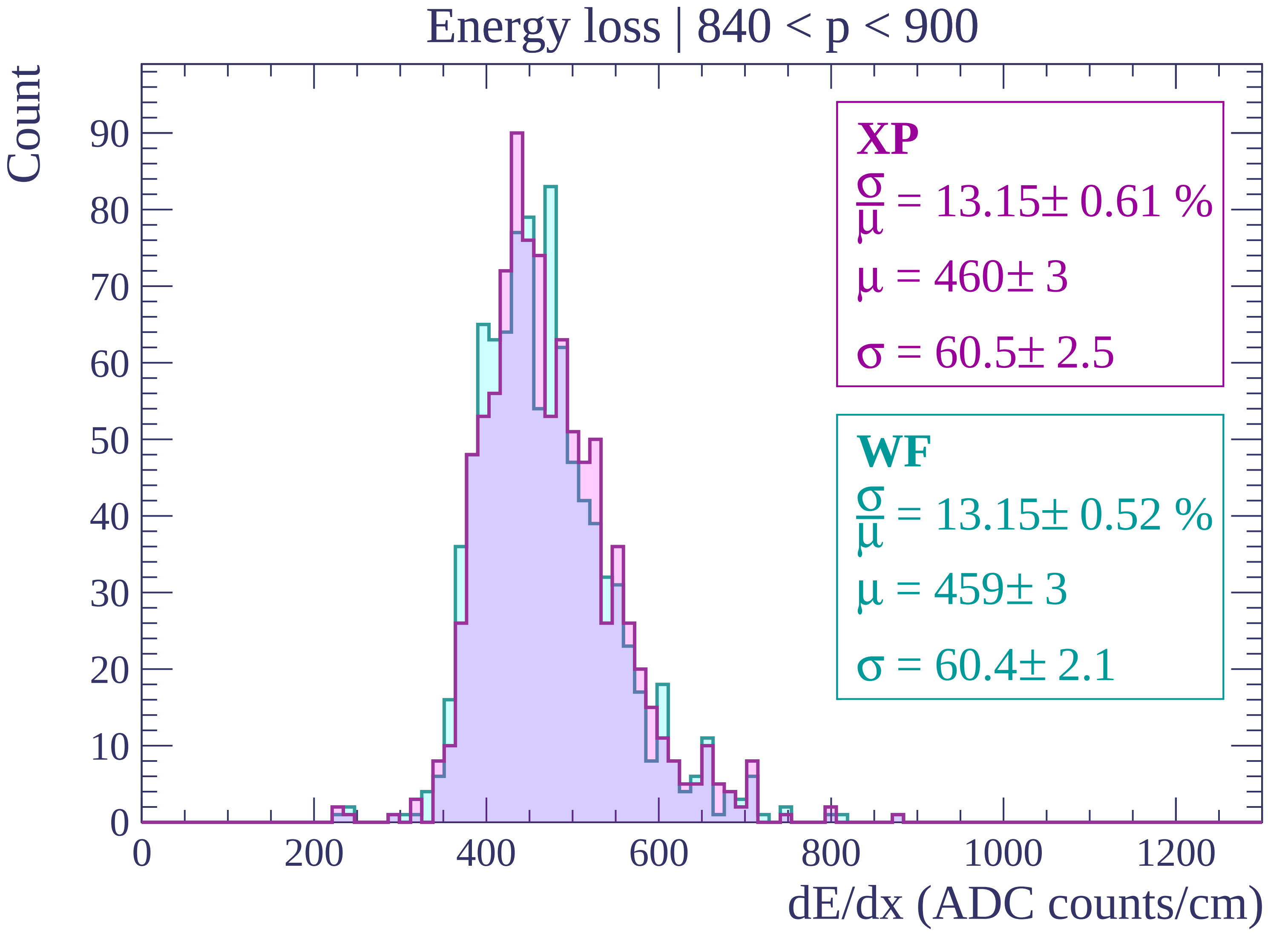
Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 11.63± 0.43 %
μ = 440± 2
σ = 51.2± 1.7
WF
σ/μ = 11.61± 0.49 %
μ = 435± 2
σ = 50.4± 1.9

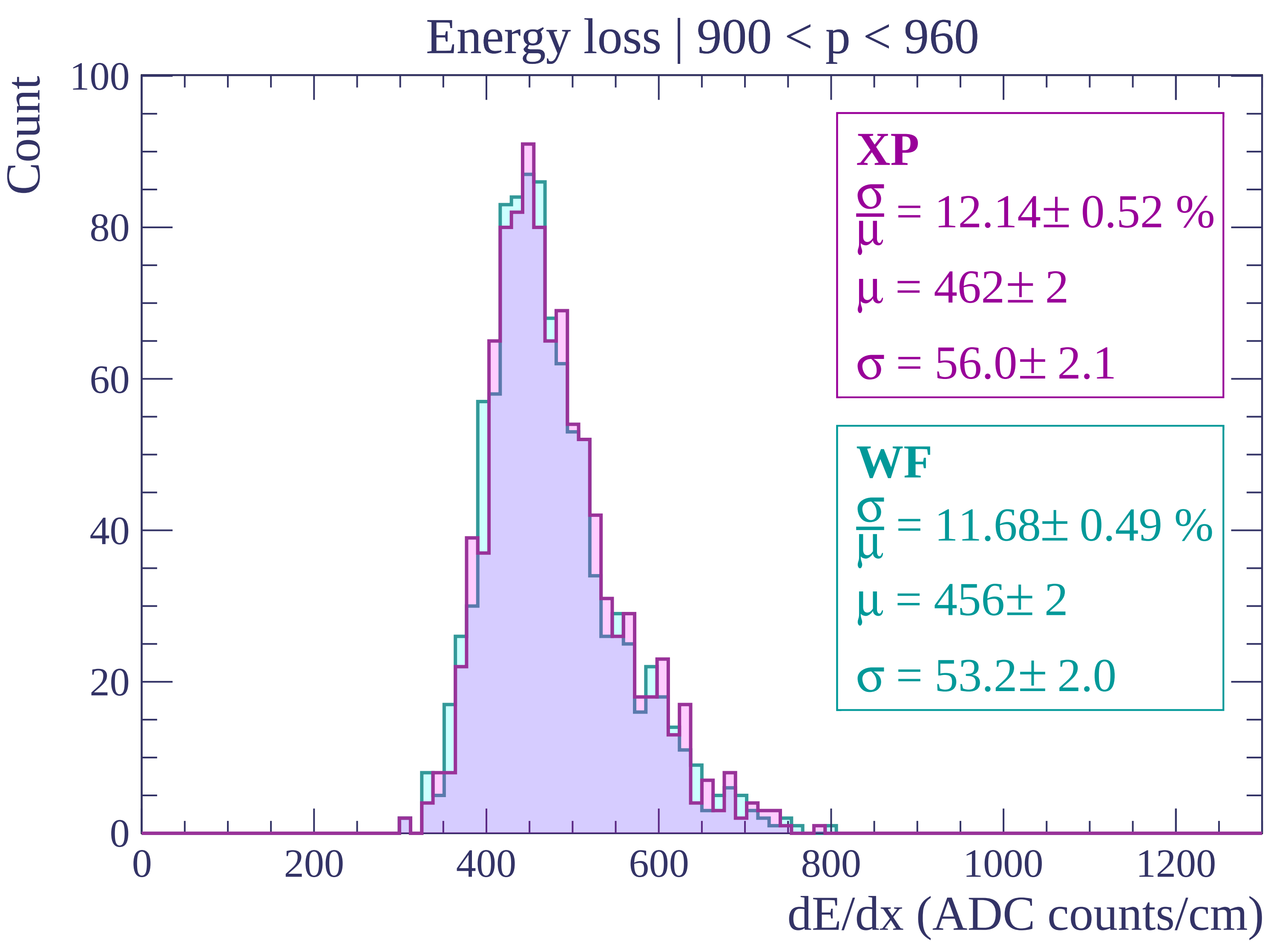

Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.42± 0.54 %
μ = 449± 2
σ = 55.8± 2.1
WF
$\frac{\sigma}{\mu}$ = 12.68± 0.51 %
μ = 446± 2
σ = 56.5± 2.0



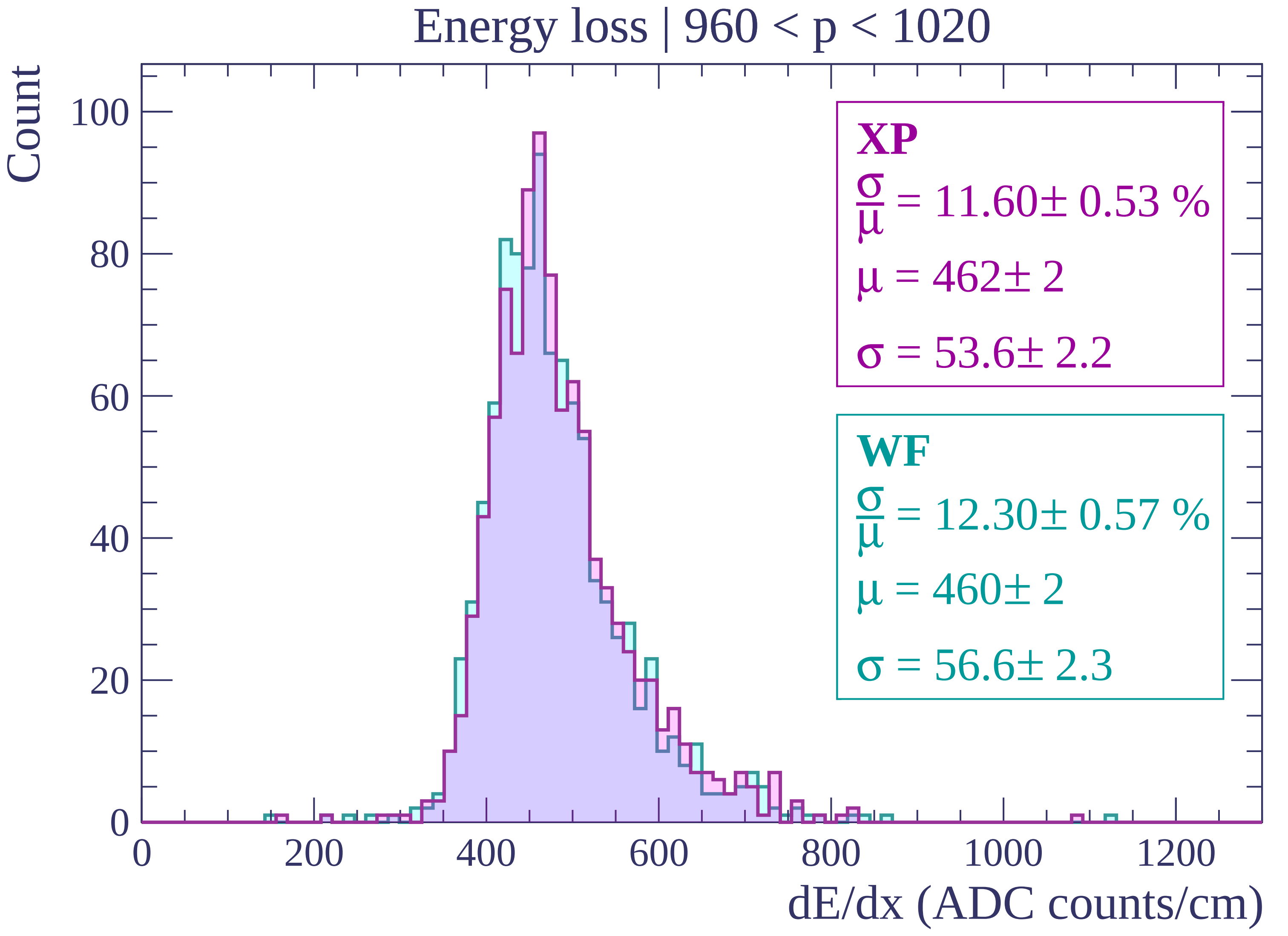
Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 11.28± 0.53 %
μ = 450± 2
σ = 50.8± 2.2
WF
σ/μ = 11.24± 0.57 %
μ = 444± 2
σ = 49.9± 2.3


Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 11.67± 0.43 %
μ = 452± 2
σ = 52.8± 1.7
WF
σ/μ = 12.45± 0.55 %
μ = 448± 2
σ = 55.8± 2.2

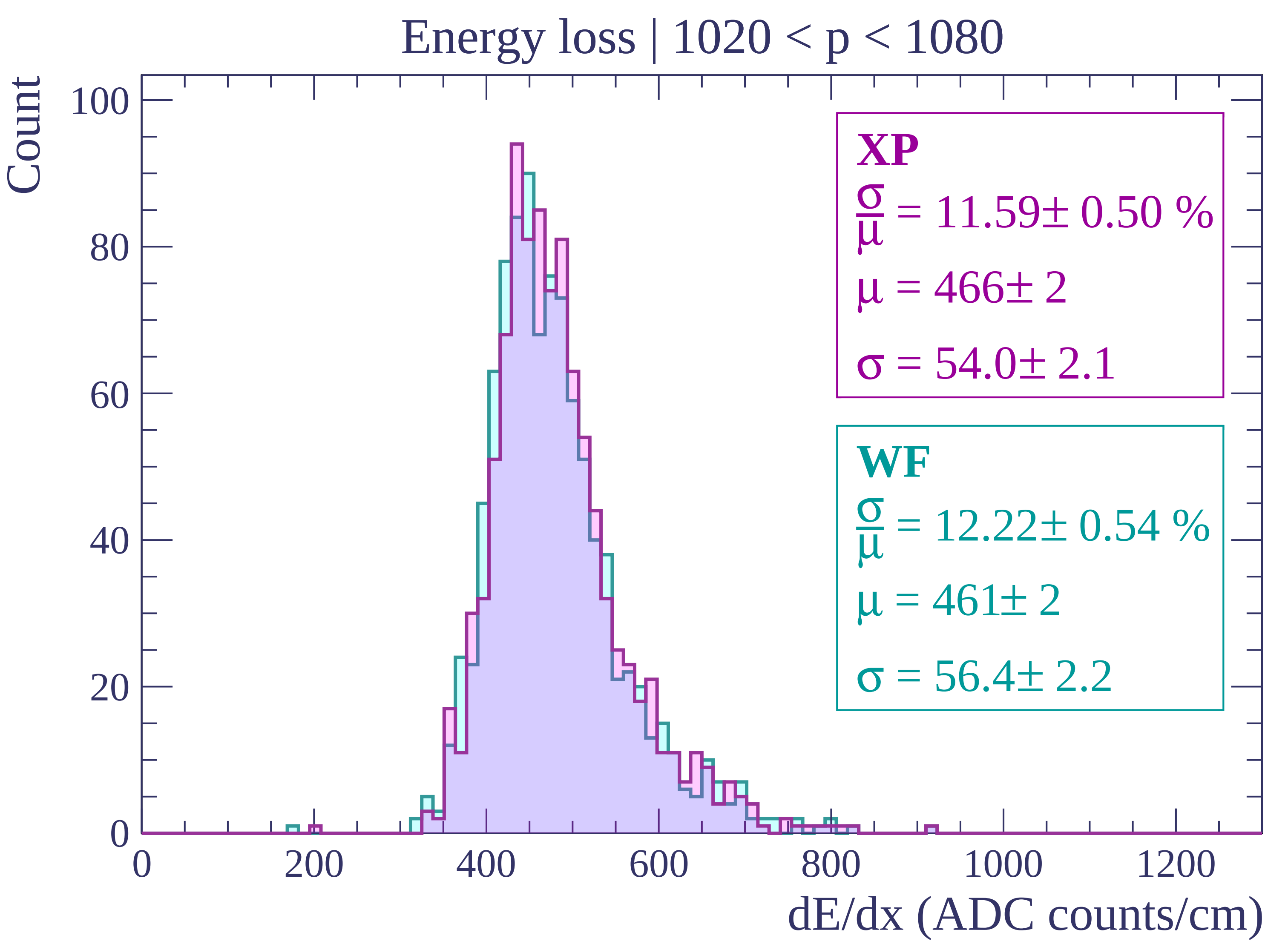

Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 13.15± 0.61 %
μ = 460± 3
σ = 60.5± 2.5
WF
σ/μ = 13.15± 0.52 %
μ = 459± 3
σ = 60.4± 2.1



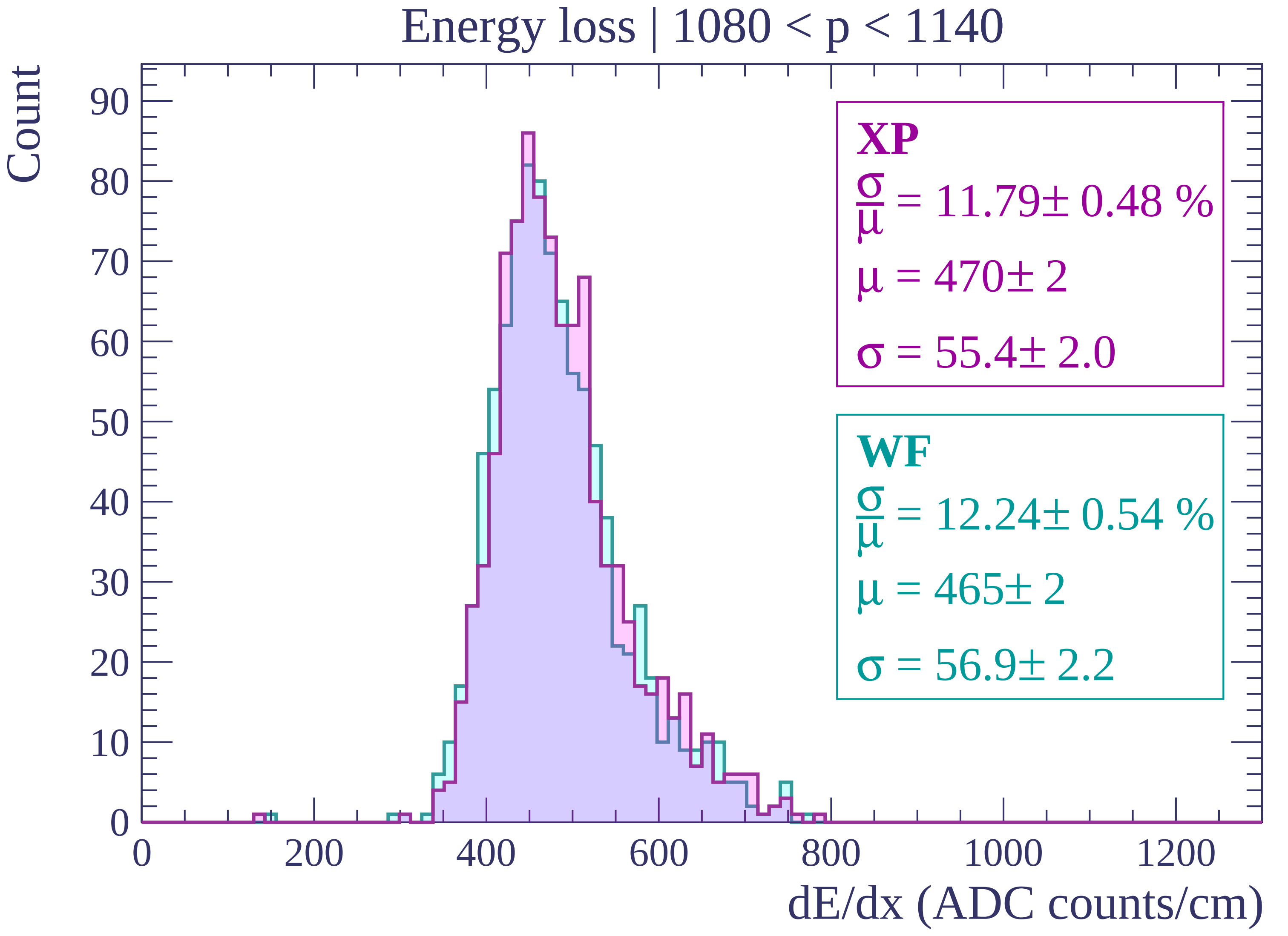
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 12.14± 0.52 %
μ = 462± 2
σ = 56.0± 2.1
WF
σ/μ = 11.68± 0.49 %
μ = 456± 2
σ = 53.2± 2.0

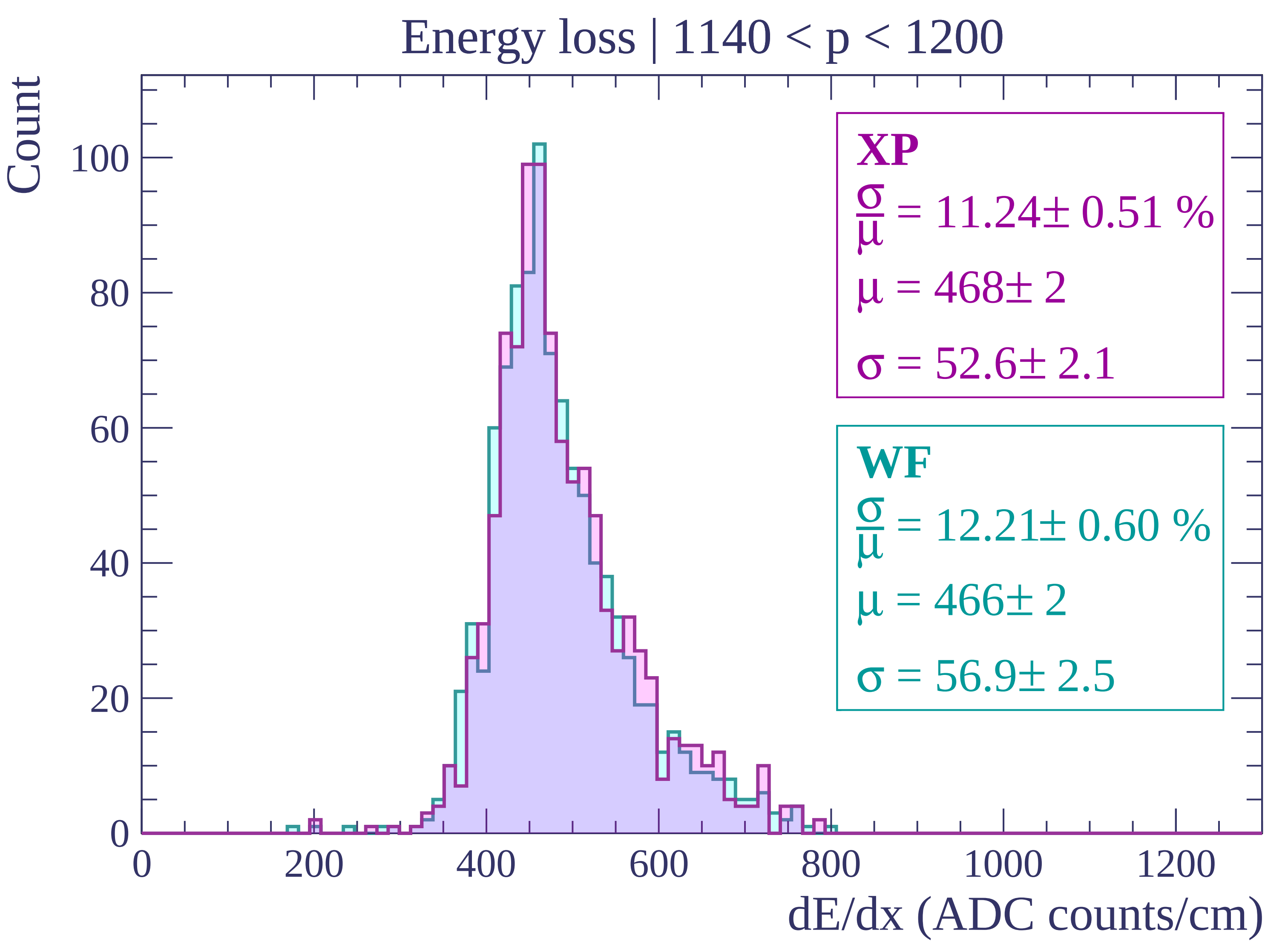


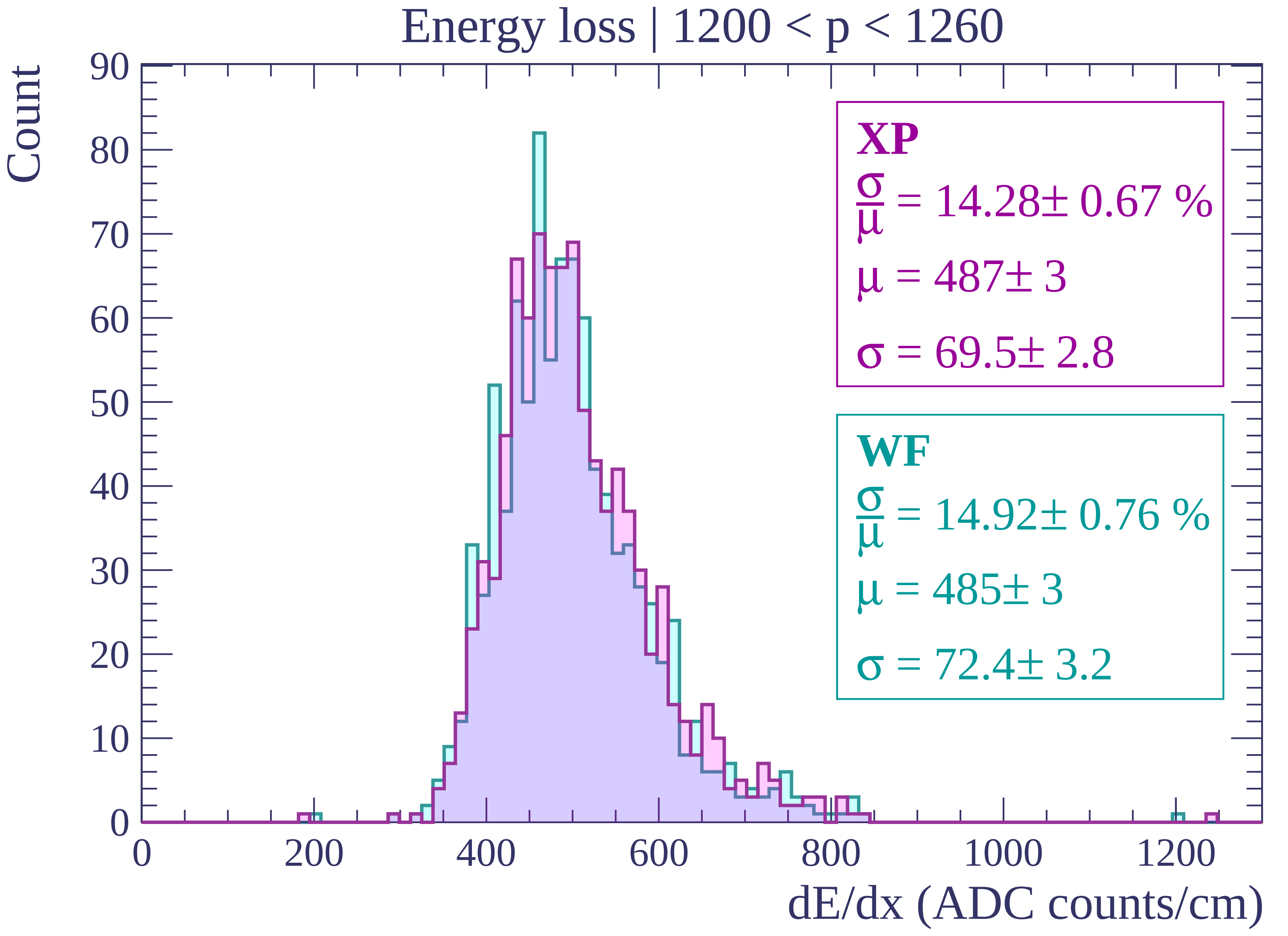
Energy loss | 960 < p < 1020
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.60± 0.53 %
μ = 462± 2
σ = 53.6± 2.2
WF
σ/μ = 12.30± 0.57 %
μ = 460± 2
σ = 56.6± 2.3



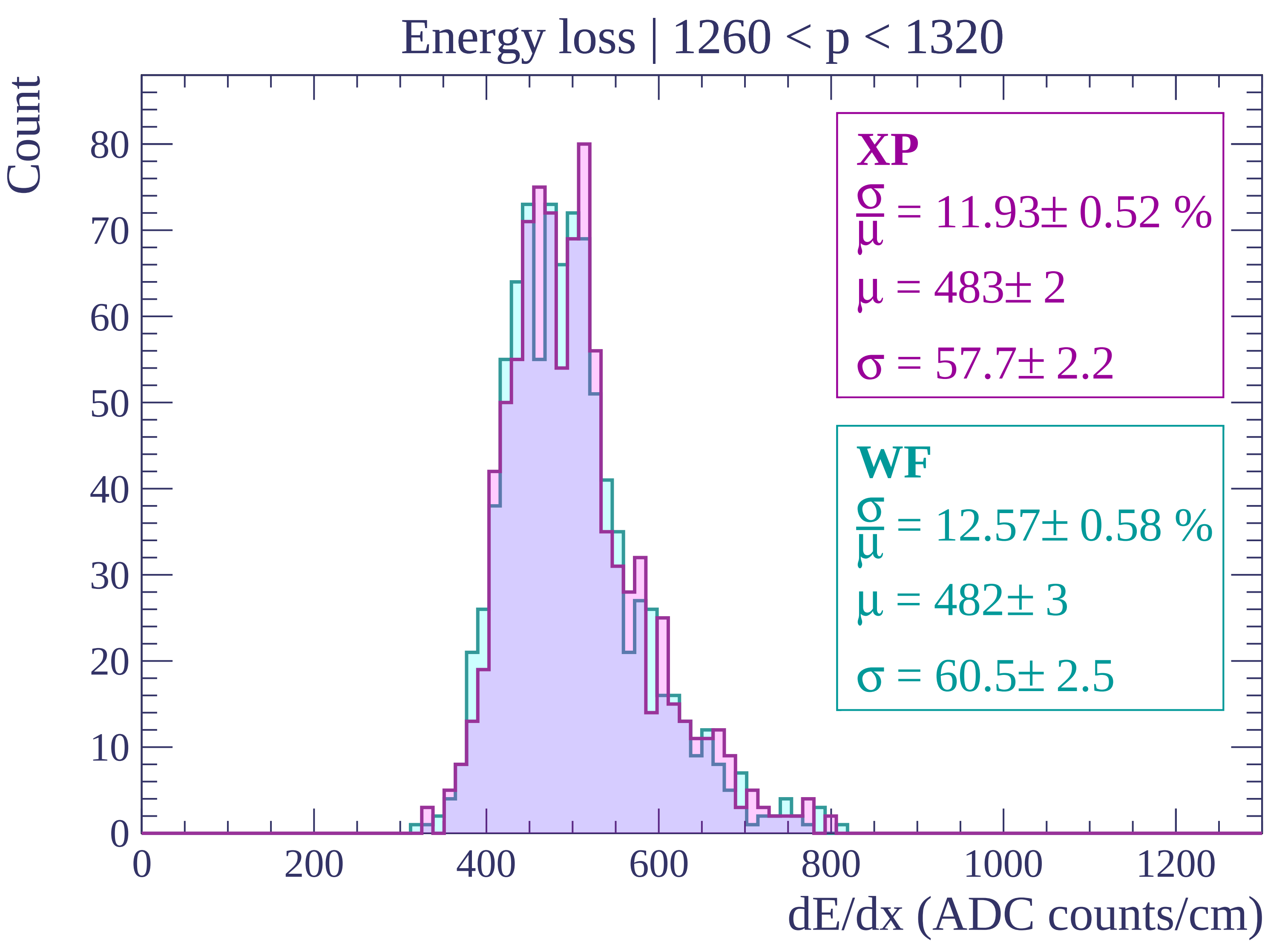
Energy loss | 1020 < p < 1080
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.59± 0.50 %
μ = 466± 2
σ = 54.0± 2.1
WF
σ/μ = 12.22± 0.54 %
μ = 461± 2
σ = 56.4± 2.2


Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 11.79± 0.48 %
μ = 470± 2
σ = 55.4± 2.0
WF
σ/μ = 12.24± 0.54 %
μ = 465± 2
σ = 56.9± 2.2

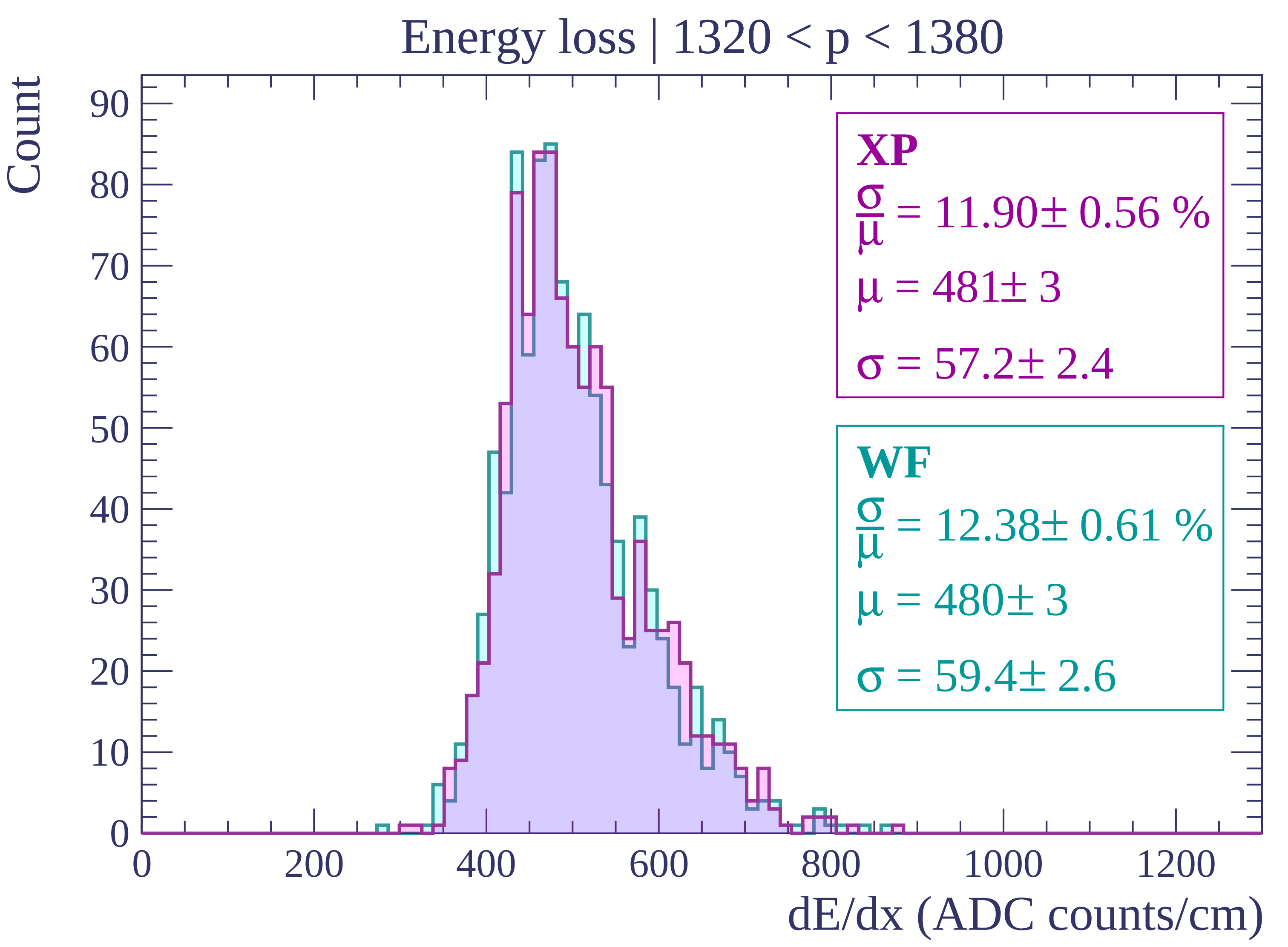

Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 11.24± 0.51 %
μ = 468± 2
σ = 52.6± 2.1
WF
σ/μ = 12.21± 0.60 %
μ = 466± 2
σ = 56.9± 2.5



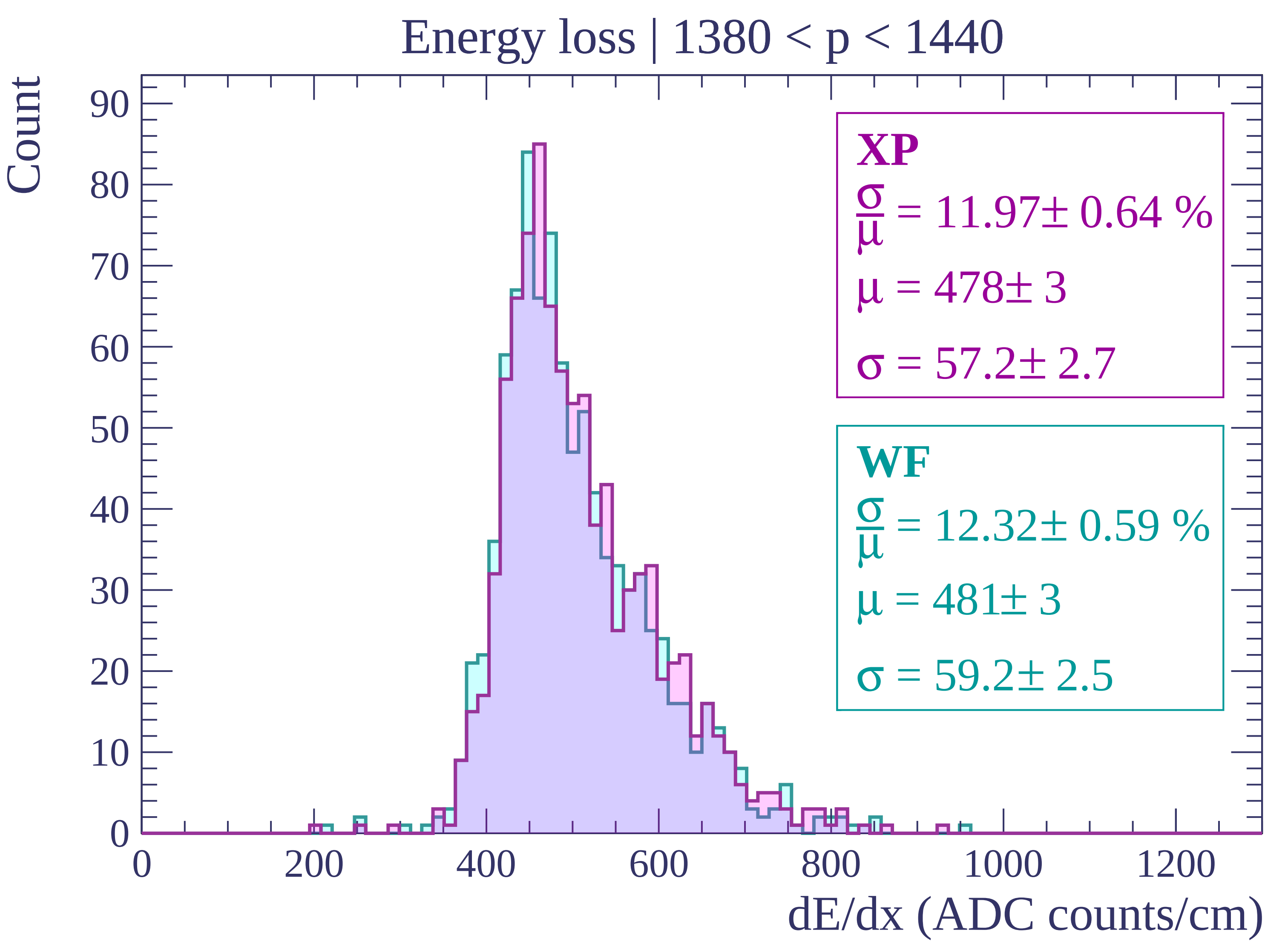
Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.28± 0.67 %
μ = 487± 3
σ = 69.5± 2.8
WF
σ/μ = 14.92± 0.76 %
μ = 485± 3
σ = 72.4± 3.2
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200



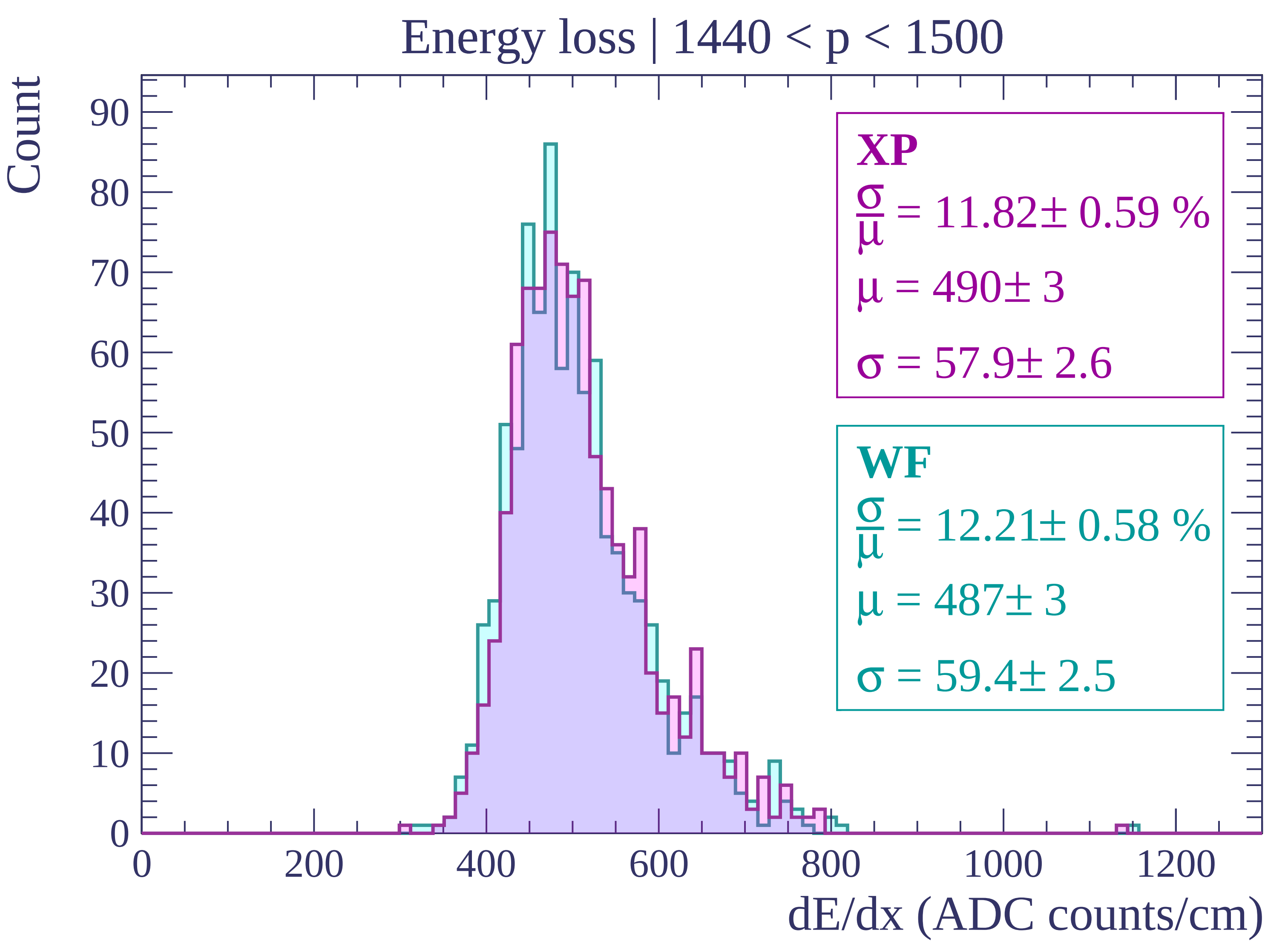
Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 11.93± 0.52 %
μ = 483± 2
σ = 57.7± 2.2
WF
σ/μ = 12.57± 0.58 %
μ = 482± 3
σ = 60.5± 2.5



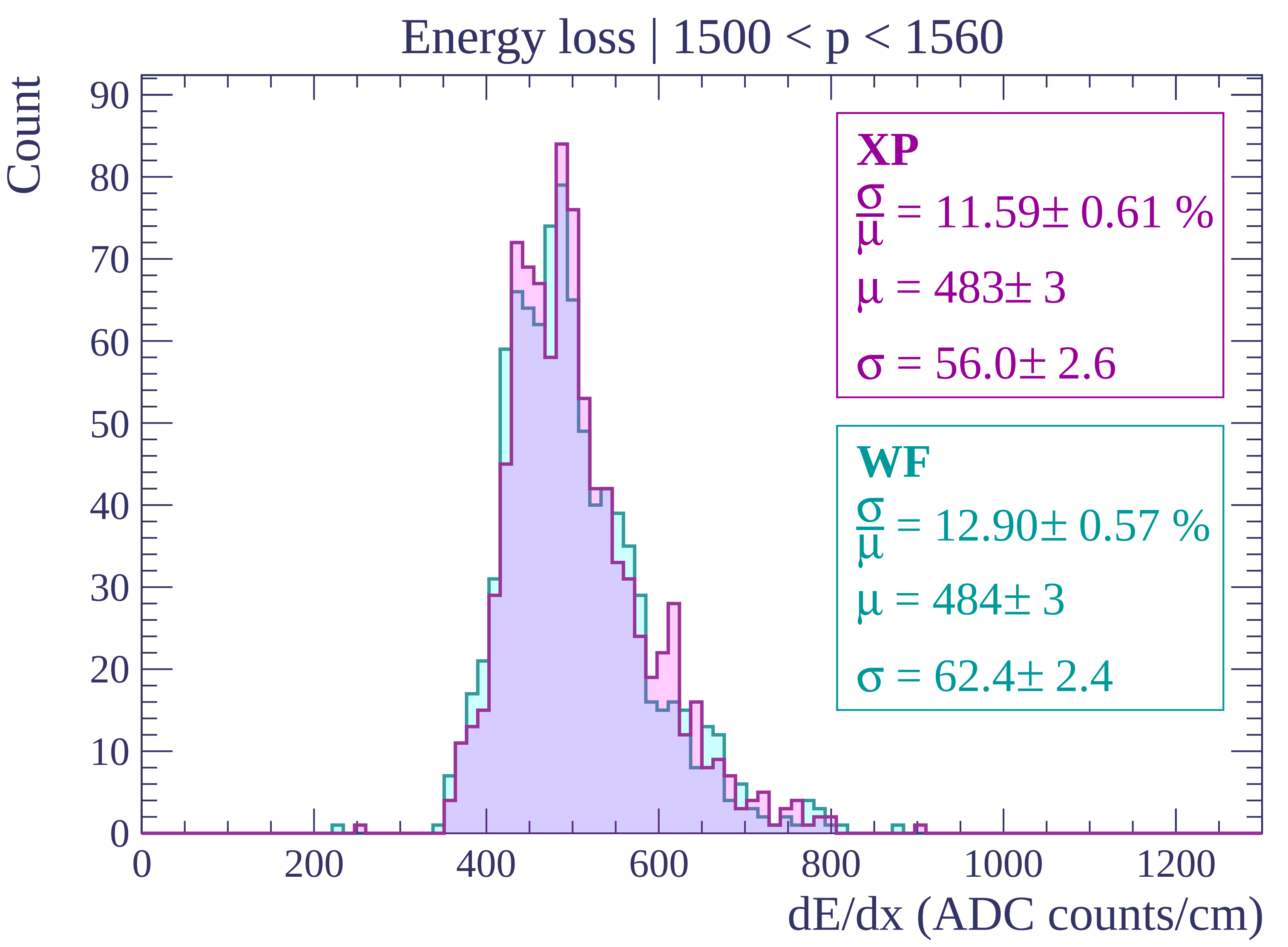
Energy loss | 1320 < p < 1380
Count
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.90± 0.56 %
μ = 481± 3
σ = 57.2± 2.4
WF
σ/μ = 12.38± 0.61 %
μ = 480± 3
σ = 59.4± 2.6

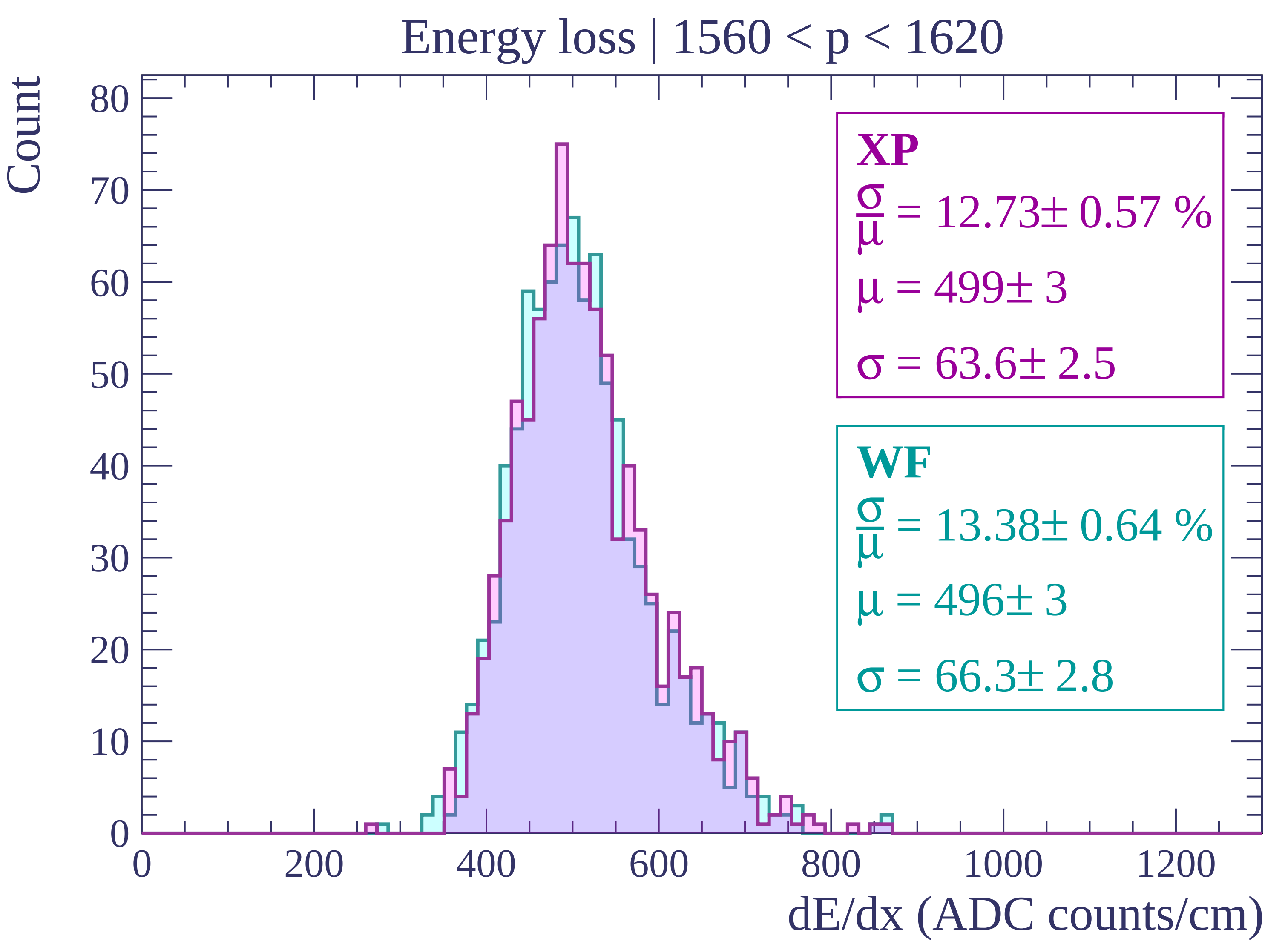


Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP
σ/μ = 11.97± 0.64 %
μ = 478± 3
σ = 57.2± 2.7
WF
σ/μ = 12.32± 0.59 %
μ = 481± 3
σ = 59.2± 2.5

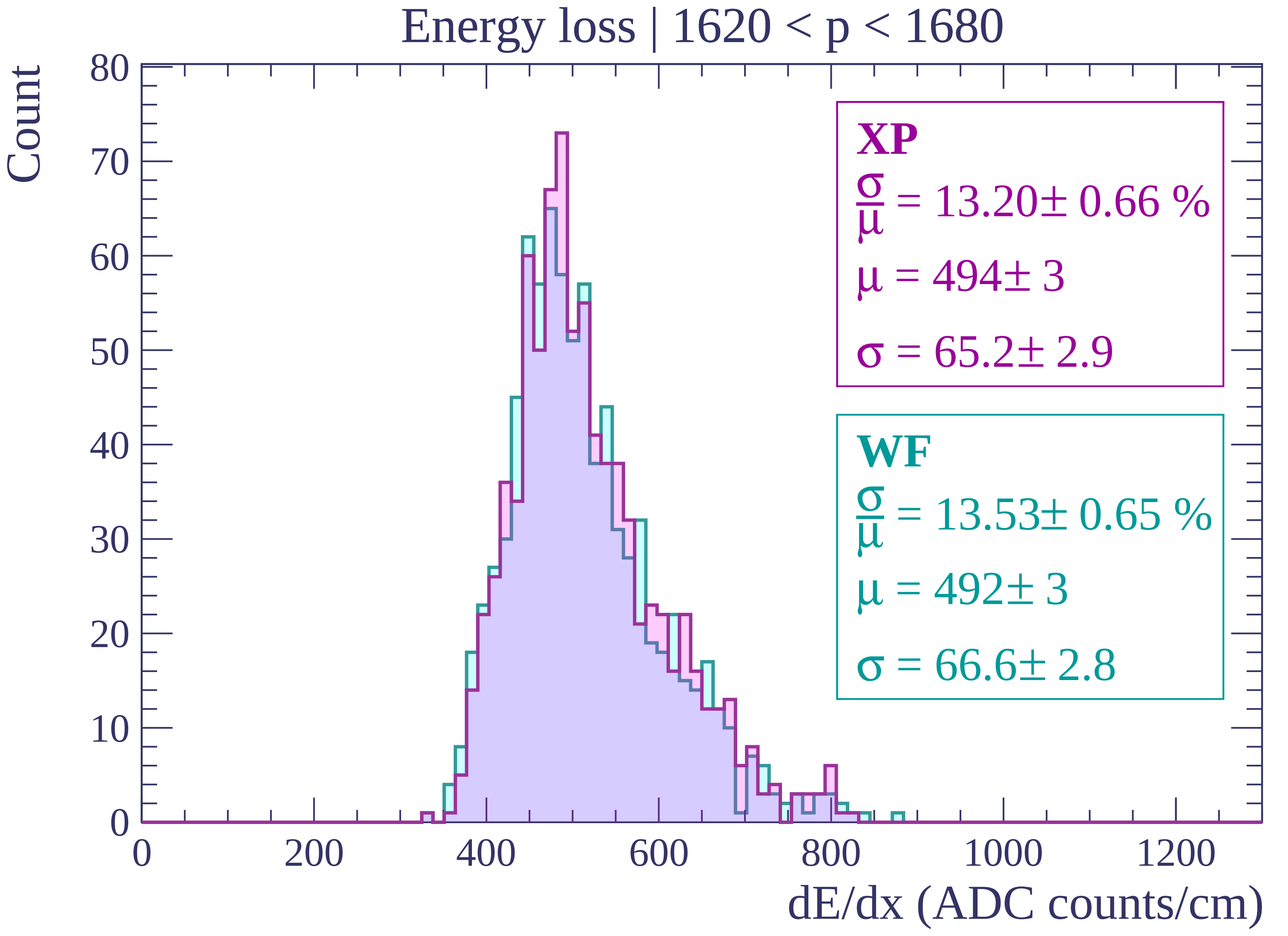


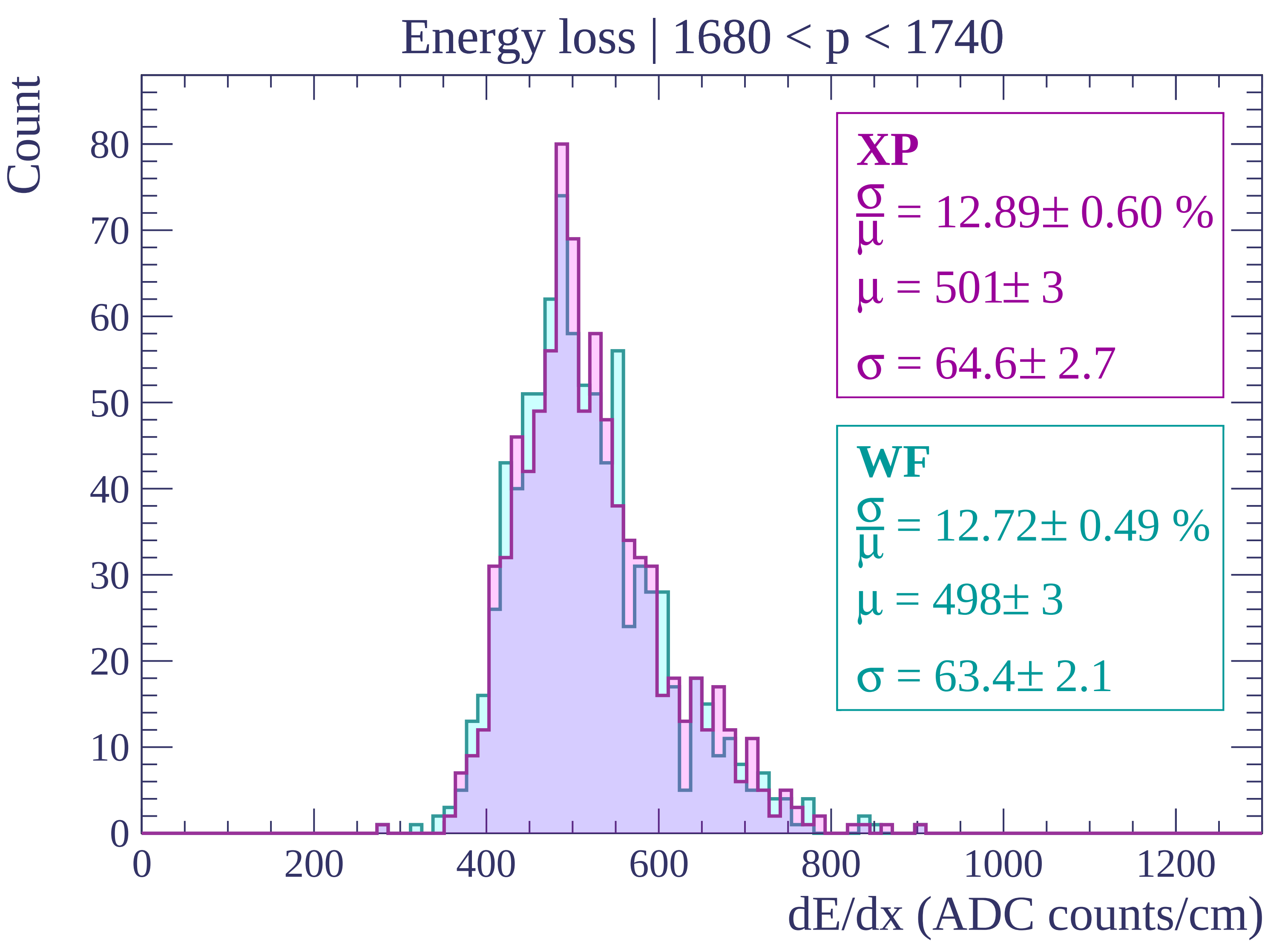
Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 11.82± 0.59 %
μ = 490± 3
σ = 57.9± 2.6
WF
σ/μ = 12.21± 0.58 %
μ = 487± 3
σ = 59.4± 2.5


Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 11.59± 0.61 %
μ = 483± 3
σ = 56.0± 2.6
WF
σ/μ = 12.90± 0.57 %
μ = 484± 3
σ = 62.4± 2.4

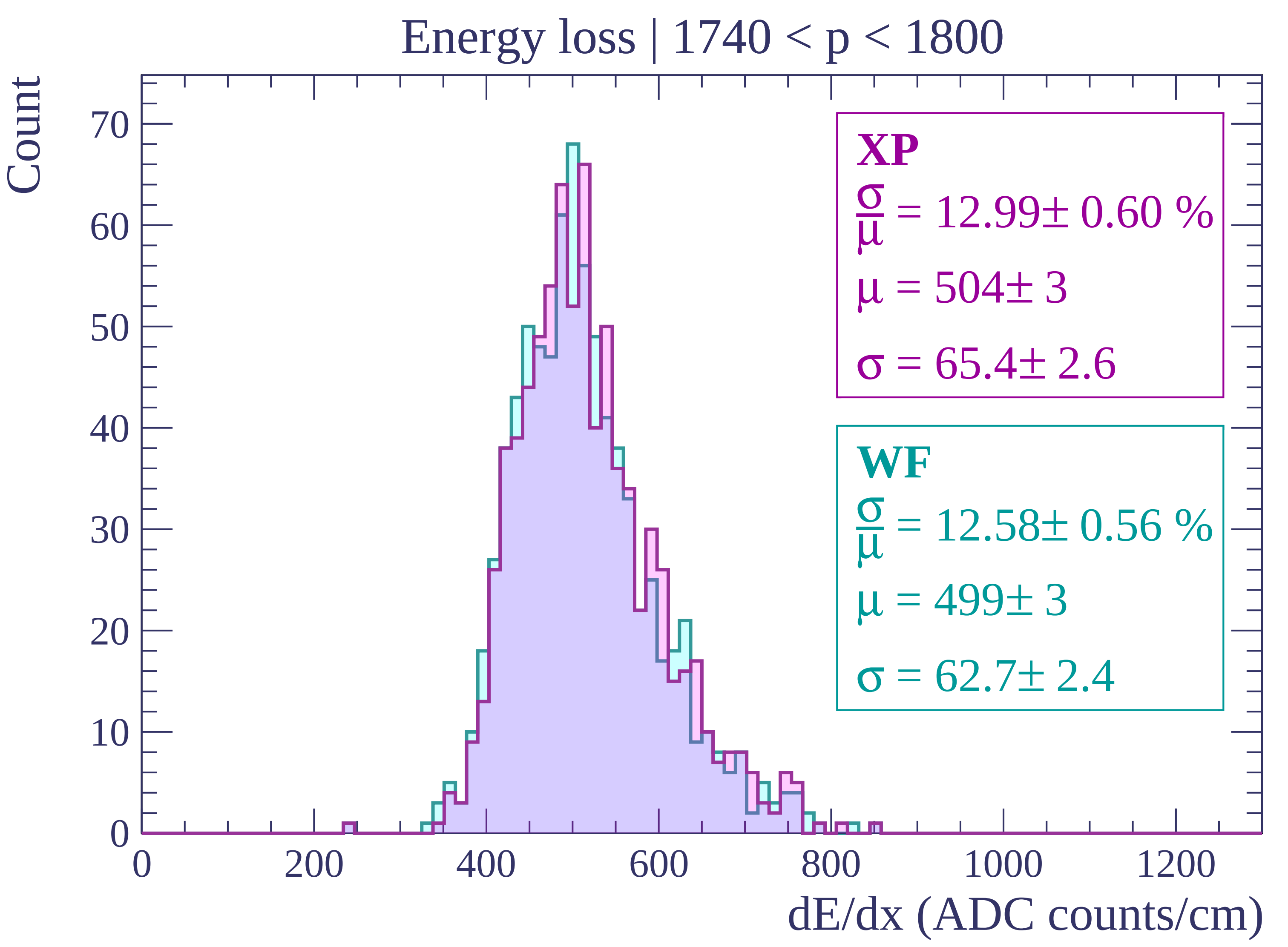


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 12.73± 0.57 %
μ = 499± 3
σ = 63.6± 2.5
WF
σ/μ = 13.38± 0.64 %
μ = 496± 3
σ = 66.3± 2.8

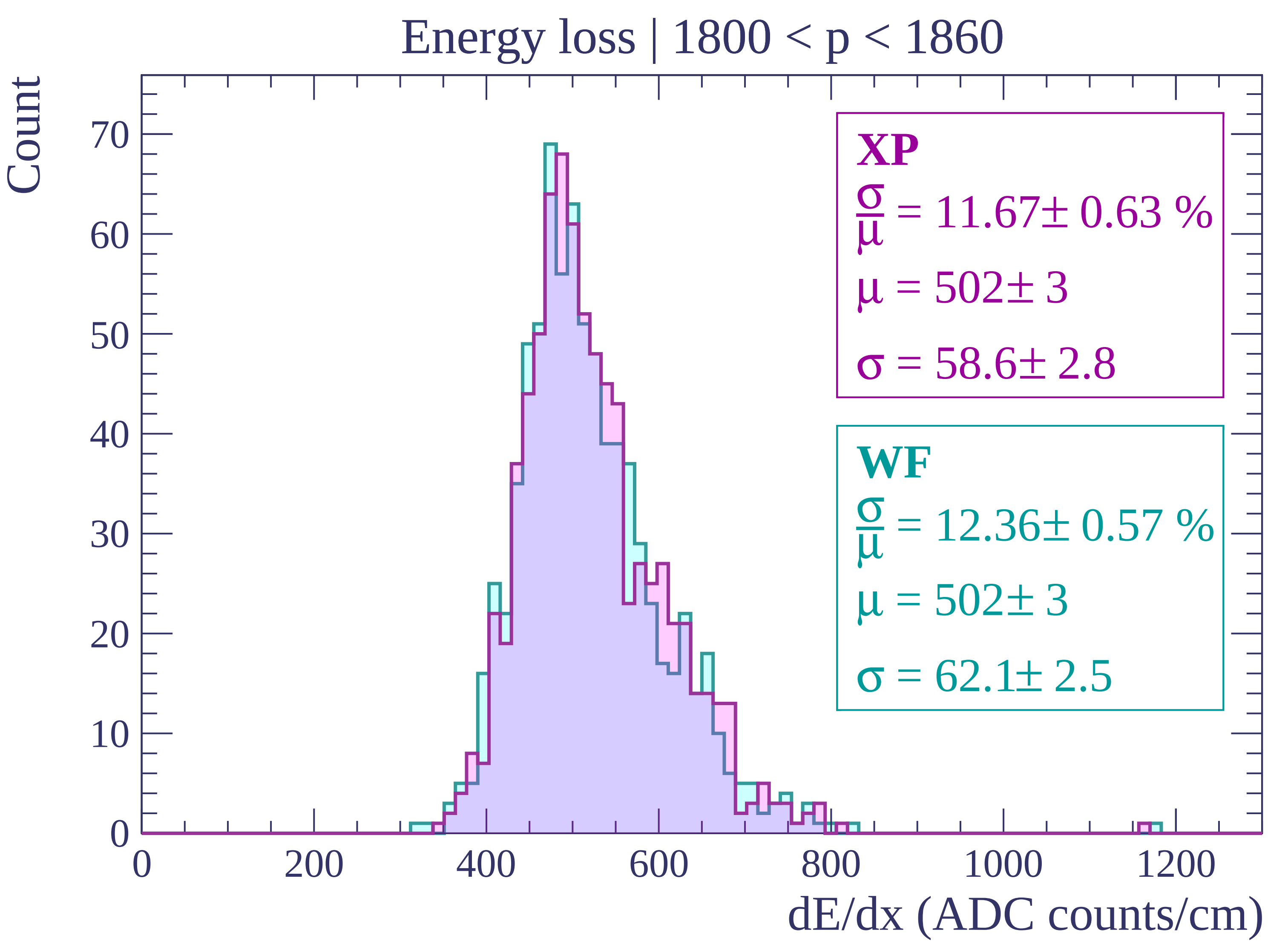

Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 13.20± 0.66 %
μ = 494± 3
σ = 65.2± 2.9
WF
σ/μ = 13.53± 0.65 %
μ = 492± 3
σ = 66.6± 2.8

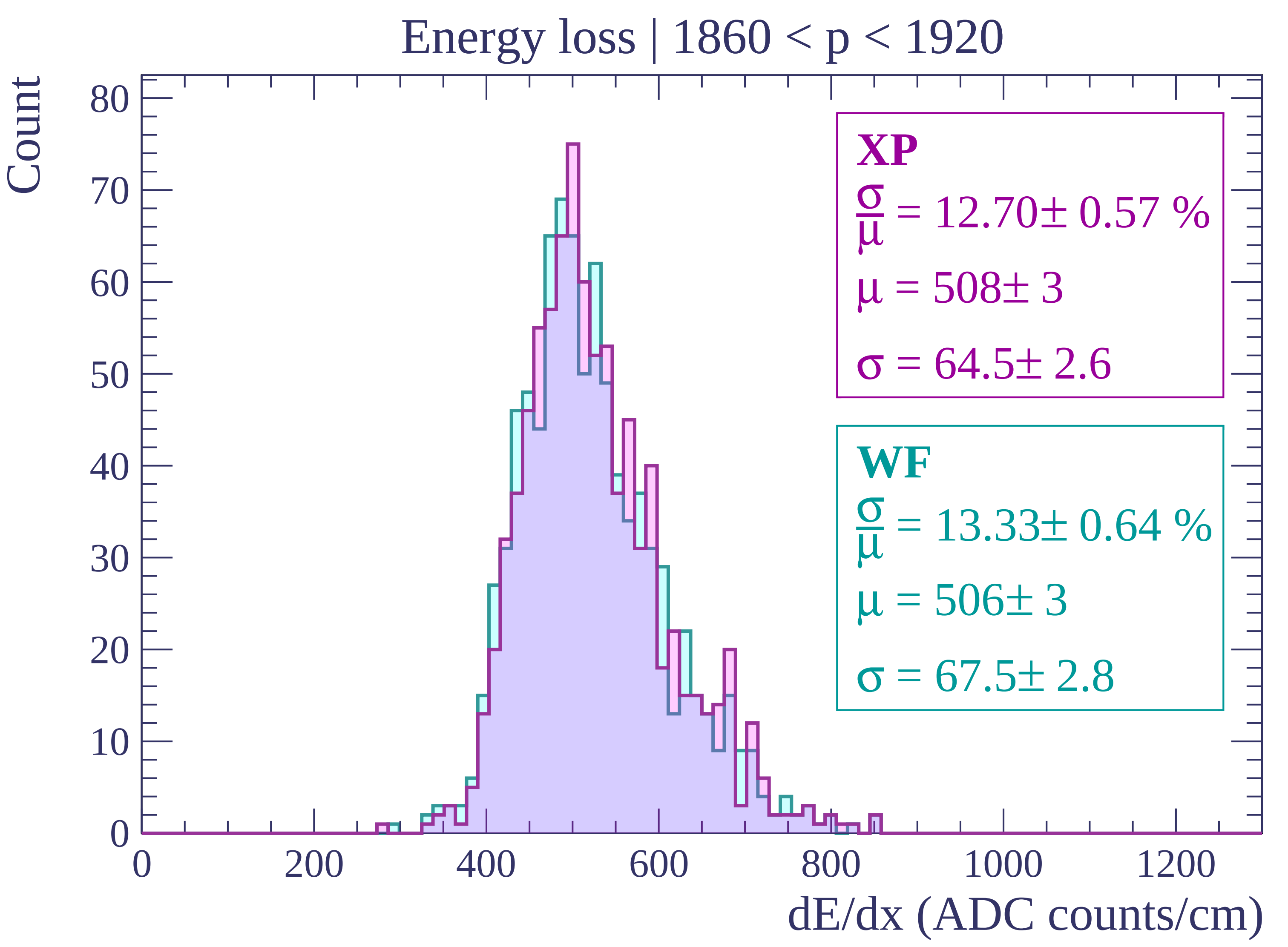


Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 12.89± 0.60 %
μ = 501± 3
σ = 64.6± 2.7
WF
σ/μ = 12.72± 0.49 %
μ = 498± 3
σ = 63.4± 2.1

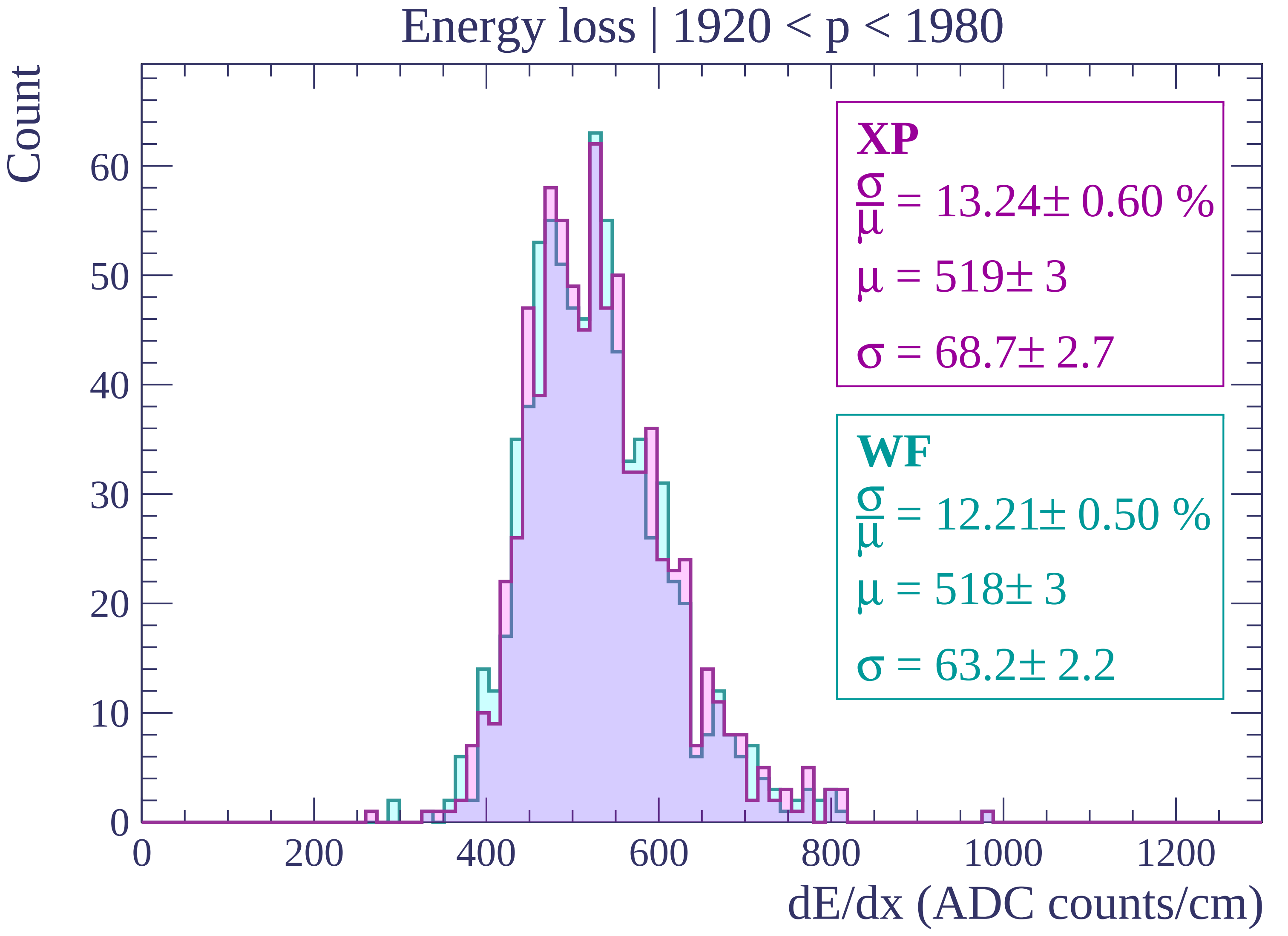


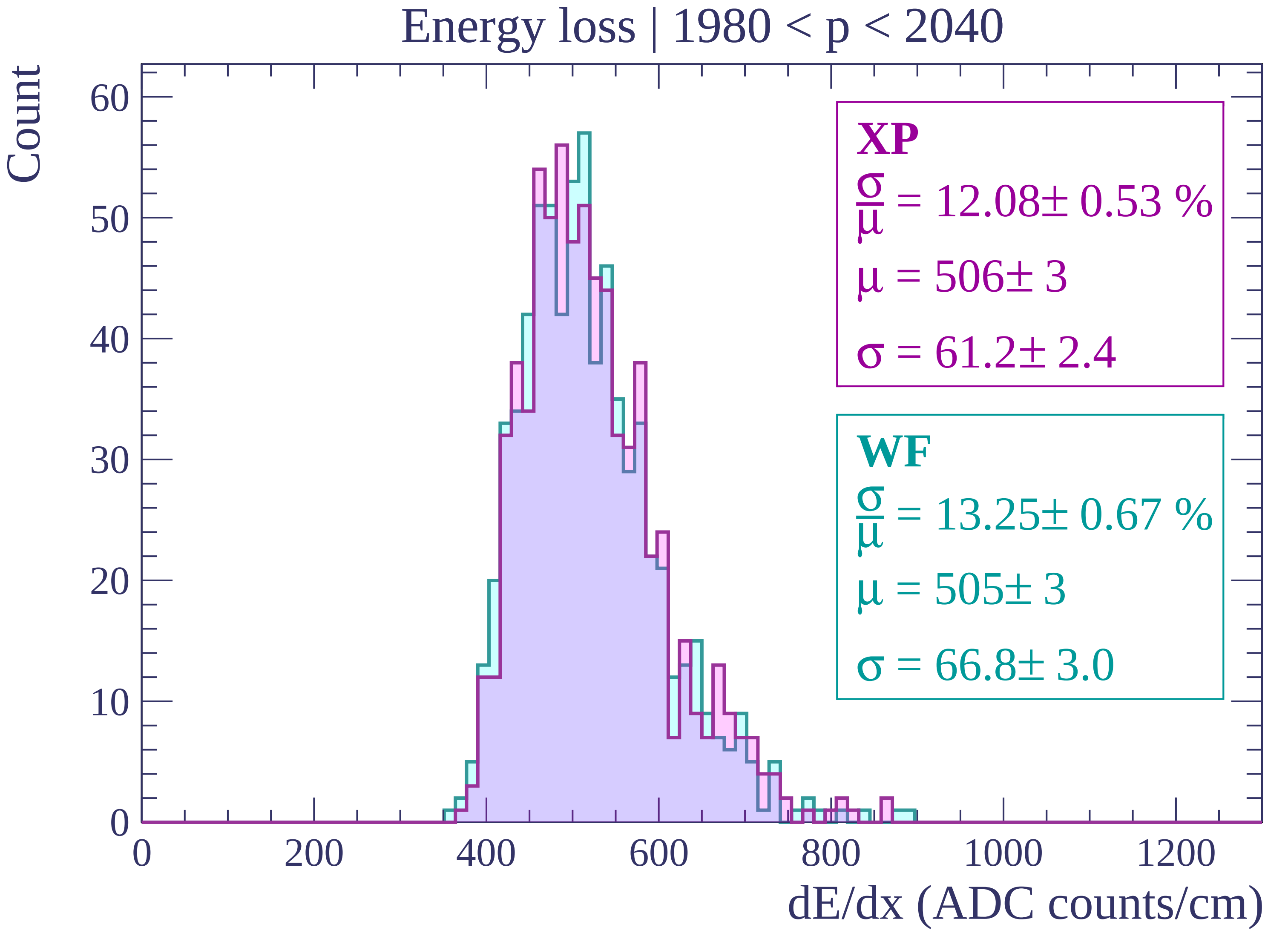
Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 12.99± 0.60 %
μ = 504± 3
σ = 65.4± 2.6
WF
σ/μ = 12.58± 0.56 %
μ = 499± 3
σ = 62.7± 2.4

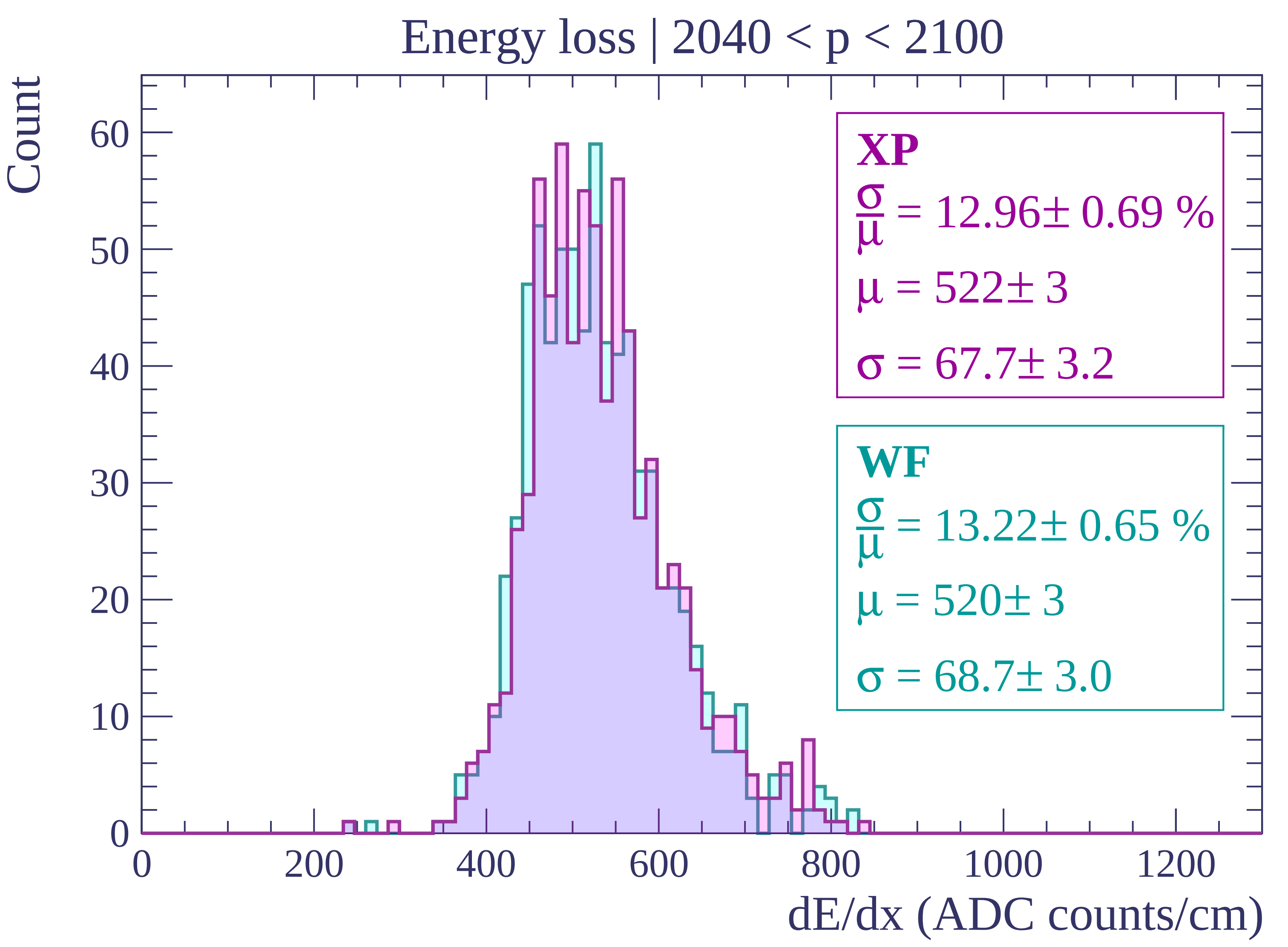

Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP
σ/μ = 11.67± 0.63 %
μ = 502± 3
σ = 58.6± 2.8
WF
σ/μ = 12.36± 0.57 %
μ = 502± 3
σ = 62.1± 2.5



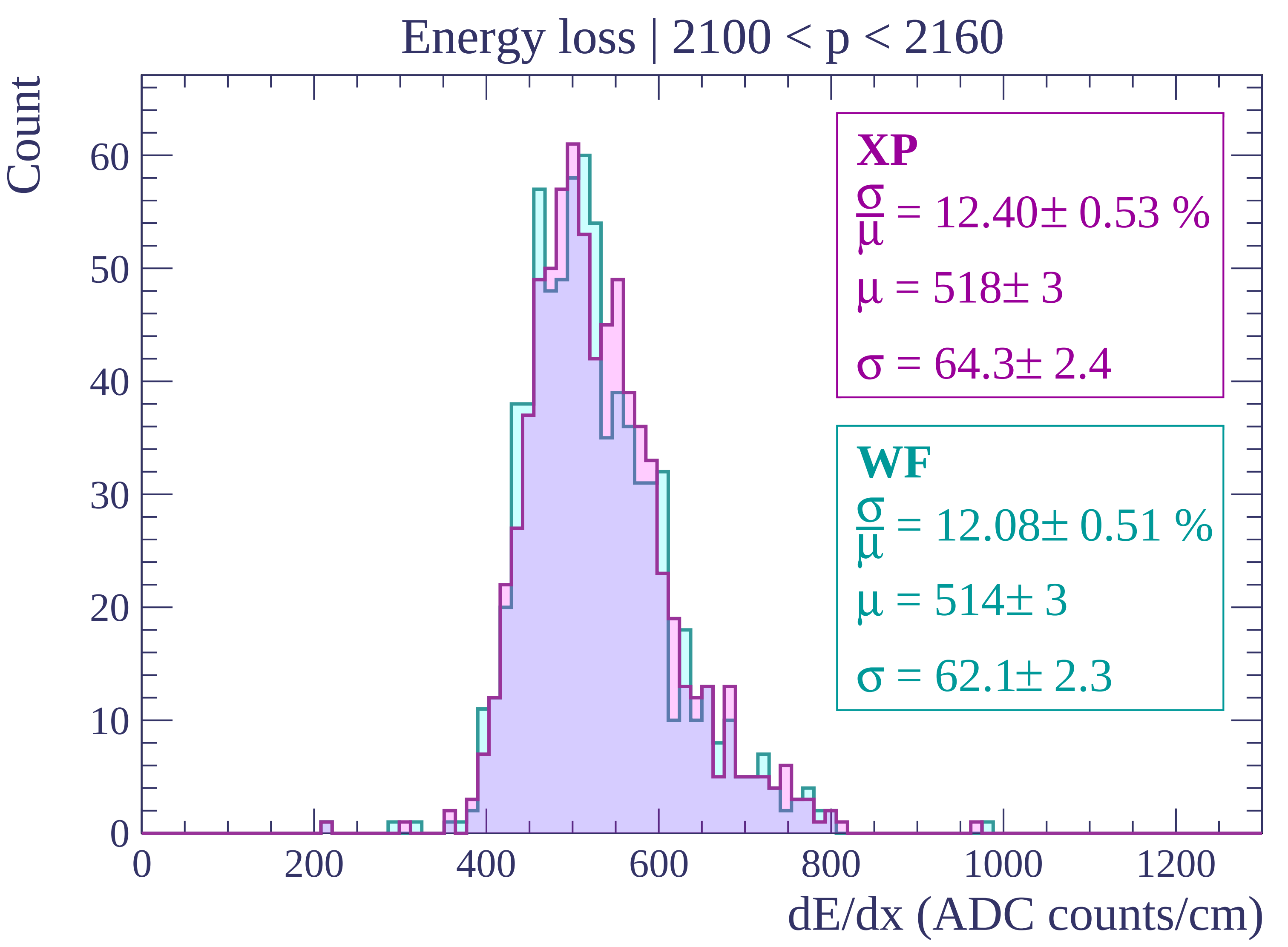
Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.70± 0.57 %
μ = 508± 3
σ = 64.5± 2.6
WF
$\frac{\sigma}{\mu}$ = 13.33± 0.64 %
μ = 506± 3
σ = 67.5± 2.8


Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.24± 0.60 %
μ = 519± 3
σ = 68.7± 2.7
WF
$\frac{\sigma}{\mu}$ = 12.21± 0.50 %
μ = 518± 3
σ = 63.2± 2.2

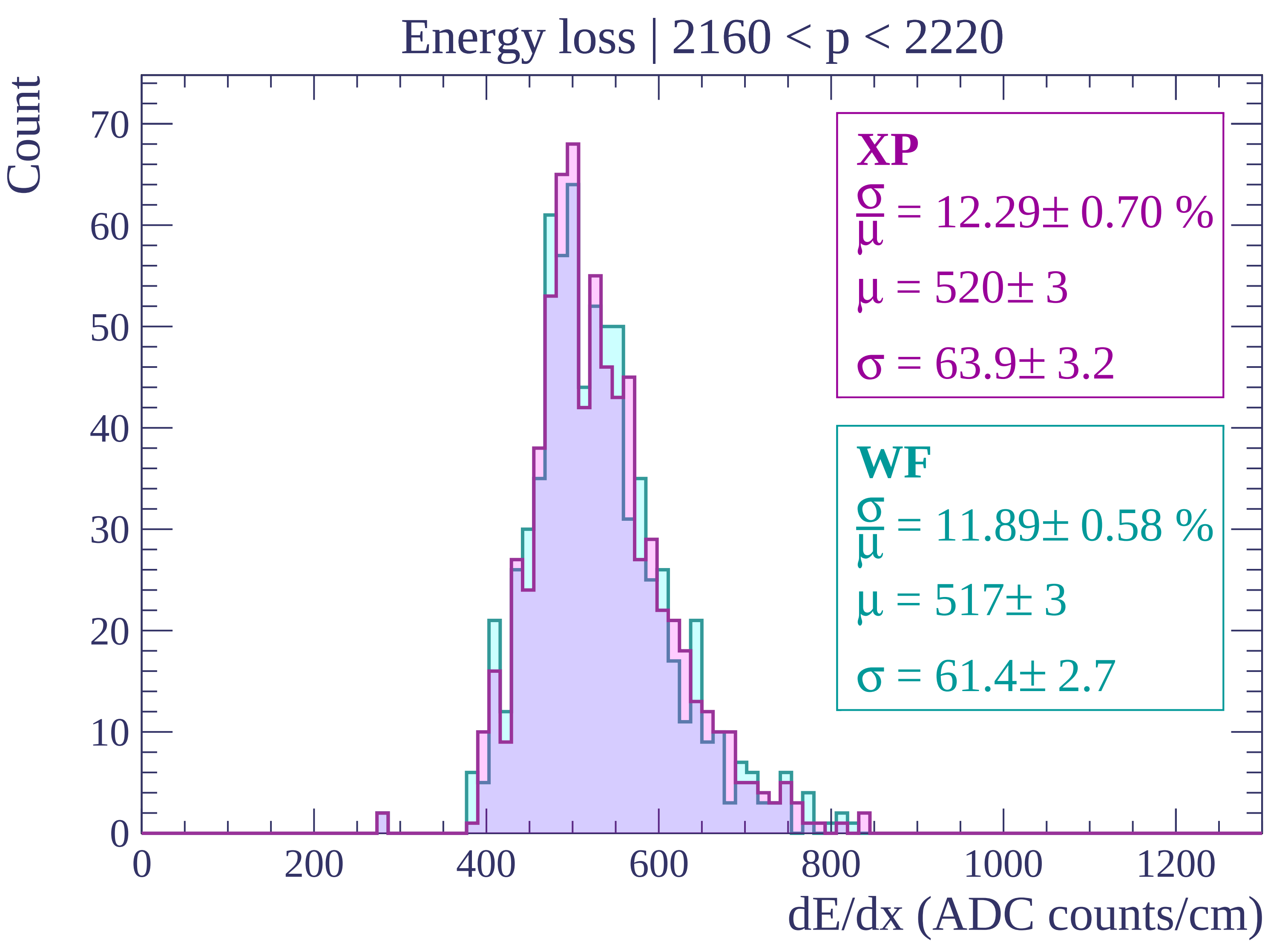

Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 12.08± 0.53 %
μ = 506± 3
σ = 61.2± 2.4
WF
σ/μ = 13.25± 0.67 %
μ = 505± 3
σ = 66.8± 3.0

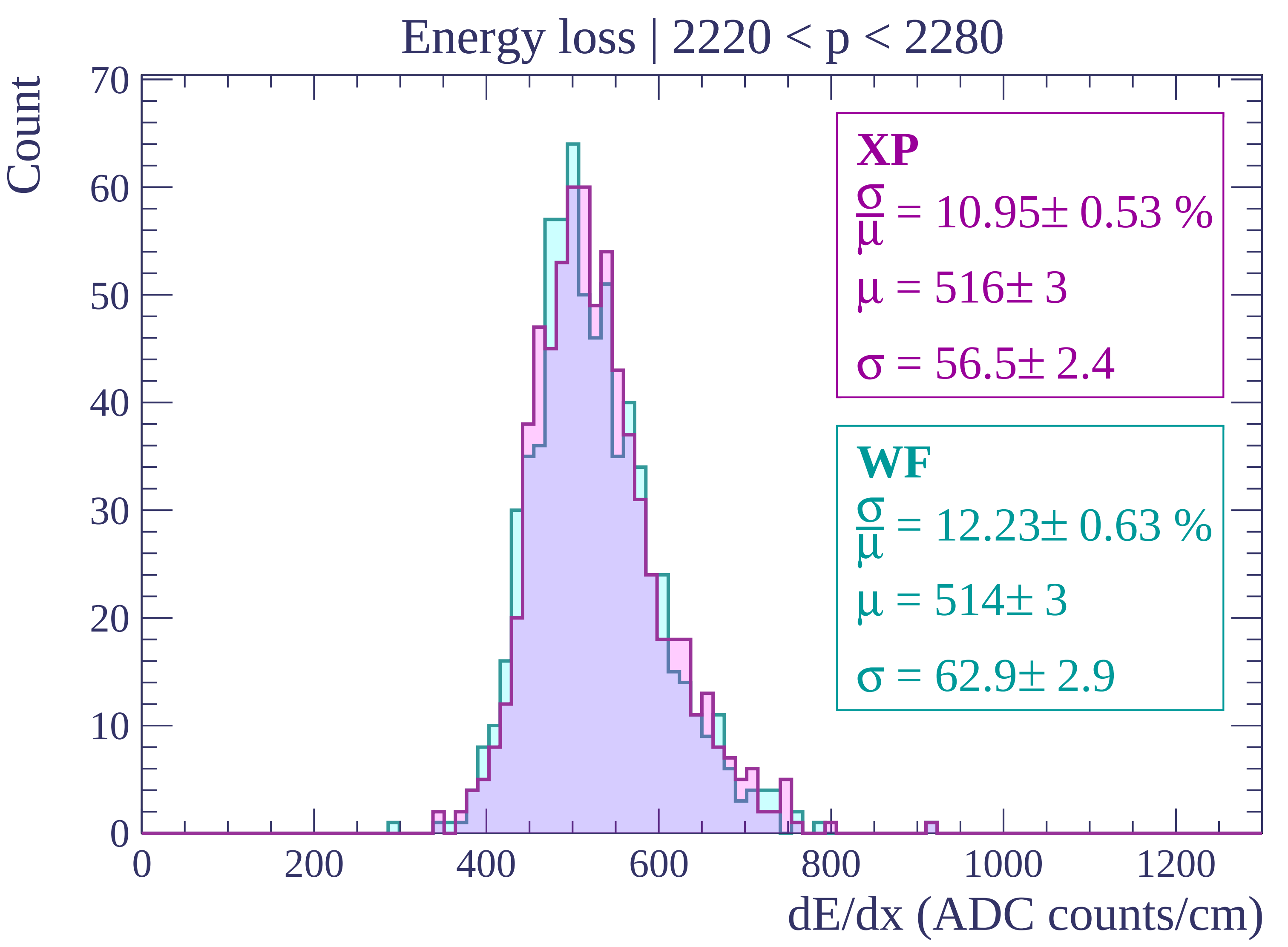


Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 12.96± 0.69 %
μ = 522± 3
σ = 67.7± 3.2
WF
σ/μ = 13.22± 0.65 %
μ = 520± 3
σ = 68.7± 3.0

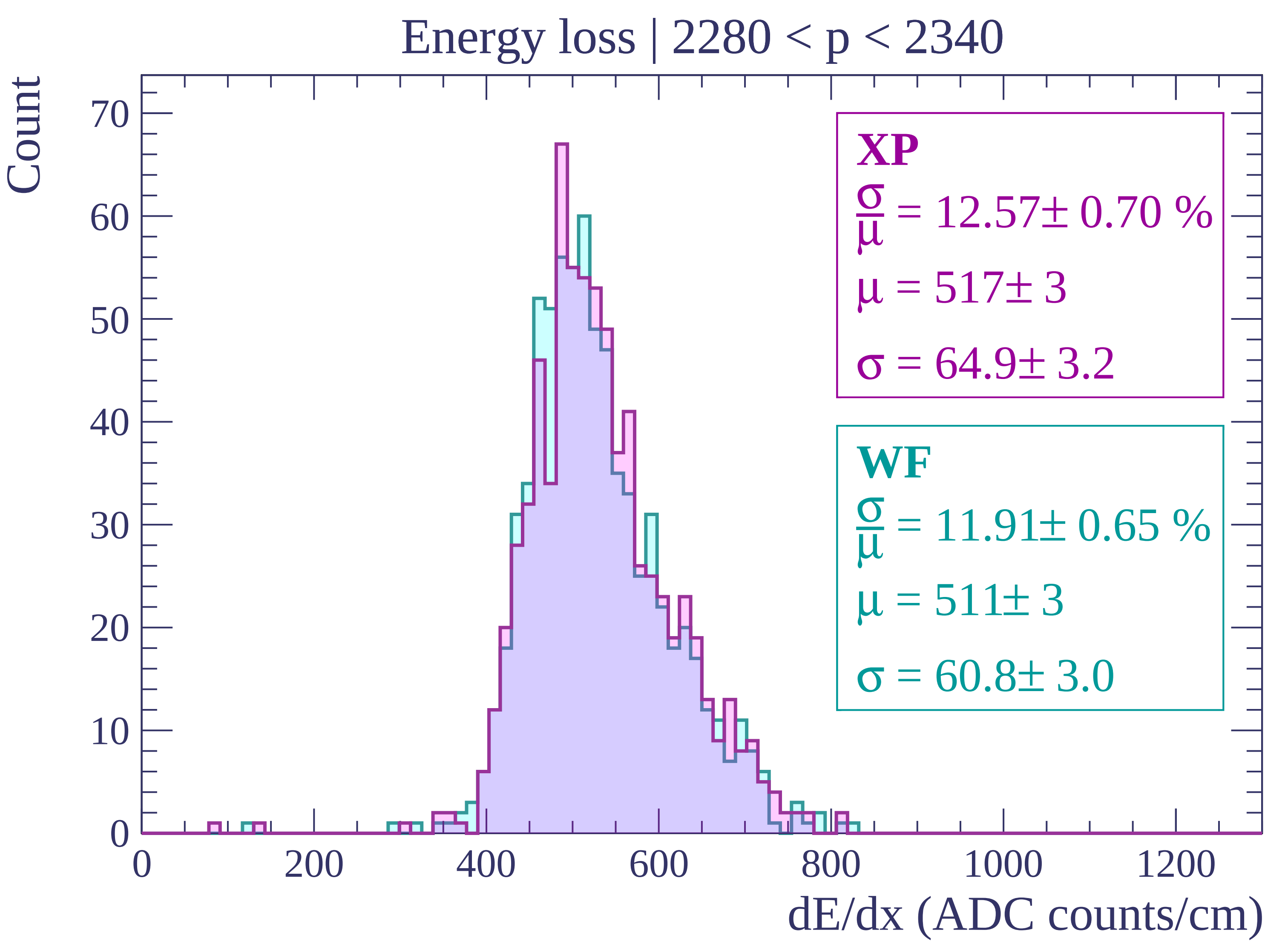


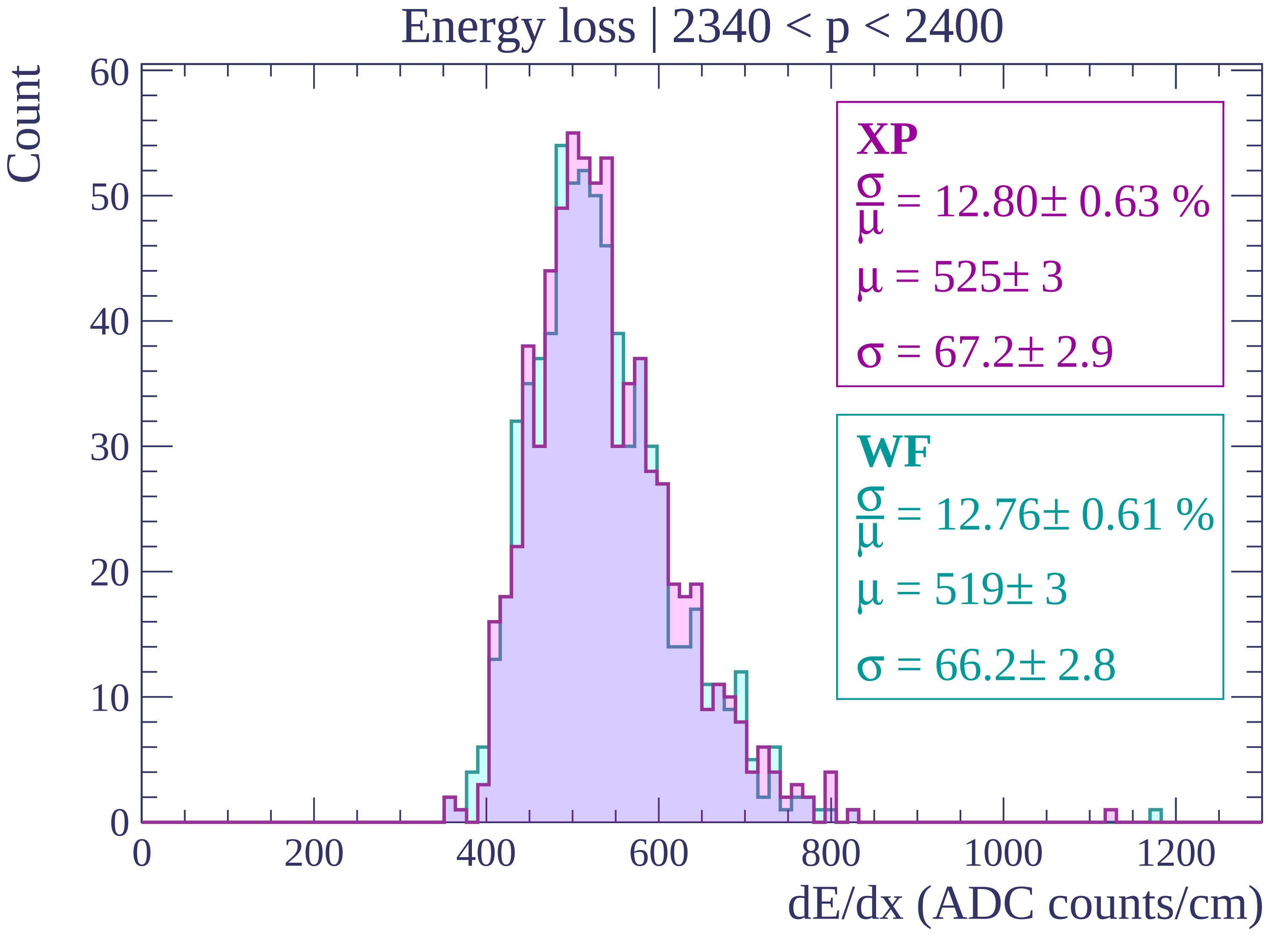
Energy loss | 2100 < p < 2160
Count
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.40± 0.53 %
μ = 518± 3
σ = 64.3± 2.4
WF
σ/μ = 12.08± 0.51 %
μ = 514± 3
σ = 62.1± 2.3

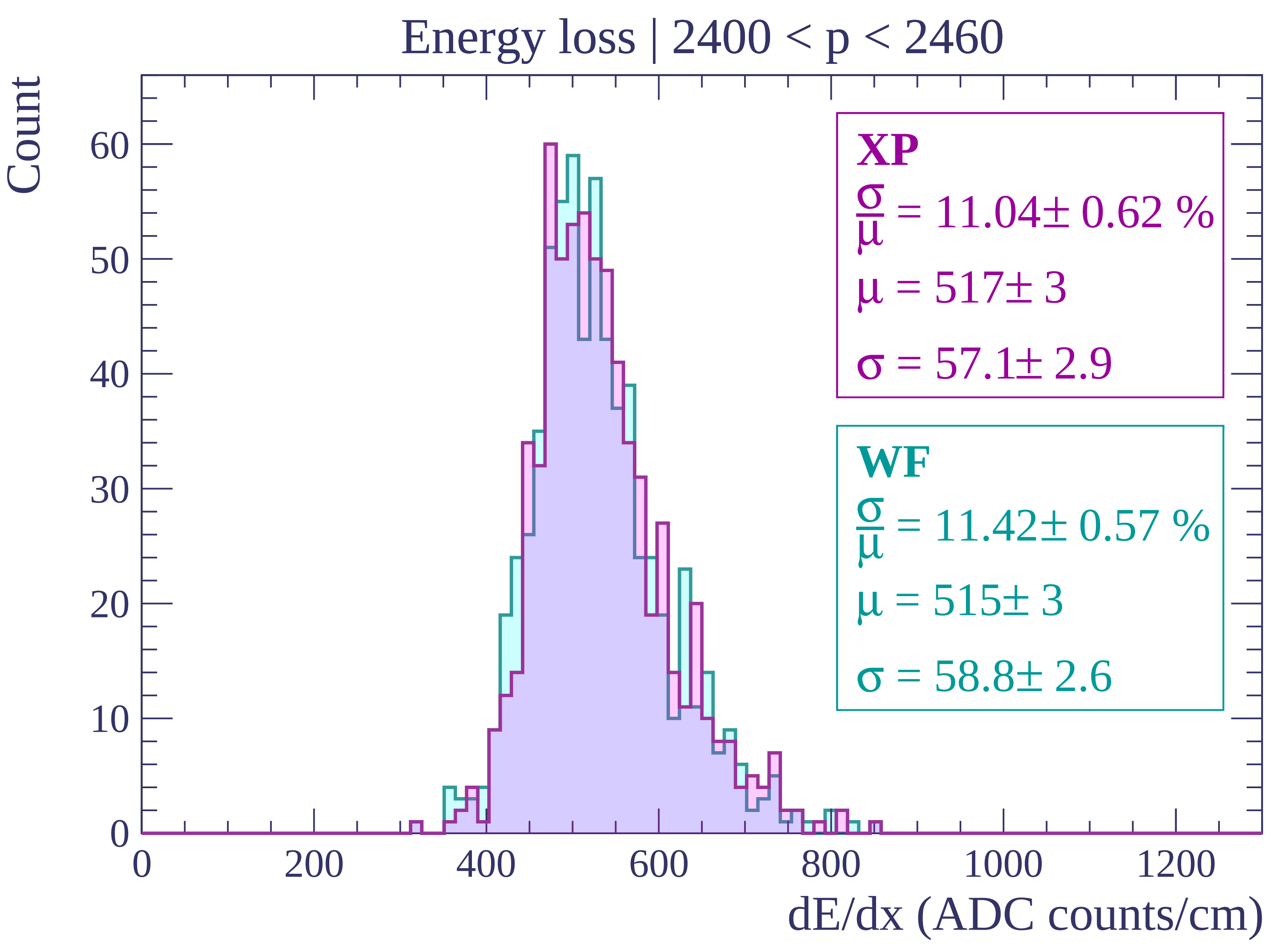

Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP
σ/μ = 12.29± 0.70 %
μ = 520± 3
σ = 63.9± 3.2
WF
σ/μ = 11.89± 0.58 %
μ = 517± 3
σ = 61.4± 2.7



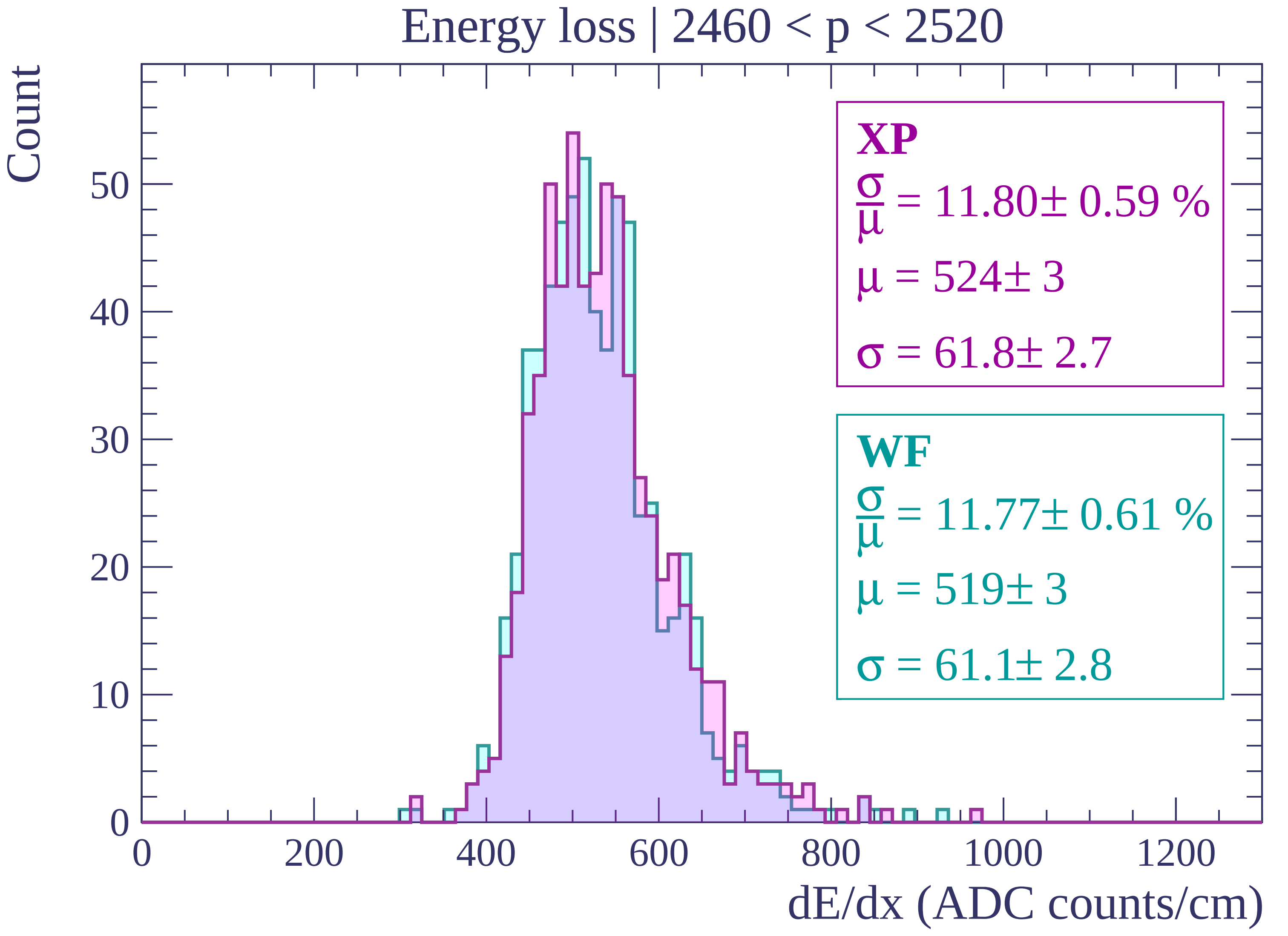
Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.95± 0.53 %
μ = 516± 3
σ = 56.5± 2.4
WF
σ/μ = 12.23± 0.63 %
μ = 514± 3
σ = 62.9± 2.9
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70

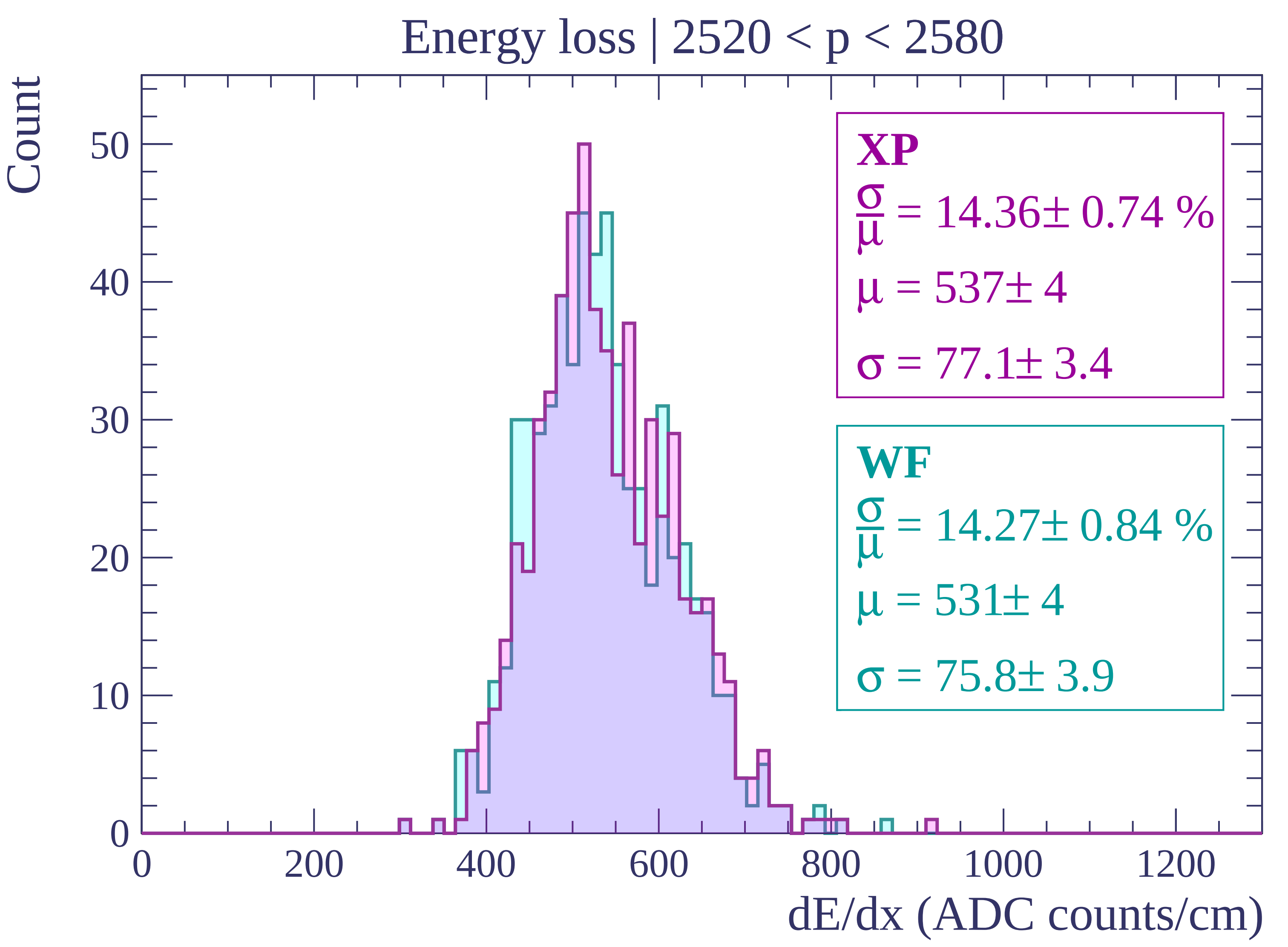

Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP
σ/μ = 12.57± 0.70 %
μ = 517± 3
σ = 64.9± 3.2
WF
σ/μ = 11.91± 0.65 %
μ = 511± 3
σ = 60.8± 3.0


Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.80± 0.63 %
μ = 525± 3
σ = 67.2± 2.9
WF
$\frac{\sigma}{\mu}$ = 12.76± 0.61 %
μ = 519± 3
σ = 66.2± 2.8

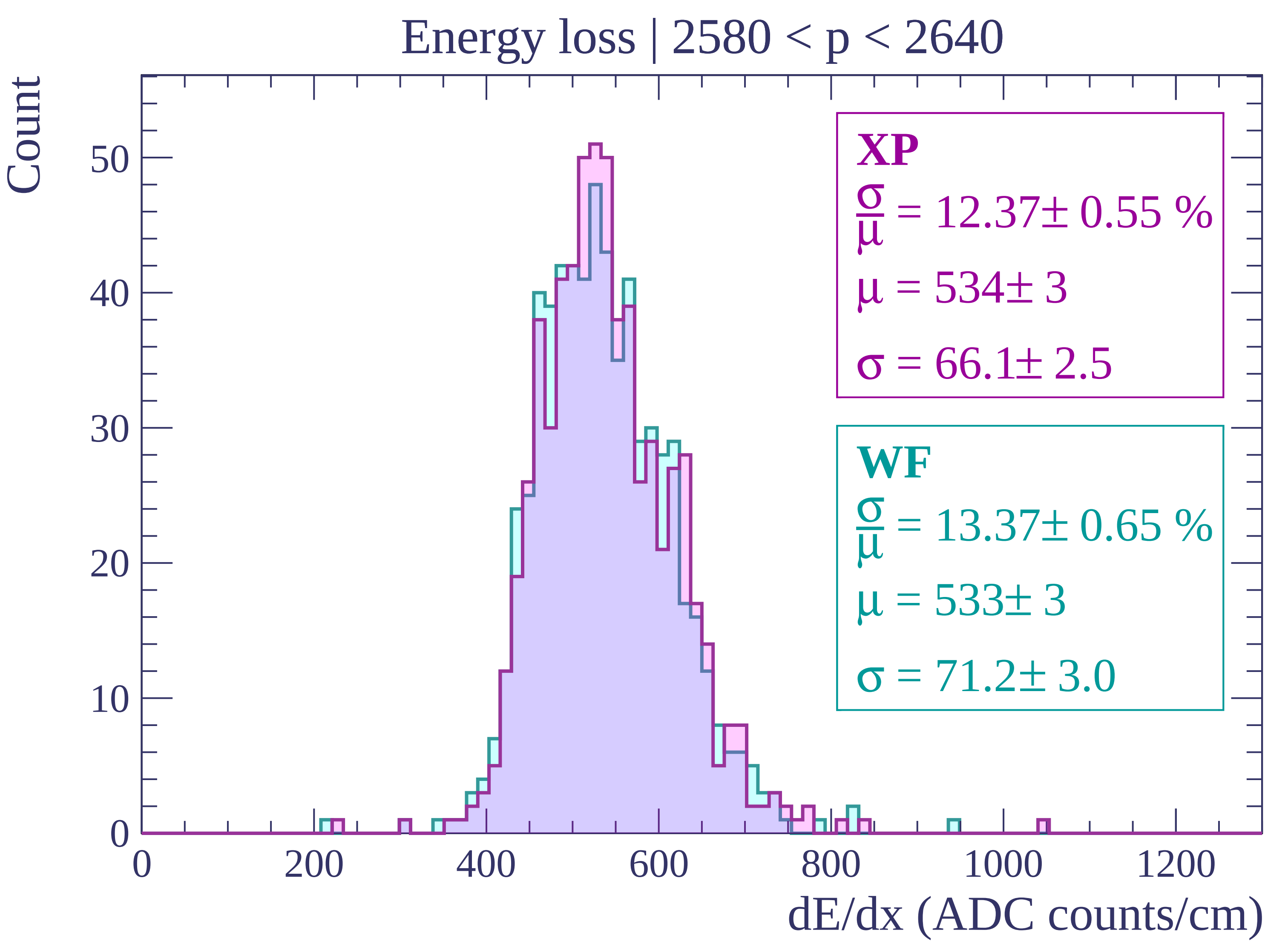


Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
$\frac{\sigma}{\mu}$ = 11.04± 0.62 %
μ = 517± 3
σ = 57.1± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.42± 0.57 %
μ = 515± 3
σ = 58.8± 2.6

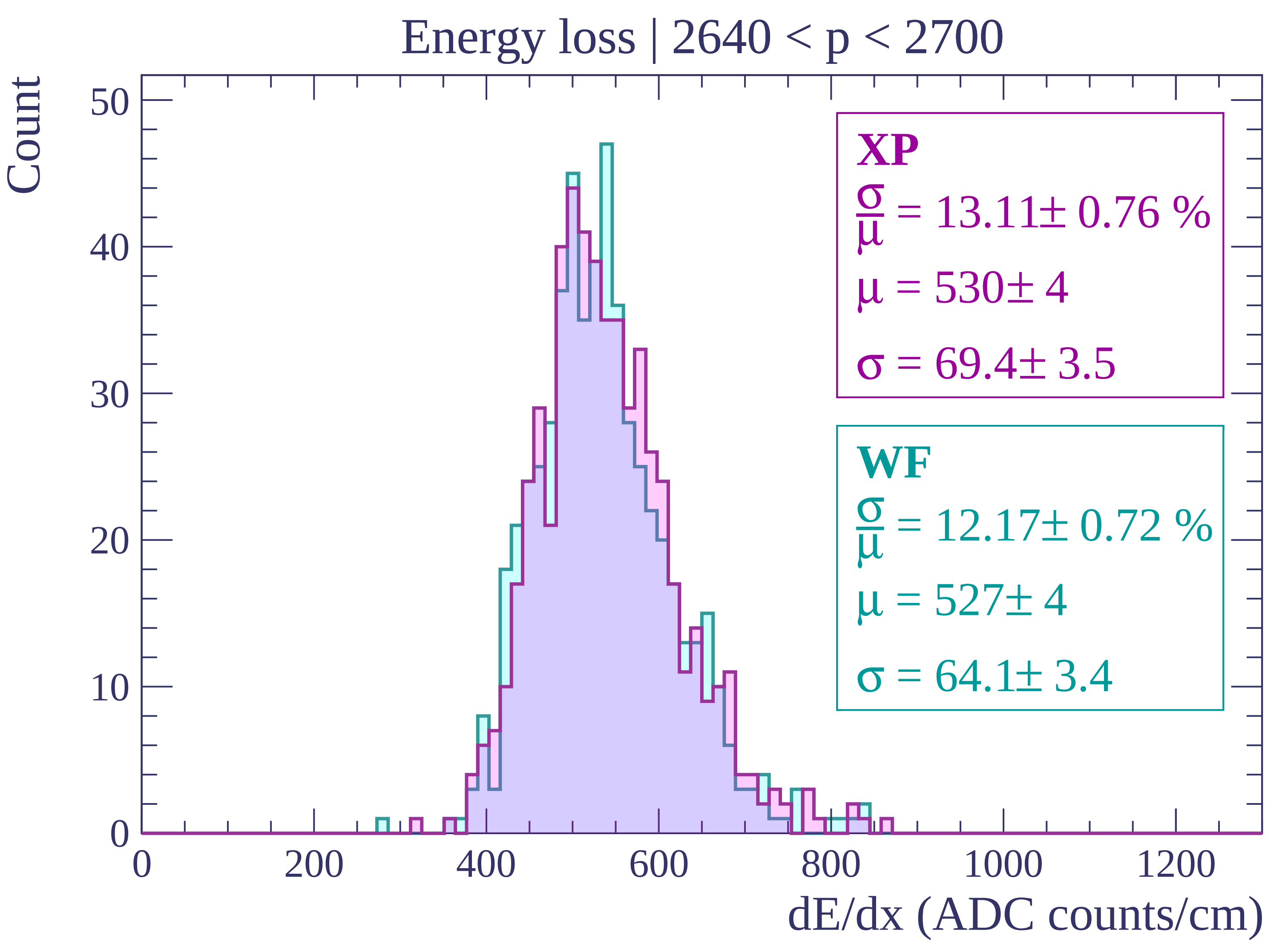

Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 11.80± 0.59 %
μ = 524± 3
σ = 61.8± 2.7
WF
σ/μ = 11.77± 0.61 %
μ = 519± 3
σ = 61.1± 2.8



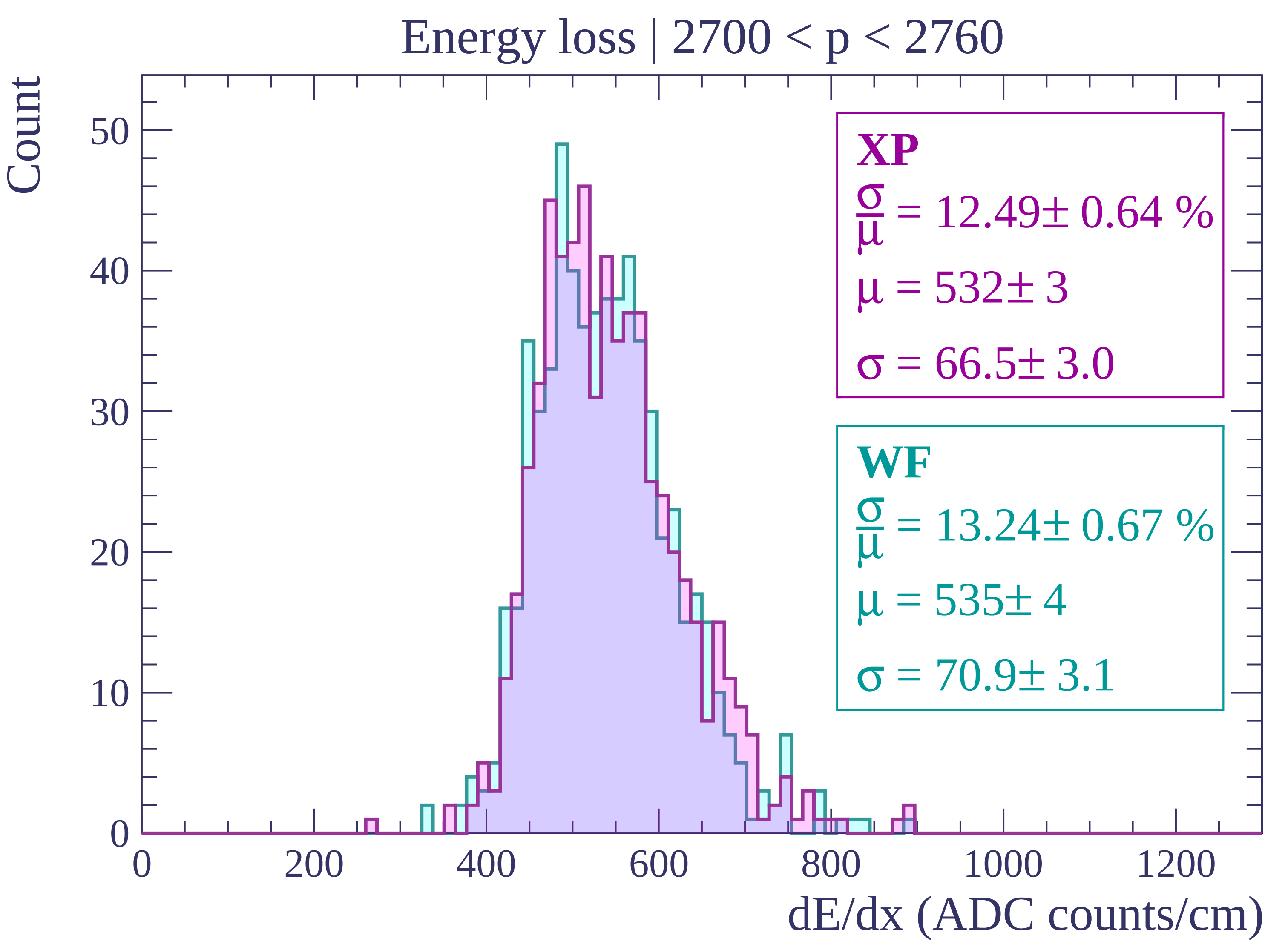
Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 14.36± 0.74 %
μ = 537± 4
σ = 77.1± 3.4
WF
σ/μ = 14.27± 0.84 %
μ = 531± 4
σ = 75.8± 3.9



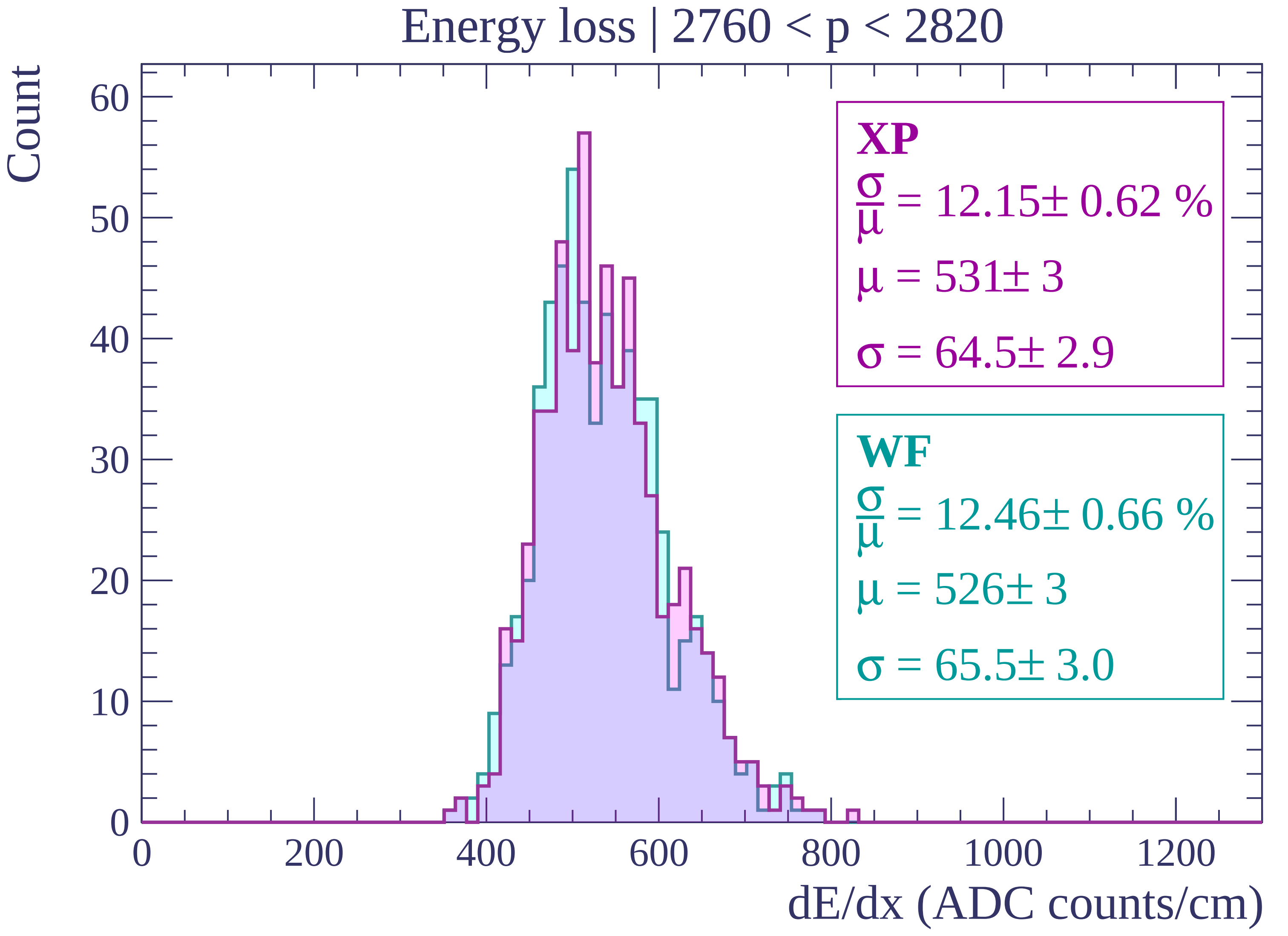
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 12.37± 0.55 %
μ = 534± 3
σ = 66.1± 2.5
WF
σ/μ = 13.37± 0.65 %
μ = 533± 3
σ = 71.2± 3.0



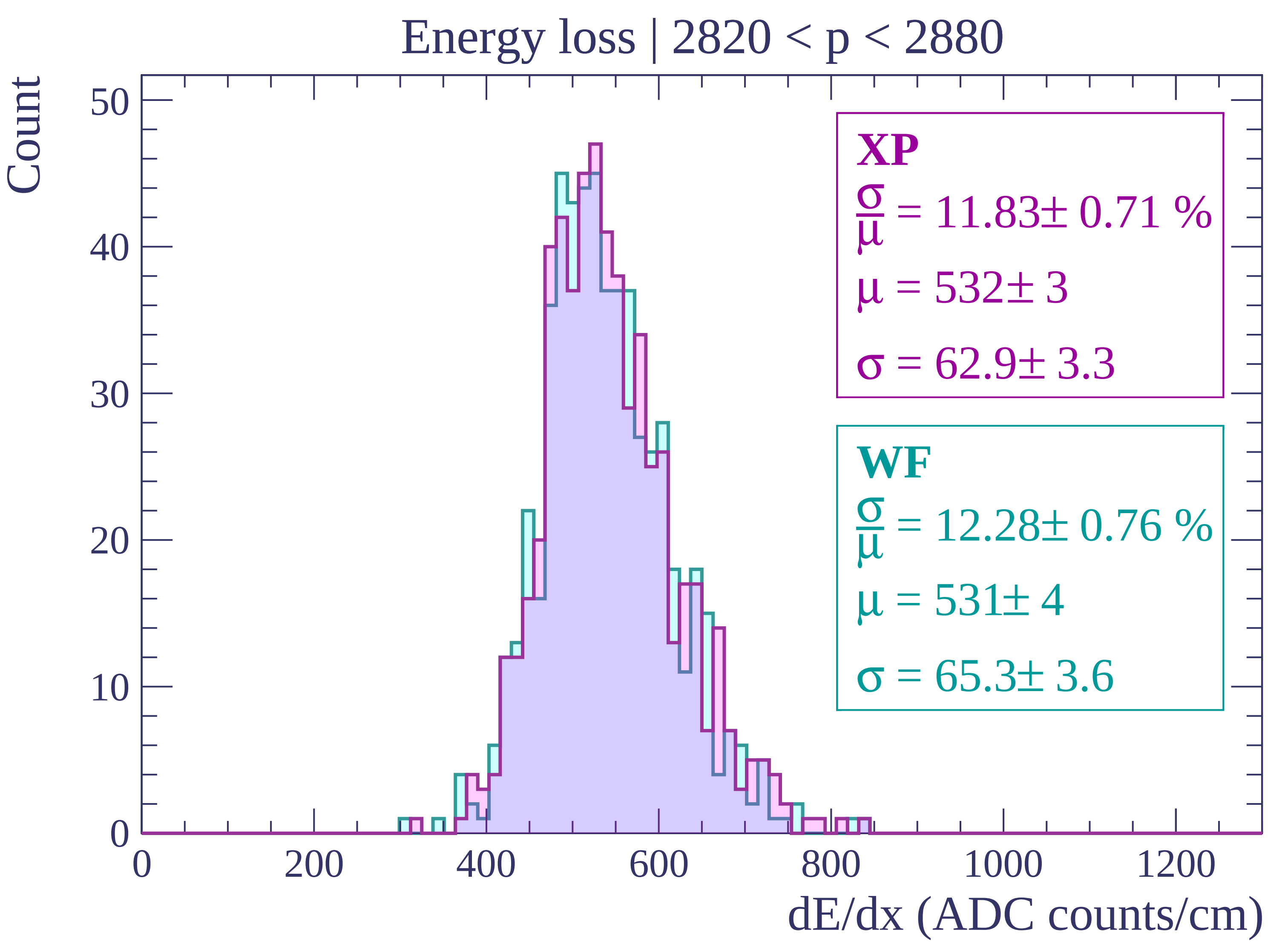
Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 13.11± 0.76 %
μ = 530± 4
σ = 69.4± 3.5
WF
σ/μ = 12.17± 0.72 %
μ = 527± 4
σ = 64.1± 3.4



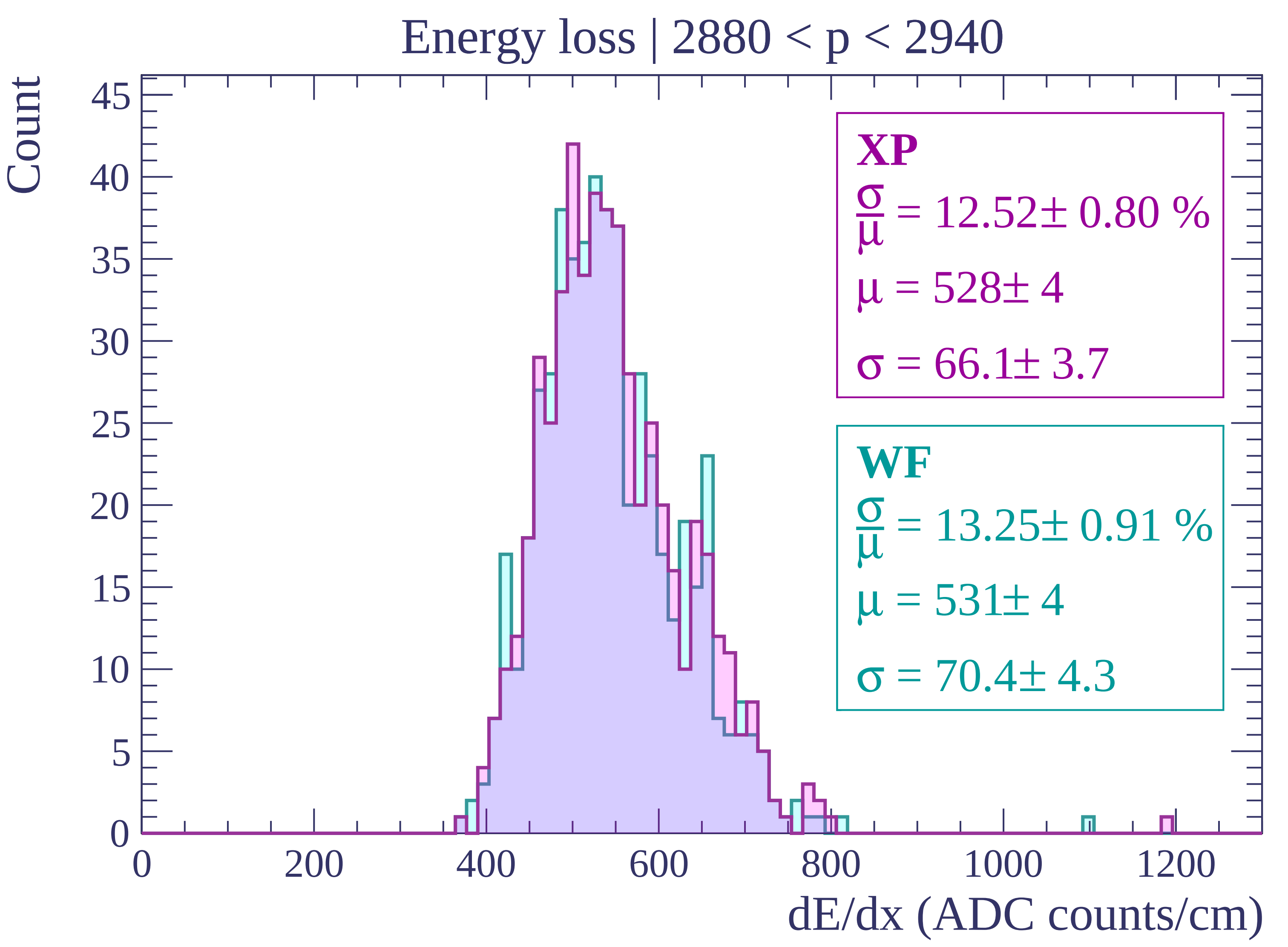
Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 12.49± 0.64 %
μ = 532± 3
σ = 66.5± 3.0
WF
σ/μ = 13.24± 0.67 %
μ = 535± 4
σ = 70.9± 3.1



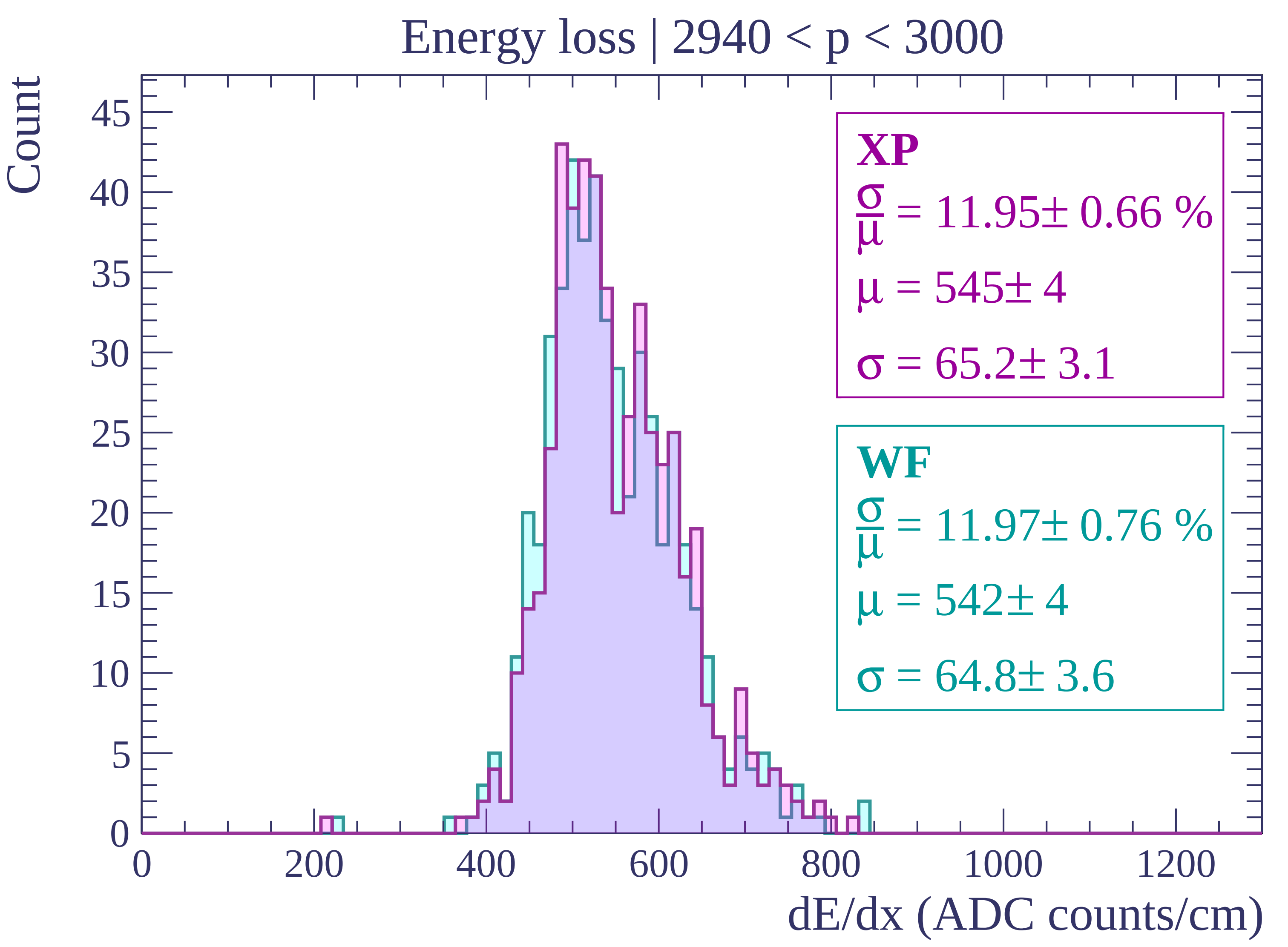
Energy loss | 2760 < p < 2820
Count
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.15± 0.62 %
μ = 531± 3
σ = 64.5± 2.9
WF
σ/μ = 12.46± 0.66 %
μ = 526± 3
σ = 65.5± 3.0



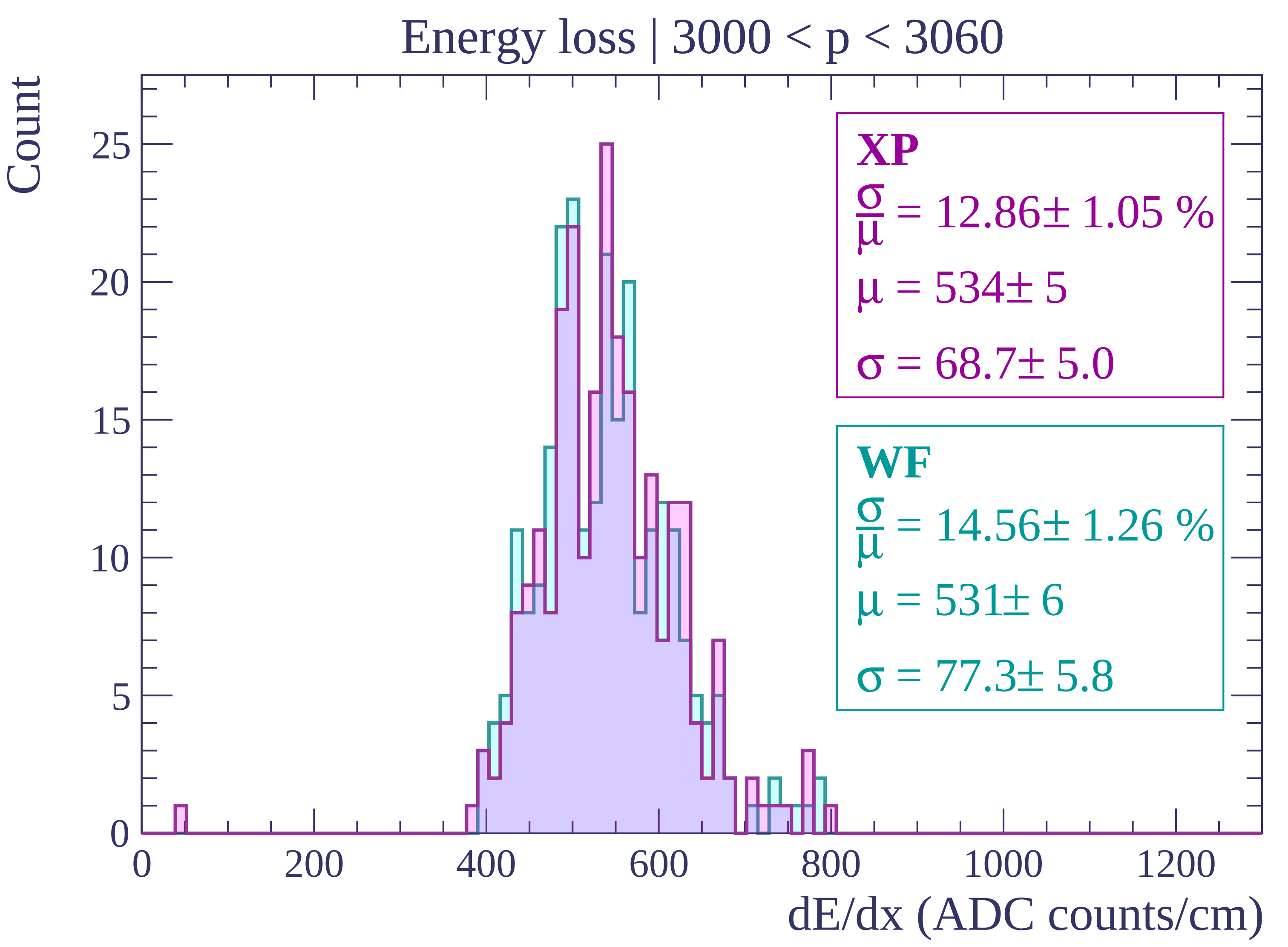
Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP
σ/μ = 11.83± 0.71 %
μ = 532± 3
σ = 62.9± 3.3
WF
σ/μ = 12.28± 0.76 %
μ = 531± 4
σ = 65.3± 3.6



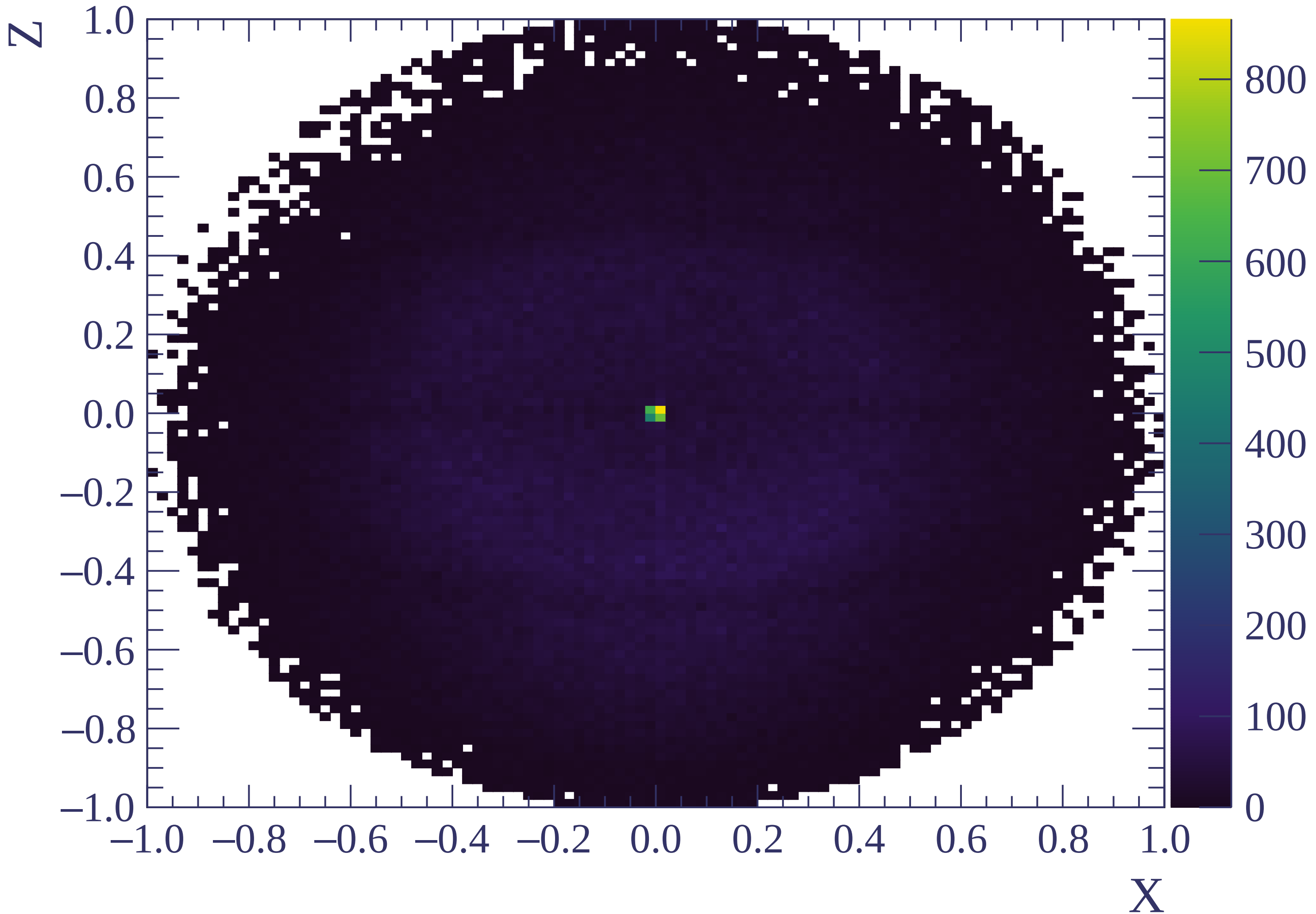
Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45
XP
σ/μ = 12.52± 0.80 %
μ = 528± 4
σ = 66.1± 3.7
WF
σ/μ = 13.25± 0.91 %
μ = 531± 4
σ = 70.4± 4.3



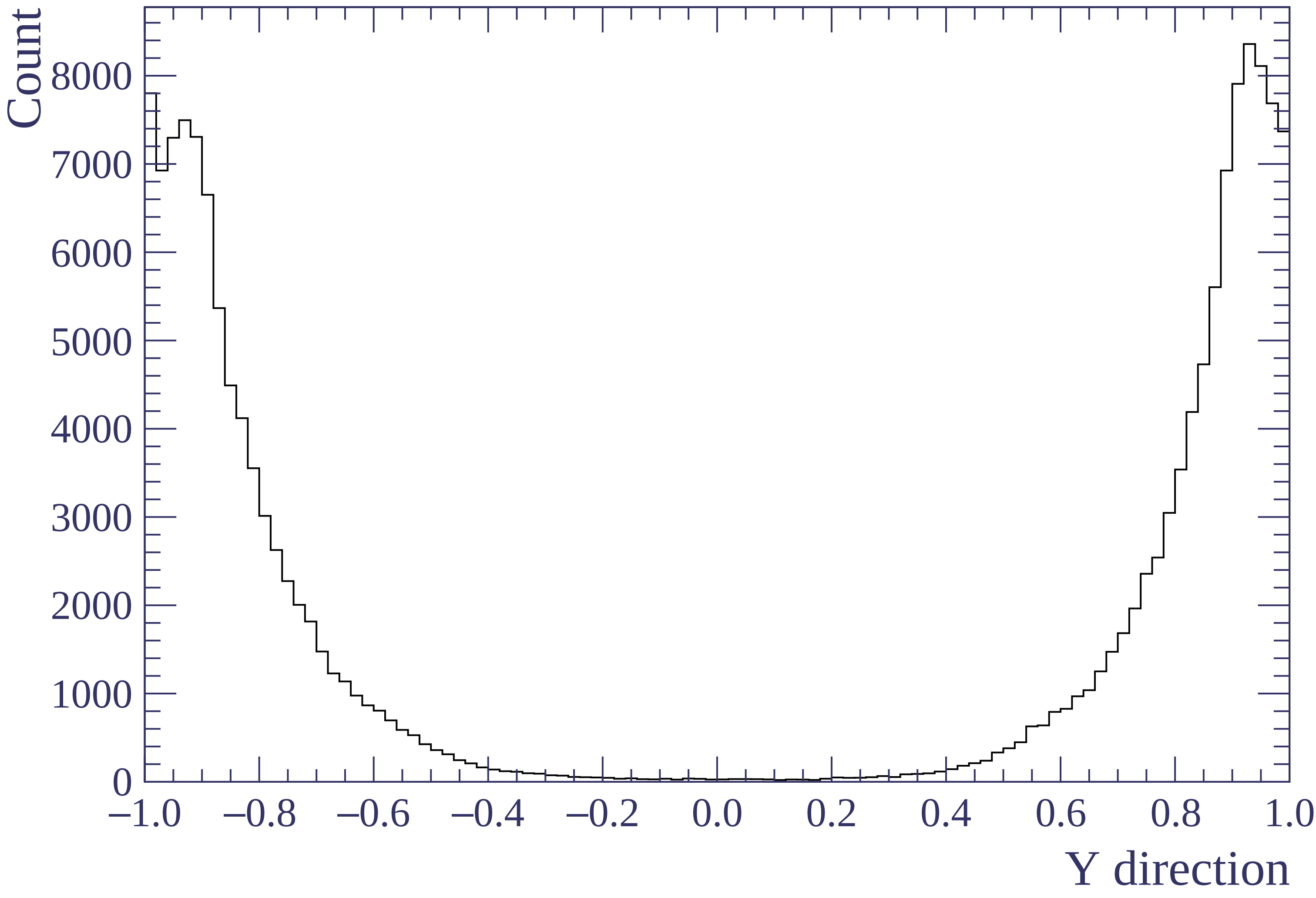
Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.95± 0.66 %
μ = 545± 4
σ = 65.2± 3.1
WF
$\frac{\sigma}{\mu}$ = 11.97± 0.76 %
μ = 542± 4
σ = 64.8± 3.6



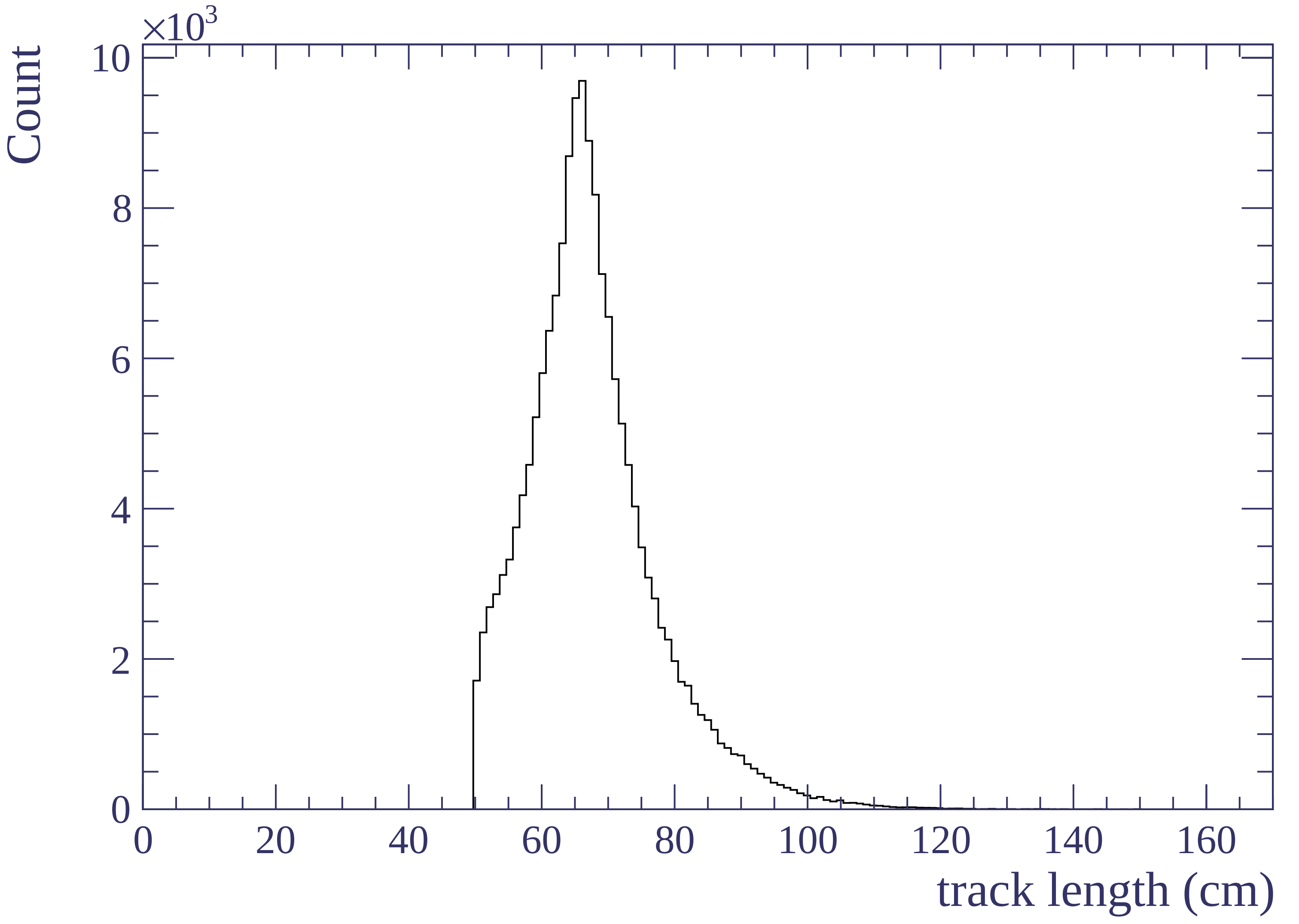
Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.86± 1.05 %
μ = 534± 5
σ = 68.7± 5.0
WF
$\frac{\sigma}{\mu}$ = 14.56± 1.26 %
μ = 531± 6
σ = 77.3± 5.8



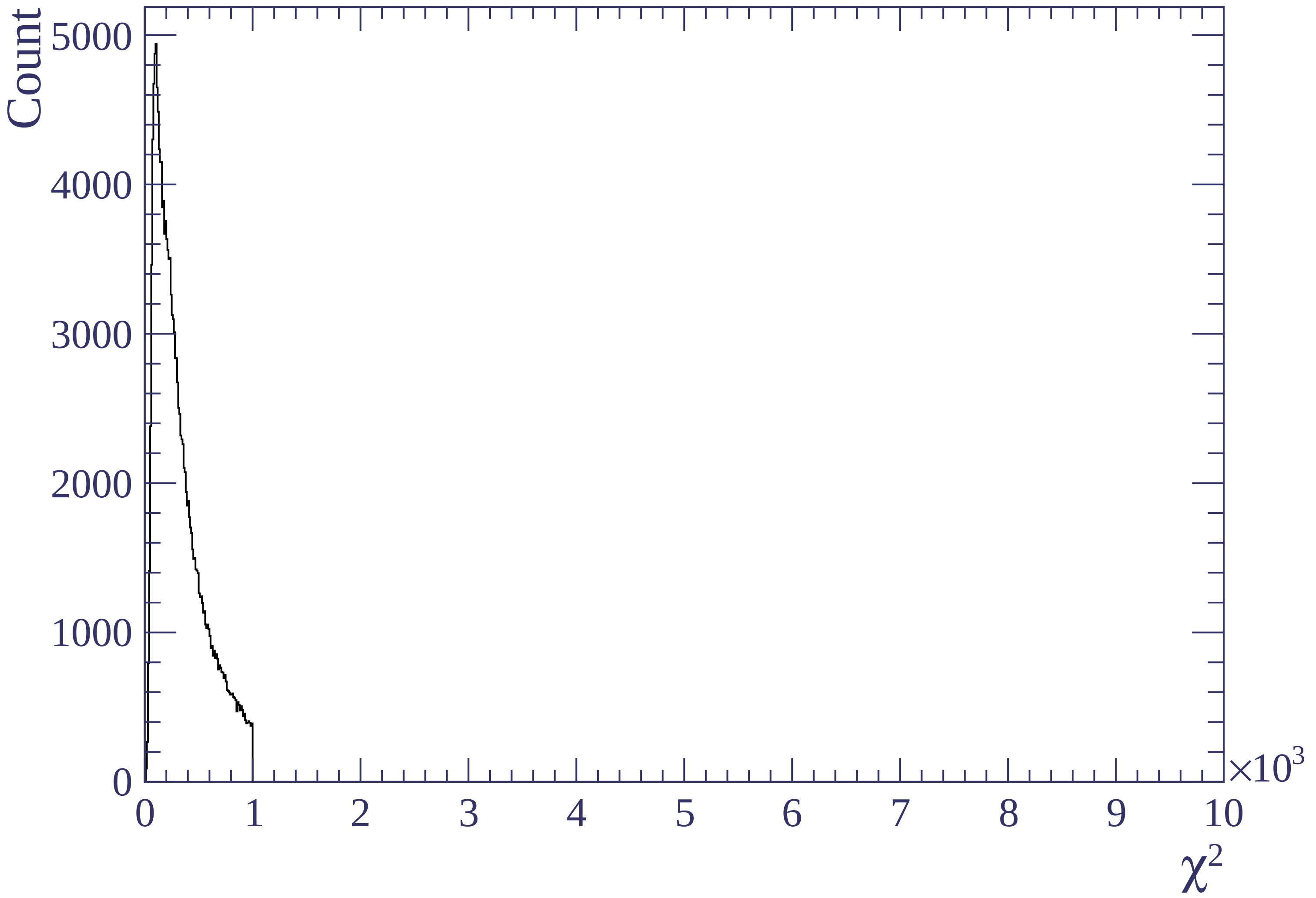
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
800
700
600
500
400
300
200
100
0

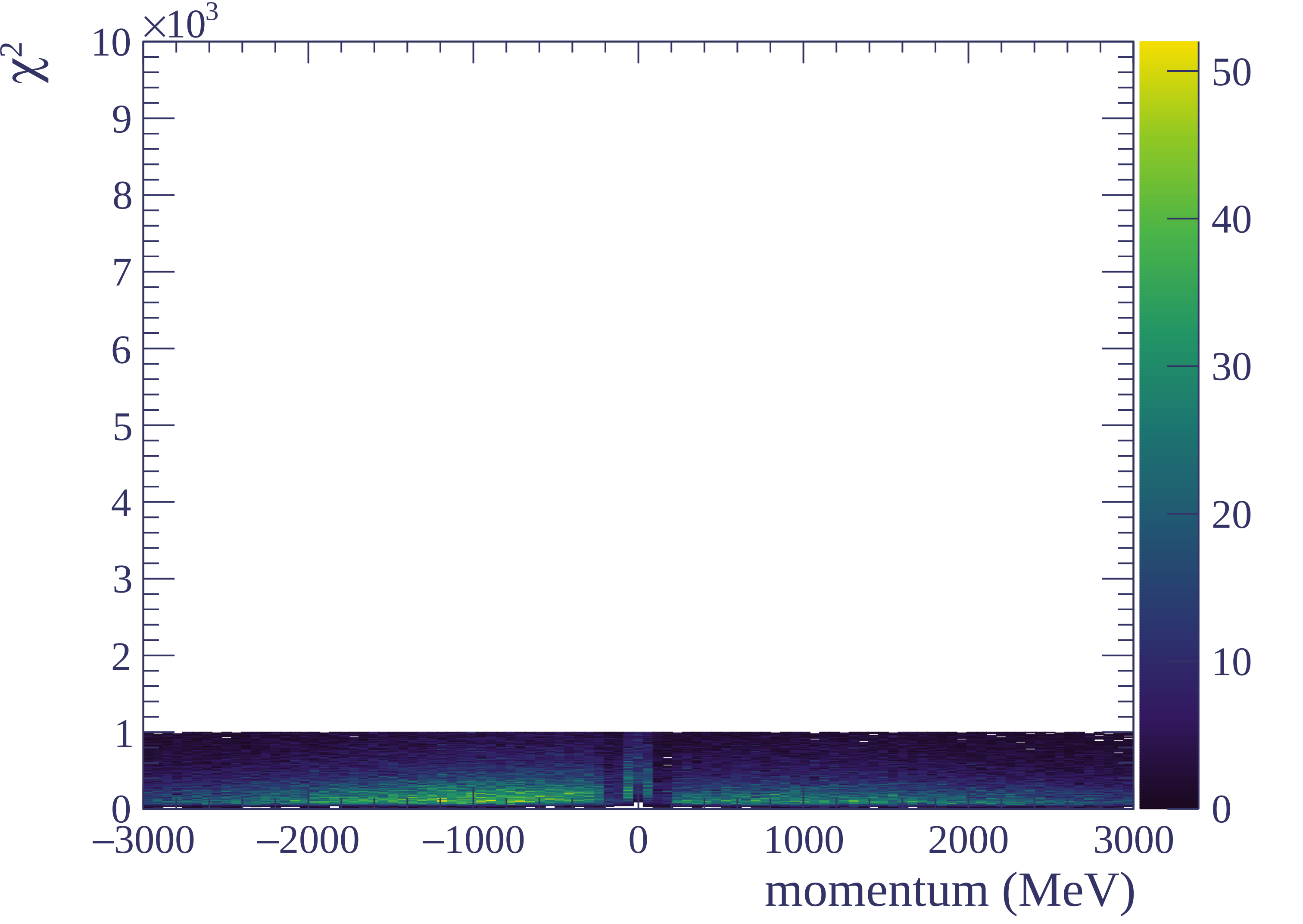


Count
0
1000
2000
3000
4000
5000
6000
7000
8000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction

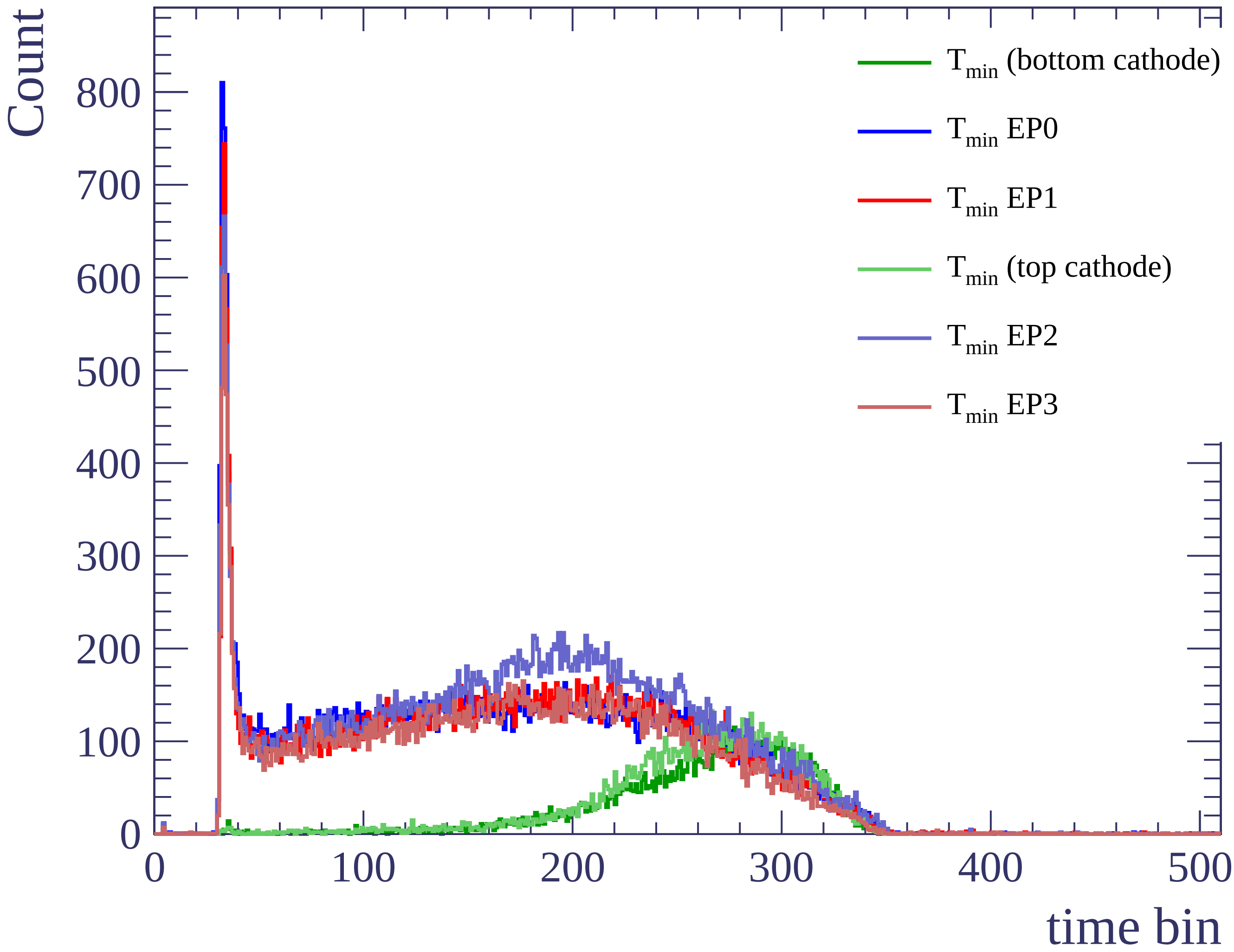


$\times10^3$
Count
10
8
6
4
2
0
0
20
40
60
80
100
120
140
160
track length (cm)

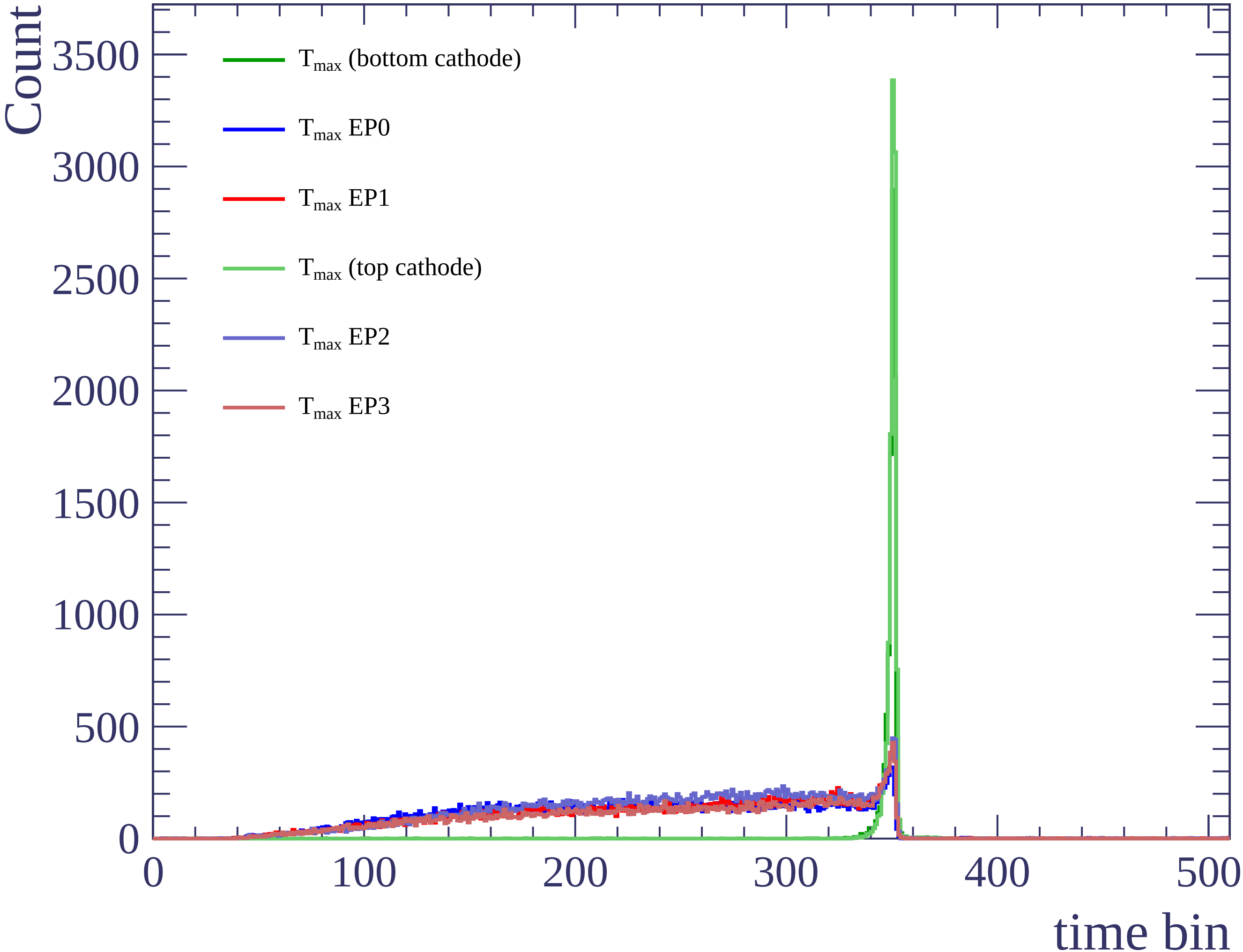

Count
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
10
$\times 10^3$
χ^2



$\times 10^3$
χ^2
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
50
40
30
20
10
0



Count
0
100
200
300
400
500
600
700
800
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
3500
3000
2500
2000
1500
1000
500
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3